Notes for Calculus II

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Colophon
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Cover image
Cushion Pine at Aoyama (Aoyama enza no matsu), from the series *Thirty-six Views of Mount Fuji (Fugaku sanjūrokkei)* by Katsushika Hokusai, ca. 1830–32.

Contents

Introduction

This book contains the notes to my Calculus II course for engineers at UC3M. They are basically summaries from Marsden and Tromba's phenomenal *Vector Calculus*. The sections in these notes indicate where in Marsden and Tromba's book the text was taken from. So, for instance, Section 4.2 says *Chain Rule (MT 2.5)*, and so this is taken from Section 2.5 of *Vector Calculus*.

The exercises are mostly taken from a problem sheet written by my colleagues Arturo de Pablo, Domingo Pestana, José Manuel Rodríguez and Elena Romera, and I want to express my gratitude to them for putting these sheets together and sharing them with the rest of the department. Yuri Martínez Ratón also sent me some problems on limits, and I would like to thank him too.

I am also grateful to several students who have pointed out mistakes in previous versions. Their input has greatly improved the manuscript.

Any mistakes are, obviously, my own. If you see something amiss, please let me know, at pcatalan [at] math [dot] uc3m [dot] com.

Pablo Catalán

Madrid, February 2025.

Review: $\mathbb{R}^n$, vectors and matrices (MT 1.5)

Definition of $\mathbb{R}^n$ and operations with vectors

We can define $\mathbb{R}^n$, where n is a positive integer, to be the set of all ordered n-tuples $(x_1, x_2, \ldots, x_n)$, where the x_i are real numbers. For instance, $(1, \sqrt{5}, 2, \sqrt[3]{3}) \in \mathbb{R}^4$.

The set $\mathbb{R}^n$ so defined is known as **euclidean n-space**, and its elements, which we write as $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, are known as vectors. By setting $n = 1, 2, 3$ we recover the line, the plane, and three-dimensional space, respectively.

We can add vectors

$$(x_1, x_2, \ldots, x_n) + (y_1, y_2, \ldots, y_n) = (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n),$$

and multiply them by an scalar $\alpha \in \mathbb{R}$:

$$\alpha(x_1, x_2, \ldots, x_n) = (\alpha x_1, \alpha x_2, \ldots, \alpha x_n).$$

Any vector $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ can be written in terms of the standard basis as

$$\mathbf{x} = \sum_{i=1}^{n} x_i \mathbf{e}_i,$$

where $\mathbf{e}_i$ is the vector that has zeroes in all components except in the i-th position.

The inner product of two vectors is defined as follows:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} x_i y_i,$$

and the length (or norm) of vector $\mathbf{x}$ is given by

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle} = \sqrt{\sum_{i=1}^{n} x_i^2}.$$

In Linear Algebra you saw some basic properties of the inner product:

Theorem 0.0.1 (Theorem 3, MT) *For* $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ *and* $\alpha, \beta \in \mathbb{R}$, *we have*

(i) $\langle (\alpha\mathbf{x} + \beta\mathbf{y}), \mathbf{z} \rangle = \alpha\langle \mathbf{x}, \mathbf{z} \rangle + \beta\langle \mathbf{x}, \mathbf{y} \rangle$
(ii) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$
(iii) $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$
(iv) $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ *iff* $\mathbf{x} = \mathbf{0}$

Proof. The proof is straightforward through the definitions. □

Matrices and linear transformations

We will also make extensive use of $m \times n$ matrices, which are arrays of mn numbers:

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

We define addition and multiplication by a scalar componentwise, just as we did for vectors[1] . Given two $m \times n$ matrices A and B, we can add them to obtain a new $m \times n$ matrix $C = A + B$, whose ij-th entry is $c_{ij} = a_{ij} + b_{ij}$. It is clear that $A + B = B + A$.

1: Because, as you will remember from Linear Algebra, matrices *are* vectors in $\mathbb{R}^{m\times n}$.

Given a scalar λ and an $m \times n$ matrix A, we can multiply A by λ to obtain $C = \lambda A$, where $c_{ij} = \lambda a_{ij}$.

Finally, we define matrix multiplication as follows. If $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times p}$, the matrix $C = AB$ is an $m \times p$ matrix which has entries given by

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}.$$

Any $m \times n$ matrix A determines a mapping of $\mathbb{R}^n$ to $\mathbb{R}^m$, resulting from multiplying the vector $\mathbf{x}$ by A. That is, we consider the vector $\mathbf{x}$ as an $m \times 1$ column matrix. The vector $A\mathbf{x}$ belongs to $\mathbb{R}^m$. This mapping is linear, that is

$$A(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha A\mathbf{x} + \beta A\mathbf{y}.$$

Remember that matrix multiplication is not commutative.

We say that an $n \times n$ matrix A is invertible if there is another (unique!) matrix, which we denote A^{-1} such that

$$AA^{-1} = A^{-1}A = I_n,$$

where I_n is the identity matrix that has ones in the diagonal and zeroes everywhere else.

One way to check if a matrix is invertible is to see if its determinant is zero.

Example 0.0.1 Calculate the determinant of the matrix

$$A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 \\ 1 & 1 & 0 & 2 \end{pmatrix}$$

The result is $\det A = -2$ and so this matrix is invertible.

Remember that invertibility is equivalent to saying the matrix has no zero eigenvalues, and that the linear transformation defined by it is one-to-one or injective.

Part I. Differential Calculus in Several Variables

Topology of $\mathbb{R}^n$. Functions of several variables 1

1.1 Neighborhoods. Closed and open sets (MT 2.2)

Let $\mathbf{x}_0 \in \mathbb{R}^n$ and let r be a positive real number. The **open disk** (or open ball) of radius r and center $\mathbf{x}_0$ is defined to be the set of all points $\mathbf{x}$ such that $\|\mathbf{x} - \mathbf{x}_0\| < r$. This set is denoted $D_r(\mathbf{x}_0)$. Notice that we include only those $\mathbf{x}$ for which strict inequality holds.

For the case $n = 1$ and $x_0 \in \mathbb{R}$, the open disk $D_r(x_0)$ is the open interval $(x_0 - r, x_0 + r)$. For the case $n = 2$, $D_r(\mathbf{x}_0)$ is the inside of the disk of radius r centered at $\mathbf{x}_0$. For the case $n = 3$, $D_r(\mathbf{x}_0)$ is the part strictly "inside" of the ball of radius r centered at $\mathbf{x}_0$.

Definition 1.1.1 (Open Sets) *Let $U \subset \mathbb{R}^n$. We call U an **open set** when for every point $\mathbf{x}_0 \in U$ there exists some $r > 0$ such that $D_r(\mathbf{x}_0)$ is contained within U.*

The number $r > 0$ can depend on the point $\mathbf{x}_0$, and generally r will shrink as $\mathbf{x}_0$ gets closer to the "edge" of U. Intuitively speaking, a set U is open when the "boundary" points of U do not lie in U.

Theorem 1.1.1 (Theorem 1, p.89 MT) *For each $\mathbf{x}_0 \in \mathbb{R}^n$ $r > 0$, $D_r(\mathbf{x}_0)$ is an open set.*

Proof. Proof can be found in MT, p. 89. □

Example 1.1.1 Prove that $A = \{(x, y) \in \mathbb{R}^2 | x > 0\}$ is an open set.

Intuitively, this set is open, because no points on the "boundary," $x = 0$, are contained in the set. Such an argument will often suffice after one becomes accustomed to the concept of openness. At first, however, we should give details. To prove that A is open, we show that for every point $(x, y) \in A$ there exists an $r > 0$ such that $D_r(x, y) \subset A$. If $(x, y) \in A$, then $x > 0$. Choose $r = x$. If $(x_1, y_1) \in D_r(x, y)$, we have

$$\|x_1 - x\| = (x_1 - x)^2 \leq (x_1 - x)^2 + (y_1 - y)^2 < r = x,$$

and so $x_1 - x < x$ and $x - x_1 < x$. The latter inequality implies $x_1 > 0$, that is, $(x_1, y_1) \in A$. Hence $D_r(x, y) \subset A$, and therefore A is open.

We call an open set containing the point $\mathbf{x} \in \mathbb{R}^n$ a **neighborhood** of $\mathbf{x}$.

Definition 1.1.2 (Boundary Points) *Let $A \subset \mathbb{R}^n$. A point $\mathbf{x} \in \mathbb{R}^n$ is called a **boundary point** of A if every neighborhood of $\mathbf{x}$ contains at least one point in A and at least one point not in A.*

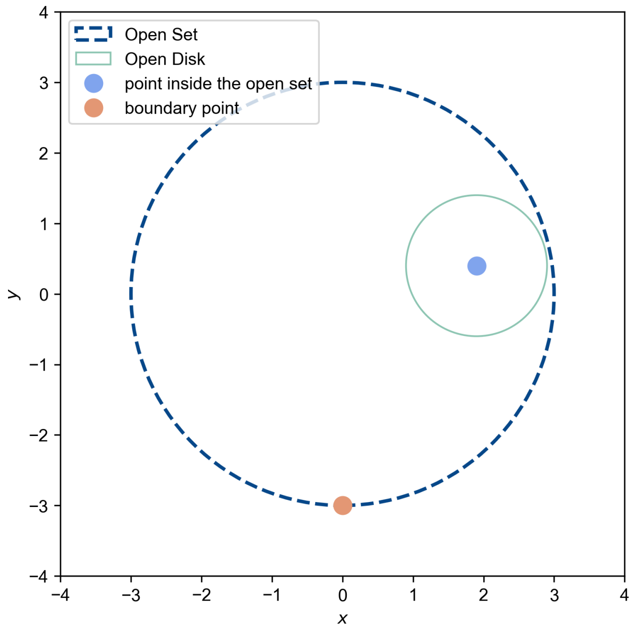


Figure 1.1: Points belonging to an open set always have a neighborhood inside the set, while boundary points don't.

Thus, a point $\mathbf{x}$ is a boundary point of an open set A if and only if $\mathbf{x}$ is not in A and every neighborhood of $\mathbf{x}$ has a nonempty intersection with A. Figure 1.1 illustrates these concepts.

Definition 1.1.3 (Closed Set) *A **closed set** is a set that includes all of its boundary points.*

Example 1.1.2 (a) Let $A = (a, b) \subset \mathbb{R}$. Then the boundary points of A consist of the points a and b.

(b) Let $A = D_r(x_0, y_0)$ be an r-disk about (x_0, y_0) in the plane. The boundary consists of points (x, y) with $\|(x - x_0)^2 + (y - y_0)^2\| = r^2$

(c) Let $A = \{(x, y) \in \mathbb{R}^2 | x > 0\}$. Then the boundary of A consists of all points on the y-axis.

(d) Let A be $D_r(\mathbf{x}_0)$ minus the point $\mathbf{x}_0$ (a *punctured* disk about $\mathbf{x}_0$). Then $\mathbf{x}_0$ is a boundary point of A.

1.2 The geometry of real-valued functions (MT 2.1)

Functions and Mappings

Let f be a function whose domain is a subset A of $\mathbb{R}^n$ and with a range contained in $\mathbb{R}^m$. By this we mean that to each $\mathbf{x} = (x_1, \ldots, x_n) \in A$, f assigns a value $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))$, an m-tuple in $\mathbb{R}^m$. Such functions f are called vector-valued functions if $m > 1$, and scalar-valued functions if $m = 1$.

Example 1.2.1 For example, the scalar-valued function $f(x, y, z) = (x^2+y^2+z^2)^{-3/2}$ maps the set $A = \{(x, y, z) \in \mathbb{R}^3 : (x, y, z) \neq (0, 0, 0)\}$ in $\mathbb{R}^3$ ($n = 3$, in this case) to $\mathbb{R}$ ($m = 1$). To denote f we sometimes write $f : (x, y, z) \longrightarrow (x^2 + y^2 + z^2)^{-3/2}$.

Note that in $\mathbb{R}^3$ we often use the notation (x, y, z) instead of (x_1, x_2, x_3). In general, the notation $\mathbf{x} \longrightarrow f(\mathbf{x})$ is useful for indicating the value to which a point $\mathbf{x} \in \mathbb{R}^n$ is sent.

We write $f : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ to signify that A is the domain of f (a subset of $\mathbb{R}^n$) and the range is contained in $\mathbb{R}^m$. We also use the expression "f maps A into $\mathbb{R}^m$". Such functions f are called functions of several variables if $A \subset \mathbb{R}^n$, $n > 1$.

Example 1.2.2 As another example we can take the vector-valued function $g : \mathbb{R}^6 \longrightarrow \mathbb{R}^2$ defined by the rule $g(\mathbf{x}) = g(x_1, x_2, x_3, x_4, x_5, x_6) = (x_1x_2x_3x_4x_5x_6, x_1^2 + x_6^2)$.

These functions arise naturally in problems studied in all the sciences. For example, to specify the temperature T in a region A of space requires a function $T : A \subset \mathbb{R}^3 \longrightarrow \mathbb{R}(n = 3, m = 1)$. Thus, $T(x, y, z)$ is the temperature at the point (x, y, z). To specify the velocity of a fluid moving in space requires a map $V : \mathbb{R}^4 \longrightarrow \mathbb{R}^3$, where $V(x, y, z, t)$ is the velocity vector of the fluid at the point (x, y, z) in space at time t. To specify the reaction rate of a solution consisting of six reacting chemicals A, B, C, D, E, F in proportions x, y, z, w, u, v requires a map $\sigma : U \subset \mathbb{R}^6 \longrightarrow \mathbb{R}$, where $\sigma(x, y, z, w, u, v)$ gives the rate when the chemicals are in the indicated proportions.

When $f : U \subset \mathbb{R}^n \longrightarrow \mathbb{R}$, we say that f is a real-valued function of n variables with domain U. The reason we say "n variables" is simply that we regard the coordinates of a point $\mathbb{x} = (x_1, \ldots, x_n) \in U$ as n variables, and $f(\mathbf{x}) = f(x_1, \ldots, x_n)$ depends on these variables. We say "real-valued" because $f(x_1, \ldots, x_n)$ is a real number. In this course we will focus on real-valued functions [1] .

1: Depending on your degree, you will have a subject called *Advanced Mathematics* in which you will study functions of complex variable. That's super fun!

Conversely, a function $c : [a, b] \longrightarrow \mathbb{R}^n$ is called a **path** in $\mathbb{R}^n$. The collection C of points $c(t)$ as t varies in $[a, b]$ is called a **curve**, and $c(a)$ and $c(b)$ are its endpoints. The path c is said to parametrize the curve C. If c is a path in $\mathbb{R}^3$, we can write $c(t) = (x(t), y(t), z(t))$, and we call $x(t)$, $y(t)$, and $z(t)$ the component functions of c.

Example 1.2.3 The unit circle $C : x^2 + y^2 = 1$ in the plane is the image of the path $c : \mathbb{R} \longrightarrow \mathbb{R}^2$, $c(t) = (\cos t, \sin t), 0 \leq t \leq 2\pi$. The unit circle is also the image of the path $\tilde{c}(t) = (\cos 2t, \sin 2t), 0 \leq t \leq \pi$. Thus, different paths may parametrize the same curve.

Graphs of Functions

For $f : U \subset \mathbb{R} \longrightarrow \mathbb{R}(n = 1)$, the graph of f is the subset of $\mathbb{R}^2$ consisting of all points $(x, f(x))$ in the plane, for $\mathbf{x} \in U$. This subset can be thought of as a curve in $\mathbb{R}^2$. We write this as

$$\operatorname{graph} f = \{(x, f(x)) \in \mathbb{R}^2 | x \in U\}.$$

We can generalize this definition to several variables:

Definition 1.2.1 *(Graph of a Function) Let $f : U \subset \mathbb{R}^n \longrightarrow \mathbb{R}$. The graph of f is the subset of $\mathbb{R}^{n+1}$ consisting of all the points $(x_1, \ldots, x_n, f(x_1, \ldots, x_n))$ in $\mathbb{R}^{n+1}$ for $(x_1, \ldots, x_n) \in U$.*

For the case $n = 1$, the graph is a curve in $\mathbb{R}^2$, while for $n = 2$, it is a surface in $\mathbb{R}^3$. For $n = 3$, it is difficult to visualize the graph, because, since we are humans living in a three-dimensional world, it is hard for us to envisage sets in $\mathbb{R}^4$. To help overcome this handicap, we introduce the idea of a level set.

Level Sets, Curves and Surfaces

Suppose $f(x, y, z) = x^2 + y^2 + z^2$. A level set is a subset of $\mathbb{R}^3$ on which f is constant; for instance, the set where $x^2 + y^2 + z^2 = 1$ is a level set for f. This we can visualize: it is just a sphere of radius 1 in $\mathbb{R}^3$.

Formally, a **level set** is the set of (x, y, z) such that $f(x, y, z) = c$, where c is a constant. The behavior or structure of a function is determined in part by the shape of its level sets; consequently, understanding these sets aids us in understanding the function in question. Level sets are also useful for understanding functions of two variables $f(x, y)$, in which case we speak of *level curves* or *level contours*.

The idea is similar to that used to prepare contour maps, where one draws lines to represent constant altitudes; walking along such a line would mean walking on a level path. In the case of a hill rising from the xy plane, a graph of all the level curves gives us a good idea of the function $h(x, y)$, which represents the height of the hill at point (x, y).

Definition 1.2.2 (Level Curves and Surfaces) *Let $f : U \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ and let $c \in \mathbb{R}$. Then the **level set** of value c is defined to be the set of those points $\mathbf{x} \in U$ at which $f(\mathbf{x}) = c$. If $n = 2$, we speak of a level curve (of value c); and if $n = 3$, we speak of a level surface. In symbols, the level set of value c is written $\{\mathbf{x} \in U | f(\mathbf{x}) = c\} \subset \mathbb{R}^n$. Note that the level set is always in the domain space.*

Example 1.2.4 Describe the graph of the quadratic function

$$f : \mathbb{R}^2 \longrightarrow \mathbb{R}, (x, y) \longrightarrow x^2 + y^2.$$

The graph is the paraboloid of revolution $z = x^2 + y^2$, oriented upward from the origin, around the z-axis. The level curve of value c is empty for $c < 0$; for $c > 0$ the level curve of value c is the set $\{(x, y) | x^2 + y^2 = c\}$, a circle of radius $\sqrt{c}$ centered at the origin. Thus, raised to height c above the xy plane, the level set is a circle of radius $\sqrt{c}$, indicating a parabolic shape.

By a **section** of the graph of f we mean the intersection of the graph and a (vertical) plane. For example, if P_1 is the xz plane in $\mathbb{R}^3$, defined by $y = 0$, then the section of f in the previous example is

$$P_1 \cap \operatorname{graph} f = \{(x, y, z) | y = 0, z = x^2\},$$

which is a parabola in the xz plane. Similarly, if P_2 denotes the yz plane, defined by $x = 0$, then the section

$$P_2 \cap \operatorname{graph} f = \{(x, y, z) | x = 0, z = y^2\}$$

is a parabola in the yz plane. It is usually helpful to compute at least one section to complement the information given by the level sets.

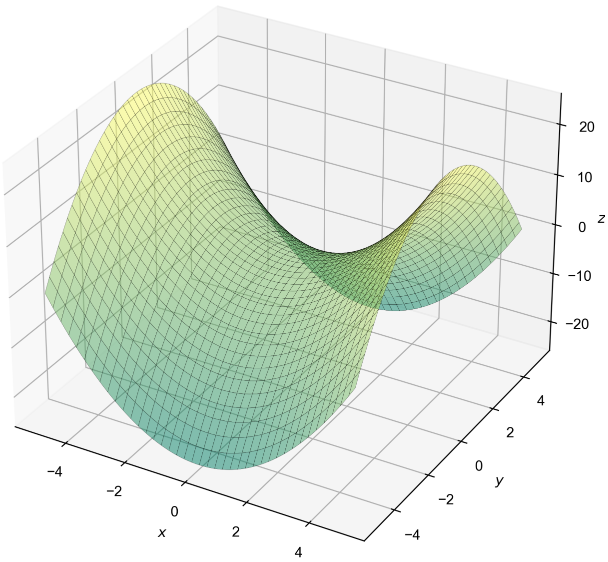


Figure 1.2: The surface plot of $z = x^2 - y^2$.

Example 1.2.5 The graph of the quadratic function

$$f : \mathbb{R}^2 \longrightarrow \mathbb{R}, (x, y) \longrightarrow x^2 - y^2.$$

is called a hyperbolic paraboloid, or saddle, centered at the origin. Let us sketch the graph (plotted in Fig. 1.2).

To visualize this surface, we first draw the level curves. To determine the level curves, we solve the equation $x^2 - y^2 = c$. Consider the values $c = 0, \pm 1, \pm 4$. For $c = 0$, we have $y^2 = x^2$, or $y = \pm x$, so that this level set consists of two straight lines through the origin. For $c = a^2 > 0$, the level curve is $x^2 - y^2 = a^2$, or $y = \pm\sqrt{x^2 - a^2}$, which is a hyperbola that passes vertically through the x-axis at the points $(\pm a, 0)$.

For $c = -a^2 < 0$, we obtain the curve $x^2 - y^2 = -a^2$, that is, $x = \pm\sqrt{y^2 - a^2}$, the hyperbola passing horizontally through the y-axis at $(0, \pm a)$.

We shall compute two sections, as in the previous example. For the section in the xz plane, we have

$$P_1 \cap \operatorname{graph} f = \{(x, y, z) | y = 0, z = x^2\},$$

which is a parabola opening upward; and for the yz plane,

$$P_2 \cap \operatorname{graph} f = \{(x, y, z) | x = 0, z = -y^2\}$$

a parabola opening downwards.

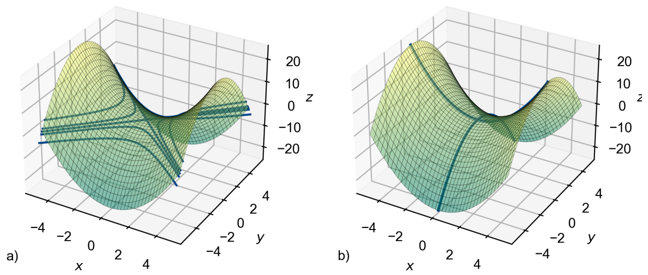


Figure 1.3: For $f(x, y) = x^2 - y^2$. a) Level curves for $c = 0, \pm 1, \pm 4$. b) Sections with xz and yz planes.

Exercises

Exercise 1.1 Decide if the following subsets are open or closed or none of the two possibilities. Describe also in each case the interior and the boundary and sketch the region.

1. $A = \{(x, y) : -1 \leq x \leq 1, y < 0\}$.
2. $B = \{(x, y) : x = 1, 1 < y < 2\}$.
3. $C = \{(x, y) : x \geq 1, 0 \leq y \leq 2x\}$.
4. $D = \{(x, y) : 2x + 3y - 5 = 0\}$.
5. $E = \{(x, y) : x^2 + 3y^2 \leq 2\}$.
6. $F = \{(x, y) : x^2 + 3y^2 < 2\}$.
7. $G = \{(x, y) : x^2 - 3y^2 = 2\}$.
8. $H = \{(x, y) : y > x^2\}$.

Exercise 1.2 Decide if the following sets are open, closed and compact:

1. $A = \{(x, y) \in \mathbb{R}^2 : 5 < x < 7, 0 < y < 1\}$,
2. $B = \{(x, y) \in \mathbb{R}^2 : x > 0\}$,
3. $C = \{(x, y) \in \mathbb{R}^2 : 5 \leq x \leq 7, 0 < y < 1\}$,
4. $D = \{(x, y) \in \mathbb{R}^2 : 5 \leq x \leq 7, 0 \leq y \leq 1\}$,
5. $E = \{x \in \mathbb{R}^n : \|x\| > 3\}$,
6. $F = \{x \in \mathbb{R}^n : \|x\| \geq 3\}$,
7. $G = \{x \in \mathbb{R}^n : \|x\| = 5\}$,
8. $H = \{x \in \mathbb{R}^n : \|x\| < 3\}$.

Exercise 1.3 Determine the interior, the closure and the boundary of the sets in the previous exercise.

Exercise 1.4

1. Find $f(1, y/x)$ if $f(x, y) = \dfrac{2xy}{x^2 + y^2}$.
2. Find $f(x, y)$ if $f(x + y, y/x) = x^2 - y^2$.
3. Find $f(x)$ and $g(x, y)$ if $f(x-y)+g(x, y) = x+y$ with the condition $g(x, 0) = x^2$.
4. Find $f(x)$ and $g(x, y)$ if now $f(\sqrt{x} - 1) + g(x, y) = \sqrt{y}$ with the condition $g(x, 1) = x$.

Exercise 1.5 Find the domain of the following functions:

1. $f(x, y) = x^2 - y^2$,
2. $f(x, y) = \sqrt{x^2 - 4y^2}$,
3. $f(x, y) = \dfrac{x^3 - y^2}{x - y}$,
4. $f(x, y) = \sqrt{x^2 + y^2 - 1}$,
5. $f(x, y) = \dfrac{1}{xy}$,
6. $f(x, y) = \arcsin(x + y)$,
7. $f(x, y) = e^{x/y}$,
8. $f(x, y) = \log(xy)$,
9. $f(x, y) = \dfrac{1}{\cos(x - y)}$,
10. $f(x, y) = \cos\left(\dfrac{1}{x - y}\right)$,
11. $f(x, y, z) = \dfrac{\sqrt{9 - y^2}}{1 + \sqrt{4 - (x^2 + z^2)}}$,

12. $f(x, y) = \dfrac{\log(1 + x) + \log(1 + y)}{\log(1 - x) + \log(1 - y)}$,
13. $f(x, y) = \log\left(\dfrac{(1 + x)(1 + y)}{(1 - x)(1 - y)}\right)$.

Exercise 1.6 Find the image of the first five functions of the preceding exercise.

Exercise 1.7 Visualize the level curves $f(x, y) = c$ of the following functions:

1. $f(x, y) = xy, \quad c = 1, -1, 3;$
2. $f(x, y) = \log(x - y), \quad c = 0, 1, -1;$
3. $f(x, y) = \dfrac{x + y}{x - y}, \quad c = 0, 2, -2.$

Exercise 1.8 Let $f(x, y) = 9x^2 + y^2$. Sketch the following.

1. The level curves for f of values $c = 0, 1, 9$
2. The sections of the graph of f in the planes $x = -1$, $x = 0$, $x = 1$
3. The sections of the graph of f in the planes $y = -1$, $y = 0$, $y = 1$
4. The graph of f

Exercise 1.9 Sketch level sets of values $c = 0, 1, 4, 9$ for both $f(x, y) = x^2 + y^2$ and $g(x, y) = \sqrt{x^2 + y^2}$. How are the graphs of f and g different? How are their sections different?

Exercise 1.10 Let S be the surface in $\mathbb{R}^3$ defined by the equation $x^2y^6-2z = 3$.

1. Find a real-valued function $f(x, y, z)$ of three variables and a constant c such that S is the level set of f of value c.
2. Find a real-valued function $g(x, y)$ of two variables such that S is the graph of g.

Exercise 1.11 Describe the behavior, as c varies, of the level curve $f(x, y) = c$ for each of these functions:

1. $f(x, y) = x^2 + y^2 + 1$
2. $f(x, y) = 1 - x^2 - y^2$
3. $f(x, y) = x^3 - x$

Exercise 1.12 Sketch or describe the surfaces in $\mathbb{R}^3$ of equations $z^2 = y^2+4$ and $z = x^2$.

Exercise 1.13 Consider the function $f(x, y) = \arctan(x^2 + y^2)$.

1. Determine the image of f. Is this function bounded?
2. Sketch the level curves of f at the values $c = 0$ and $c = \frac{\pi}{4}$.

Exercise 1.14 Consider the function $f(x, y) = e^{\frac{y^2}{1+x^2}}$. Sketch the level curves at the values $c = 1$ and $c = 2$.

2 Limits and continuity

2.1 Limits (MT 2.2)

We now turn our attention to the concept of a limit. Throughout the following discussions the domain of definition of the function f will be an open set A. We are interested in finding the limit of f as $x \in A$ approaches either a point of A or a boundary point of A.

You should appreciate the fact that the limit concept is a basic and useful tool for the analysis of functions; it enables us to study derivatives, and hence maxima and minima, asymptotes, improper integrals, and other important features of functions, as well as being useful for infinite series and sequences. We will present a theory of limits for functions of several variables that includes the theory for functions of one variable as a special case.

Definition 2.1.1 (Limit) *Let $f : A \subset \mathbb{R}^n \to \mathbb{R}^m$, where A is an open set. Let $\mathbf{x}_0$ be in A or be a boundary point of A, and let N be a neighborhood of $\mathbf{b} \in \mathbb{R}^m$. We say f is eventually in N as $\mathbf{x}$ approaches $\mathbf{x}_0$ if there exists a neighborhood U of $\mathbf{x}_0$ such that $\mathbf{x} \neq \mathbf{x}_0, \mathbf{x} \in U$, and $\mathbf{x} \in A$ imply $f(\mathbf{x}) \in N$. Note that $\mathbf{x}_0$ need not be in the set A, so that $f(\mathbf{x}_0)$ is not necessarily defined.*

We say $f(\mathbf{x})$ approaches $\mathbf{b}$ as $\mathbf{x}$ approaches $\mathbf{x}_0$, or that $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}$, when, given any neighborhood N of $\mathbf{b}$, f is eventually in N as $\mathbf{x}$ approaches $\mathbf{x}_0$. It may be that as $\mathbf{x}$ approaches $\mathbf{x}_0$, the values $f(\mathbf{x})$ do not get close to any particular vector. In this case, we say that $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x})$ does not exist.

You can check that you understand the definition by proving the obvious limit

$$\lim_{\mathbf{x}\to\mathbf{x}_0} \mathbf{x} = \mathbf{x}_0.$$

Example 2.1.1 This example illustrates a limit that does not exist. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x \leq 0. \end{cases}$$

The limit $\lim_{x\to 0} f(x)$ does not exist, since there are points x_1 arbitrarily close to 0 with $f(x_1) = 1$ and also points x_2 arbitrarily close to 0 with $f(x_2) = -1$.

Example 2.1.2 This example illustrates a function whose limit does exist, but whose limiting value does not equal its value at the limiting point. Consider the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x \neq 0. \end{cases}$$

In this case, $\lim_{x \to 0} = 0$ since for any neighborhood U of 0, $x \in U$ and $x \neq 0$ imply $f(x) = 0$. In other words, f approaches 0 as $x \longrightarrow 0$: we do not care that f happens to take on some other value at 0.

Definition 2.1.2 (Limit, Second Definition) *We say that the limit of f when $\mathbf{x} \longrightarrow \mathbf{x}_0$ exists if there is a $\mathbf{b} \in \mathbb{R}^m$ such that for every sequence $\mathbf{x}_n \in U \subset \mathbb{R}^n$ (U is a neighborhood of $\mathbf{x}_0$), with $\lim_{n\to\infty} \mathbf{x}_n = \mathbf{x}_0$, the sequence $f(\mathbf{x}_n)$ has as limit $\mathbf{b}$.*

Example 2.1.3 Prove that the function $f(x, y) = \dfrac{x^2y^2}{x^2y^2 + (x-y)^2}$ verifies the condition

$$\lim_{x\to 0}\left(\lim_{y\to 0} f(x,y)\right) = \lim_{y\to 0}\left(\lim_{x\to 0} f(x,y)\right) = 0$$

and yet $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

If $x \neq 0$, we have

$$\lim_{y\to 0} f(x,y) = \lim_{y\to 0} \frac{x^2y^2}{x^2} = 0.$$

If $x = 0$ then $\lim_{y\to 0} f(x,y) = \lim_{y\to 0} \frac{0}{y^2} = 0$.

Therefore,

$$\lim_{x\to 0}\left(\lim_{y\to 0} f(x,y)\right) = \lim_{x\to 0} 0 = 0.$$

Similarly, we can show that

$$\lim_{y\to 0}\left(\lim_{x\to 0} f(x,y)\right) = 0.$$

Now, let us choose the lines $x = y$ and $x = -y$, and approach the origin through both of them. This is analogous to choosing the sequences $(x_n, y_n) = (1/n, 1/n)$ and $(x'_n, y'_n) = (1/n, -1/n)$ and taking the limit when $n \longrightarrow \infty$. Then we have

$$\lim_{(x,x)\to(0,0)} f(x,x) = \lim_{x\to 0} \frac{x^4}{x^4} = 1,$$

but

$$\lim_{(x,-x)\to(0,0)} f(x,-x) = \lim_{x\to 0} \frac{x^4}{x^4 + 4x^2} = 0.$$

These two expressions are different, and therefore the limit does not exist.

Properties of Limits

To properly speak of the limit, we should establish that f can have at most one limit as $x \longrightarrow x_0$. This is intuitively clear and we now state it formally.

Theorem 2.1.1 (Uniqueness of Limits) *If* $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}_1$ *and also* $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}_2$*, then* $\mathbf{b}_1 = \mathbf{b}_2$*.*

Proof. If $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}_1$, then every sequence $\mathbf{x}_n \longrightarrow \mathbf{x}_0$ must fulfil that $\lim_{n\to\infty} f(\mathbf{x}_n) = \mathbf{b}_1$. If $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}_2$ then, by the same reasoning, using the same sequence $\mathbf{x}_n$, $\lim_{n\to\infty} f(\mathbf{x}_n) = \mathbf{b}_2$. But a convergent sequence can have only one limit, and therefore $\mathbf{b}_1 = \mathbf{b}_2$. $\square$

Theorem 2.1.2 (Properties of Limits) *Let* $f : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$, $g : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$, $\mathbf{x}_0$ *be in* A *or be a boundary point of* A, $\mathbf{b} \in \mathbb{R}^m$, *and* $c \in \mathbb{R}$*; then*

(i) If $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}$*, then* $\lim_{\mathbf{x}\to\mathbf{x}_0} cf(\mathbf{x}) = c\mathbf{b}$*, where* $cf : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ *is defined by* $\mathbf{x} \longrightarrow cf(\mathbf{x})$*.*

(ii) If $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b}_1$ *and* $\lim_{\mathbf{x}\to\mathbf{x}_0} g(\mathbf{x}) = \mathbf{b}_2$*, then* $\lim_{\mathbf{x}\to\mathbf{x}_0}(f+g)(\mathbf{x}) = \mathbf{b}_1 + \mathbf{b}_2$*, where* $f + g : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ *is defined by* $\mathbf{x} \longrightarrow f(\mathbf{x}) + g(\mathbf{x})$*.*

(iii) In the special case $m = 1$*, if* $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = b_1$ *and* $\lim_{\mathbf{x}\to\mathbf{x}_0} g(\mathbf{x}) = b_2$*, then* $\lim_{\mathbf{x}\to\mathbf{x}_0}(fg)(\mathbf{x}) = b_1 b_2$*, where* $fg : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ *is defined by* $\mathbf{x} \longrightarrow f(\mathbf{x})g(\mathbf{x})$*.*

(iv) If $m = 1$*,* $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = b \neq 0$ *and* $f(\mathbf{x}) \neq 0$ *for all* $x \in A$*, then* $\lim_{\mathbf{x}\to\mathbf{x}_0} 1/f(\mathbf{x}) = 1/b$*, where* $1/f : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ *is defined by* $\mathbf{x} \longrightarrow 1/f(\mathbf{x})$*.*

(v) If $f(x) = (f_1(x), \ldots, f_m(x))$*, where* $f_i : A \longrightarrow \mathbb{R}$*,* $i = 1, \ldots, m$ *are the component functions of* f*, then* $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = \mathbf{b} = (b_1, \ldots, b_m)$ *if and only if* $\lim_{\mathbf{x}\to\mathbf{x}_0} f_i(\mathbf{x}) = b_i$ *for each* $i = 1, \ldots, m$*.*

Proof. The proof follows directly from the definition of limit. $\square$

Example 2.1.4 Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$, $(x, y) \longrightarrow x^2 + y^2 + 2$. Compute the limit

$$\lim_{(x,y)\to(0,1)} f(x, y).$$

Here f is the sum of the three functions $(x, y) \longrightarrow x^2$, $(x, y) \longrightarrow y^2$ and $(x, y) \longrightarrow 2$. The limit of a sum is the sum of the limits, and the limit of a product is the product of the limits. Hence, using the fact that $\lim_{(x,y)\to(0,1)} \mathbf{x} = \mathbf{x}_0$, we obtain

$$\lim_{(x,y)\to(x_0,y_0)} x^2 = \left(\lim_{(x,y)\to(0,1)} x\right)^2 = x_0^2$$

and, using the same reasoning, $\lim_{(x,y)\to(x_0,y_0)} y^2 = y_0^2$. Consequently,

$$\lim_{(x,y)\to(0,1)} f(x,y) = 0^2 + 1^2 + 2 = 3.$$

2.2 Continuous Functions (MT 2.2)

In single-variable calculus we learned that the idea of a continuous function is based on the intuitive notion of a function whose graph is an unbroken curve; that is, a curve that has no jumps, or the kind of curve that would be traced by a particle in motion or by a moving pencil point that is not lifted from the paper. In the case of several variables we just need to extend this idea intuitively: the continuity of f thus means that its graph has no "breaks" in it. The formal definition of continuity is as follows:

Definition 2.2.1 (Continuity) *Let $f : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be a given function with domain A. Let $\mathbf{x}_0 \in A$. We say f is continuous at $\mathbf{x}_0$ if and only if*

$$\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x}) = f(\mathbf{x}_0).$$

If we just say that f is continuous, we shall mean that f is continuous at each point $\mathbf{x}_0$ of A. If f is not continuous at $\mathbf{x}_0$, we say f is discontinuous at $\mathbf{x}_0$. If f is discontinuous at some point in its domain, we say f is discontinuous.

Any polynomial $p(x) = a_0 + a_1x + \cdots + a_nx^n$ is continuous from $\mathbb{R}$ to $\mathbb{R}$. Indeed, from Theorem 2.1.2 and Example 2.1.4,

$$\lim_{x\to x_0} (a_0 + a_1x + \cdots + a_nx^n) = a_0 + a_1x_0 + \cdots + a_nx_0^n,$$

Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$, $f(x,y) = xy$. Then f is continuous, because, by the limit theorems and Example 2.1.4 again,

$$\lim_{(x,y)\to(x_0,y_0)} xy = x_0y_0.$$

We can see by the same method that any polynomial $p(x,y)$ in x and y is continuous.

Theorem 2.2.1 (Properties of Continuous Functions) *Suppose that $f : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$, $g : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$, and let c be a real number.*

(i) If f is continuous at $\mathbf{x}_0$, so is cf, where $(cf)(\mathbf{x}) = c[f(\mathbf{x})]$.
(ii) If f and g are continuous at $\mathbf{x}_0$, so is $f+g$, where the sum of f and g is defined by $(f+g)(\mathbf{x}) = f(\mathbf{x}) + g(\mathbf{x})$.
(iii) If f and g are continuous at $\mathbf{x}_0$ and $m = 1$, then the product function fg defined by $(fg)(\mathbf{x}) = f(\mathbf{x})g(\mathbf{x})$ is continuous at $\mathbf{x}_0$.
(iv) If $f : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ is continuous at $\mathbf{x}_0$ and nowhere zero on A, then the quotient $1/f$ is continuous at $\mathbf{x}_0$, where $(1/f)(\mathbf{x}) = 1/f(\mathbf{x})$.
(v) If $f : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ and $f(\mathbf{x}) = (f_1(\mathbf{x}), \ldots, f_m(\mathbf{x}))$, then f is continuous at $\mathbf{x}_0$ if and only if each of the real-valued functions

$f_1, \ldots, f_m$ is continuous at $\mathbf{x}_0$.

Proof. The proof follows from applying the properties of the limits in Theorem 2.1.2 to the definition of continuity. □

A variant of (iv) is often used: If $f(\mathbf{x}_0) \neq 0$ and f is continuous, then $f(\mathbf{x}) \neq 0$ in a neighborhood of $\mathbf{x}_0$ and so $1/f$ is defined in that neighborhood, and $1/f$ is continuous at $\mathbf{x}_0$.

Example 2.2.1 Let $f : \mathbb{R}^2 \to \mathbb{R}^2$, $(x, y) \mapsto (x^2y, (y + x^3)/(1 + x^2))$. Show that f is continuous.

To see this, it is sufficient, by property (v) of Theorem 2.2.1, to show that each component is continuous. As we have mentioned, any polynomial in two variables is continuous; thus, the map $(x, y) \mapsto x^2y$ is continuous. Because $1 + x^2$ is continuous and nonzero, by property (iv), we know that $1/(1 + x^2)$ is continuous; hence, $(y + x^3)/(1 + x^2)$ is a product of continuous functions, and by (iii) is continuous.

Similar reasoning applies to examples like the function $c : \mathbb{R} \to \mathbb{R}^3$ given by $c(t) = (t^2, 1, t^3/(1 + t^2))$ to show they are continuous as well.

Composition

Next we discuss composition, another basic operation that can be performed on functions. If g maps A to B and f maps B to C, the composition of g with f, or of f on g, denoted by $f \circ g$, maps A to C by sending $\mathbf{x} \mapsto f(g(\mathbf{x}))$. For example, $\sin(x^2)$ is the composition of $x \mapsto x^2$ with $y \mapsto \sin y$.

Theorem 2.2.2 (Continuity of Compositions) *Let $g : A \subset \mathbb{R}^n \to \mathbb{R}^m$ and let $f : B \subset \mathbb{R}^m \to \mathbb{R}^p$. Suppose $g(A) \subset B$, so that $f \circ g$ is defined on A. If g is continuous at $\mathbf{x}_0 \in A$ and f is continuous at $y_0 = g(\mathbf{x}_0)$, then $f \circ g$ is continuous at $\mathbf{x}_0$.*

Proof. Take a sequence $\mathbf{x}_n \to \mathbf{x}_0$ such that $\mathbf{x}_n \in A$. Then $\lim_{n\to\infty} g(\mathbf{x}_n) = g(\mathbf{x}_0)$, because g is continuous, with $g(\mathbf{x}_n) \in B$. Now $\lim_{n\to\infty} f(g(\mathbf{x}_n)) = f(g(\mathbf{x}))$, because f is continuous, and so $\lim_{\mathbf{x}\to\mathbf{x}_0} f \circ g(\mathbf{x}) = f \circ g(\mathbf{x}_0)$, which completes the proof. □

Example 2.2.2 Let $f(x, y, z) = (x^2 + y^2 + z^2)^{30} + \sin(z^3)$. Show that f is continuous. Here we can write f as a sum of the two functions $(x^2+y^2+z^2)^{30}$ and $\sin(z^3)$, so it suffices to show that each is continuous. The first is the composite of $(x, y, z) \mapsto (x^2 + y^2 + z^2)$ with $u \mapsto u^{30}$, and the second is the composite of $(x, y, z) \mapsto z^3$ with $u \mapsto \sin u$, and so we have continuity by Theorem 2.2.2.

Exercises

Exercise 2.1 Study the following limits:

i) $\lim_{(x,y)\to(0,0)} \frac{x}{x^2+y^2}$

ii) $\lim_{(x,y)\to(2,4)} \frac{x+y}{x-y}$

iii) $\lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^2}$

iv) $\lim_{(x,y)\to(0,0)} \frac{\tan x}{y}$

v) $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$

vi) $\lim_{(x,y)\to(0,0)} \frac{xy^2}{(x^2+y^2)^{3/2}}$

vii) $\lim_{(x,y)\to(0,0)} \frac{y^2}{x^2-y^2}$

viii) $\lim_{(x,y)\to(0,0)} \frac{\sin(xy)}{xy}$

ix) $\lim_{(x,y)\to(0,0)} \frac{x^3-y^2}{x^2+y^2}$

x) $\lim_{(x,y)\to(0,0)} \frac{x^4+y^4}{x^2y^2+(x-y)^2}$

Exercise 2.2 Consider the function $f(x,y) = \dfrac{x^4-y}{x^4-y^3}$. The points $(0,0)$ and $(1,1)$ are excluded from its domain. Decide if it is possible to define $f(0,0)$ and $f(1,1)$ in such a way that f is continuous at these points.

Exercise 2.3 Study the continuity of the following functions:

i) $f(x,y) = \begin{cases} \dfrac{xy}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

ii) $f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

iii) $f(x,y) = \begin{cases} \dfrac{x^4+x^2y-y^3}{\sqrt{x^2+y^2}} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

iv) $f(x,y) = \begin{cases} \dfrac{x^2+y^3}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

v) $f(x,y) = \begin{cases} \dfrac{x^4-y^4}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

vi) $f(x,y) = \begin{cases} \dfrac{x+y^2}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

vii) $f(x,y) = \dfrac{\cos(xy)}{e^x+e^y}$

viii) $f(x,y) = \begin{cases} \dfrac{\sin(x-y)}{e^x-e^y} & \text{if } x \neq y \\ 1 & \text{if } x = y \end{cases}$

ix) $f(x,y) = \begin{cases} \dfrac{x^2y}{x^4-y^2} & \text{if } y \neq \pm x^2 \\ 0 & \text{if } y = \pm x^2 \end{cases}$

x) $f(x,y) = \begin{cases} (x+y)\sin\left(\dfrac{1}{x}\right)\sin\left(\dfrac{1}{y}\right) & \text{if } xy \neq 0 \\ 0 & \text{if } xy = 0 \end{cases}$

Exercise 2.4 Consider the function $f(x,y) = \begin{cases} \dfrac{x^3y}{x^6+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

i) Compute the limit of f when $(x,y) \longrightarrow (0,0)$ along the lines $y = \lambda x$.
ii) Compute the limit of f when $(x,y) \longrightarrow (0,0)$ along the curve $y = x^3$.
iii) Is the function f continuous at $(0,0)$?

Exercise 2.5 Calculate the following limits:

i) $\displaystyle\lim_{x\to\infty, y\to\infty} \frac{x+y}{x^2-xy+y^2}$
ii) $\displaystyle\lim_{x\to\infty, y\to\infty} \frac{x^2+y^2}{x^4+y^4}$
iii) $\displaystyle\lim_{x\to 0, y\to a} \frac{\sin xy}{x}$
iv) $\displaystyle\lim_{x\to\infty, y\to\infty} (x^2+y^2)e^{-(x+y)}$
v) $\displaystyle\lim_{x\to\infty, y\to\infty} \left(\frac{xy}{x^2+y^2}\right)^{x^2}$

Exercise 2.6 Find the directions in which the following functions have finite limits, where $r^2 = x^2 + y^2$:

i) $\displaystyle\lim_{r\to 0} e^{\frac{x}{x^2+y^2}}$
ii) $\displaystyle\lim_{r\to\infty} e^{x^2-y^2} \sin 2xy$
iii) $\displaystyle\lim_{x\to 0, y\to 0} (x^2+y^2)^{x^2y^2}$

Derivatives in higher dimensions 3

For a function $f : \mathbb{R} \to \mathbb{R}$, the derivative gives the slope of the tangent line at a given point. A function is differentiable at a point x_0 if the tangent line is well-defined and has a finite slope. How can we extend this idea for higher dimensions? In other words, what does it mean for a function of several variables to be differentiable at a point $\mathbf{x}_0$? For a function $g : \mathbb{R}^2 \to \mathbb{R}$ the tangent line becomes a tangent plane, and this generalizes into a tangent hyperplane for higher dimensions. But first, we need to introduce partial derivatives.

3.1 Partial Derivatives (MT 2.3)

Definition 3.1.1 (Partial Derivatives) *Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$, with U an open set. Then $\frac{\partial f}{\partial x_j} : \mathbb{R}^n \to \mathbb{R}$, the partial derivative of f with respect to x_j ($j = 1, \ldots, n$), is defined by*

$$\begin{aligned}\frac{\partial f}{\partial x_j}(x_1, \ldots, x_n) &= \lim_{h \to 0} \frac{f(x_1, x_2, \ldots, x_j + h, \ldots, x_n) - f(x_1, \ldots, x_n)}{h} \\ &= \lim_{h \to 0} \frac{f(\mathbf{x} + h\mathbf{e}_j) - f(\mathbf{x})}{h}\end{aligned}$$

if the limit exists, where $\mathbf{e}_j$ is the j-th vector of the canonical basis, with 1 in the j-th component and 0 elsewhere. The domain of the function $\frac{\partial f}{\partial x_j}$ is the set of $x \in \mathbb{R}^n$ for which the limit exists.

In other words, the partial derivative is the derivative of f with respect to x_j when we consider the other variables as parameters or constants. If $f : \mathbb{R}^3 \to \mathbb{R}$, we often use the notation $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}$ instead of $\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \frac{\partial f}{\partial x_3}$.

Example 3.1.1 If $f(x, y) = x^2y + y^3$, find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

To find $\frac{\partial f}{\partial x}$ we hold y constant and differentiate only with respect to x; this yields

$$\frac{\partial}{\partial x}(x^2y + y^3) = \frac{\partial f}{\partial x} = 2xy.$$

Similarly, to find $\frac{\partial f}{\partial y}$ we hold x constant and differentiate only with respect to y:

$$\frac{\partial}{\partial y}(x^2y + y^3) = \frac{\partial f}{\partial y} = x^2 + 3y^2.$$

Example 3.1.2 Find $\frac{\partial f}{\partial x}$ if $f(x,y) = \frac{xy}{x^2+y^2}$. By the quotient rule,

$$\frac{\partial f}{\partial x} = \frac{y(x^2+y^2) - 2x^2 y}{(x^2+y^2)^2} = \frac{y(y^2-x^2)}{(x^2+y^2)^2}.$$

We can write all partial derivatives together in vector form: the gradient.

Definition 3.1.2 (Gradient) *Let $f : U \subset \mathbb{R}^n \longrightarrow \mathbb{R}$. The vector*

$$\left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n}\right)$$

is called the gradient of f and denoted by ∇f, or grad f.

For functions $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$, we can obtain partial derivatives for each of its components: $\frac{\partial f_i}{\partial x_j}$ is the partial derivative of the i-th component of f with respect to x_j. We can join all these partial derivatives in matrix form to obtain the Jacobian matrix.

Definition 3.1.3 (Jacobian matrix) *Let $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$. The Jacobian matrix of f is the $m \times n$ matrix of all partial derivatives of f, given by*

$$J_f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{pmatrix},$$

defined wherever the partial derivatives exist. The Jacobian matrix is sometimes denoted by $Df(\mathbf{x})$. Note that the gradient is just the particular case of the Jacobian matrix when $m = 1$.

Having introduced the partial derivatives, we can now continue our discussion of differentiability.

3.2 Tangent Plane (MT 2.3)

To "motivate" our definition of differentiability, let us compute the equation of the plane tangent to the graph of $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$, $(x,y) \mapsto f(x,y)$ at (x_0, y_0). In $\mathbb{R}^3$, a nonvertical plane has an equation of the form

$$z = ax + by + c.$$

If it is to be the plane tangent to the graph of f, the slopes along the x and y axes must be equal to $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$. Thus, $a = \frac{\partial f}{\partial x}(x_0, y_0)$, $b = \frac{\partial f}{\partial y}(x_0, y_0)$. Finally, we obtain the constant c from the fact that $z = f(x_0, y_0)$ when $x = x_0$, $y = y_0$. Thus, we get the linear approximation[1] given by

$$z = f(x_0, y_0) + \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) + \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0).$$

1: More accurately, it should be the *affine approximation*

Our definition of differentiability will mean in effect that this tangent plane is a good approximation of f near (x_0, y_0). What do we mean by this? For a differentiable function $f : \mathbb{R} \longrightarrow \mathbb{R}$, we have that

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0),$$

and that this approximation becomes exact in the limit

$$\lim_{x \longrightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0).$$

We can rewrite the previous expression as

$$\lim_{x \longrightarrow x_0} \frac{f(x) - f(x_0) - f'(x_0)(x - x_0)}{x - x_0} = 0.$$

Thus, a one-dimensional function is differentiable at a point x_0 if there exists a value $f'(x_0)$ such that the previous limit is zero. The idea here is that the error made by this approximation is very small compared with $x - x_0$. We can extend this idea for functions of two variables:

Definition 3.2.1 (Differentiability: Two Variables) *Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$. We say f is differentiable at (x_0, y_0) if $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist at (x_0, y_0) and if*

$$\frac{f(x,y) - f(x_0, y_0) - \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) - \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0)}{\|(x,y) - (x_0, y_0)\|} \longrightarrow 0$$

as $(x, y) \longrightarrow (x_0, y_0)$.

The plane defined by

$$z = f(x_0, y_0) + \left[\frac{\partial f}{\partial x}(x_0, y_0)\right](x - x_0) + \left[\frac{\partial f}{\partial y}(x_0, y_0)\right](y - y_0)$$

is called the ***tangent plane*** *of f at (x_0, y_0).*

Example 3.2.1 Compute the plane tangent to the graph of $z = x^2 + y^4 + e^{xy}$ at the point $(1, 0, 2)$.

The partial derivatives are

$$\frac{\partial z}{\partial x} = 2x + ye^{xy}, \quad \frac{\partial z}{\partial y} = 4y^3 + xe^{xy}.$$

At $(1, 0, 2)$, these partial derivatives are 2 and 1, respectively. The tangent plane is then

$$z = 2(x - 1) + 1(y - 0) + 2,$$

that is,

$$z = 2x + y.$$

Writing $\nabla f(x_0, y_0) = \left(\frac{\partial f}{\partial x}(x_0, y_0), \frac{\partial f}{\partial y}(x_0, y_0)\right)$, we can rewrite the definition of the tangent plane as

$$z = f(x_0, y_0) + \nabla f(x_0, y_0) \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}$$

3.3 The General Case (MT 2.3)

Now we are ready to give a definition of differentiability for maps f of $\mathbb{R}^n$ to $\mathbb{R}^m$, using the preceding discussion as motivation. The derivative $J_f(\mathbf{x}_0)$ of $f = (f_1, \ldots, f_m)$ is defined as follows.

Definition 3.3.1 (Differentiability: n Variables, m Functions) *Let $f : U \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$, U an open set. We say that f is differentiable at $\mathbf{x}_0 \in U$ if the partial derivatives of f exist at $\mathbf{x}_0$ and if*

$$\lim_{\mathbf{x} \longrightarrow \mathbf{x}_0} \frac{f(\mathbf{x}) - f(\mathbf{x}_0) - J_f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)}{\|\mathbf{x} - \mathbf{x}_0\|} = 0,$$

where $J_f(\mathbf{x}_0)$ is the Jacobian matrix of f evaluated at $\mathbf{x}_0$ and $J_f(\mathbf{x}_0)(\mathbf{x} - \mathbf{x}_0)$ means the product of $J_f(\mathbf{x}_0)$ with $\mathbf{x} - \mathbf{x}_0$ (regarded as a column matrix). We call $J_f(\mathbf{x}_0)$ the derivative of f at $\mathbf{x}_0$.

Example 3.3.1 Calculate the matrices of partial derivatives for these functions.

(a) $f(x, y) = (e^{x+y} + y, y^2 x)$
(b) $f(x, y) = (x^2 + \cos y, ye^x)$
(c) $f(x, y, z) = (ze^x, -ye^z)$

Using the definition, we have:

a) J_f will be a 2×2 matrix given by

$$J_f = \begin{pmatrix} e^{x+y} & e^{x+y} + 1 \\ y^2 & 2xy \end{pmatrix}$$

b) J_f will be a 2×2 matrix given by

$$J_f = \begin{pmatrix} 2x & -\sin y \\ ye^x & e^x \end{pmatrix}$$

a) J_f will be a 2×3 matrix given by

$$J_f = \begin{pmatrix} ze^x & 0 & e^x \\ 0 & -e^z & -ye^z \end{pmatrix}$$

3.4 Differentiability and Continuity (MT 2.3)

From one-variable calculus we know that if f is differentiable, then f is continuous. Is this true for functions of several variables? Yes:

Theorem 3.4.1 *Let $f : U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at $\mathbf{x}_0 \in U$. Then f is continuous at $\mathbf{x}_0$.*

This result is very reasonable, because "differentiability" means that there is enough smoothness to have a tangent plane, which is stronger than just being continuous.

A different question is, what are the conditions that ensure f is differentiable? Is it enough to check that the partial derivatives exist? The answer is no:

Example 3.4.1 Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} 1 & \text{if } x = 0 \text{ or if } y = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Because f is constant on the x and y axes, where it equals 1,

$$\frac{\partial f}{\partial x}(0,0) = 0 \quad \text{and} \quad \frac{\partial f}{\partial y}(0,0) = 0.$$

But f is not continuous at $(0,0)$, because $\lim_{(x,y)\to(0,0)} f(x,y)$ does not exist.

Here, the partial derivatives exist at a given point, but f is not continuous, so it can't be differentiable. The following theorem gives a sufficient condition for f to be differentiable:

Theorem 3.4.2 *Let $f : U \subset \mathbb{R}^n \to \mathbb{R}^m$. Suppose the partial derivatives $\frac{\partial f_i}{\partial x_j}$ of f all exist and are continuous in a neighborhood of a point $\mathbf{x}_0 \in U$. Then f is differentiable at $\mathbf{x}_0$.*

Example 3.4.2 Let

$$f(x,y) = \frac{\cos x + e^{xy}}{x^2 + y^2}.$$

Show that f is differentiable at all points $(x,y) \neq (0,0)$.

Observe that the partial derivatives

$$\frac{\partial f}{\partial x} = \frac{(x^2+y^2)(ye^{xy} - \sin x) - 2x(\cos x + e^{xy})}{(x^2+y^2)^2}$$

and

$$\frac{\partial f}{\partial y} = \frac{(x^2+y^2)xe^{xy} - 2y(\cos x + e^{xy})}{(x^2+y^2)^2}$$

are continuous except when $x = 0$ and $y = 0$. Thus, f is differentiable at those points.

Having continuous partial derivatives implies differentiability, but the converse is not true:

Example 3.4.3 The function

$$f(x,y) = \begin{cases} (x^2+y^2)\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) & \text{if } x^2+y^2 \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

is differentiable but the partial derivatives are not continuous at 0.

Indeed, the partial derivatives exist at the origin and are equal to zero, as

$$\frac{\partial f}{\partial x}(0,0) = \lim_{x\to 0} x\sin(1/|x|) = 0,$$

and $\partial f/\partial y = 0$ due to the symmetry of the function with respect to x and y.

Using the definition of derivative, we have

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y)-0}{\sqrt{x^2+y^2}} = \lim_{(x,y)\to(0,0)} \sqrt{x^2+y^2}\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) = 0,$$

as the limit is identical to

$$\lim_{x\to 0} |x| \sin(1/|x|).$$

So the function is differentiable at the origin, but

$$\frac{\partial f}{\partial x} = 2x\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right) - \frac{x\cos\left(\frac{1}{\sqrt{x^2+y^2}}\right)}{\sqrt{x^2+y^2}}$$

is not continuous at $(0,0)$.

And differentiability implies the existence of partial derivatives, but the converse is not true either:

Example 3.4.4 The function

$$f(x,y) = x^{1/3}y^{1/3}$$

has partial derivatives but is not differentiable at $(x,y) = (0,0)$.

Again, both partial derivatives are zero at the origin (check this using the definition). But, using the definition of differentiability,

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y)-0}{\sqrt{x^2+y^2}} = \frac{x^{1/3}y^{1/3}}{\sqrt{x^2+y^2}}$$

does not exist, as taking $y = x$ we obtain

$$\lim_{x\to 0} \frac{x^{2/3}}{\sqrt{2}|x|} = \pm\infty.$$

However, if a function is not differentiable at a given point, as in the previous example, it must mean that its partial derivatives are not continuous there.

Exercises

Exercise 3.1 Compute the gradient of the following functions:

1. $f(x, y) = 3x^2 - xy + y$
2. $f(x, y) = x^3 e^{-y}$
3. $f(x, y) = \log(x^2 + y^4)$
4. $f(x, y) = (g(x))^2 h(y)$
5. $f(x, y) = \sqrt{1 - (x^2 + y^2)}$
6. $f(x, y, z) = xe^{y^2} + ye^z$
7. $f(x, y, z) = x \sin y + y \sin z + z \sin x$
8. $f(x, y, z) = \sin(x + xy + z^2)$
9. $f(x, y, z) = z^{xy^2}$
10. $f(x, y, z) = (g(x, y))^2 (h(x, z))^3$
11. $f(r, \theta) = r^2 \sin\theta + \cos^2\theta$
12. $f(\rho, \phi, \theta) = \rho^3 \sin\phi \cos\theta$

Exercise 3.2 Consider the function

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^4 + y^4}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

1. Show that the partial derivatives $\frac{\partial f}{\partial x}(0, 0)$ and $\frac{\partial f}{\partial y}(0, 0)$ exist.
2. Show that f is not continuous at $(0, 0)$.

Exercise 3.3 Show that the following functions are differentiable in the indicated sets:

1. $f(x, y) = x^2 + y^2$ in $\mathbb{R}^2$.
2. $f(x, y) = xy$ in $\mathbb{R}^2$.
3. $f(x, y, z) = x^2 + y^2 + z^2$ in $\mathbb{R}^3$.
4. $f(x, y, z) = \sin(x + y + z)$ in $\mathbb{R}^3$.
5. $f(x, y) = e^x \sin y$ in $\mathbb{R}^2$.
6. $f(x, y) = (x^2 + y^2)e^{-xy}$ in $\mathbb{R}^2$.
7. $f(x, y) = \frac{x}{x^2 + y^2}$ at $(x, y) \neq (0, 0)$.

Exercise 3.4 Consider the function

$$f(x, y) = \begin{cases} \frac{2xy}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0, & \text{if } (x, y) = (0, 0). \end{cases}$$

1. Show that the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist at $(0, 0)$.
2. Show that $\frac{\partial f}{\partial x}$ is not continuous at $(0, 0)$.
3. Show that f is not differentiable at $(0, 0)$.

Exercise 3.5 Consider the function $f(x, y) = \sqrt{x^2 + y^2}$. Show that $\frac{\partial f}{\partial x}$ is not defined at $(0, 0)$.

Exercise 3.6

1. Study the continuity of the function

$$f(x,y) = \begin{cases} \frac{x^2y^4}{x^2+y^2}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

2. Compute the partial derivatives at $(0,0)$ and study the differentiability at this point.

Exercise 3.7 Consider the function

$$g(x,y) = \begin{cases} \frac{x^2+y}{\sqrt{x^2+y^2}}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$$

1. Study the continuity of g on $\mathbb{R}^2$.
2. Compute the partial derivatives of $g(x,y)$ at the origin $(0,0)$ if it is possible.
3. Can g be differentiable at $(0,0)$?

Exercise 3.8 Consider the function $f(x,y) = |xy|^\alpha$.

1. If $\alpha > \frac{1}{2}$, show that f is differentiable at $(0,0)$.
2. If $\alpha = \frac{1}{2}$, compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(0,0)$.
3. Show that, in this second case, f is not differentiable at $(0,0)$.

Exercise 3.9 Find the equation of the tangent plane for the graphs of the following functions at the prescribed points $(x_0, y_0, f(x_0,y_0))$:

1. $f(x,y) = x - y + 2, \quad (x_0,y_0) = (1,3)$.
2. $f(x,y) = x^2 + 4y^2, \quad (x_0,y_0) = (2,-1)$.
3. $f(x,y) = \log(x+y) + x\cos y, \quad (x_0,y_0) = (1,0)$.
4. $f(x,y) = \sqrt{x^2+y^2}, \quad (x_0,y_0) = (1,\frac{1}{2})$.

Exercise 3.10 Compute the gradient of $f(x,y) = \arctan(x^2+y^2)$ and the equation of the tangent plane to the graph of f at $(x_0,y_0) = (1,0)$.

Exercise 3.11 Compute the Jacobian matrix of the following functions:

1. $A(x,y,z) = (x^y, z)$.
2. $b(x,y) = \sin(x \sin y)$.
3. $C(x) = (x + e^x, x^2, \cos x)$.
4. $D(x,y) = (xe^y + \cos y, x, x + e^y)$.
5. $E(x,y,z) = (x + e^z + y, yx^2)$.
6. $F(x,y,z) = (xye^{xy}, x\sin y, 5xy^2)$.
7. $g(x,y,z,t) = x^2 \log t + x\sqrt{z} - ty$.

Exercise 3.12 1. Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by means of $T(x,y,z) := (x - 3y + 2z, 2x + y - 2z)$. Compute the Jacobian matrix DT.
2. Show that any linear mapping

$$T_A : \mathbb{R}^n \longrightarrow \mathbb{R}^m, \quad \mathbf{x} \mapsto A\mathbf{x}$$

with an $m \times n$-matrix A, is differentiable at any point $\mathbf{a} \in \mathbb{R}^n$ and that $DT_A(\mathbf{a}) = A$.

Exercise 3.13 Define the following transformations and compute the Jacobian (i.e., the determinant of the matrix of derivatives):

1. From polar coordinates to Cartesian coordinates (in $\mathbb{R}^2$).
2. From cylindrical coordinates to Cartesian coordinates (in $\mathbb{R}^3$).
3. From spherical coordinates to Cartesian coordinates (in $\mathbb{R}^3$).
4. From cylindrical coordinates to spherical coordinates (in $\mathbb{R}^3$).

4 Properties of the Derivative. The Chain Rule. Directional Derivatives

In elementary calculus, we learn how to differentiate sums, products, quotients, and composite functions. We now generalize these ideas to functions of several variables, paying particular attention to the differentiation of composite functions. The rule for differentiating composites, called the chain rule, takes on a more profound form for functions of several variables than for those of one variable.

4.1 Properties of the Derivative (MT 2.5)

Theorem 4.1.1 (Sums, Products, Quotients) *Let $f : U \subset \mathbb{R}^n \to \mathbb{R}^m$ and $g : U \subset \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at $\mathbf{x}_0$, and $c \in \mathbb{R}$. Then:*

(i) ***Constant Multiple Rule.*** *$h(\mathbf{x}) = cf(\mathbf{x})$ is differentiable at $\mathbf{x}_0$ and $J_h(\mathbf{x}_0) = cJ_f(\mathbf{x}_0)$.*

(ii) ***Sum Rule.*** *$h(\mathbf{x}) = f(\mathbf{x}) + g(\mathbf{x})$ is differentiable at $\mathbf{x}_0$ and $J_h(\mathbf{x}_0) = J_f(\mathbf{x}_0) + J_g(\mathbf{x}_0)$.*

(iii) ***Product Rule.*** *If $m = 1$, $h(\mathbf{x}) = g(\mathbf{x})f(\mathbf{x})$ is differentiable at $\mathbf{x}_0$ and $J_h(\mathbf{x}_0) = g(\mathbf{x}_0)J_f(\mathbf{x}_0) + f(\mathbf{x}_0)J_g(\mathbf{x}_0)$.*

(iv) ***Quotient Rule.*** *If $m = 1$ and $g(\mathbf{x}) \neq 0$ for all $\mathbf{x} \in U$, $h(\mathbf{x}) = f(\mathbf{x})/g(\mathbf{x})$ is differentiable at $\mathbf{x}_0$ and*

$$J_h(\mathbf{x}_0) = \frac{g(\mathbf{x}_0)J_f(\mathbf{x}_0) - f(\mathbf{x}_0)J_g(\mathbf{x}_0)}{[g(\mathbf{x}_0)]^2}.$$

Example 4.1.1 Find the derivative of $h(x, y, z) = \frac{x^2+y^2+z^2}{x^2+1}$.

Let us first do it by direct differentiation:

$$J_h(x, y, z) = \left(\frac{2x(1 - y^2 - z^2)}{(x^2 + 1)^2}, \frac{2y}{x^2 + 1}, \frac{2z}{x^2 + 1} \right).$$

Now, $h(x, y, z) = f(x, y, z)/g(x, y, z)$, where $f(x, y, z) = x^2 + y^2 + z^2$ and $g(x, y, z) = x^2 + 1$. We can apply the quotient rule to obtain

$$J_h(x, y, z) = \frac{(x^2 + 1)(2x, 2y, 2z) - (x^2 + y^2 + z^2)(2x, 0, 0)}{(x^2 + 1)^2},$$

which is the same as what we obtained directly.

4.2 Chain Rule (MT 2.5)

If f is a real-valued function of one variable, written as $z = f(y)$, and y is a function of x, written $y = g(x)$, then z becomes a function of x through substitution, namely, $z = f(g(x))$, and we have the familiar chain rule:

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx} = f'(g(x))g'(x). \tag{4.1}$$

We now write the generalization of the chain rule for functions of several variables:

Theorem 4.2.1 (Chain Rule) *Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ be open sets. Let $g : U \subset \mathbb{R}^n \longrightarrow \mathbb{R}^m$ and $f : V \subset \mathbb{R}^m \longrightarrow \mathbb{R}^p$ be given functions such that g maps U into V, so that $f \circ g$ is defined. Suppose g is differentiable at $\mathbf{x}_0$ and f is differentiable at $\mathbf{y}_0 = g(\mathbf{x}_0)$. Then $f \circ g$ is differentiable at $\mathbf{x}_0$ and*

$$J_{f \circ g}(\mathbf{x}_0) = J_f(\mathbf{y}_0)J_g(\mathbf{x}_0).$$

In order to prove this theorem in full, we will first see two special cases.

Example 4.2.1 (First special case of the Chain Rule) Suppose $c : \mathbb{R} \longrightarrow \mathbb{R}^3$ is a differentiable path and $f : \mathbb{R}^3 \longrightarrow \mathbb{R}$. Let $h(t) = f(c(t)) = f(x(t), y(t), z(t))$, where $c(t) = (x(t), y(t), z(t))$. Then

$$\frac{dh}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt}.$$

That is,

$$\frac{dh}{dt} = \nabla f(c(t)) \cdot c'(t) = J_f(c(t))J_c(t),$$

where $c'(t) = (x'(t), y'(t), z'(t))$.

Proof. By definition,

$$\frac{dh}{dt}(t_0) = \lim_{t \to t_0} \frac{h(t) - h(t_0)}{t - t_0} = \lim_{t \to t_0} \frac{f(x(t), y(t), z(t)) - f(x(t_0), y(t_0), z(t_0))}{t - t_0}.$$

With some algebra, we have

$$\begin{aligned} \frac{f(x(t), y(t), z(t)) - f(x(t_0), y(t_0), z(t_0))}{t - t_0} = & \frac{f(x(t), y(t), z(t)) - f(x(t_0), y(t), z(t))}{t - t_0} \\ + & \frac{f(x(t_0), y(t), z(t)) - f(x(t_0), y(t_0), z(t))}{t - t_0} \\ + & \frac{f(x(t_0), y(t_0), z(t)) - f(x(t_0), y(t_0), z(t_0))}{t - t_0} \end{aligned}$$

Now, applying the mean-value theorem for one-variable functions[1] we obtain

$$\begin{aligned} f(x(t), y(t), z(t)) - f(x(t_0), y(t), z(t)) &= \left[\frac{\partial f}{\partial x}(c, y(t), z(t))\right](x(t) - x(t_0)), \\ f(x(t_0), y(t), z(t)) - f(x(t_0), y(t_0), z(t)) &= \left[\frac{\partial f}{\partial y}(x(t_0), d, z(t))\right](y(t) - y(t_0)), \\ f(x(t_0), y(t_0), z(t)) - f(x(t_0), y(t_0), z(t_0)) &= \left[\frac{\partial f}{\partial z}(x(t_0), y(t_0), e)\right](z(t) - z(t_0)), \end{aligned}$$

where $c \in (x(t_0), x(t))$, $d \in (y(t_0), y(t))$ and $e \in (z(t_0), z(t))$.

1: The mean-value theorem from one-variable calculus states that if $f : [a, b] \to \mathbb{R}$ is continuous in $[a, b]$ and is differentiable in the open interval (a, b), then there is a point c in (a, b) such that $f(b) - f(a) = f'(c)(b - a)$.

Assuming that the partial derivatives are continuous and taking the limit, we have

$$\begin{aligned}\frac{dh}{dt}(t_0) &= \frac{\partial f}{\partial x}(c, y(t), z(t)) \lim_{t\to t_0} \frac{x(t)-x(t_0)}{t-t_0} + \\ &+ \frac{\partial f}{\partial y}(x(t_0), d, z(t)) \lim_{t\to t_0} \frac{y(t)-y(t_0)}{t-t_0} + \\ &+ \frac{\partial f}{\partial z}(x(t_0), y(t_0), e) \lim_{t\to t_0} \frac{z(t)-z(t_0)}{t-t_0} \\ &= \left[\frac{\partial f}{\partial x}(x,y,z)\right] x'(t_0) + \left[\frac{\partial f}{\partial y}(x,y,z)\right] y'(t_0) + \left[\frac{\partial f}{\partial z}(x,y,z)\right] z'(t_0),\end{aligned}$$

as desired. □

Example 4.2.2 (Second Special Case of the Chain Rule) Let $f : \mathbb{R}^3 \longrightarrow \mathbb{R}$ and let $g : \mathbb{R}^3 \longrightarrow \mathbb{R}^3$. Write

$$g(x,y,z) = (u(x,y,z), v(x,y,z), w(x,y,z))$$

and define $h = f \circ g : \mathbb{R}^3 \longrightarrow \mathbb{R}$ by setting

$$h(x,y,z) = f(u(x,y,z), v(x,y,z), w(x,y,z)).$$

In this case, the chain rule states that

$$\begin{pmatrix} \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} & \frac{\partial h}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} & \frac{\partial f}{\partial w} \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}$$

or, in other words, that

$$J_h(x,y,z) = J_f(u,v,w) J_g(x,y,z).$$

Proof. By definition, $\frac{\partial h}{\partial x}$ is obtained by differentiating h with respect to x, holding y and z fixed. But then $(u(x,y,z), v(x,y,z), w(x,y,z))$ may be regarded as a vector function of the single variable x. The first special case applies to this situation and, after the variables are renamed, gives

$$\frac{\partial h}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial x}$$

$$\frac{\partial h}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial y}$$

$$\frac{\partial h}{\partial z} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial z} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial z} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial z}$$

as desired. □

Proof of the Chain Rule. The general case may be proved in two steps. First, we generalize the first special case to m variables; that is, for $f(x_1, \ldots, x_m)$ and $c(t) = (x_1(t), \ldots, x_m(t))$, we have

$$\frac{dh}{dt} = \sum_{i=1}^{m} \frac{\partial f}{\partial x_i}\frac{dx_i}{dt}, \tag{4.2}$$

where $h(t) = f(x_1(t), \dots, x_m(t))$. Second, the result obtained in the first step is used to obtain the formula

$$\frac{\partial h_j}{\partial x_i} = \sum_{k=1}^{m} \frac{\partial f_j}{\partial y_k} \frac{\partial y_k}{\partial x_i}, \tag{4.3}$$

where $f = (f_1, \dots, f_p)$ is a vector function of arguments $y_1, \dots, y_m$; $g(x_1, \dots, x_n) = (y_1(x_1, \dots, x_n), \dots, y_m(x_1, \dots, x_n))$; and $h_j(x_1, \dots, x_n) = f_j(y_1(x_1, \dots, x_n), \dots, y_m(x_1, \dots, x_n))$. (Using the letter y for both functions and arguments is an abuse of notation, but it can help us remember the formula.) □

Example 4.2.3 Verify the chain rule for $f(u, v, w) = u^2 + v^2 - w$, where $u(x, y, z) = x^2 y$, $v(x, y, z) = y^2$, and $w(x, y, z) = e^{-xz}$.

Here,

$$\begin{aligned} h(x, y, z) &= f(u(x, y, z), v(x, y, z), w(x, y, z)) \\ &= (x^2 y)^2 + y^4 - e^{-xz} = x^4 y^2 + y^4 - e^{-xz}. \end{aligned}$$

Thus, differentiating directly,

$$\frac{\partial h}{\partial x} = 4x^3 y^2 + z e^{-xz}.$$

On the other hand, using the chain rule,

$$\begin{aligned} \frac{\partial h}{\partial x} &= \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial f}{\partial w}\frac{\partial w}{\partial x} \\ &= 2u(2xy) + 2v \cdot 0 + (-1)(-z e^{-xz}) \\ &= (2x^2 y)(2xy) + z e^{-xz}, \end{aligned}$$

which is the same as the preceding equation.

Example 4.2.4 Given $g(x, y) = (x^2 + 1, y^2)$ and $f(u, v) = (u + v, u, v^2)$, compute the derivative of $f \circ g$.

The matrices of partial derivatives are

$$J_f(u, v) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 2v \end{pmatrix}, \qquad J_g(x, y) = \begin{pmatrix} 2x & 0 \\ 0 & 2y \end{pmatrix}.$$

Now,

$$J_{f \circ g}(x, y) = J_f(g(x, y)) J_g(x, y) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 2y^2 \end{pmatrix} \begin{pmatrix} 2x & 0 \\ 0 & 2y \end{pmatrix} = \begin{pmatrix} 2x & 2y \\ 2x & 0 \\ 0 & 4y^3 \end{pmatrix},$$

which is the same matrix we would obtain by directly differentiating $f \circ g = (x^2 + y^2 + 1, x^2 + 1, y^4)$.

4.3 Directional Derivatives (MT 2.6)

Suppose $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ is a real-valued function. Let $\mathbf{v}$ and $\mathbf{x} \in \mathbb{R}^n$ be fixed vectors and consider the function from $\mathbb{R}$ to $\mathbb{R}$ defined by $t \mapsto f(\mathbf{x} + t\mathbf{v})$. The set of points of the form $\mathbf{x} + t\mathbf{v}$, $t \in \mathbb{R}$, is the line L through the point $\mathbf{x}$ parallel to the vector $\mathbf{v}$.

The function $t \mapsto f(\mathbf{x} + t\mathbf{v})$ represents the function f restricted to the line L. We may ask: How fast are the values of f changing along the line L at the point $\mathbf{x}$? The answer to this question is the value of the derivative of this function of t at $t = 0$. This would be the derivative of f at the point $\mathbf{x}$ in the direction of L; that is, of $\mathbf{v}$. We can formalize this concept as follows.

Definition 4.3.1 (Directional Derivatives) *If $f : \mathbb{R}^n \longrightarrow \mathbb{R}$, the directional derivative of f at $\mathbf{x}$ along the vector $\mathbf{v}$, $f'_{\mathbf{v}}(\mathbf{x})$, is given by*

$$f'_{\mathbf{v}}(\mathbf{x}) = \frac{d}{dt} f(\mathbf{x} + t\mathbf{v})\bigg|_{t=0} = \lim_{t \to 0} \frac{f(\mathbf{x} + t\mathbf{v}) - f(\mathbf{x})}{t}$$

if this exists.

In the definition of a directional derivative, we normally choose $\mathbf{v}$ to be a unit vector, as this will give us a better sense of how fast f is changing. Suppose that f measures the temperature in degrees and that we are interested in how fast the temperature changes as we move in a particular direction. If we are measuring distance in meters, then the rate of change of temperature will be measured in degrees per meter. Such a relation is going to hold only when $\mathbf{v}$ is a unit vector, reflecting the fact that we are going ahead by one meter.

If you think about it, the directional derivative is a generalization of the partial derivatives but taking the limit along a direction $\mathbf{v}$ instead of taking only the directions $\mathbf{e}_j = (0, \ldots, 1, \ldots, 0)$.

Theorem 4.3.1 *If $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ is differentiable, then all directional derivatives exist. The directional derivative at $\mathbf{x}$ in the direction $\mathbf{v} \in \mathbb{R}^n$ is given by*

$$J_f(\mathbf{x})\mathbf{v} = \nabla f(\mathbf{x}) \cdot \mathbf{v}.$$

Proof. Let $\mathbf{c}(t) = \mathbf{x} + t\mathbf{v}$, so that $f(\mathbf{x} + t\mathbf{v}) = f(\mathbf{c}(t))$. By the chain rule, $\frac{d}{dt} f(\mathbf{c}(t)) = \nabla f(\mathbf{c}(t)) \cdot \mathbf{c}'(t)$. However, $\mathbf{c}(0) = \mathbf{x}$ and $\mathbf{c}'(0) = \mathbf{v}$, and so

$$\frac{d}{dt} f(\mathbf{x} + t\mathbf{v})\bigg|_{t=0} = \nabla f(\mathbf{x}) \cdot \mathbf{v},$$

as desired. □

Note that we don't have to restrict ourselves to straight lines: for a general path $\mathbf{c}(t)$ with $\mathbf{c}(0) = \mathbf{x}$ and $\mathbf{c}'(0) = \mathbf{v}$, we have from the chain rule,

$$\frac{d}{dt} f(\mathbf{c}(t))\bigg|_{t=0} = \nabla f(\mathbf{c}(t)) \cdot \mathbf{c}'(t) = \nabla f(\mathbf{x}) \cdot \mathbf{v}.$$

Example 4.3.1 Let $f(x,y,z) = x^2 e^{-yz}$. Compute the rate of change of f in the direction of the unit vector

$$\mathbf{v} = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

at the point $(1,0,0)$.

The required rate of change is, using Theorem 4.3.1,

$$\nabla f \cdot \mathbf{v} = (2xe^{-yz}, -x^2 z e^{-yz}, -x^2 y e^{-yz}) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right),$$

which, at the point $(1,0,0)$, becomes

$$f'_{\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)}(1,0,0) = ((2,0,0) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = \frac{2}{\sqrt{3}}.$$

Gradients and Directional Derivatives

The definition of directional derivatives is helpful to better understand what the gradient means.

Theorem 4.3.2 *Assume $\nabla f(\mathbf{x}) \neq 0$. Then $\nabla f(\mathbf{x})$ points in the direction along which f is increasing the fastest.*

Proof. If $\mathbf{n}$ is a unit vector, the rate of change of f in direction $\mathbf{n}$ is given by $\nabla f(\mathbf{x}) \cdot \mathbf{n} = \|\nabla f(\mathbf{x})\| \cos\theta$, where θ is the angle between $\mathbf{n}$ and $\nabla f(\mathbf{x})$. This is maximum when $\theta = 0$; that is, when $\mathbf{n}$ and ∇f are parallel. [If $\nabla f(\mathbf{x}) = 0$ this rate of change is 0 for any $\mathbf{n}$.] $\square$

Theorem 4.3.3 (The Gradient is Normal to Level Sets) *Let $f : \mathbb{R}^n \longrightarrow \mathbb{R}$ be differentiable and let $\mathbf{x}_0$ lie on the level set S defined by $f(\mathbf{x}) = k \in \mathbb{R}$. Then $\nabla f(\mathbf{x}_0)$ is normal to the level set in the following sense: if $\mathbf{v}$ is the tangent vector at $t = 0$ of a path $\mathbf{c}(t)$ in S with $\mathbf{c}(0) = \mathbf{x}_0$, then $\nabla f(\mathbf{x}_0) \cdot \mathbf{v} = 0$.*

Proof. Let $\mathbf{c}(t)$ lie in S; then $f(\mathbf{c}(t)) = k$. Let $\mathbf{v} = \mathbf{c}'(0)$ be the tangent vector to $\mathbf{c}(t)$ at $\mathbf{x}_0 = \mathbf{c}(0)$. Hence, the fact that $f(\mathbf{c}(t))$ is constant in t and the chain rule give

$$0 = \frac{d}{dt} f(\mathbf{c}(t))\bigg|_{t=0} = \nabla f(\mathbf{c}(0)) \cdot \mathbf{v}.$$

$\square$

As a consequence to the previous theorem, we can define the tangent plane to a level set S as the orthogonal plane to the gradient.

Definition 4.3.2 (Tangent Planes to Level Sets) *Let S be the set consisting of those $\mathbf{x}$ such that $f(\mathbf{x}) = k$. The tangent plane of S at a point $\mathbf{x}_0 \in S$ is* *defined by the equation*

$$\nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0) = 0$$

if $\nabla f(\mathbf{x}_0) \neq 0$. *That is, the tangent plane is the set of points* $\mathbf{x}$ *that satisfy this equation.*

Example 4.3.2 Compute the equation of the plane tangent to the surface defined by $3xy + z^2 = 4$ at $(1, 1, 1)$.

Here $f(x, y, z) = 3xy + z^2$ and $\nabla f = (3y, 3x, 2z)$, which at $(1, 1, 1)$ is the vector $(3, 3, 2)$. Thus, the tangent plane is

$$(3, 3, 2) \cdot (x - 1, y - 1, z - 1) = 0;$$

that is,

$$3x + 3y + 2z = 8.$$

The relationship between this way to calculate the tangent plane and our previous definition is the following. Suppose we have a level surface $f(x, y, z) = 0$[2] and we can solve for z[3] then we will have an equation of the type $z = g(x, y)$ where $f(x, y, z) = g(x, y) - z$. Therefore,

2: If $c \neq 0$ we can always redefine $h = f - c$ to have $h = 0$.

3: This is not always possible.

$$\nabla f(x, y, z) = \left(\frac{\partial g}{\partial x}(x, y), \frac{\partial g}{\partial y}(x, y), -1\right)$$

and

$$\nabla f(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0) = \frac{\partial g}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial g}{\partial x}(x, y)(y - y_0) - (z - z_0),$$

which implies

$$z = z_0 + \nabla g(x_0, y_0) \cdot (x - x_0, y - y_0).$$

Since $z_0 = g(x_0, y_0)$ this is exactly the definition of tangent plane we had given when a surface is written as a graph of a function $g(x, y)$.

Exercises

Exercise 4.1 Find the directional derivative at the given point and along the given direction:

1. $f(x, y) = x^2 + y^3$ at $(1, 1)$ along the direction $(1, -1)$.
2. $f(x, y) = x + \sin(x + y)$ at $(0, 0)$ along the direction $(2, 1)$.
3. $f(x, y) = xe^y - ye^x$ at $(1, 0)$ along the direction $(3, 4)$.
4. $f(x, y) = \frac{3x}{x-y}$ at $(1, 0)$ along the direction $(1, -\sqrt{3})$.
5. $f(x, y, z) = x^2y + y^2z + z^2x$ at $(1, -1, 1)$ along the direction $(1, -1, 2)$.
6. $f(x, y) = x^2y + xy^2$ at $(1, 1)$ along the direction $(-1, 3)$.
7. $f(x, y, z) = \log\sqrt{x^2 + y^2 + z^2}$ at $(2, 0, 1)$ along the direction $(1, 2, 0)$.
8. $f(x, y, z) = e^x \cos(yz)$ at $(0, 0, 0)$ along the direction $(2, 1, -2)$.
9. $f(x, y) = 2x^3y - 3y^2$ at $(2, 1)$ along the direction $(a, \sqrt{1 - a^2})$. Find a in order to get a maximum.
10. $f(x, y, z) = xy + yz + zx$ at $(1, 1, 2)$ along the direction $(10, -1, 2)$.

Exercise 4.2 Consider the function $f(x, y) = e^{x+2y}$. Find the set of points $(x, y) \in \mathbb{R}^2$ such that the directional derivative of f at the point (x, y) along the direction $(4, 3)$ equals the value $2e$.

Exercise 4.3 Consider the function $f(x, y) = 1 + \sin(3x + y)$. Is there any direction $v \in \mathbb{R}^2$ such that the directional derivative of f at $(0, 0)$ along the direction v has the value 1?

Exercise 4.4 Consider the function $f(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$.

1. Determine the direction along which the directional derivative of f at $(1, 1)$ vanishes.
2. The same question as before for an arbitrary point (x_0, y_0) in the first quadrant.
3. Use the results in the preceding item to describe the level curves of f.

Exercise 4.5 Find the equation of the tangent plane for the following surfaces at the prescribed points (x_0, y_0, z_0):

1. $x^2 + y^2 + z^2 = 3$, $(x_0, y_0, z_0) = (1, 1, 1)$.
2. $x^3 - 2y^3 + z^3 = 0$, $(x_0, y_0, z_0) = (1, 1, 1)$.
3. $e^z \cos x \cos y = 0$, $(x_0, y_0, z_0) = (\pi/2, 1, 0)$.
4. $e^{xyz} = 1$, $(x_0, y_0, z_0) = (1, 2, 0)$.

Exercise 4.6 Consider the following functions:

$$
\begin{aligned}
f(x, y) &= \left(\tan\frac{y}{x} - x + y, \log\frac{y+1}{x}\right), \\
g(t, s) &= (t\cos s, e^t, s - 2t), \\
h(u, v, w) &= uv^2e^w, \\
F &= h \circ g \circ f.
\end{aligned}
$$

1. Compute Df, Dg, Dh, and $DF(1, 0)$.
2. Compute the tangent plane to the graph of F at the point $(1, 0, F(1, 0))$.

3. Compute the directional derivative of F at that point along the direction $\alpha = (3, -4)$.

Exercise 4.7 The temperature of a metal plate is given by the function $T(x, y) = e^x \cos y + e^y \cos x$.

1. In what direction from $(0, 0)$ does the temperature increase the fastest?
2. In what direction from $(0, 0)$ does the temperature decrease the fastest?

Exercise 4.8 The mass density of a metal ball centered at the origin is defined in terms of the function

$$\rho(x, y, z) = ke^{-(x^2+y^2+z^2)}, \quad k \text{ a positive constant.}$$

1. In what direction from the point (x, y, z) does the density increase the fastest?
2. Describe the level surfaces of the density function.

Exercise 4.9 1. The function $h(x, y) = 2e^{-x^2} + e^{-3y^2}$ describes the height of a mountain at $(x, y) \in \mathbb{R}^2$. In what direction from $(1, 0)$ must one start walking in order to climb the mountain as fast as possible?
2. Assume that the temperature at a point $(x, y, z) \in \mathbb{R}^3$ is given by the function $T(x, y, z) = e^{-x} + e^{-2y} + e^{-3z}$. In what direction from $(1, 1, 1)$ must one start walking in order to cool down as fast as possible?

Exercise 4.10 1. Given the functions $F(x, y) = (x^2+1, y^2)$ and $G(u, v) = (u + v, u, v^2)$, compute $D(G \circ F)(1, 1)$.
2. Compute using the chain rule:
   a) $\dfrac{dh}{dx}$ where $h(x) = f(x, u(x), v(x))$.
   b) $\dfrac{\partial h}{\partial x}$ where $h(x, y, z) = f(u(x, y, z), v(x, y), w(x))$.
   c) $\dfrac{\partial h}{\partial z}$ where $h(x, y, z) = f(u(x, w(y, z)), v(w(y, z), z))$.
3. Given $f \in C^1(\mathbb{R})$ we define $z(x, y) = f(x + y) + f(x - y)$. Compute $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
4. Given the functions $u = \log(x + y)$, $v = \arctan(x/y)$, and $z = e^{uv}$, compute $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$.
5. Compute $h'(0)$ if $h = f \circ s$, where

$$f(x, y, z) = \frac{\log(1 + x^2 + 2z^2)}{1 + y^2}, \quad s(t) = (t + 1, 1 - t^2, \sin t)$$

.

Exercise 4.11 Write in polar coordinates the following expression:

$$x\frac{\partial u}{\partial x} - y\frac{\partial u}{\partial y}.$$

Exercise 4.12 The position of a particle in $\mathbb{R}^3$ is given by $c : \mathbb{R} \to \mathbb{R}^3$,

$$c(t) = (e^{t^2}, e^{-t^2}, \log(1 + t^2)).$$

Consider also the potential function $V : \mathbb{R}^3 \to \mathbb{R}$ given by $V(x, y, z) = (xy)^2$ and its restriction to the trajectory, $h(t) = (V \circ c)(t)$. Compute $h'(t)$ directly and also using the chain rule.

Exercise 4.13 Consider a particle of mass m which moves along the trajectory $r(t)$ in $\mathbb{R}^3$ according to Newton's law in a force field $F = -\nabla V$ for a given potential energy V.

1. Show that the total energy (i.e., potential + kinetic energy)

$$E(t) = \frac{m}{2}||r'(t)||^2 + V(r(t))$$

   is constant in time.
2. Show that if the particle moves within an equipotential surface, then the modulus of the velocity is constant.

Exercise 4.14 Show that the following trajectories satisfy the relation $s'(t) = F(s(t))$ for the corresponding fields (that is, they are flux lines):

1. $s(t) = (t^2, 2t - 1, \sqrt{t})$, $\quad F(x, y, z) = (y + 1, 2, 1/2z)$.
2. $s(t) = \left(e^{2t}, 1/t, \log t\right)$, $\quad F(x, y, z) = (2x, -y^2, y)$.
3. $s(t) = \left(\dfrac{1}{1-t}, \log t, \dfrac{e^t}{1-t}\right)$, $\quad F(x, y, z) = \left(x^2, \dfrac{x}{x-1}, z(1 + x)\right)$.

Part II. Local Properties of Functions

Higher-order derivatives. Differential Operators

5

5.1 Iterated Partial Derivatives (MT 3.1)

In this section, we proceed to study higher-order derivatives, with the goal of proving the equality of the "mixed second partial derivatives" of a function.

Let $f : \mathbb{R}^n \to \mathbb{R}$. If $\partial f / \partial x_j$ exist and are continuous for all $j = 1, \dots, n$ we say f is of class C^1. If these derivatives, in turn, have continuous partial derivatives, we say that f is of class C^2, or is twice continuously differentiable, and so on with class C^3 and further. If a function is infinitely differentiable we say it is of class C^∞.

Some examples of iterated partial derivatives are:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right), \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right), \quad \frac{\partial^3 f}{\partial z \partial y \partial x} = \frac{\partial}{\partial z}\left[\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)\right],$$

Example 5.1.1 Find all second partial derivatives of $f(x, y) = xy + (x + 2y)^2$.

The first partials are

$$\frac{\partial f}{\partial x} = y + 2(x + 2y), \quad \frac{\partial f}{\partial y} = x + 4(x + 2y).$$

Now differentiate each of these expressions with respect to x and y:

$$\frac{\partial^2 f}{\partial x^2} = 2, \quad \frac{\partial^2 f}{\partial x \partial y} = 5, \quad \frac{\partial^2 f}{\partial y^2} = 8, \quad \frac{\partial^2 f}{\partial y \partial x} = 5.$$

Note that $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$. It is a basic and perhaps surprising fact that this is always the case for C^2 functions:

Theorem 5.1.1 (Equality of Mixed Partials) *If $f : \mathbb{R}^n \to \mathbb{R}$ is of class C^2 (is twice continuously differentiable), then the mixed partial derivatives are equal; that is,*

$$\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i},$$

for all i and j.

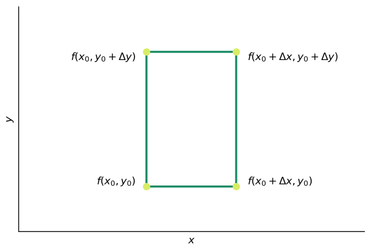


Figure 5.1: The algebra behind the equality of mixed partials: writing the difference of differences in two ways.

Proof. For simplicity of notation, we will use x and y instead of x_i and x_j. This is the idea behind the proof. We can approximate $\partial f / \partial x$ by

$$\frac{\partial f}{\partial x}(x_0, y_0) \approx \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x}$$

and use this to approximate $\partial^2 f/\partial y\partial x$:

$$\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}(x_0,y_0)\right) \approx \frac{\partial f/\partial x(x_0,y_0+\Delta y)-\partial f/\partial x(x_0,y_0)}{\Delta y}$$
$$\approx \frac{f(x_0+\Delta x,y_0+\Delta y)-f(x_0,y_0+\Delta y)-f(x_0+\Delta x,y_0)+f(x_0,y_0)}{\Delta x\Delta y}.$$

We could also approximate first $\partial f/\partial y$ by

$$\frac{\partial f}{\partial y}(x_0,y_0) \approx \frac{f(x_0,y_0+\Delta y)-f(x_0,y_0)}{\Delta y}$$

and use this to approximate $\partial^2 f/\partial x\partial y$:

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}(x_0,y_0)\right) \approx \frac{\partial f/\partial y(x_0+\Delta x,y_0)-\partial f/\partial y(x_0,y_0)}{\Delta x}$$
$$\approx \frac{f(x_0+\Delta x,y_0+\Delta y)-f(x_0+\Delta x,y_0)-f(x_0,y_0+\Delta y)+f(x_0,y_0)}{\Delta x\Delta y},$$

which is clearly the same expression. Intuitively, if we make Δx and Δy very small, the approximatiomn will get better and better and the two mixed partial derivatives will become equal. The formal proof is as follows.

Let us define (see Figure 5.1)

$$u(\Delta x,\Delta y) = f(x_0+\Delta x,y_0+\Delta y)-f(x_0+\Delta x,y_0),$$
$$v(\Delta x,\Delta y) = f(x_0+\Delta x,y_0+\Delta y)-f(x_0,y_0+\Delta y),$$
$$w(\Delta x,\Delta y) = f(x_0+\Delta x,y_0+\Delta y)-f(x_0+\Delta x,y_0)-f(x_0,y_0+\Delta y)+f(x_0,y_0).$$

Notice that $w(\Delta x,\Delta y) = u(\Delta x,\Delta y)-u(0,\Delta y)$ and also $w(\Delta x,\Delta y) = v(\Delta x,\Delta y)-v(\Delta x,0)$. Now, using the mean value for functions of one variable, assuming that f is twice continuously differentiable, and taking $h\in(0,\Delta x)$, we can write

$$w(\Delta x,\Delta y) = u(\Delta x,\Delta y)-u(0,\Delta y) = \Delta x\frac{\partial u}{\partial x}(h,\Delta y)$$
$$= \Delta x\left(\frac{\partial f}{\partial x}(x_0+h,y_0+\Delta y)-\frac{\partial f}{\partial x}(x_0+h,y_0)\right),$$

but we can apply the mean value theorem again to $\partial f/\partial x$, taking $k\in(0,\Delta y)$, to obtain

$$w(\Delta x,\Delta y) = \Delta x\Delta y\frac{\partial^2 f}{\partial y\partial x}(x_0+h,y_0+k).$$

Similarly, we obtain

$$w(\Delta x,\Delta y) = v(\Delta x,\Delta y)-v(\Delta x,0) = \Delta y\frac{\partial v}{\partial y}(\Delta x,k)$$
$$= \Delta y\left(\frac{\partial f}{\partial y}(x_0+\Delta x,y_0+k)-\frac{\partial f}{\partial x}(x_0,y_0+k)\right)$$
$$= \Delta x\Delta y\frac{\partial^2 f}{\partial x\partial y}(x_0+h,y_0+k).$$

That is, we have

$$\Delta x \Delta y \frac{\partial^2 f}{\partial y \partial x}(x_0 + h, y_0 + k) = \Delta x \Delta y \frac{\partial^2 f}{\partial x \partial y}(x_0 + h, y_0 + k)$$

and, since $\Delta x, \Delta y \neq 0$, this is equal to

$$\frac{\partial^2 f}{\partial y \partial x}(x_0 + h, y_0 + k) = \frac{\partial^2 f}{\partial x \partial y}(x_0 + h, y_0 + k).$$

Taking the limit $(h, k) \to (0, 0)$ yields the desired result. □

Example 5.1.2 Verify the equality of the mixed second partial derivatives for the function $f(x, y) = xe^y + yx^2$.

Here

$$\frac{\partial f}{\partial x} = e^y + 2xy, \quad \frac{\partial f}{\partial y} = xe^y + x^2,$$
$$\frac{\partial^2 f}{\partial y \partial x} = e^y + 2x, \quad \frac{\partial^2 f}{\partial x \partial y} = e^y + 2x,$$

and so we have

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}.$$

Sometimes the notation f_x, f_y, f_z is used for the partial derivatives: $f_x = \frac{\partial f}{\partial x}$, and so on. With this notation, we write $f_{xy} = (f_x)_y$, and so equality of the mixed partials is denoted by $f_{xy} = f_{yx}$. Notice that $f_{xy} = \frac{\partial^2 f}{\partial y \partial x}$, so the order of x and y is reversed in the two notations. The following example illustrates this subscript notation.

Example 5.1.3 Let $k(s, t) = f(x(s, t), y(s, t))$ for some functions f, x and y. Calculate k_{st}.

By the chain rule,

$$k_s = f_x x_s + f_y y_s.$$

Differentiating in t using the product rules gives

$$k_{st} = (f_x)_t x_s + f_x (x_s)_t + (f_y)_t y_s + f_y (y_s)_t.$$

Applying the chain rule again to $(f_x)_t$ and $(f_y)_t$ gives

$$(f_x)_t = f_{xx} x_t + f_{xy} y_t$$

and

$$(f_y)_t = f_{yx} x_t + f_{yy} y_t,$$

and so k_{st} becomes

$$\begin{aligned} k_{st} &= (f_{xx} x_t + f_{xy} y_t) x_s + f_x x_{st} + (f_{yx} x_t + f_{yy} y_t) y_s + f_y y_{st} \\ &= f_{xx} x_t x_s + f_{xy}(y_t x_s + y_s x_t) + f_{yy} y_t y_s + f_x x_{st} + f_y y_{st}. \end{aligned}$$

Notice that this last formula is symmetric in (s, t), verifying the equality $k_{st} = k_{ts}$.

5.2 Vector Fields (MT 4.3)

We now turn our attention to some very common differential operators that are important for physical applications: the divergence, the curl and the laplacian. First, we need to introduce vector fields:

Definition 5.2.1 (Vector Fields) *A vector field in $\mathbb{R}^n$ is a map $F : A \subset \mathbb{R}^n \longrightarrow \mathbb{R}^n$ that assigns to each point $\mathbf{x} \in A$ a vector $\mathbf{F}(\mathbf{x})$.*

Note that the gradient ∇f is a vector field.

Example 5.2.1 Some forms of rotary motion (such as the motion of particles on a turntable) can be described by the vector field

$$V(x, y) = -y\mathbf{i} + x\mathbf{j},$$

where $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$. See Figure 5.2.

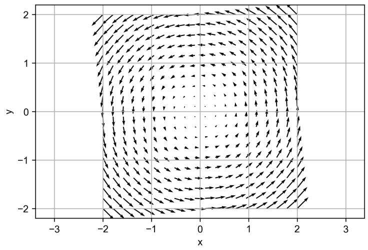


Figure 5.2: A rotary vector field.

5.3 Divergence (MT 4.4)

Definition 5.3.1 (Divergence) *Let $\mathbf{F} = (F_1, \dots, F_n)$ be a vector field on $\mathbb{R}^n$. We define its divergence as*

$$div\mathbf{F} = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i} = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} + \cdots + \frac{\partial F_n}{\partial x_n}.$$

Example 5.3.1 Compute the divergence of $\mathbf{F} = x^2y\mathbf{i} + z\mathbf{j} + xyz\mathbf{k}$.

$$\text{div}\mathbf{F} = \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(xyz) = 2xy + 0 + xy = 3xy.$$

The divergence has an important physical interpretation. If we imagine $\mathbf{F}$ to be the velocity field of a fluid, then div$\mathbf{F}$ represents the rate of expansion per unit volume under the flow of the fluid. If div$\mathbf{F} < 0$, the fluid is compressing. We will see a detailed proof of this when we see the Gauss's theorem in Chapter 14.

Example 5.3.2 The flow lines of the vector field $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$ are straight lines directed away from the origin.

If these flow lines are those of a fluid, the fluid is expanding as it moves out from the origin, so div$\mathbf{F}$ should be positive. In fact,

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}x + \frac{\partial}{\partial y}y = 2.$$

Example 5.3.3 As we saw in Example 5.2.1, the flow lines of $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$ are concentric circles about the origin, moving counterclockwise (see Figure 5.2). From this figure, it appears that the fluid is neither compressing nor expanding. This is confirmed by calculating

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(-y) + \frac{\partial}{\partial y}(x) = 0 + 0 = 0.$$

5.4 Curl (MT 4.4)

Definition 5.4.1 (Curl of a Vector Field) *If* $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$*, the curl of* $\mathbf{F}$ *is the vector field*

$$\begin{aligned} \text{curl } \mathbf{F} = \nabla \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix} \\ &= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right)\mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\mathbf{k}. \end{aligned}$$

Example 5.4.1 Find the curl of $\mathbf{F}(x, y, z) = x\mathbf{i} + xy\mathbf{j} + \mathbf{k}$.

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & xy & 1 \end{vmatrix} = (0-0)\mathbf{i} - (0-0)\mathbf{j} + (y-0)\mathbf{k}.$$

Thus, $\nabla \times \mathbf{F} = y\mathbf{k}$.

The physical significance of the curl will be discussed when we study Stokes' theorem in Chapter 13.

Gradients are Curl Free

Theorem 5.4.1 (Curl of a Gradient) *For any* C^2 *function* f*,*

$$\nabla \times (\nabla f) = \mathbf{0}.$$

That is, the curl of any gradient is the zero vector.

Proof. Because $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z}\right)$ we have, by definition,

$$\begin{aligned} \nabla \times \nabla f &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} \\ &= \left(\frac{\partial^2 f}{\partial y \partial z} - \frac{\partial^2 f}{\partial z \partial y}\right)\mathbf{i} + \left(\frac{\partial^2 f}{\partial z \partial x} - \frac{\partial^2 f}{\partial x \partial z}\right)\mathbf{j} + \left(\frac{\partial^2 f}{\partial x \partial y} - \frac{\partial^2 f}{\partial y \partial x}\right)\mathbf{k}. \end{aligned}$$

Each component is zero because of the equality of mixed partial derivatives. □

Example 5.4.2 Let $\mathbf{V}(x, y, z) = y\mathbf{i} - x\mathbf{j}$. Show that $\mathbf{V}$ is not a gradient field.

If $\mathbf{V}$ were a gradient field, then it would satisfy curl $\mathbf{V} = \mathbf{0}$ by Theorem 5.4.1. But

$$\text{curl } \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & -x & 0 \end{vmatrix} = -2\mathbf{k} \neq \mathbf{0},$$

so $\mathbf{V}$ cannot be a gradient.

Divergence and Curl

Theorem 5.4.2 (Divergence of a Curl) *For any C^2 vector field* $\mathbf{F}$,

$$\textit{div curl } \mathbf{F} = \nabla \cdot (\nabla \times \mathbf{F}) = 0.$$

That is, the divergence of any curl is zero.

As with the curl of a gradient, the proof rests on the equality of the mixed partial derivatives.

Example 5.4.3 Show that the vector field $\mathbf{V}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ cannot be the curl of some vector field $\mathbf{F}$; that is, there is no $\mathbf{F}$ with $\mathbf{V} = \text{curl } \mathbf{F}$.

If this were so, then div $\mathbf{V}$ would be zero by Theorem 5.4.2. But

$$\text{div } \mathbf{V} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3,$$

so $\mathbf{V}$ cannot be curl $\mathbf{F}$ for any $\mathbf{F}$.

5.5 Laplacian (MT 4.4)

Definition 5.5.1 (Laplacian) *Let $f : \mathbb{R}^n \to \mathbb{R}$. Then the Laplacian of f, denoted by Δf, $\nabla^2 f$ o $\nabla \cdot \nabla f$, is given by*

$$\Delta f = \nabla^2 f = \nabla \cdot (\nabla f) = \sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2}.$$

This operator plays an important role in many physical laws.

Example 5.5.1 Show that $\nabla^2 f = 0$ for

$$f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = \frac{1}{r}$$

and $(x, y, z) \neq (0, 0, 0)$.

The first derivatives are

$$\frac{\partial f}{\partial x} = \frac{-x}{(x^2+y^2+z^2)^{3/2}},$$
$$\frac{\partial f}{\partial y} = \frac{-y}{(x^2+y^2+z^2)^{3/2}},$$
$$\frac{\partial f}{\partial z} = \frac{-z}{(x^2+y^2+z^2)^{3/2}}.$$

Computing the second derivatives, we find that

$$\frac{\partial^2 f}{\partial x^2} = \frac{3x^2}{(x^2+y^2+z^2)^{5/2}} - \frac{1}{(x^2+y^2+z^2)^{3/2}},$$
$$\frac{\partial^2 f}{\partial y^2} = \frac{3y^2}{(x^2+y^2+z^2)^{5/2}} - \frac{1}{(x^2+y^2+z^2)^{3/2}},$$
$$\frac{\partial^2 f}{\partial z^2} = \frac{3z^2}{(x^2+y^2+z^2)^{5/2}} - \frac{1}{(x^2+y^2+z^2)^{3/2}}.$$

Thus,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = \frac{3(x^2+y^2+z^2)}{(x^2+y^2+z^2)^{5/2}} - \frac{3}{(x^2+y^2+z^2)^{3/2}} = 0.$$

Exercises

Exercise 5.1 Consider the function

$$f(x,y)=\begin{cases}\frac{xy(x^2-y^2)}{x^2+y^2}, & \text{if } (x,y)\neq(0,0),\\ 0, & \text{if } (x,y)=(0,0).\end{cases}$$

1. Compute the partial derivatives $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, and $\frac{\partial^2 f}{\partial x\partial y}$.
2. Show that $\frac{\partial^2 f}{\partial x\partial y}$ is not continuous at the origin.
3. Show that $\frac{\partial^2 f}{\partial x\partial y}(0,0)=1$, while $\frac{\partial^2 f}{\partial y\partial x}(0,0)=-1$, and explain these results.

Exercise 5.2 Prove the following classical identities in vector analysis, assuming that f and $\mathbf{F}$ are of class C^2 (this implies that the mixed partial derivatives are equal, i.e., $\frac{\partial^2 f}{\partial x_i\partial x_j}=\frac{\partial^2 f}{\partial x_j\partial x_i}$):

1. $\operatorname{curl}(\nabla f)=0$;
2. $\operatorname{div}(\operatorname{curl}\mathbf{F})=0$;
3. $\nabla(fg)=f\nabla g+g\nabla f$;
4. $\operatorname{div}(f\mathbf{F})=f\operatorname{div}\mathbf{F}+\mathbf{F}\cdot\nabla f$;
5. $\operatorname{div}(f\nabla g-g\nabla f)=f\Delta g-g\Delta f$.

Exercise 5.3 Consider $\mathbf{x}\in\mathbb{R}^n$, $\mathbf{x}\neq 0$, $r=||\mathbf{x}||$, and a constant k:

1. If $f(\mathbf{x})=r^k$, compute ∇f.
2. If $\mathbf{F}(\mathbf{x})=r^k\mathbf{x}$, compute $\operatorname{div}\mathbf{F}$.
3. If $\mathbf{F}(\mathbf{x})=r^k\mathbf{x}$, and $n=3$, compute $\operatorname{curl}\mathbf{F}$.
4. If $f(\mathbf{x})=r^k$, compute Δf.

Exercise 5.4 Consider a particle of mass $m=3$ which moves along the trajectory $s(t)=(e^t,e^{-t},\cos t)$ according to Newton's law (force = mass × acceleration).

1. Compute the force that acts on the particle at the time $t=0$.
2. At the time $t=1$ the force field disappears and the particle continues its trajectory along the tangent line. At which point is the particle at $t=2$?

Exercise 5.5 Compute the divergence and the curl of the following fields:

1. $F(x,y)=(3x^2y,x^3+y^3)$.
2. $F(x,y,z)=(yz,xz,xy)$.
3. $F(x,y,z)=\dfrac{(yz,-xz,xy)}{x^2+y^2+z^2}$.

Exercise 5.6 Compute the divergence and the curl of the following fields:

1. $\mathbf{v}(x,y,z)=x\mathbf{i}+y\mathbf{j}+z\mathbf{k}$.
2. $\mathbf{v}(x,y,z)=x\mathbf{i}+2y\mathbf{j}+3z\mathbf{k}$.
3. $\mathbf{v}(x,y,z)=\frac{1}{x^2+y^2+z^2}(x\mathbf{i}+y\mathbf{j}+z\mathbf{k})$.

Exercise 5.7 Show that the following functions on $\mathbb{R}^2$ are harmonic, i.e., they satisfy the Laplace equation $\Delta u=0$ on the corresponding domain:

1. $u(x,y)=\arctan\left(\frac{y}{x}\right)$.
2. $u(x,y)=\log(x^2+y^2)$.

Exercise 5.8 1. If $u(x,y) = f(r)$, with $r = \sqrt{x^2+y^2}$ (i.e., u is a radial function), show that

$$\Delta u = f''(r) + \frac{1}{r} f'(r).$$

2. If $u(x,y) = f(r,\theta)$ (in polar coordinates), show that

$$\Delta u = \frac{\partial^2 f}{\partial r^2} + \frac{1}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2}.$$

3. Work out the previous problem again using these formulas.

Exercise 5.9 1. Show that the following functions are harmonic ($\Delta u = 0$) on the corresponding domain:
   a) $u_1(x,y,z) = (x^2+y^2+z^2)^{-1/2}$ on $\mathbb{R}^3$.
   b) $u_2(x,y,z,w) = (x^2+y^2+z^2+w^2)^{-1}$ on $\mathbb{R}^4$.

2. If $u(\mathbf{x}) = f(r)$ (where u is a radial function) is a function defined on $\mathbb{R}^n$, show that

$$\Delta u = f''(r) + \frac{n-1}{r} f'(r).$$

3. Calculate $k \in \mathbb{R}$ such that the function $u(x) = r^k$, with $x \in \mathbb{R}^n$ and $r = ||x||$, is harmonic for $x \neq 0$.

Exercise 5.10 Find solutions of the form $f(x,t) = \cos(kx - \omega t)$ for the following partial differential equations (that is, find the relation between the constants k and ω):

1. $\dfrac{\partial^2 f}{\partial t^2} = c^2 \dfrac{\partial^2 f}{\partial x^2}$ (wave equation with constant c).
2. $\dfrac{\partial^2 f}{\partial t^2} = c^2 \dfrac{\partial^2 f}{\partial x^2} - h^2 f$ (Klein-Gordon equation with constants c, h).

Exercise 5.11 Find solutions of the form $u(x,t) = e^{-\lambda t}\sin kx$ for the equation (where λ and k are constants):

$$\frac{\partial u}{\partial t} = \mu \frac{\partial^2 u}{\partial x^2} \quad \text{(heat equation, with constant } \mu\text{).}$$

Taylor expansions. Maxima and minima

6

6.1 Taylor's Theorem for functions of several variables (MT 3.2)

For smooth functions of one variable we know that

$$f(x_0 + h) = \sum_{i=0}^{k} \frac{f^{(i)}(x_0)}{i!} h^i + R_{k,x_0}(h),$$

where $f^{(k)}(x_0)$ is the kth derivative of f at x_0 and

$$R_{k,x_0}(h) = \frac{1}{k!} \int_{x_0}^{x_0+h} (x-t)^n f^{(k+1)}(t)\, dt$$

is called the remainder[1] . That is, we can approximate any smooth function $f(x)$ by a polynomial of the desired degree, controlling the error of the approximation. Importantly, $R_{k,x_0}(x) \to 0$ as $h \to 0$.

1: This is the integral form of the remainder, and we recall it here because it will be useful when we prove Taylor's theorem for functions of several variables. Other forms of the remainder exist, such as Lagrange's or Cauchy's. You can look them up in any book on Calculus on one variable.

We wish to derive a similar result for functions of several variables. In fact, we already know a first-order version ($k = 1$). Indeed, if $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at x_0 and we define

$$R_{1,\mathbf{x}_0}(\mathbf{h}) = f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) - J_f(\mathbf{x}_0)\mathbf{h},$$

so that

$$f(\mathbf{x}_0 + h) = f(\mathbf{x}_0) + J_f(\mathbf{x}_0)\mathbf{h} + R_{1,\mathbf{x}_0}(\mathbf{h}),$$

then by the definition of differentiability,

$$\frac{|R_{1,\mathbf{x}_0}(\mathbf{h})|}{\|\mathbf{h}\|} \to 0$$

as $\mathbf{h} \to \mathbf{0}$. That is, if f is differentiable, we can approximate it (to first-order) with the tangent plane.

Let's see the second-order approximation:

Theorem 6.1.1 *(Second-order Taylor's Theorem) Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$ have continuous partial derivatives of third order. Then we may write*

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(\mathbf{x}_0) + \frac{1}{2} \sum_{i,j=1}^{n} h_i h_j \frac{\partial^2 f}{\partial x_i \partial x_j}(x_0) + R_{2,\mathbf{x}_0}(\mathbf{h}),$$

where $\frac{R_{2,\mathbf{x}_0}(\mathbf{h})}{\|\mathbf{h}\|^2} \to 0$ as $\mathbf{h} \to \mathbf{0}$ and the second sum is over all possible pairs i, j between 1 and n (so there are n^2 terms).

Notice that this result can be written in matrix form as

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + J_f(\mathbf{x}_0)\mathbf{h} + \frac{1}{2}\mathbf{h}^T H_f(\mathbf{x}_0)\mathbf{h},$$

where $J_f(\mathbf{x}_0)$ is the Jacobian matrix of f and

$$H_f(\mathbf{x}_0) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix}$$

*is called the **Hessian matrix** of f.*

Proof. Let $g(t) = f(\mathbf{x}_0 + t\mathbf{h})$ with $\mathbf{x}_0$ and $\mathbf{h}$ fixed. Note that g is a C^3 function of t and so we can write

$$g(1) = g(0) + g'(0) + \frac{g''(0)}{2!} + R_{2,0}(1),$$

where

$$R_{2,0}(1) = \frac{1}{2!}\int_0^1 (1-t)^2 g'''(t)\,dt.$$

By the chain rule,

$$g'(t) = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(\mathbf{x}_0 + t\mathbf{h})h_i,$$

$$g''(t) = \sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0 + t\mathbf{h})h_i h_j,$$

and

$$g'''(t) = \sum_{i,j,k=1}^{n} \frac{\partial^3 f}{\partial x_i \partial x_j \partial x_k}(\mathbf{x}_0 + t\mathbf{h})h_i h_j h_k.$$

Writing $R_{2,0} = R_{2,\mathbf{x}_0}(\mathbf{h})$, we have thus proved:

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(\mathbf{x}_0) + \frac{1}{2}\sum_{i,j=1}^{n} h_i h_j \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0) + R_{2,\mathbf{x}_0}(h),$$

where

$$R_{2,\mathbf{x}_0}(\mathbf{h}) = \frac{1}{2}\int_0^1 (1-t)^2 \sum_{i,j,k=1}^{n} \frac{\partial^3 f}{\partial x_i \partial x_j \partial x_k}(\mathbf{x}_0 + t\mathbf{h})h_i h_j h_k\,dt.$$

The integrand is a continuous function of t and is therefore bounded by a positive constant C on a small neighborhood of $\mathbf{x}_0$ (because it has to be close to its value at $\mathbf{x}_0$). Also note that $|h_i| \leq \|\mathbf{h}\|$, and so

$$|R_{2,\mathbf{x}_0}(\mathbf{h})| \leq \|\mathbf{h}\|^3 C.$$

In particular,

$$\frac{|R_{2,\mathbf{x}_0}(\mathbf{h})|}{\|\mathbf{h}\|^2} \leq \|\mathbf{h}\| C \to 0 \quad \text{as} \quad \mathbf{h} \to \mathbf{0},$$

as required by the theorem. □

The third-order Taylor formula is given by

$$f(\mathbf{x}_0+\mathbf{h}) = f(\mathbf{x}_0) + \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(\mathbf{x}_0) + \frac{1}{2}\sum_{i,j=1}^{n} h_i h_j \frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{x}_0) + \\ + \frac{1}{3!}\sum_{i,j,k=1}^{n} h_i h_j h_k \frac{\partial^3 f}{\partial x_i \partial x_j \partial x_k}(\mathbf{x}_0) + R_{3,\mathbf{x}_0}(\mathbf{h}),$$

where $\frac{R_{3,\mathbf{x}_0}(\mathbf{h})}{\|\mathbf{h}\|^3} \to 0$ as $\mathbf{h} \to \mathbf{0}$, and so on. The general formula can be proved by induction, using the method of proof already given.

Example 6.1.1 Compute the second-order Taylor formula for the function $f(x,y) = \sin(x+2y)$, about the point $(0,0)$.

Notice that

$$f(0,0) = 0,$$
$$\frac{\partial f}{\partial x}(0,0) = \cos(0+2\cdot 0) = 1,$$
$$\frac{\partial f}{\partial y}(0,0) = 2\cos(0+2\cdot 0) = 2,$$
$$\frac{\partial^2 f}{\partial x^2}(0,0) = 0, \quad \frac{\partial^2 f}{\partial y^2}(0,0) = 0, \quad \frac{\partial^2 f}{\partial x \partial y}(0,0) = 0.$$

Thus,

$$f(\mathbf{h}) = f(h_1,h_2) = h_1 + 2h_2 + R_{2,\mathbf{0}}(\mathbf{h}),$$

where

$$\frac{R_{2,\mathbf{0}}(\mathbf{h})}{\|\mathbf{h}\|^2} \to 0 \quad \text{as} \quad \mathbf{h} \to \mathbf{0}.$$

Example 6.1.2 Find linear and quadratic approximations to the expression $(3.98-1)^2/(5.97-3)^2$. Compare with the exact value.

Let $f(x,y) = \frac{(x-1)^2}{(y-3)^2}$. The desired expression is close to $f(4,6) = 1$, with $(h_1,h_2) = (-0.02,-0.03)$. To find the approximations, we differentiate:

$$f_x = \frac{2(x-1)}{(y-3)^2}, \quad f_y = -\frac{2(x-1)^2}{(y-3)^3},$$

$$f_{xy} = f_{yx} = -\frac{4(x-1)}{(y-3)^3}, \quad f_{xx} = \frac{2}{(y-3)^2}, \quad f_{yy} = \frac{6(x-1)^2}{(y-3)^4}.$$

At the point of approximation, we have

$$f_x(4,6) = \frac{2}{3}, \quad f_y(4,6) = -\frac{2}{3}, \quad f_{xy}(4,6) = -\frac{4}{9}, \quad f_{yy}(4,6) = \frac{2}{3}.$$

The linear approximation is then

$$1 + (-0.02)\,\frac{2}{3} - (-0.03)\,\frac{2}{3} = 1.00666.$$

The quadratic approximation is

$$1.00666 + \frac{2}{3}(-0.02)^2 - \frac{4}{9}(-0.02)(-0.03) + \frac{2}{3}(-0.03)^2 = 1.00674.$$

The "exact" value using a calculator is 1.00675.

6.2 Extrema (MT 3.3)

In many applications it is very useful to find the maxima or minima of a given function, what we usually call *extrema*:

Definition 6.2.1 *If $f : U \subset \mathbb{R}^n \longrightarrow \mathbb{R}$, a point $\mathbf{x}_0 \in U$ is called a* ***local minimum*** *(respectively, maximum) of f if there is a neighborhood V of $\mathbf{x}_0$ such that for all points $\mathbf{x} \in V$, $f(\mathbf{x}) \geq f(\mathbf{x}_0)$ (or $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ in the case of the maximum). If the inequality is strict, we say the minimum (or maximum) is strict.*

The point $\mathbf{x}_0 \in U$ is said to be a ***local (or relative) extremum*** *if it is either a local minimum or a local maximum. A point $\mathbf{x}_0$ is a* ***critical point*** *of f if either f is not differentiable at $\mathbf{x}_0$, or if it is, and $J_f(\mathbf{x}_0) = 0$. A critical point that is not a local extremum is called a* ***saddle point****.*

First-Derivative Test for Local Extrema

The location of extrema is based on the following fact, which should be familiar from one-variable calculus (the case $n = 1$): Every extremum is a critical point.

Theorem 6.2.1 (First-Derivative Test for Local Extrema) *If $U \subset \mathbb{R}^n$ is open, the function $f : U \longrightarrow \mathbb{R}$ is differentiable, and $\mathbf{x}_0 \in U$ is a local extremum, then $J_f(\mathbf{x}_0) = 0$; that is, $\mathbf{x}_0$ is a critical point of f.*

Proof. Suppose that f achieves a local maximum at $\mathbf{x}_0$. Then for any $\mathbf{h} \in \mathbb{R}^n$, the function $g(t) = f(\mathbf{x}_0 + t\mathbf{h})$ has a local maximum at $t = 0$. Thus, from one-variable calculus $g'(0) = 0$. On the other hand, by the chain rule, $g'(0) = J_f(\mathbf{x}_0)\mathbf{h}$. Thus, $J_f(\mathbf{x}_0)\mathbf{h} = 0$ for every $\mathbf{h}$, and so $J_f(\mathbf{x}_0) = 0$. The case in which f achieves a local minimum at $\mathbf{x}_0$ is entirely analogous. □

That $J_f(\mathbf{x}_0) = 0$ means that all partial derivatives of f are zero at $\mathbf{x}_0$. In other words, $\nabla f(\mathbf{x}_0) = 0$.

Example 6.2.1 Find all the critical points of $z = x^2y + y^2x$.

Differentiating, we obtain

$$\frac{\partial z}{\partial x} = 2xy + y^2, \quad \frac{\partial z}{\partial y} = x^2 + 2yx.$$

Equating the partial derivatives to zero yields

$$2xy + y^2 = 0, \quad x^2 + 2yx = 0.$$

Subtracting, we obtain $x^2 = y^2$. Thus, $x = \pm y$. Substituting $x = +y$ in the first of the two preceding equations, we find that

$$2y^2 + y^2 = 3y^2 = 0,$$

so that $y = 0$ and thus $x = 0$. If $x = -y$, then

$$-2y^2 + y^2 = -y^2 = 0,$$

so $y = 0$ and therefore $x = 0$. Hence, the only critical point is $(0, 0)$. For $x = y$, $z = 2x^3$, which is both positive and negative for x near zero. Thus, $(0, 0)$ is not a relative extremum.

When is a critical point a local extremum? In what follows we will derive a useful criterion based on the second derivative, analogous to the criterion used for functions of one variable.

Second-Derivative Test for Local Extrema

Definition 6.2.2 (Quadratic functions) *We say that a function $g : \mathbb{R}^n \to \mathbb{R}$ is quadratic if it has the form*

$$g(h_1, \dots, h_n) = \sum_{i,j=1}^{n} a_{ij} h_i h_j = \begin{pmatrix} h_1 & \cdots & h_n \end{pmatrix} \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}.$$

We can, if we wish, assume that $[a_{ij}]$ is symmetric; in fact, g is unchanged if we replace $[a_{ij}]$ by the symmetric matrix $[b_{ij}]$, where $b_{ij} = \frac{1}{2}(a_{ij} + a_{ji})$, because $h_i h_j = h_j h_i$ and the sum is over all i and j. The quadratic nature of g is reflected in the identity

$$g(\lambda h_1, \dots, \lambda h_n) = \lambda^2 g(h_1, \dots, h_n),$$

which follows from the definition.

Example 6.2.2 The function $g(h_1, h_2, h_3) = h_1^2 - 2h_1 h_2 + h_2^2$ is quadratic, as we can write it as

$$g(h_1, h_2, h_3) = \begin{pmatrix} h_1 & h_2 & h_3 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ h_3 \end{pmatrix}.$$

We can use the Hessian matrix introduced in Taylor's theorem to define a quadratic function:

Definition 6.2.3 (Hessian function) *If $f : U \subset \mathbb{R}^n \to \mathbb{R}$ has continuous* *second partial derivatives at* $\mathbf{x}_0 \in U$*, we define the* ***Hessian*** *of* f *at* $\mathbf{x}_0$ *as*

$$H_f(\mathbf{x}_0)(\mathbf{h}) = \frac{1}{2}\mathbf{h}^T \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{pmatrix} \mathbf{h}$$

Note that the Hessian matrix is symmetric due to the equality of mixed partial derivatives.

The Hessian function is usually used at critical points $\mathbf{x}_0 \in U$. In this case, $J_f(\mathbf{x}_0) = 0$, so the Taylor expansion for f around $\mathbf{x}_0$ becomes

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + H_f(\mathbf{x}_0)(\mathbf{h}) + R_{2,\mathbf{x}_0}(\mathbf{h}).$$

Thus, at a critical point the Hessian equals the first nonconstant term in the Taylor series of f.

Definition 6.2.4 (Positive-definite) *A quadratic function* $g : \mathbb{R}^n \to \mathbb{R}$ *is called* ***positive-definite*** *if* $g(\mathbf{h}) \geq 0$ *for all* $\mathbf{h} \in \mathbb{R}^n$ *and* $g(\mathbf{h}) = 0$ *only for* $\mathbf{h} = \mathbf{0}$*. Similarly,* g *is* ***negative-definite*** *if* $g(\mathbf{h}) \leq 0$ *and* $g(\mathbf{h}) = 0$ *for* $\mathbf{h} = \mathbf{0}$ *only.*

Note that if $n = 1$, $H_f(\mathbf{x}_0)(\mathbf{h}) = \frac{1}{2} f''(x_0)h^2$, which is positive-definite if and only if $f''(x_0) > 0$.

Theorem 6.2.2 (Second-Derivative Test for Local Extrema) *If* $f : U \subset \mathbb{R}^n \to \mathbb{R}$ *is of class* C^3*,* $\mathbf{x}_0 \in U$ *is a critical point of* f*, and the Hessian* $H_f(\mathbf{x}_0)(\mathbf{h})$ *is positive-definite, then* $\mathbf{x}_0$ *is a (strict) relative minimum of* f*. Similarly, if* $H_f(\mathbf{x}_0)(\mathbf{h})$ *is negative-definite, then* $\mathbf{x}_0$ *is a (strict) relative maximum.*

The proof of Theorem 6.2.2 requires Taylor's theorem and the following result from linear algebra.

Lemma 6.2.3 *If* $B = [b_{ij}]$ *is an* $n \times n$ *real matrix, and if the associated quadratic function*

$$H : \mathbb{R}^n \to \mathbb{R}, \quad (h_1, \ldots, h_n) \mapsto \frac{1}{2}\sum_{i,j=1}^{n} b_{ij}h_i h_j$$

is positive-definite, then there is a constant $M > 0$ *such that for all* $\mathbf{h} \in \mathbb{R}^n$*;* $H(\mathbf{h}) \geq M\|\mathbf{h}\|^2$.

Proof. Restricting ourselves to the set of all $\mathbf{h}$ such that $\|\mathbf{h}\| = 1$, H is a continuous function defined on a compact set, and so it reaches a minimum value[2] M. Now, take a general $\mathbf{h}$. Because H is quadratic, we have

$$H(\mathbf{h}) = H\left(\frac{\mathbf{h}}{\|\mathbf{h}\|}\|\mathbf{h}\|\right) = H\left(\frac{\mathbf{h}}{\|\mathbf{h}\|}\right)\|\mathbf{h}\|^2 \geq M\|\mathbf{h}\|^2$$

for any $\mathbf{h} \neq 0$. (The result is obviously valid if $\mathbf{h} = \mathbf{0}$.) □

2: See Theorem 6.2.7 later in this chapter.

Proof of Theorem 6.2.2. Recall that if $f : U \subset \mathbb{R}^n \longrightarrow \mathbb{R}$ is of class C^3 and $\mathbf{x}_0 \in U$ is a critical point, Taylor's theorem may be expressed in the form

$$f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) = H_f(\mathbf{x}_0)(\mathbf{h}) + R_{2,\mathbf{x}_0}(\mathbf{h}),$$

where $R_{2,\mathbf{x}_0}(\mathbf{h})/\|\mathbf{h}\|^2 \longrightarrow 0$ as $\mathbf{h} \longrightarrow \mathbf{0}$. Because $H_f(\mathbf{x}_0)$ is positive-definite, Lemma 6.2.3 assures us of a constant $M > 0$ such that for all $\mathbf{h} \in \mathbb{R}^n$

$$H_f(\mathbf{x}_0)(\mathbf{h}) \geq M\|\mathbf{h}\|^2.$$

Because $R_{2,\mathbf{x}_0}(\mathbf{h})/\|\mathbf{h}\|^2 \longrightarrow 0$ as $\mathbf{h} \longrightarrow \mathbf{0}$, there is a $\delta > 0$ such that for $0 < \|\mathbf{h}\| < \delta$

$$|R_{2,\mathbf{x}_0}(\mathbf{h})| < M\|\mathbf{h}\|^2.$$

Thus, $0 < H_f(\mathbf{x}_0)(\mathbf{h}) + R_{2,\mathbf{x}_0}(\mathbf{h}) = f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0)$ for $0 < \|\mathbf{h}\| < \delta$, so that $\mathbf{x}_0$ is a relative minimum; in fact, a strict relative minimum. The proof in the negative-definite case is similar, or else follows by applying the preceding to $-f$. □

Example 6.2.3 Consider again the function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$, $(x, y) \mapsto x^2 + y^2$. Then $(0, 0)$ is a critical point, and f is already in the form of Taylor's theorem:

$$f((0,0) + (h_1, h_2)) = f(0,0) + (h_1^2 + h_2^2) + 0.$$

We can see directly that the Hessian at $(0, 0)$ is

$$H_f(0)(\mathbf{h}) = h_1^2 + h_2^2,$$

which is clearly positive-definite. Thus, $(0, 0)$ is a relative minimum. This simple case can, of course, be done without calculus. Indeed, it is clear that $f(x, y) > 0$ for all $(x, y) \neq (0, 0)$.

If the Hessian matrix is neither positive- nor negative-definite, but its determinant is nonzero, it is of saddle type (it is neither a maximum nor a minimum). If the determinant of the Hessian is zero, it is said to be of degenerate type and nothing can be said about the nature of the critical point without further analysis.

But when is a quadratic function positive-definite? The answer lies in the following theorem.

Theorem 6.2.4 *Let $B = [b_{ij}]$ an $n \times n$ symmetric matrix, and $H : \mathbb{R}^n \longrightarrow \mathbb{R}$ given by $H(\mathbf{h}) = \frac{1}{2}\sum_{i,j}^{n} b_{ij} h_i h_j$ its associated quadratic function. Then H is positive-definite if and only if all the (real) eigenvalues of B are positive, and it is negative-definite if and only if all the eigenvalues of B are negative.*

Proof. Since B is a symmetric real matrix, it is diagonalizable. Write $B = PDP^T$, D a diagonal with the (real) eigenvalues of B in the diagonal, and P an orthogonal matrix whose columns (the eigenvectors of B) form a basis of $\mathbb{R}^n$. The matrix D may be regarded as the same matrix as B but expressed in terms of the basis defined by the eigenvectors in P.

Define the quadratic function $L(\mathbf{h}) = \frac{1}{2}\mathbf{h}^T D\mathbf{h}$. For any $\mathbf{z} \in \mathbb{R}^n$, and writing $\mathbf{y} = P^T\mathbf{z}$, we have that

$$\begin{aligned} H(\mathbf{z}) = \frac{1}{2}\mathbf{z}^T B\mathbf{z} > 0, \text{ for all } \mathbf{z} \in \mathbb{R}^n &\iff \frac{1}{2}\mathbf{z}^T PDP^T\mathbf{z} > 0, \text{ for all } \mathbf{z} \in \mathbb{R}^n \\ &\iff \frac{1}{2}\mathbf{y}^T D\mathbf{y} > 0, \text{ for all } \mathbf{y} \in \mathbb{R}^n, \end{aligned}$$

which means that H is positive-definite if and only if L is positive-definite. But

$$L(\mathbf{y}) = \frac{1}{2}\mathbf{y}^T D\mathbf{y} = \sum_i^n d_i y_i^2$$

and so this is positive for all $\mathbf{y} \in \mathbb{R}^n$ if and only if $d_i > 0$ for all i. That is, if and only if the eigenvalues of B are positive. The proof for the negative-definite case is identical. □

Having this in mind, we shall give a useful criterion for when a quadratic function defined by a 2×2 matrix is positive-definite. This will be useful in conjunction with Theorem 6.2.2.

Lemma 6.2.5 *Let* $B = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ *and* $H(\mathbf{h}) = \frac{1}{2}\mathbf{h}^T B\mathbf{h}$. *Then* $H(\mathbf{h})$ *is positive-definite if and only if* $a > 0$ *and* $\det B = ac - b^2 > 0$.

Proof. For a 2×2 symmetric matrix B, the two eigenvalues λ_1, λ_2 are given by the system of equations $\lambda_1\lambda_2 = \det B = ac - b^2$ and $\lambda_1 + \lambda_2 = \operatorname{tr} B = a + c$. If $ac - b^2 > 0$ this means that the two eigenvalues are either both positive or both negative. Requiring that $a > 0$ ensures that they are both positive, and by Theorem 6.2.4, H is positive-definite. Alternatively, if $ac - b^2 > 0$ and $a < 0$, then H is negative-definite. □

Note that if $\det B < 0$ (for 2×2 matrices) this means that H is neither positive nor negative-definite. We can combine everything into a theorem for functions of two variables:

Theorem 6.2.6 (Second-Derivative Maximum-Minimum Test for Functions of Two Variables) *Let $f(x, y)$ be of class C^2 on an open set $U \subset \mathbb{R}^2$. A point $(x_0, y_0) \in U$ is a (strict) local minimum of f if*

1. $\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$
2. $\frac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$
3. $D = \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2 > 0$ *at* (x_0, y_0).

*(D is called the **discriminant** of the Hessian). If $\partial^2 f/\partial x^2 < 0$ at (x_0, y_0) and the other conditions are satisfied, then the point is a (strict) local maximum. If $D < 0$ then (x_0, y_0) is a **saddle point**.*

Example 6.2.4 Classify the critical points of the function $f : \mathbb{R}^2 \to \mathbb{R}$, defined by $(x, y) \mapsto x^2 - 2xy + 2y^2$.

We have that $f(0, 0) = 0$, the origin is the only critical point, and the

Hessian is

$$H_f(\mathbf{0})(\mathbf{h}) = h_1^2 - 2h_1h_2 + 2h_2^2 = (h_1 - h_2)^2 + h_2^2,$$

which is clearly positive-definite. Thus, f has a relative minimum at $(0,0)$. Alternatively, we can apply Theorem 6.2.6. At $(0,0)$, $\frac{\partial^2 f}{\partial x^2} = 2$, $\frac{\partial^2 f}{\partial y^2} = 4$, and $\frac{\partial^2 f}{\partial x \partial y} = -2$. Since $\partial^2 f/\partial x^2 > 0$ and the determinant is positive, f has a relative minimum at $(0,0)$.

Example 6.2.5 Locate the relative maxima, minima, and saddle points of the function $f(x,y) = \log(x^2 + y^2 + 1)$.

We must first locate the critical points of this function. The gradient is

$$\nabla f(x,y) = \frac{2x}{x^2 + y^2 + 1}\mathbf{i} + \frac{2y}{x^2 + y^2 + 1}\mathbf{j}.$$

Thus, $\nabla f(x,y) = 0$ if and only if $(x,y) = (0,0)$, and so the only critical point of f is $(0,0)$. Now we must determine whether this is a maximum, a minimum, or a saddle point. The second partial derivatives are

$$\frac{\partial^2 f}{\partial x^2} = \frac{2(x^2 + y^2 + 1) - (2x)(2x)}{(x^2 + y^2 + 1)^2},$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{2(x^2 + y^2 + 1) - (2y)(2y)}{(x^2 + y^2 + 1)^2},$$

and

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{-2x(2y)}{(x^2 + y^2 + 1)^2}.$$

Therefore,

$$\frac{\partial^2 f}{\partial x^2}(0,0) = \frac{\partial^2 f}{\partial y^2}(0,0) = 2$$

and

$$\frac{\partial^2 f}{\partial x \partial y}(0,0) = 0,$$

which yields

$$D = 2 \cdot 2 = 4 > 0.$$

Because $\left(\frac{\partial^2 f}{\partial x^2}\right)(0,0) > 0$, we conclude by Theorem 6.2.6 that $(0,0)$ is a local minimum. (Can you show this just from the fact that $\log t$ is an increasing function of $t > 0$?)

Global Maxima and Minima

We end this section with a discussion of the theory of absolute, or global, maxima and minima of functions of several variables. Unfortunately, the location of absolute maxima and minima for functions on $\mathbb{R}^n$ is, in general, a more difficult problem than for functions of one variable.

Definition 6.2.5 *Let* $f : A \subset \mathbb{R}^n \to \mathbb{R}$. *A point* $\mathbf{x}_0 \in A$ *is said to be an* *absolute maximum (or absolute minimum) point of f if $f(\mathbf{x}) \leq f(\mathbf{x}_0)$ [or $f(\mathbf{x}) \geq f(\mathbf{x}_0)$] for all $\mathbf{x} \in A$.*

In one-variable calculus, we leart that every continuous function on a closed interval I assumes its absolute maximum (or minimum) value at some point x_0 in I. A generalization of this theoretical fact also holds in $\mathbb{R}^n$. Such theorems guarantee that the maxima or minima one is seeking actually exist; therefore, the search for them is not in vain.

Definition 6.2.6 *A set $D \in \mathbb{R}^n$ is said to be bounded if there is a number $M > 0$ such that $\|\mathbf{x}\| < M$ for all $\mathbf{x} \in D$.*

Thus, a set is bounded if it can be strictly contained in some (large) ball. The appropriate generalization of the one-variable theorem on maxima and minima is the following result:

Theorem 6.2.7 (Global Existence Theorem for Maxima and Minima) *Let D be closed and bounded in $\mathbb{R}^n$ and let $f : D \longrightarrow \mathbb{R}$ be continuous. Then f assumes its absolute maximum and minimum values at some points $\mathbf{x}_0$ and $\mathbf{x}_1$ of D.*

Suppose now that $D = U \cup \partial U$, where U is open and ∂U is its boundary. If $D \subset \mathbb{R}^2$, we suppose that ∂U is a piecewise smooth curve; that is, D is a region bounded by a collection of smooth curves. If $\mathbf{x}_0$ and $\mathbf{x}_1$ are in U, we know from Theorem 6.2.1 that they are critical points of f. If they are in ∂U, and ∂U is a smooth curve (i.e., the image of a smooth path c with $c' \neq 0$), then they are maximum or minimum points of f viewed as a function on ∂U. These observations provide a method of finding the absolute maximum and minimum values of f on a region D.

Strategy for Finding the Absolute Maxima and Minima on a Region with Boundary

Let f be a continuous function of two variables defined on a closed and bounded region D in $\mathbb{R}^n$, which is bounded by a smooth closed curve. To find the absolute maximum and minimum of f on D:

1. Locate all critical points for f in U.
2. Find all the critical points of f viewed as a function only on ∂U.
3. Compute the value of f at all of these critical points.
4. Compare all these values and select the largest and the smallest.

If $D \subset \mathbb{R}^2$ is a region bounded by a collection of smooth curves (such as a square), then we follow a similar procedure, but including in the third step the points where the curves meet (such as the corners of the square).

To carry out the second step in the plane, you can find a smooth parametrization of ∂U[3] and consider the function of one variable $t \longrightarrow f(c(t))$, where $t \in I$, and locate the maximum and minimum points $t_0, t_1 \in I$ (remember to check the endpoints). Then $c(t_0), c(t_1)$ will be maximum and minimum points for f as a function on ∂U. Another method for dealing with this step is the Lagrange multiplier method, to be presented in the next chapter.

3: That is, a path $c : I \longrightarrow \partial U$, where I is some interval, such that $c(I)$ covers ∂U

Exercises

Exercise 6.1 Compute the Hessian matrix for the functions:

1. $f(x, y) = \log(x^2 + y^3)$.
2. $f(x, y) = x^2 \cos y + y^2 \sin x$.
3. $f(x, y) = x^y$.
4. $f(x, y) = \arctan(xy)$.
5. $f(x, y) = \arctan\left(\frac{x}{y}\right)$.
6. $f(x, y, z) = \sin(x + z^y)$.
7. $f(x, y, z) = (x + y)(y + z)(z + x)$.

Exercise 6.2 Find the critical points and the values of the local extrema of the following two-variable functions:

1. $f(x, y) = x^2 + 2y^2 - 4y$.
2. $g(x, y) = x^2 - xy + y^2 + 2x + 2y - 6$.
3. $h(x, y) = 3x^2 + 2xy + 2x + y^2 - y + 4$.
4. $k(x, y) = 8x^3 - 24xy + y^3$.
5. $m(x, y) = x^3 + y^3 + 3x^2 - 18y^2 + 81y + 5$.

Exercise 6.3 Determine the extrema of the function $f(x, y) = e^{-x^2+\varepsilon y^2}$ for $\varepsilon = 0, 1, -1$.

Exercise 6.4 Find the extrema of the following functions:

1. $f(x, y) = x^3y^3$.
2. $f(x, y) = x^4y^4$.
3. $f(x, y) = x^2 + xy + y^2 - 3x - 6y$.
4. $f(x, y) = \frac{x-y}{1+x^2+y^2}$.

Exercise 6.5 Find the extrema of the following functions on the specified domains:

1. $f(x, y) = |x| + |y|$ on $A = \{(x, y) : |x| \leq 1, |y| \leq 1\}$.
2. $f(x, y) = x^2 + 2xy + y^2$ on $B = \{(x, y) : |x| \leq 2, |y| \leq 1\}$.
3. $f(x, y) = x^2 + 2xy + y^2$ on $C = \{(x, y) : x^2 + y^2 \leq 8\}$.

Exercise 6.6 Study if the following functions have a (local or global) extremum at the origin:

1. $f(x, y) = x^4 + 2xy^3 + y^4 + 2xy$.
2. $g(x, y) = \begin{cases} xy + xy^3 \sin\left(\frac{x}{y}\right), & y \neq 0, \\ 0, & y = 0. \end{cases}$

Hint: In item (2), observe that one may find two straight lines through the origin along which the function has different signs.

Exercise 6.7 Consider the set of pairs of points $\{(x_i, y_i), i = 1, \ldots, N\}$, and define the function

$$f(m, b) = \sum_{i=1}^{N} (y_i - mx_i - b)^2.$$

Find the values of m and b that minimize f. (This is the *least squares method* to approximate the set of points $\{(x_i, y_i)\}$ by means of the line $y = mx + b$).

Exercise 6.8 Classify all the critical points of the following functions:

1. $f(x, y) = \sin x \cos y$.
2. $g(x, y) = \sin x^2 - \sin y^2$.

Lagrange multipliers 7

7.1 Finding extrema of a function satisfying one constraint

Often we are required to maximize or minimize a function subject to certain constraints or side conditions. For example, we might need to maximize $f(x, y)$ subject to the condition that $x^2 + y^2 = 1$. More generally, we might need to maximize or minimize $f(x, y)$ subject to the side condition that (x, y) also satisfies an equation $g(x, y) = c$, where g is some function and c equals a constant. The set of such (x, y) is a level curve for g.

Example 7.1.1 Let $S \subset \mathbb{R}^2$ be the line through $(-1, 0)$ inclined at $45°$, and let $f : \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto x^2 + y^2$. Find the extrema of $f|_S$.

The line S is given by the equation $y = x + 1$. The path $c : \mathbb{R} \to \mathbb{R}^2$ given by $c(t) = (-1, 0) + t(1, 1) = (-1 + t, t)$ maps the line S. The restriction of f to S can be studied through the composition $h(t) = f \circ c(t) = f(c(t)) = (t - 1)^2 + t^2$, where $h : \mathbb{R} \to \mathbb{R}$ is a differentiable function in one variable.

The maxima and minima of $f|_S$ will therefore be given by $h(t)$. The derivative $h'(t) = 4t - 2$ is zero at $t = 1/2$, and the second derivative $h''(t) = 4$ is positive at that point, so $t = 1/2$ is a minimum of $h(t)$. Since the function is the sum of two quadratic terms, this is an absolute minimum in the whole line. There are no maxima, since the line is unbounded.

Since $c(1/2) = (-1/2, 1/2)$ we have that $f|_S$ has an absolute minimum at $(-1/2, 1/2)$ with value $1/2$.

Notice that $\nabla f(x, y) = (2x, 2y)$ and that $\nabla f(-1/2, 1/2) = (-1, 1)$, which is perpendicular to S. This also means that $(-1/2, 1/2)$ is the only point where S is tangent to a level curve of f, in particular the curve $x^2 + y^2 = 1/2$. But this also means that ∇f is parallel to ∇g, where $g(x, y) = y - x$. In this case S is just the level curve $g(x, y) = -1$, and $\nabla g(x, y) = (-1, 1)$ is perpendicular to the level curves of g.

Finding a function c that maps the desired restriction, obtaining the composition $f \circ c$ and finding the relative extrema of that function seems cumbersome in general. In this chapter we will derive methods to solve this problem using the geometric intuition that we have noted in the previous example:

Theorem 7.1.1 (The Method of Lagrange Multipliers) *Suppose that $f : U \subset \mathbb{R}^n \to \mathbb{R}$ and $g : U \subset \mathbb{R}^n \to \mathbb{R}$ are given C^1 real-valued functions. Let $\mathbf{x}_0 \in U$ and $g(\mathbf{x}_0) = c$, and let S be the level set for g with* *value c [recall that this is the set of points $\mathbf{x} \in \mathbb{R}^n$ satisfying $g(\mathbf{x}) = c$]. Assume $\nabla g(\mathbf{x}_0) \neq 0$. If $f|_S$, which denotes "f restricted to S," has a local maximum or minimum on S at $\mathbf{x}_0$, then there is a real number λ (which might be zero) such that*

$$\nabla f(\mathbf{x}_0) = \lambda \nabla g(\mathbf{x}_0).$$

Proof. We have not developed enough techniques to give a complete proof, but we can provide the essential points.

Why would ∇f and ∇g be proportional?

First, since $g(\mathbf{x}) = k$ is a level curve of g, ∇g is perpendicular to the tangent vector to that curve at any point (remember that gradients are normal to level sets, as seen in Theorem 4.3.3).

Also, if $f|_S$ has a maximum (or minimum) at $\mathbf{x}_0$, and $c(t)$ is a path that maps S where $c(0) = \mathbf{x}_0$, then $f(c(t))$ has a maximum (or minimum) at $t = 0$ (see Example 7.1.1). By one-variable calculus, $\frac{d}{dt} f(c(t))\Big|_{t=0} = 0$. Hence, by the chain rule,

$$0 = \frac{d}{dt} f(c(t))\Big|_{t=0} = \nabla f(x_0) \cdot c'(0).$$

Thus, $\nabla f(x_0)$ is perpendicular to the tangent of every curve in S and so is perpendicular to the whole tangent space to S at $\mathbf{x}_0$. Because the space perpendicular to this tangent space is a line, $\nabla f(\mathbf{x}_0)$ and $\nabla g(\mathbf{x}_0)$ are parallel. Because $\nabla g(\mathbf{x}_0) \neq 0$, it follows that $\nabla f(\mathbf{x}_0)$ is a multiple of $\nabla g(\mathbf{x}_0)$, which is the conclusion of the theorem. □

In order to apply this method, we look for a point $\mathbf{x}_0$ and a constant λ, called a Lagrange multiplier[1] , such that $\nabla f(\mathbf{x}_0) = \lambda \nabla g(\mathbf{x}_0)$. That is, we need to solve the system of equations

1: Although they receive this name, Euler introduced these multipliers in 1744, some 40 years before Lagrange!

$$\begin{aligned}
\frac{\partial g}{\partial x_1}(x_1, \ldots, x_n) &= \lambda \frac{\partial f}{\partial x_1}(x_1, \ldots, x_n) \\
\frac{\partial g}{\partial x_2}(x_1, \ldots, x_n) &= \lambda \frac{\partial f}{\partial x_2}(x_1, \ldots, x_n) \\
&\vdots \\
\frac{\partial g}{\partial x_n}(x_1, \ldots, x_n) &= \lambda \frac{\partial f}{\partial x_n}(x_1, \ldots, x_n) \\
g(x_1, \ldots, x_n) &= c
\end{aligned}$$

for $x_1, \ldots, x_n$ and λ.

Another way of looking at these equations is as follows: Think of λ as an additional variable and form the auxiliary function

$$h(x_1, \ldots, x_n, \lambda) = f(x_1, \ldots, x_n) - \lambda[g(x_1, \ldots, x_n) - c].$$

The Lagrange multiplier theorem says that to find the extreme points of $f|_S$, we should examine the critical points of h. These are found by solving the equations:

$$\begin{aligned}
0 &= \frac{\partial h}{\partial x_1} = \frac{\partial f}{\partial x_1} - \lambda \frac{\partial g}{\partial x_1}, \\
&\vdots \\
0 &= \frac{\partial h}{\partial x_n} = \frac{\partial f}{\partial x_n} - \lambda \frac{\partial g}{\partial x_n}, \\
0 &= \frac{\partial h}{\partial \lambda} = g(x_1, \ldots, x_n) - c
\end{aligned}$$

which are the same as the previous system.

Example 7.1.2 Let $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$, $(x, y) \mapsto x^2 - y^2$, and let S be the circle of radius 1 around the origin. Find the extrema of $f|_S$.

The set S is the level curve for g with value 1, where $g : \mathbb{R}^2 \longrightarrow \mathbb{R}$, $(x, y) \mapsto x^2 + y^2$. Let us use the method of Lagrange multipliers. Clearly,

$$\nabla f(x, y) = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) = (2x, -2y) \quad \text{and} \quad \nabla g(x, y) = (2x, 2y).$$

Note that $\nabla g(x, y) \neq 0$ if $x^2 + y^2 = 1$. Thus, according to the Lagrange multiplier theorem, we must find a λ such that

$$(2x, -2y) = \lambda(2x, 2y) \quad \text{and} \quad (x, y) \in S, \quad \text{i.e.,} \quad x^2 + y^2 = 1.$$

These conditions yield three equations, which can be solved for the three unknowns x, y, and λ. From $2x = \lambda 2x$, we conclude that either $x = 0$ or $\lambda = 1$. If $x = 0$, then $y = \pm 1$ and $-2y = \lambda 2y$ implies $\lambda = -1$. If $\lambda = 1$, then $y = 0$ and $x = \pm 1$. Thus, we get the points $(0, \pm 1)$ and $(\pm 1, 0)$. As we have mentioned, this method only locates potential extrema; whether they are maxima, minima, or neither must be determined by other means.

Example 7.1.3 Maximize the function $f(x, y, z) = x + z$ subject to the constraint $x^2 + y^2 + z^2 = 1$. By Theorem 6.2.7 we know that the function f restricted to the unit sphere $x^2 + y^2 + z^2 = 1$ has a maximum (and also a minimum). To find the maximum, we again use the Lagrange multiplier theorem. We seek λ and (x, y, z) such that

$$\begin{aligned}
1 &= 2x\lambda, \\
0 &= 2y\lambda, \\
1 &= 2z\lambda, \\
\text{and} \quad x^2 + y^2 + z^2 &= 1.
\end{aligned}$$

From the first or the third equation, we see that $\lambda \neq 0$. Thus, from the second equation, we get $y = 0$. From the first and third equations, $x = z$, and so from the fourth equation, $x = \pm\frac{1}{\sqrt{2}} = z$. Hence, our points are $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$ and $\left(-\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}\right)$.

Comparing the values of f at these points, we can see that the first point yields the maximum of f (restricted to the constraint) and the second the minimum.

7.2 Several Constraints

If a surface S is defined by a number of constraints, namely,

$$\begin{gathered} g_1(x_1, \dots, x_n) = c_1, \\ g_2(x_1, \dots, x_n) = c_2, \\ \vdots \\ g_k(x_1, \dots, x_n) = c_k, \end{gathered}$$

then the Lagrange multiplier theorem may be generalized as follows: If f has a maximum or a minimum at x_0 on S, there must exist constants $\lambda_1, \dots, \lambda_k$ such that

$$\nabla f(x_0) = \lambda_1 \nabla g_1(x_0) + \cdots + \lambda_k \nabla g_k(x_0). \tag{7.1}$$

Example 7.2.1 Find the extreme points of $f(x, y, z) = x + y + z$ subject to the two conditions $x^2 + y^2 = 2$ and $x + z = 1$.

Here there are two constraints:

$$\begin{aligned} g_1(x, y, z) &= x^2 + y^2 - 2 = 0, \\ g_2(x, y, z) &= x + z - 1 = 0. \end{aligned}$$

Thus, we must find x, y, z, λ_1, and λ_2 such that

$$\begin{gathered} \nabla f(x, y, z) = \lambda_1 \nabla g_1(x, y, z) + \lambda_2 \nabla g_2(x, y, z), \\ g_1(x, y, z) = 0, \quad \text{and} \quad g_2(x, y, z) = 0. \end{gathered}$$

Computing the gradients and equating components, we get

$$\begin{gathered} 1 = \lambda_1 \cdot 2x + \lambda_2 \cdot 1, \\ 1 = \lambda_1 \cdot 2y + \lambda_2 \cdot 0, \\ 1 = \lambda_1 \cdot 0 + \lambda_2 \cdot 1, \\ x^2 + y^2 = 2, \quad \text{and} \quad x + z = 1. \end{gathered}$$

These are five equations for x, y, z, λ_1, and λ_2. From the third equation, $\lambda_2 = 1$, and so $2x\lambda_1 = 0$, $2y\lambda_1 = 1$. Because the second implies $\lambda_1 \neq 0$, we have $x = 0$. Thus, $y = \pm\sqrt{2}$ and $z = 1$. Hence, the possible extrema are $(0, \sqrt{2}, 1)$ and $(0, -\sqrt{2}, 1)$. By inspection, $(0, \sqrt{2}, 1)$ gives a relative maximum, and $(0, -\sqrt{2}, 1)$ a relative minimum.

The condition $x^2 + y^2 = 2$ implies that x and y must be bounded. The condition $x + z = 1$ implies that z is also bounded. It follows that the constraint set S is closed and bounded. By Theorem 6.2.7 it follows that f has a maximum and minimum on S that must therefore occur at $(0, \sqrt{2}, 1)$ and $(0, -\sqrt{2}, 1)$, respectively.

The geometric interpretation of the method for multiple constraints is as follows. Suppose we have M constraints and are walking along the set of points satisfying $g_i(\mathbf{x}) = 0, i = 1, \ldots, M$. Every point $\mathbf{x}$ on the contour of a given constraint function g_i has a space of allowable directions: the space of vectors perpendicular to $\nabla g_i(\mathbf{x})$. The set of directions that are allowed by all constraints is thus the space of directions perpendicular to all of the constraints' gradients. Denote this space of allowable moves by A and denote the span of the constraints' gradients by S. Then $A = S^\perp$, the space of vectors perpendicular to every element of S.

We are still interested in finding points where f does not change as we walk, since these points might be (constrained) extrema. We therefore seek $\mathbf{x}$ such that any allowable direction of movement away from $\mathbf{x}$ is perpendicular to $\nabla f(\mathbf{x})$ (otherwise we could increase f by moving along that allowable direction). In other words, $\nabla f(\mathbf{x}) \in A^\perp = S$. Thus there are scalars $\lambda_1, \lambda_2, \ldots, \lambda_M$ such that

$$\nabla f(\mathbf{x}) = \sum_{k=1}^{M} \lambda_k \nabla g_k(\mathbf{x}) \quad \Longleftrightarrow \quad \nabla f(\mathbf{x}) - \sum_{k=1}^{M} \lambda_k \nabla g_k(\mathbf{x}) = 0.$$

These scalars are the Lagrange multipliers. We now have M of them, one for every constraint.

7.3 Global Maxima and Minima

The method of Lagrange multipliers enhances our techniques for finding global maxima and minima. In this respect, the following is useful.

Definition 7.3.1 *Let U be an open region in $\mathbb{R}^n$ with boundary ∂U. We say that ∂U is **smooth** if ∂U is the level set of a smooth function g whose gradient ∇g never vanishes (i.e., $\nabla g \neq 0$).*

Then we have the following strategy. Let f be a differentiable function on a closed and bounded region $D = U \cup \partial U$, U open in $\mathbb{R}^n$, with smooth boundary ∂U.

To find the absolute maximum and minimum of f on D:

1. Locate all critical points of f in U.
2. Use the method of Lagrange multiplier to locate all the critical points of $f|\partial U$.
3. Compute the values of f at all these critical points.
4. Select the largest and the smallest.

Example 7.3.1 Find the absolute maximum and minimum of $f(x, y) = \frac{1}{2}x^2 + \frac{1}{2}y^2$ in the elliptical region D defined by $\frac{1}{2}x^2 + y^2 \leq 1$.

Again by Theorem 6.2.7, the absolute maximum exists. We first locate the critical points of f in U, the set of points (x, y) with $\frac{1}{2}x^2 + y^2 < 1$.

Because

$$\frac{\partial f}{\partial y} = y,$$
$$\frac{\partial f}{\partial x} = x,$$

the only critical point is the origin $(0, 0)$.

We now find the maximum and minimum of f on C, the boundary of U, which is the level curve $g(x, y) = 1$, where $g(x, y) = \frac{1}{2}x^2 + y^2$. The Lagrange multiplier equations are

$$\nabla f(x, y) = (x, y) = \lambda \nabla g(x, y) = \lambda(x, 2y),$$
$$\frac{x^2}{2} + y^2 = 1.$$

If $x = 0$, then $y = \pm 1$ and $\lambda = \frac{1}{2}$. If $y = 0$, then $x = \pm\sqrt{2}$ and $\lambda = 1$. If $x \neq 0$ and $y \neq 0$, we get both $\lambda = 1$ and $\frac{1}{2}$, which is impossible.

Thus, the candidates for the maxima and minima of f on C are $(0, \pm 1)$, $(\pm\sqrt{2}, 0)$ and for f inside D, the candidate is $(0, 0)$. The value of f at $(0, \pm 1)$ is $\frac{1}{2}$, at $(\pm\sqrt{2}, 0)$ it is 1, and at $(0, 0)$ it is 0. Thus, the absolute minimum of f occurs at $(0, 0)$ and is 0. The absolute maximum of f on D is thus 1 and occurs at the points $(\pm\sqrt{2}, 0)$.

Example 7.3.2 Find the absolute maximum and minimum of the function $f(x, y, z) = x^2 + y^2 + z^2 - x + y$ on the set $D = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$.

As in the previous examples, we know the absolute maximum and minimum exists. Now $D = U \cup \partial U$, where $U = \{(x, y, z) \mid x^2+y^2+z^2 < 1\}$ and $\partial U = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$. The gradient of f is $\nabla f(x, y, z) = (2x - 1, 2y + 1, 2z)$. Thus, $\nabla f = 0$ at $(\frac{1}{2}, -\frac{1}{2}, 0)$ which is in U, the interior of D.

Let $g(x, y, z) = x^2 + y^2 + z^2$. Then ∂U is the level set $g(x, y, z) = 1$. By the method of Lagrange multipliers, the maximum and minimum must occur at a critical point of $f|\partial U$; that is, at a point x_0 where $\nabla f(x_0) = \lambda \nabla g(x_0)$ for some scalar λ. Thus,

$$(2x - 1, 2y + 1, 2z) = \lambda(2x, 2y, 2z)$$

or

$$(i) \quad 2x - 1 = 2\lambda x$$
$$(ii) \quad 2y + 1 = 2\lambda y$$
$$(iii) \quad 2z = 2\lambda z$$

If $\lambda = 1$, then we would have $2x - 1 = 2x$ or $-1 = 0$, which is impossible. We may assume that $\lambda \neq 0$ since if $\lambda = 0$, we only get an interior point as above. Thus (iii) implies that $z = 0$ and

$$(iv) \quad x^2 + y^2 = 1.$$

Now, from (i) and (ii) we get $\lambda = 1 - 1/2x$ and $\lambda = 1 + 1/2y$ and therefore $x = -y$. Substituting in (iv) we get (remembering that $z = 0$) the points $(1/\sqrt{2}, -1/\sqrt{2}, 0)$ and $(-1/\sqrt{2}, 1/\sqrt{2}, 0)$.

Finally, we find the values of f in the three critical points obtained:

$$f(1/2, -1/2, 0) = -\frac{1}{2},$$

$$f(1/\sqrt{2}, -1/\sqrt{2}, 0) = 1 - \sqrt{2},$$

$$f(-1/\sqrt{2}, 1/\sqrt{2}, 0) = 1 + \sqrt{2}.$$

Therefore, the absolute maximum is at $(-1/\sqrt{2}, 1/\sqrt{2}, 0)$ and the absolute minimum is at $(1/2, -1/2, 0)$.

Exercises

Exercise 7.1 1. Find the minimum of the function $f(x, y) = x^2 + y^2$ on the set $A = \{xy = 1\}$.
2. Find the extrema of the function $f(x, y) = xy$ on the set $A = \{x^2 + 4y^2 = 4\}$.

Exercise 7.2 Compute the absolute maxima and minima of the function

$$f(x, y) = x^2 + y^2 + 6x - 8y + 25$$

on the set $D = \{x^2 + y^2 \leq 16\}$.

Exercise 7.3 Compute the extrema of the following functions with the specified restrictions:

1. $f(x, y) = xy$ restricted to $2x + 3y - 5 = 0$.
2. $g(x, y, z) = xyz$ restricted to $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
3. $h(x, y, z) = x^2y^4z^6$ restricted to $x + y + z = 1$, $x, y, z > 0$.

Exercise 7.4 Compute the extrema of the following functions on the given subsets:

1. $f(x, y) = x^2 + y^2 - 2x - 2y + 2$ on $T = \{y/2 \leq x \leq 3 - \sqrt{2y}, 0 \leq y \leq 2\}$.
2. $f(x, y, z) = x + y + z$ on $S = \{2x^2 + 3y^2 + 6z^2 = 1\}$.
3. $f(x, y, z) = x^2 + y^2 + z^2$ on $U = \{z \geq x^2 + y^2 - 2\}$.

Exercise 7.5 Maximize the expression xyz on the sphere $x^2 + y^2 + z^2 = 1$.

Exercise 7.6 Minimize the expression $x + 2y + 4z$ on the sphere $x^2 + y^2 + z^2 = 1$.

Exercise 7.7 Find the maximal and minimal values of the function $f(x, y, z) = x + 2y + 3z$ taking into account the two restrictions $x^2 + y^2 = 2$, $x + z = 1$.

Exercise 7.8 Compute the distance of the point $(4, 4, 10)$ to the sphere $(x - 1)^2 + y^2 + (z + 2)^2 = 25$ in two ways:

1. Geometrically.
2. Using Lagrange multipliers.

Exercise 7.9 Express a positive number A as a product of four positive factors with minimal sum.

Exercise 7.10 The production of a company is described in terms of the function $f(K, L) = K^\alpha L^{1-\alpha}$, where $0 < \alpha < 1$, K is the amount of capital, and L is the amount of manpower used. The price of the capital is p and the price of the manpower is q. Compute the proportion between the capital and manpower needed to maximize the production using a budget B.

Exercise 7.11 A company produces three different products in quantities Q_1, Q_2, Q_3 and generates a profit given by the expression

$$P(Q_1, Q_2, Q_3) = 2Q_1 + 8Q_2 + 24Q_3.$$

Find the values of Q_1, Q_2, Q_3 that maximize the profit if the production is constrained to $Q_1^2 + 2Q_2^2 + 4Q_3^2 = 4.5 \times 10^9$.

Exercise 7.12 Find all maxima and minima of the scalar function $f(x, y) = x^3 + y^3$ restricted to the curve $g(x, y) = x^4 + y^4 - 1 = 0$.

Exercise 7.13 Find all points on the hyperbola $x^2 + 8xy + 7y^2 = 225$ that have minimal distance to the origin.

Part III. Integral Calculus in $\mathbb{R}^n$

Calculating volumes: Double Integrals 8

8.1 Double Integrals as Volumes

Consider a continuous function of two variables $f : R \subset \mathbb{R}^2 \to \mathbb{R}$ whose domain R is a rectangle with sides parallel to the coordinate axes. The rectangle R can be described in terms of the two closed intervals $[a, b]$ and $[c, d]$, representing the sides of R along the x and y axes, respectively. In this case, we say that R is the Cartesian product of $[a, b]$ and $[c, d]$ and write $R = [a, b] \times [c, d]$.

Assume that $f(x, y) \geq 0$ on R, so that the graph of $z = f(x, y)$ is a surface lying above the rectangle R. This surface, the rectangle R, and the four planes $x = a$, $x = b$, $y = c$, and $y = d$ form the boundary of a region V in space.

The volume of the region above R and under the graph of a nonnegative function f is called the (double) integral of f over R and is denoted by[1]

$$\iint_R f(x, y)\, dA \quad \text{or} \quad \iint_R f(x, y)\, dx\, dy.$$

1: These ideas are similar to those for a single integral

$$\int_a^b f(x)\, dx,$$

which represents the area under the graph of f if $f \geq 0$. Single integrals $\int_a^b f(x)\, dx$ can be rigorously defined, without recourse to the area concept, as a limit of Riemann sums. The idea is to approximate $\int_a^b f(x)\, dx$ by choosing a partition $a = x_0 < x_1 < \cdots < x_n = b$ of $[a, b]$, selecting points $c_i \in [x_i, x_{i+1}]$, and forming the Riemann sum

$$\sum_{i=0}^{n-1} f(c_i)(x_{i+1} - x_i) \approx \int_a^b f(x)\, dx$$

Let's define the double integral rigorously. For the rectangle R, consider a regular partition of the intervals $[a, b]$ and $[c, d]$, that is, two ordered collections of $n+1$ equally spaced points $\{x_j\}_{j=0}^n$ and $\{y_k\}_{k=0}^n$ satisfying

$$a = x_0 < x_1 < \cdots < x_n = b,$$

$$c = y_0 < y_1 < \cdots < y_n = d$$

and

$$x_{j+1} - x_j = \frac{b-a}{n},$$

$$y_{k+1} - y_k = \frac{d-c}{n}$$

Let R_{jk} be the rectangle $[x_j, x_{j+1}] \times [y_k, y_{k+1}]$, and let c_{jk} be any point in R_{jk}. Suppose $f : R \to \mathbb{R}$ is a bounded real-valued function[2] . Form the sum

$$S_n = \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} f(c_{jk}) \Delta x \Delta y = \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} f(c_{jk}) \Delta A,$$

2: That is, there exists a real number $M > 0$ such that $|f(x, y)| \leq M$ for all (x, y) in the domain of f.

where

$$\Delta x = x_{j+1} - x_j = \frac{b-a}{n}, \quad \Delta y = y_{k+1} - y_k = \frac{d-c}{n},$$

and

$$\Delta A = \Delta x \Delta y.$$

This sum is taken over all j's and k's from 0 to $n-1$, and so there are n^2 terms. A sum of this type is called a Riemann sum for f.

Definition 8.1.1 (Double Integral) *If the sequence $\{S_n\}$ converges to a limit S as $n \longrightarrow \infty$ and if the limit S is the same for any choice of points c_{jk} in the rectangles R_{jk}, then we say that f is integrable over R and we write*

$$\iint_R f(x,y)\,dA, \qquad \iint_R f(x,y)\,dx\,dy, \quad \text{or} \quad \iint_R f\,dx\,dy$$

for the limit S.

Thus, we can rewrite integrability in the following way:

$$\lim_{n\longrightarrow\infty} \sum_{j=0}^{n-1}\sum_{k=0}^{n-1} f(c_{jk})\Delta x \Delta y = \iint_R f\,dx\,dy.$$

Example 8.1.1 Evaluate the integral

$$\iint_R (x^2+y^2)\,dx\,dy,$$

where $R = [-1,1]\times[0,1]$.

Consider a partition $x_i = -1+\frac{2i}{n}$, $i=0,\ldots,n$ and $y_j = \frac{j}{n}$, $j=0,\ldots,n$. Then $\Delta x = \frac{2}{n}$ and $\Delta y = \frac{1}{n}$ and the Riemann sum (taking the values of the upper right-hand corner of each rectangle) is

$$S_n = \sum_{i=1}^{n}\sum_{j=1}^{n} f(x_i,y_i)\Delta x\Delta y = \sum_{i=1}^{n}\sum_{j=1}^{n}\left[\left(-1+\frac{2i}{n}\right)^2+\left(\frac{j}{n}\right)^2\right]\frac{2}{n^2}.$$

Expanding we obtain

$$S_n = \frac{2}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left[1+\frac{4}{n}^2 i^2-\frac{4}{n}i+\frac{j^2}{n^2}\right] = \frac{2}{n^2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left[1+\frac{5}{n^2}i^2-\frac{4}{n}i\right],$$

since the sum is symmetric with respect to i and j. Also, since the summand in brackets does not depend on j, we have

$$S_n = \frac{2}{n}\sum_{i=1}^{n}\left[1+\frac{5}{n^2}i^2-\frac{4}{n}i\right].$$

Finally, we need to use

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and therefore

$$S_n = \frac{2}{n}\left[n+\frac{5}{n^2}\frac{n(n+1)(2n+1)}{6}-\frac{4}{n}\frac{n(n+1)}{2}\right] = \frac{4}{3}+\frac{1}{n}+\frac{5}{3n^2}.$$

If we take the limit $n \to \infty$ we see that $\iint_R (x^2 + y^2)\, dx\, dy = \frac{4}{3}$.

A simple condition for functions to be integrable is the following, that we state without proof:

Theorem 8.1.1 (Integrability of Bounded Functions) *Let $f : R \to \mathbb{R}$ be a bounded real-valued function on the rectangle R, and suppose that the set of points where f is discontinuous lies on a finite union of graphs of continuous functions. Then f is integrable over R.*

Note that this theorem implies that continuous functions are integrable.

8.2 Properties of the Integral

From the definition of the integral as a limit of sums and the limit theorems, we can deduce some fundamental properties of the integral

$$\iint_R f(x, y)\, dA;$$

these properties are essentially the same as for the integral of a real-valued function of a single variable.

Theorem 8.2.1 (Properties of the Integral) *Let f and g be integrable functions on the rectangle R, and let c be a constant. Then $f + g$ and cf are integrable, and*

(i) Linearity

$$\iint_R [af(x, y) + bg(x, y)]\, dA = a \iint_R f(x, y)\, dA + b \iint_R g(x, y)\, dA.$$

(ii) Monotonicity *If $f(x, y) \geq g(x, y)$, then*

$$\iint_R f(x, y)\, dA \geq \iint_R g(x, y)\, dA.$$

(iii) Additivity *If R_i, $i = 1, \ldots, m$, are pairwise disjoint rectangles such that f is bounded and integrable over each R_i and if $Q = R_1 \cup R_2 \cup \cdots \cup R_m$ is a rectangle, then $f : Q \to \mathbb{R}$ is integrable over Q and*

$$\iint_Q f(x, y)\, dA = \sum_{i=1}^{m} \iint_{R_i} f(x, y)\, dA.$$

(iv)

$$\left| \iint_R f\, dA \right| \leq \iint_R |f|\, dA.$$

Proof. Property (i) is a consequence of the definition of the integral as a limit of a sum and the following facts for convergent sequences $\{S_n\}$ and $\{T_n\}$:

$$\lim_{n\to\infty} (T_n + S_n) = \lim_{n\to\infty} T_n + \lim_{n\to\infty} S_n$$

$$\lim_{n\to\infty} (cS_n) = c \lim_{n\to\infty} S_n.$$

To demonstrate monotonicity (property (ii)), we first observe that if $h(x, y) \geq 0$ and $\{S_n\}$ is a sequence of Riemann sums that converges to $\iint_R h(x, y)\, dA$, then $S_n \geq 0$ for all n, so that $\iint_R h(x, y)\, dA = \lim_{n\to\infty} S_n \geq 0$. If $f(x, y) \geq g(x, y)$ for all $(x, y) \in R$, then $(f - g)(x, y) \geq 0$ for all (x, y), and, using properties (i) and (ii), we have

$$\iint_R f(x, y)\, dA - \iint_R g(x, y)\, dA = \iint_R [f(x, y) - g(x, y)]\, dA \geq 0.$$

The proof of property (iii) is more technical, but it should be intuitively obvious.

To see why property (iv) is true, note that, by the definition of absolute value,

$$-|f| \leq f \leq |f|;$$

therefore, from the monotonicity and homogeneity of integration (with $c = -1$),

$$\iint_R |f|\, dA \leq -\iint_R f\, dA \leq \iint_R |f|\, dA.$$

□

8.3 How to calculate double integrals: Fubini's Theorem

Computing double integrals by Riemann sums is impractical in most cases. Therefore, we wish to find a result that simplifies this calculation. This is given by Fubini's theorem.

Theorem 8.3.1 (Fubini's Theorem) *Let f be a continuous function with a rectangular domain $R = [a, b] \times [c, d]$. Then*

$$\int_a^b \left(\int_c^d f(x, y)\, dy \right) dx = \int_c^d \left(\int_a^b f(x, y)\, dx \right) dy = \iint_R f(x, y)\, dA. \tag{4}$$

Proof. We shall first show that

$$\int_a^b \int_c^d f(x, y)\, dy\, dx = \iint_R f(x, y)\, dA.$$

Let $c = y_0 < y_1 < \cdots < y_n = d$ be a partition of $[c, d]$ into n equal parts. Define

$$F(x) = \int_c^d f(x, y)\, dy.$$

Then

$$F(x) = \sum_{k=0}^{n-1} \int_{y_k}^{y_{k+1}} f(x, y)\, dy.$$

Using the integral version of the mean-value theorem, for each fixed x and for each k we have

$$\int_{y_k}^{y_{k+1}} f(x,y)\,dy = f(x, Y_k(x))(y_{k+1} - y_k),$$

where the point $Y_k(x)$ belongs to $[y_k, y_{k+1}]$ and may depend on x, k, and n. We have thus shown that

$$F(x) = \sum_{k=0}^{n-1} f(x, Y_k(x))(y_{k+1} - y_k).$$

By the definition of the integral in one variable as a limit of Riemann sums,

$$\int_a^b F(x)\,dx = \lim_{n\to\infty} \sum_{j=0}^{n-1} F(p_j)(x_{j+1} - x_j),$$

where $a = x_0 < x_1 < \cdots < x_n = b$ is a partition of the interval $[a,b]$ into n equal parts and p_j is any point in $[x_j, x_{j+1}]$. Setting $c_{jk} = (p_j, Y_k(p_j)) \in R_{jk}$, we have

$$F(p_j) = \sum_{k=0}^{n-1} f(c_{jk})(y_{k+1} - y_k).$$

Therefore,

$$\begin{aligned}
\int_a^b \left(\int_c^d f(x,y)\,dy \right) dx = \int_a^b F(x)\,dx &= \lim_{n\to\infty} \sum_{j=0}^{n-1} F(p_j)(x_{j+1} - x_j) \\
&= \lim_{n\to\infty} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} f(c_{jk})(y_{k+1} - y_k)(x_{j+1} - x_j) \\
&= \iint_R f(x,y)\,dA.
\end{aligned}$$

Thus, we have proved that

$$\int_a^b \left(\int_c^d f(x,y)\,dy \right) dx = \iint_R f(x,y)\,dA.$$

By the same reasoning we can show that

$$\int_c^d \left(\int_a^b f(x,y)\,dx \right) dy = \iint_R f(x,y)\,dA.$$

These two conclusions are exactly what we wanted to prove. □

Fubini's theorem can be generalized to the case where f is not continuous, but its discontinuities lie on a finite union of graphs of continuous functions. If the integral $\int_c^d f(x,y)\,dy$ exists for each $x \in [a,b]$, then

$$\int_a^b \left(\int_c^d f(x,y)\,dy \right) dx = \iint_R f(x,y)\,dA.$$

Similarly, if $\int_a^b f(x,y)\,dx$ exists for each $y \in [c,d]$, then

$$\int_c^d \int_a^b f(x,y)\,dx\,dy = \iint_R f(x,y)\,dA.$$

Example 8.3.1 Compute

$$\iint_R (x^2+y)\,dA, \quad \text{where } R \text{ is the square } [0,1]\times[0,1].$$

By Fubini's theorem,

$$\iint_R (x^2+y)\,dA = \int_0^1 \int_0^1 (x^2+y)\,dx\,dy = \int_0^1 \int_0^1 (x^2+y)\,dy\,dx.$$

By the fundamental theorem of calculus, the x integration may be performed:

$$\int_0^1 (x^2+y)\,dx = \left[\frac{x^3}{3}+yx\right]_{x=0}^{x=1} = \frac{1}{3}+y.$$

Thus,

$$\iint_R (x^2+y)\,dA = \int_0^1 \left(\frac{1}{3}+y\right)\,dy = \int_0^1 \left(\frac{y^2}{2}+\frac{y}{3}\right)\,dy = \frac{5}{6}.$$

A consequence of Fubini's theorem is that interchanging the order of integration in the iterated integrals does not change the answer:

$$\int_0^1 \int_0^1 (x^2+y)\,dy\,dx = \int_0^1 \left[\frac{y^2}{2}+xy\right]_{y=0}^{y=1}\,dx = \int_0^1 \left(\frac{1}{2}+x\right)\,dx = \frac{5}{6}.$$

We have seen that when $f(x,y)\geq 0$ on $R=[a,b]\times[c,d]$, the integral $\iint_R f(x,y)\,dA$ can be interpreted as a volume. If the function also takes on negative values, then the double integral can be thought of as the sum of all volumes lying between the surface $z=f(x,y)$ and the plane $z=0$, bounded by the planes $x=a$, $x=b$, $y=c$, and $y=d$; here the volumes above $z=0$ are counted as positive and those below as negative. However, Fubini's theorem as stated remains valid in the case where $f(x,y)$ is negative or changes sign on R; that is, there is no restriction on the sign of f in the hypotheses of the theorem.

8.4 The Double Integral Over More General Regions

Our goal in this section is twofold: First, we wish to define the double integral of a function $f(x,y)$ over regions D more general than rectangles; second, we want to develop a technique for evaluating this type of integral. To accomplish this, we shall define three special types of subsets of the xy-plane, and then extend the notion of the double integral to them.

Elementary Regions

Let D be the set of all points (x, y) such that $x \in [a, b]$ and $\varphi_1(x) \leq y \leq \varphi_2(x)$ for some functions $\varphi_1 : [a, b] \to \mathbb{R}$ and $\varphi_2 : [a, b] \to \mathbb{R}$ ($\varphi_1(x) \leq \varphi_2(x)$ for all $x \in [a, b]$). This region D is said to be y-simple. The curves and straight-line segments that bound the region together constitute the boundary of D, ∂D. We use the phrase "y-simple" because the region is described using y as a function of x.

We say that a region D is x-simple if there are continuous functions ψ_1 and ψ_2 defined on $[c, d]$ such that D is the set of points (x, y) satisfying $y \in [c, d]$ and $\psi_1(y) \leq x \leq \psi_2(y)$, where $\psi_1(y) \leq \psi_2(y)$ for all $y \in [c, d]$. Again, the curves that bound the region D con- stitute its boundary ∂D.

Finally, a simple region is one that is both x- and y-simple; that is, a simple region can be described as both an x-simple region and a y-simple region. An example of a simple region is a unit disk.

Sometimes we will refer to any of these regions as elementary regions.

The Integral over an Elementary Region

We can now use an interesting "trick" to extend the definition of the integral from rectangles to elementary regions.

Definition 8.4.1 (Integral over an Elementary Region) *If D is an elementary region in the plane, choose a rectangle R that contains D. Given $f : D \to \mathbb{R}$, where f is continuous (and hence bounded), define $\iint_D f(x, y)\, dA$, the integral of f over the set D, as follows: Extend f to a function f^* defined on all of R by*

$$f^*(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \in D \\ 0 & \text{if } (x, y) \notin D \text{ and } (x, y) \in R. \end{cases}$$

Note that f^* is bounded (because f is) and continuous except possibly on the boundary of D. The boundary of D consists of graphs of continuous functions, and so f^* is integrable over R by Theorem 8.1.1. Therefore, we can define

$$\iint_D f(x, y)\, dA = \iint_R f^*(x, y)\, dA.$$

Reduction to Iterated Integrals

If $R = [a, b] \times [c, d]$ is a rectangle containing D, we have

$$\iint_D f^*(x, y)\, dA = \int_a^b \int_c^d f^*(x, y)\, dy\, dx = \int_c^d \int_a^b f^*(x, y)\, dx\, dy,$$

where f^* equals f in D and zero outside D, as before. Assume that D is a y-simple region determined by functions $\varphi_1 : [a, b] \to \mathbb{R}$ and

$\varphi_2 : [a, b] \longrightarrow \mathbb{R}$. Consider the iterated integral

$$\int_a^b \int_c^d f^*(x, y)\, dy\, dx$$

and, in particular, the inner integral $\int_c^d f^*(x, y)\, dy$ for some fixed x. By definition, $f^*(x, y) = 0$ if $y < \varphi_1(x)$ or $y > \varphi_2(x)$, so we obtain

$$\int_c^d f^*(x, y)\, dy = \int_{\varphi_1(x)}^{\varphi_2(x)} f^*(x, y)\, dy = \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y)\, dy.$$

Theorem 8.4.1 (Reduction to Iterated Integrals) *If D is a y-simple region determined by functions $\varphi_1 : [a, b] \longrightarrow \mathbb{R}$ and $\varphi_2 : [a, b] \longrightarrow \mathbb{R}$, then*

$$\iint_D f(x, y)\, dA = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y)\, dy\, dx. \tag{1}$$

Similarly, if D is a x-simple region determined by functions $\psi_1 : [c, d] \longrightarrow \mathbb{R}$ and $\psi_2 : [c, d] \longrightarrow \mathbb{R}$, then

$$\iint_D f(x, y)\, dA = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)\, dx\, dy. \tag{2}$$

In the case $f(x, y) = 1$ for all $(x, y) \in D$, $\iint_D f(x, y)\, dA$ is the area of D.

Example 8.4.1 Find $\iint_T (x^3 y + \cos x)\, dA$, where T is the triangle consisting of all points (x, y) such that $0 \le x \le \frac{\pi}{2}, 0 \le y \le x$.

We have

$$\iint_T (x^3 y + \cos x)\, dA = \int_0^{\frac{\pi}{2}} \int_0^x (x^3 y + \cos x)\, dy\, dx.$$

Now

$$\int_0^x (x^3 y + \cos x)\, dy = \frac{x^5}{2} + x \cos x$$

and so

$$\int_0^{\frac{\pi}{2}} \frac{x^5}{2} + x \cos x\, dx = \frac{x^6}{12} + x \sin x + \cos x \Big|_0^{\frac{\pi}{2}} = \frac{\pi^6}{768} + \frac{\pi}{2} - 1.$$

Example 8.4.2 Find the volume of the tetrahedron bounded by the planes $y = 0$, $z = 0$, $x = 0$, and $y - x + z = 1$.

We first note that the given tetrahedron has a triangular base D whose points (x, y) satisfy $-1 \le x \le 0$ and $0 \le y \le 1 + x$; hence, D is a y-simple region. In fact, D is a simple region. For any point (x, y) in D, the height of the surface z above (x, y) is $1 - y + x$. Thus, the volume we seek is given by the integral

$$\iint_D (1 - y + x)\, dA.$$

Using $\varphi_1(x) = 0$ and $\varphi_2(x) = x + 1$, we have

$$\begin{aligned}
\iint_D (1 - y + x)\, dA &= \int_{-1}^{0} \int_{0}^{1+x} (1 - y + x)\, dy\, dx \\
&= \int_{-1}^{0} \left[\frac{(1 + x)y^2}{2} \right]_{y=0}^{y=1+x} dx \\
&= \int_{-1}^{0} \frac{(1 + x)^3}{2}\, dx \\
&= \left[\frac{(1 + x)^4}{6} \right]_{-1}^{0} = \frac{1}{6}.
\end{aligned}$$

Exercises

Exercise 8.1 Compute the double integral $\iint_Q f(x,y)\,dx\,dy$ for the following functions:

1. $f(x,y) = xy(x+y)$, $Q = [0,1]\times[0,1]$.
2. $f(x,y) = x^3 + 3x^2y + y^3$, $Q = [0,1]\times[0,1]$.
3. $f(x,y) = \sin^2 x \sin^2 y$, $Q = [0,\pi]\times[0,\pi]$.
4. $f(x,y) = \sin(x+y)$, $Q = [0,\pi/2]\times[0,\pi/2]$.
5. $f(x,y) = x\sin y - ye^x$, $Q = [-1,1]\times[0,\pi/2]$.

Exercise 8.2 Sketch the integration region Q and compute $\iint_Q f(x,y)\,dx\,dy$ in the following cases:

1. $f(x,y) = x^2 - y$, $Q = \{(x,y)\in\mathbb{R}^2 \mid x\in[-1,1], -x^2 \le y \le x^2\}$.
2. $f(x,y) = xy - x^3$, $Q = \{(x,y)\in\mathbb{R}^2 \mid x\in[0,1], -1 \le y \le x\}$.
3. $f(x,y) = 2x - \sin(x^2y)$, $Q = \{(x,y)\in\mathbb{R}^2 \mid x\in[-2,2], |y| \le |x|\}$.
4. $f(x,y) = y\sin x$, $Q = \{(x,y)\in\mathbb{R}^2 \mid |x| + |y| \le 1\}$.

Exercise 8.3
1. Prove the following inequalities without solving the integral explicitly:

$$4\pi \le \iint_D (x^2+y^2+1)\,dx\,dy \le 20\pi,$$

where D is the disc with radius 2 centered at the origin.
2. Let A be the square $[0,2]\times[1,3]$ and $f(x,y) = x^2y$. Prove the following inequalities without solving the integral explicitly:

$$0 \le \iint_A f(x,y)\,dx\,dy \le 48.$$

3. Using a partition of A into four equal squares, improve the preceding estimations and show that

$$3 \le \iint_A f(x,y)\,dx\,dy \le 25.$$

Exercise 8.4 Compute the double integral

$$\int_0^1 \int_0^1 \max(|x|,|y|)\,dx\,dy.$$

Changing the order of integration and Triple Integrals 9

9.1 Changing the Order of Integration

Suppose that D is a simple region—that is, it is both x-simple and y-simple. Thus, it can be given as the set of points (x, y) such that

$$a \leq x \leq b, \quad \varphi_1(x) \leq y \leq \varphi_2(x),$$

and also as the set of points (x, y) such that

$$c \leq y \leq d, \quad \psi_1(y) \leq x \leq \psi_2(y).$$

Hence, we have the formulas

$$\iint_D f(x,y)\,dA = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x,y)\,dy\,dx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x,y)\,dx\,dy.$$

If we are required to compute one of the preceding iterated integrals, we may do so by evaluating the other iterated integral; this technique is called changing the order of integration. It can be useful to make such a change when evaluating iterated integrals, because one of the iterated integrals may be more difficult to compute than the other.

Example 9.1.1 By changing the order of integration, evaluate

$$\int_0^a \int_0^{(a^2-x^2)^{1/2}} (a^2-y^2)^{1/2}\,dy\,dx.$$

Note that x varies between 0 and a, and for each such fixed x, we have $0 \leq y \leq (a^2-x^2)^{1/2}$. Thus, the iterated integral is equivalent to the double integral

$$\iint_D (a^2-y^2)^{1/2}\,dy\,dx,$$

where D is the set of points (x, y) such that $0 \leq x \leq a$ and $0 \leq y \leq (a^2-x^2)^{1/2}$. But this is the representation of one-quarter (the positive quadrant portion) of the disk of radius a; hence, D can also be described as the set of points (x, y) satisfying

$$0 \leq y \leq a, \quad 0 \leq x \leq (a^2-y^2)^{1/2}.$$

Thus,

$$\int_0^a \int_0^{(a^2-y^2)^{1/2}} (a^2-y^2)^{1/2}\,dx\,dy = \int_0^a \left[x(a^2-y^2)^{1/2}\right]_{x=0}^{x=(a^2-y^2)^{1/2}} dy$$
$$= \int_0^a (a^2-y^2)\,dy = \frac{2a^3}{3}.$$

Compare this with the alternative $\int_0^a \int_0^{(a^2-x^2)^{1/2}} (a^2-y^2)^{1/2}\,dy\,dx$. For the inner integral we can use the change of variable $y = a\sin t$, which leads to (note the change in the limits of integration):

$$\begin{aligned}\int_0^{(a^2-x^2)^{1/2}} (a^2-y^2)^{1/2}\,dy &= \int_0^{\arccos\frac{x}{a}} a^2\cos^2 t\,dt\\ &= \frac{a^2}{2}\left[t+\frac{1}{2}\sin 2t\right]_0^{\arccos\frac{x}{a}}\\ &= \frac{a^2}{2}\left[\arccos\frac{x}{a}+\frac{x}{a}\sin\left(\arccos\frac{x}{a}\right)\right]\\ &= \frac{a^2}{2}\left[\arccos\frac{x}{a}+\frac{x}{a}\sqrt{1-\left(\frac{x}{a}\right)^2}\right]\end{aligned}$$

where we have used the facts that $\cos^2 t = (1+\cos 2t)/2$, $\sin 2t = 2\sin t\cos t$ and $\sin(\arccos x) = \sqrt{1-x^2}$. So now we have to compute $\int_0^a \int_0^{(a^2-x^2)^{1/2}} (a^2-y^2)^{1/2}\,dy\,dx = \frac{a^2}{2}\int_0^a \left[\arccos\frac{x}{a}+\frac{x}{a}\sqrt{1-\left(\frac{x}{a}\right)^2}\right]dx.$

The first integral is, using the fact $\int \arccos x\,dx = x\arccos x - \sqrt{1-x^2} + c$ (you can prove this using integration by parts) and the change of variable $u = x/a$:

$$\int_0^a \arccos\frac{x}{a}\,dx = a\int_0^1 \arccos u\,du = a\left[u\arccos u - \sqrt{1-u^2}\right]_0^1 = a.$$

The second integral, using the same change of variable $u = x/a$, becomes

$$\int_0^a \frac{x}{a}\sqrt{1-\left(\frac{x}{a}\right)^2}\,dx = a\int_0^1 u\sqrt{1-u^2}\,du = \frac{a}{2}\left[-\frac{2}{3}(1-u^2)^{3/2}\right]_0^1 = \frac{a}{3}.$$

Summing all together, we have

$$\int_0^a \int_0^{(a^2-x^2)^{1/2}} (a^2-y^2)^{1/2}\,dy\,dx = \frac{a^2}{2}\left(a+\frac{a}{3}\right) = \frac{2}{3}a^3.$$

Example 9.1.2 Evaluate

$$\int_1^2 \int_0^{\log x} (x-1)\sqrt{1+e^{2y}}\,dy\,dx.$$

This integral is equal to $\iint_D (x-1)\sqrt{1+e^{2y}}\,dA$, where D is the set of (x,y) such that $1\le x\le 2$ and $0\le y\le \log x$. The region D is simple and can also be described by $0\le y\le \log 2$ and $e^y\le x\le 2$. Thus, the given iterated integral is equal to

$$\int_0^{\log 2} \int_{e^y}^2 (x-1)\sqrt{1+e^{2y}}\,dx\,dy.$$

Thus, we have

$$\int_0^{\log 2}\left[\int_{e^y}^2 (x-1)\,dx\right]\sqrt{1+e^{2y}}\,dy = \int_0^{\log 2}\left[\frac{x^2}{2}-x\right]_{e^y}^2\sqrt{1+e^{2y}}\,dy$$
$$= -\int_0^{\log 2}\left(\frac{e^{2y}}{2}-e^y\right)\sqrt{1+e^{2y}}\,dy$$
$$= -\frac{1}{2}\int_0^{\log 2} e^{2y}\sqrt{1+e^{2y}}\,dy +$$
$$+\int_0^{\log 2} e^y\sqrt{1+e^{2y}}\,dy.$$

In the first integral we substitute $u = e^{2y}$, and in the second, $v = e^y$. Hence, we obtain

$$-\frac{1}{4}\int_1^4 \sqrt{1+u}\,du + \int_1^2 \sqrt{1+v^2}\,dv.$$

For the first integral, we get

$$-\frac{1}{4}\int_1^4 \sqrt{1+u}\,du = -\frac{1}{6}\left[(1+u)^{3/2}\right]_1^4 = -\frac{1}{6}[5^{3/2}-2^{3/2}].$$

For the second integral, we need to introduce the change of variable $v = \tan t$ which, since $1+v^2 = \sec^2 t$, yields

$$\int_1^2 \sqrt{1+v^2}\,dv = \int_{\arctan 1}^{\arctan 2} \sec^3 t\,dt.$$

Let's obtain a primitive for the second integrand. Integrating by parts ($u = \sec t, dv = \sec^2 t\,dt$), we obtain

$$\int \sec^3 t\,dt = \sec t\tan t - \int \sec t\tan^2 t\,dt$$
$$= \sec t\tan t - \int \sec^3 t\,dt + \int \sec t\,dt,$$

where we have used the fact that $\tan^2 t = \sec^2 t - 1$. Now, solving for $\int \sec^3 t\,dt$, we get

$$\int \sec^3 t\,dt = \frac{1}{2}\sec t\tan t + \frac{1}{2}\log|\sec t + \tan t| + c,$$

where we have used $\int \sec t\,dt = \log|\sec t + \tan t| + c$ (which you can obtain by multiplying and dividing by $\sec t + \tan t$ and realizing that the numertaor is the derivative of the denominator).

Now, in order to undo the change of variable for our original integral, note that $\sec t = \sqrt{1+v^2}$ and so

$$\int_1^2 \sqrt{1+v^2}\,dv = \frac{1}{2}\left[v\sqrt{1+v^2} + \log(v+\sqrt{1+v^2})\right]_1^2$$
$$= \frac{1}{2}\left[2\sqrt{5} + \log(2+\sqrt{5}) - \sqrt{2} - \log(\sqrt{2}+1)\right].$$

Finally, the answer is

$$\frac{1}{2}\left[2\sqrt{5}-\sqrt{2}+\log\left(\frac{\sqrt{5}+2}{\sqrt{2}+1}\right)\right]-\frac{1}{6}\left[5^{3/2}-2^{3/2}\right].$$

9.2 The Triple Integral

Our objective now is to define the triple integral of a function $f(x,y,z)$ over a box (rectangular parallelepiped) $B=[a,b]\times[c,d]\times[p,q]$. Proceeding as in double integrals, we partition the three sides of B into n equal parts and form the sum

$$S_n=\sum_{i=0}^{n-1}\sum_{j=0}^{n-1}\sum_{k=0}^{n-1}f(c_{ijk})\Delta V,$$

where c_{ijk} is a point in B_{ijk}, the ijk-th rectangular parallelepiped (or box) in the partition of B, and ΔV is the volume of B_{ijk}.

Definition 9.2.1 (Triple Integrals) *Let f be a bounded function of three variables defined on B. If $\lim_{n\to\infty}S_n=S$ exists and is independent of any choice of c_{ijk}, we call f integrable and call S the triple integral (or simply the integral) of f over B and denote it by*

$$\iiint_B f\,dV,\quad \iiint_B f(x,y,z)\,dV\quad \text{or}\quad \iiint_B f(x,y,z)\,dx\,dy\,dz.$$

Properties of Triple Integrals

As before, we can prove that continuous functions defined on B are integrable. Moreover, bounded functions whose discontinuities are confined to graphs of continuous functions [such as $x=\alpha(y,z)$, $y=\beta(x,z)$, or $z=\gamma(x,y)$] are integrable. The other basic properties (such as the fact that the integral of a sum is the sum of the integrals) for double integrals also hold for triple integrals. Especially important is the reduction to iterated integrals:

Theorem 9.2.1 (Reduction to Iterated Integrals) *Let $f(x,y,z)$ be integrable on the box $B=[a,b]\times[c,d]\times[p,q]$. Then any iterated integral that exists is equal to the triple integral; that is,*

$$\begin{aligned}\iiint_B f(x,y,z)\,dx\,dy\,dz &= \int_p^q\int_c^d\int_a^b f(x,y,z)\,dx\,dy\,dz\\ &= \int_p^q\int_a^b\int_c^d f(x,y,z)\,dy\,dx\,dz\\ &= \int_a^b\int_p^q\int_c^d f(x,y,z)\,dy\,dz\,dx,\end{aligned}$$

and so on. (There are six possible orders altogether.)

Example 9.2.1 Let B be the box $[0,1] \times [-\frac{1}{2}, 0] \times [0, \frac{1}{3}]$. Evaluate

$$\iiint_B (x + 2y + 3z)^2 \, dx \, dy \, dz.$$

Verify that we get the same answer if the integration is done in the order y first, then z, and then x.

This integral may be evaluated as

$$\begin{aligned}
&\int_0^{\frac{1}{3}} \int_{-\frac{1}{2}}^0 \int_0^1 (x + 2y + 3z)^2 \, dx \, dy \, dz = \\
&= \int_0^{\frac{1}{3}} \int_{-\frac{1}{2}}^0 \frac{1}{3} \left[(1 + 2y + 3z)^3 - (2y + 3z)^3 \right] \, dy \, dz \\
&= \int_0^{\frac{1}{3}} \frac{1}{24} \left[(3z + 1)^4 - 2(3z)^4 + (3z - 1)^4 \right] \, dz \\
&= \frac{1}{360} \left[2^5 - 2 \right] = \frac{1}{12}.
\end{aligned}$$

and also as

$$\begin{aligned}
&\int_0^1 \int_0^{\frac{1}{3}} \int_{-\frac{1}{2}}^0 (x + 2y + 3z)^2 \, dy \, dz \, dx = \\
&= \int_0^1 \int_0^{\frac{1}{3}} \frac{1}{6} \left[(x + 3z)^3 - (x + 3z - 1)^3 \right] \, dz \, dx \\
&= \int_0^1 \frac{1}{72} \left[(x + 1)^4 + (x - 1)^4 - 2x^4 \right] dx \\
&= \frac{1}{360} \left[2^5 - 2 \right] = \frac{1}{12}.
\end{aligned}$$

Example 9.2.2 Integrate e^{x+y+z} over the box $[0,1] \times [0,1] \times [0,1]$.

We perform the integrations in the standard order:

$$\begin{aligned}
\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} \, dx \, dy \, dz &= \int_0^1 \int_0^1 (e^{1+y+z} - e^{y+z}) \, dy \, dz \\
&= \int_0^1 (e^{2+z} - 2e^{1+z} + e^z) \, dz \\
&= e^3 - 3e^2 + 3e - 1 = (e - 1)^3.
\end{aligned}$$

Notice that $e^{x+y+z} = e^x e^y e^z$ and so we could have calculated

$$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} \, dx \, dy \, dz = \left(\int_0^1 e^x \, dx \right)^3 = (e - 1)^3.$$

Elementary Regions

As before, we restrict our attention to particularly simple regions. An elementary region in three-dimensional space is one defined by restricting one of the variables to be between two functions of the remaining variables, the domains of these functions being an elementary (i.e., an x-simple or a y-simple) region in the plane. For example, if D is an elementary region in the xy plane and if $\gamma_1(x, y)$ and $\gamma_2(x, y)$ are two functions with $\gamma_2(x, y) \geq \gamma_1(x, y)$, an elementary region consists of all (x, y, z) such that (x, y) lies in D and $\gamma_1(x, y) \leq z \leq \gamma_2(x, y)$.

Triple Integrals by Iterated Integration

Suppose that W is an elementary region described by bounding z between two functions of x and y. Then either

$$\iiint_W f(x, y, z)\, dx\, dy\, dz = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \int_{\gamma_1(x,y)}^{\gamma_2(x,y)} f(x, y, z)\, dz\, dy\, dx$$

or

$$\iiint_W f(x, y, z)\, dx\, dy\, dz = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} \int_{\gamma_1(x,y)}^{\gamma_2(x,y)} f(x, y, z)\, dz\, dx\, dy.$$

Example 9.2.3 Verify that the volume formula for the ball of radius 1:

$$\iiint_W dx\, dy\, dz = \frac{4\pi}{3},$$

where W is the set of (x, y, z) with $x^2 + y^2 + z^2 \leq 1$.

The integral is

$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz\, dy\, dx = 2\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \sqrt{1 - x^2 - y^2}\, dy\, dx.$$

The last integral can be expressed as $\int_{-a}^{a} \sqrt{a^2 - y^2}\, dy$, which is equal to $a^2\pi/2$ (you can try to solve this by using the change of variable $y = a \sin t$). Carrying on,

$$\begin{aligned}\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{-\sqrt{1-x^2-y^2}}^{\sqrt{1-x^2-y^2}} dz\, dy\, dx &= 2\int_{-1}^{1} \pi \frac{1 - x^2}{2}\, dx \\ &= 2\pi \int_0^1 (1 - x^2)\, dx = \frac{4}{3}\pi.\end{aligned}$$

Example 9.2.4 Let W be the region bounded by the planes $x = 0$, $y = 0$, and $z = 2$, and the surface $z = x^2 + y^2$ and lying in the quadrant $x \geq 0$, $y \geq 0$. Compute $\iiint_W x\, dx\, dy\, dz$.

Method 1. We can describe this region by the inequalities

$$0 \leq y \leq \sqrt{2 - x^2}, \quad x^2 + y^2 \leq z \leq 2.$$

Therefore,

$$\begin{aligned}\iiint_W x\,dx\,dy\,dz &= \int_0^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} \int_{x^2+y^2}^{2} x\,dz\,dy\,dx\\ &= \int_0^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} x(2-x^2-y^2)\,dy\,dx.\\ &= \int_0^{\sqrt{2}} \left[\frac{2x(2-x^2)^{3/2}}{3}\right] dx = \frac{8\sqrt{2}}{15}.\end{aligned}$$

Method 2. We can also place limits on x first and describe W by $0 \le x \le \sqrt{z-y^2}$ and (y,z) in D, where D is the subset of the yz plane with $0 \le z \le 2$ and $0 \le y \le \sqrt{z}$.

Therefore,

$$\begin{aligned}\iiint_W x\,dx\,dy\,dz &= \int_0^{2} \int_0^{\sqrt{z}} \int_0^{\sqrt{z-y^2}} x\,dx\,dy\,dz\\ &= \int_0^{2} \int_0^{\sqrt{z}} \frac{z-y^2}{2}\,dy\,dz\\ &= \frac{1}{2}\int_0^{2} \frac{2z^{3/2}}{3}\,dz = \frac{8\sqrt{2}}{15},\end{aligned}$$

which agrees with our previous answer.

Exercises

Exercise 9.1 Describe the domain of integration and change the order of integration in the following integrals:

1. $\int_0^3 \int_{4x/3}^{\sqrt{25-x^2}} f(x,y)\,dy\,dx$.
2. $\int_0^1 \int_0^y f(x,y)\,dx\,dy$.
3. $\int_0^{\pi/2} \int_{-\sin(x/2)}^{\sin(x/2)} f(x,y)\,dy\,dx$.
4. $\int_1^e \int_0^{\log x} f(x,y)\,dy\,dx$.

Exercise 9.2 Consider the functions $f(x,y) = \frac{1}{\sqrt{1-x^2}}$ and $g(x,y) = \sin(y-1)$ on $R = \{(x,y) \in \mathbb{R}^2, x^2 + (y-1)^2 \leq 1, x \geq 0\}$. Apply Fubini's Theorem to write the integrals $\iint_R f(x,y)\,dA$ and $\iint_R g(x,y)\,dA$ in two different orders. Compute the integrals in the most convenient way.

Exercise 9.3 Compute the integral

$$\int_0^\pi \int_x^\pi \frac{\sin y}{y}\,dy\,dx.$$

Exercise 9.4 Compute the following triple integrals:

1. $\iiint_{[0,1]\times[0,1]\times[0,1]} (x^2+y^2+z^2)\,dx\,dy\,dz$.
2. $\iiint_{[0,1]\times[0,1]\times[0,1]} (x+y+z)^2\,dx\,dy\,dz$.

Exercise 9.5 Compute the following integrals on the specified regions:

1. $\iiint_W x^3\,dV$, with $W = [0,1]\times[0,1]\times[0,1]$.
2. $\iiint_W e^{-xy} y\,dV$, with $W = [0,1]\times[0,1]\times[0,1]$.
3. $\iiint_W (2x+3y+z)\,dV$, with $W = [1,2]\times[-1,1]\times[0,1]$.
4. $\iiint_W ze^{x+y}\,dV$, with $W = [0,1]\times[0,1]\times[0,1]$.

Exercise 9.6 Compute the integral

$$\iiint_W x^2 \cos x\,dV,$$

where W is the region defined by the planes $z = 0$, $z = \pi$, $y = 0$, $y = 1$, $x = 0$, and $x + y = 1$. Sketch the integration region.

Change of Variables 10

For functions of one variable, it is sometimes useful to change the variable of integration, for instance in order to make its calculation easier. If we have the integral $\int_a^b f(x)\,dx$, then the change of variable $x = g(u)$ transforms small increments in u into small increments in x and our integral has to reflect these changes, as the area of the infinitesimal rectangles we are summing will not be the same. If $\Delta u = (u + \Delta u) - u$, how is Δx related to Δu? Using Taylor expansions

$$\Delta x = g(u + \Delta u) - g(u) \approx g(u) + g'(u)\Delta u - g(u) = g'(u)\Delta u,$$

and so, when $\Delta u \longrightarrow 0$, we obtain $dx = g'(u)du$. Note also that the limits of integration change: calling $c = g^{-1}(a)$ and $d = g^{-1}(b)$ we have

$$\int_a^b f(x)\,dx = \int_c^d f(g(u))g'(u)\,du.$$

Note that this works only if g is injective, and otherwise we have to be careful[1] .

1: This is easy to see with the following example. The integral

$$\int_{-1}^{1} 3x^2\,dx = x^3\Big|_{-1}^{1} = 2$$

can be transformed with the change of variable $x^2 = y$, where $2xdx = dy$, and so the integral becomes $\int_1^1 \frac{1}{2\sqrt{y}}\,dy = 0$, as $y = 1$ both when $x = -1$ and $x = 1$. Or does it? Since $y' = 0$ at $x = 0$, we must actually divide the integral in two parts: one from $x = -1$ to $x = 0$ and the other from $x = 0$ to $x = 1$. Note also that for the first integral, we must choose the negative branch of the square root, as x is negative. The integral becomes

$$\int_{-1}^{1} x^2\,dx = \int_1^0 -\frac{1}{2\sqrt{y}}\,dy + \int_0^1 \frac{1}{2\sqrt{y}}\,dy$$
$$= -\sqrt{y}\Big|_1^0 + \sqrt{y}\Big|_0^1 = 2,$$

as we wanted. This example shows why we must have great care when doing a change of variable.

10.1 The Change of Variables Theorem for Double Integrals

We turn now to extending this formula for functions of several variables, and will give some intuition for the extension using two-dimensional functions.

Given two regions D and D^* in $\mathbb{R}^2$, a differentiable map $T : D^* \longrightarrow \mathbb{R}^2$ given by $T(u, v) = (x(u, v), y(u, v))$ and such that $T(D^*) = D$, and any real-valued integrable function $f : D \longrightarrow \mathbb{R}$, we would like to express $\iint_D f(x, y)\,dA$ as an integral over D^* of the composite function $f \circ T$. In order to do that, we need a measure of how the transformation T distorts the area of a region. It is not easy to prove this rigorously, but we will give some intuition in what follows.

Recall that $A(D) = \iint_D dx\,dy$ was obtained by dividing up D into little rectangles, summing their areas, and then taking the limit of this sum as the size of the subrectangles tended to zero. The challenge is that a transformation T may map rectangles into regions whose area is not easily computed. The solution is to approximate these images by simpler regions whose area we can compute. The derivative of T provides the best linear approximation to T.

Consider a small rectangle D^* in the uv plane. Let $J_T(u_0, v_0)$ denote the derivative of T evaluated at (u_0, v_0), so $J_T(u_0, v_0)$ is a 2×2 matrix. A good approximation to $T(u, v)$ is given by $T(u_0, v_0) + J_T(u_0, v_0)\begin{pmatrix} u - u_0 \\ v - v_0 \end{pmatrix}$. This mapping J_T takes D^* into a parallelogram with a vertex at $T(u_0, v_0)$ and with adjacent sides given by the vectors

$$J_T(u_0,v_0)\begin{pmatrix} u-u_0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}\begin{pmatrix} u-u_0 \\ 0 \end{pmatrix} = (u-u_0)\begin{pmatrix} \frac{\partial x}{\partial u} \\ \frac{\partial y}{\partial u} \end{pmatrix}$$

$$J_T(u_0,v_0)\begin{pmatrix} 0 \\ v-v_0 \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix}\begin{pmatrix} 0 \\ v-v_0 \end{pmatrix} = (v-v_0)\begin{pmatrix} \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial v} \end{pmatrix}$$

where the partial derivatives are evaluated at (u_0, v_0).

Recall from Linear Algebra that the area of the parallelogram with sides equal to the vectors (a, b) and (c, d) is equal to the absolute value of the determinant $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$. Thus, the area of $T(D^*)$ is approximately equal to the absolute value of

$$\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}_{(u_0,v_0)} (u-u_0)(v-v_0).$$

This determinant is called the *Jacobian determinant* or, simply, the **Jacobian**.

To get a sense of how this approximation works, look at the following example. Take the rectangle $(0,0), (0,0.2), (0.1,0), (0.1,0.2)$ and the transformation $f(x,y) = (y^2+2x, x^2+3y)$. Figure 10.1 shows how this rectangle (in black) is transformed into a new figure (in blue) which is approximated to a good extent by the linear transformation

$$T(x,y) = (2x, 3y) = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

where the matrix $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ is the Jacobian matrix of $f(x,y)$ evaluated at $(0,0)$.

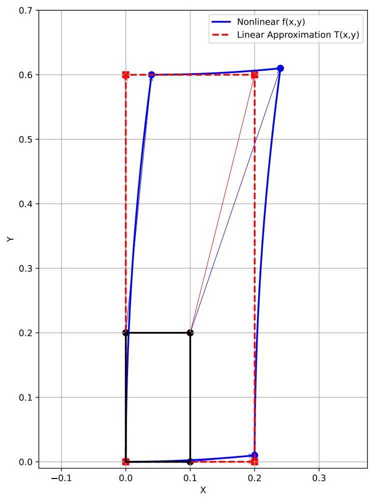


Figure 10.1: The linear transformation given by the Jacobian matrix of f approximates the action of f close to the point (x_0, y_0).

In the limit when $u \longrightarrow u_0$ and $v \longrightarrow v_0$, this approximation becomes exact and we obtain the change of variable formula:

Theorem 10.1.1 (Change of Variables: Double Integrals) *Let D and D^* be elementary regions in the plane and let $T : D^* \longrightarrow D$ be of class C^1; suppose that T is one-to-one on D^*. Furthermore, suppose that $D = T(D^*)$. Then for any integrable function $f : D \longrightarrow \mathbb{R}$, we have*

$$\iint_D f(x,y)\,dx\,dy = \iint_{D^*} f(x(u,v), y(u,v))|J_T|\,du\,dv,$$

where $|J_T|$ is the Jacobian determinant of T evaluated at (u, v).

Sometimes the Jacobian is written $\frac{\partial(x,y)}{\partial(u,v)}$ which is helpful to understand the direction of the derivatives.

One of the purposes of the change of variables theorem is to provide a method by which some double integrals can be simplified. We might encounter an integral $\iint_D f\,dA$ for which either the integrand f or the region D is complicated, making direct evaluation difficult. Therefore, a mapping T is chosen so that the integral is easier to evaluate with the new integrand $f \circ T$ and with the new region D^* [defined by $T(D^*) = D$]. However, the problem may actually become more complicated if T is not selected carefully.

Example 10.1.1 Use the change of variables

$$T(u, v) = (x, y) = (u^2 - v^2, 2uv)$$

to evaluate the integral $\iint_R y \, dA$ where $R = T(S)$ and $S = \{0 \le u \le 1, 0 \le v \le 1\}$.

The region S is the unit square in the uv plane, and we need to study what happens to its sides (see Figure 10.2):

1. In S_1, we have $v = 0, u \in [0, 1]$. In terms of x and y this transforms into $y = 0, x = u^2$, or $y = 0, x \in [0, 1]$.
2. In S_2, we have $u = 1, v \in [0, 1]$ and then $x = 1 - v^2, y = 2v \implies x = 1 - y^2/4$ with $x \in [0, 1]$.
3. In S_3 we have $v = 1, u \in [0, 1]$. In terms of x and y this transforms into $y = 2u, x = u^2 - 1$, or $x = y^2/4 - 1, x \in [0, 1]$.
4. Finally, in S_4 we have $u = 0, v \in [0, 1]$ which turns into $y = 0, x \in [-1, 0]$.

The region R is plotted in Figure 10.2.

Now, in order to calculate the integral we need the Jacobian of T, which is

$$|J_T| = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4(u^2 + v^2).$$

Note that this is always positive except when $u = v = 0$, but this is not problematic since it is only a point in the uv plane and it has no effect on the integral.

Finally,

$$\begin{aligned} \iint_R y \, dA &= \iint_S 2uv \cdot (4u^2 + 4v^2) \, du \, dv \\ &= 8 \int_0^1 \int_0^1 (u^3 v + uv^3) \, du \, dv \\ &= 16 \int_0^1 \int_0^1 u^3 v \, du \, dv \\ &= 16 \int_0^1 \frac{1}{4} v \, dv \\ &= 4\frac{1}{2} = 2. \end{aligned}$$

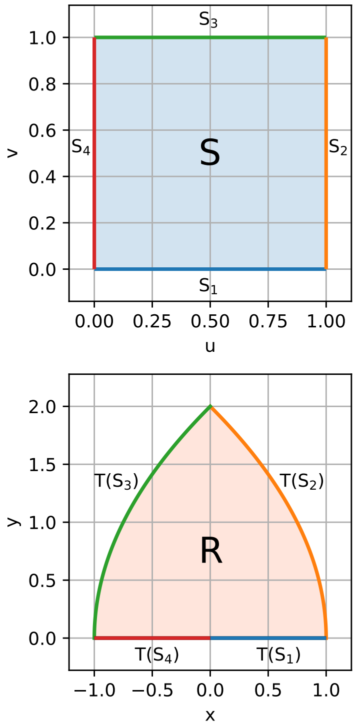


Figure 10.2: Regions S and R in Example 10.1.1.

Integrals in Polar Coordinates

Suppose we consider the rectangle D^* defined by $0 \le \theta \le 2\pi, 0 \le r \le a$ in the $r\theta$-plane. The transformation T given by $T(r, \theta) = (r \cos\theta, r \sin\theta)$ takes D^* onto the disk D with equation $x^2 + y^2 \le a^2$ in the xy-plane. This transformation represents the change from Cartesian coordinates to polar coordinates. However, T does not satisfy the requirements of the change of variables theorem, because it is not one-to-one on D^*. In particular, T sends all points with $r = 0$ to $(0, 0)$. Nevertheless, the change of variables theorem is valid in this case, as the set of points where T is not one-to-one lies on an edge of D^*, which can be neglected for integration purposes.

The function from $\mathbb{R}^2$ to $\mathbb{R}^2$ that transforms polar coordinates into Cartesian coordinates is given by

$$x = r\cos\theta, \quad y = r\sin\theta,$$

and its Jacobian determinant is

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r(\cos^2\theta + \sin^2\theta) = r.$$

Example 10.1.2 Under suitable restrictions on the function T, the area of $D = T(D^*)$ is obtained by integrating the absolute value of the Jacobian $\frac{\partial(x,y)}{\partial(u,v)}$ over D^*; that is, we have the equation

$$A(D) = \iint_D dx\, dy = \iint_{D^*} \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\, dv.$$

To illustrate, take $T : D^* \to D$, where $D = T(D^*)$ is the set of (x, y) with $x^2 + y^2 \le 1$ and $D^* = [0, 1] \times [0, 2\pi]$, and $T(r, \theta) = (r\cos\theta, r\sin\theta)$. Then the area of D is

$$A(D) = \iint_{D^*} r\, dr\, d\theta = \pi.$$

For a general function f, the change of variables formula in polar coordinates is given by

$$\iint_D f(x, y)\, dx\, dy = \iint_{D^*} f(r\cos\theta, r\sin\theta) r\, dr\, d\theta.$$

Example 10.1.3 Evaluate $\iint_D \log(x^2 + y^2)\, dx\, dy$, where D is the region in the first quadrant lying between the arcs of the circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$, where $0 < a < b$.

In polar coordinates, these circles have the simple equations $r = a$ and $r = b$. The polar coordinate transformation $x = r\cos\theta$, $y = r\sin\theta$ sends the rectangle D^* given by $a \le r \le b$, $0 \le \theta \le \frac{\pi}{2}$ onto the region D. We have

$$\iint_D \log(x^2 + y^2)\, dx\, dy = \int_a^b \int_0^{\frac{\pi}{2}} r \log r^2\, d\theta\, dr = \frac{\pi}{2} \int_a^b 2r \log r\, dr.$$

Applying integration by parts, or using the formula

$$\int x \log x\, dx = \frac{x^2}{2} \log x - \frac{x^2}{4},$$

we obtain

$$\frac{\pi}{2}\int_a^b 2r\log r\,dr = \frac{\pi}{2}\left[\frac{1}{2}b^2\log b - \frac{a^2}{2}\log a - \frac{1}{2}(b^2-a^2)\right].$$

Example 10.1.4 (The Gaussian Integral) One of the most beautiful applications of the change of variables formula, polar coordinates, and the reduction to iterated integrals is their application to the Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

This formula is not only attractive in its own right but also useful in areas such as statistics. It illustrates the unity of the transcendental numbers e and π nearly as well as the classic formula $e^{i\pi} = -1$.

To evaluate the Gaussian integral, consider the double integral over the disk D_a with $x^2 + y^2 \leq a^2$:

$$\iint_{D_a} e^{-(x^2+y^2)}\,dx\,dy.$$

Using polar coordinates where $r^2 = x^2 + y^2$ and $dx\,dy = r\,dr\,d\theta$, the change of variables formula gives

$$\begin{aligned}\iint_{D_a} e^{-(x^2+y^2)}\,dx\,dy &= \int_0^{2\pi}\int_0^a e^{-r^2}r\,dr\,d\theta \\ &= \int_0^{2\pi}\left(\frac{1}{2}\left[1-e^{-a^2}\right]\right)d\theta = \pi(1-e^{-a^2}).\end{aligned}$$

Letting $a \to \infty$, we get

$$\iint_{\mathbb{R}^2} e^{-(x^2+y^2)}\,dx\,dy = \pi.$$

Assuming that this improper integral can be evaluated as the limit of the integrals over the rectangles $R_a = [-a,a]\times[-a,a]$ as $a \to \infty$, we get

$$\lim_{a\to\infty}\iint_{R_a} e^{-(x^2+y^2)}\,dx\,dy = \pi.$$

By reduction to iterated integrals, this is

$$\lim_{a\to\infty}\left(\int_{-a}^{a} e^{-x^2}\,dx\right)^2 = \pi.$$

Thus,

$$\left(\int_{-\infty}^{\infty} e^{-x^2}\,dx\right)^2 = \pi.$$

Taking square roots, we arrive at the desired result.

We can use this technique to evaluate modifications of this integral:

$$\int_{-\infty}^{\infty} e^{-cx^2}\,dx.$$

Using the change of variables $y = \sqrt{c}x$, this reduces to

$$\int_{-\infty}^{\infty} e^{-cx^2}\,dx = \lim_{a\to\infty} \frac{1}{\sqrt{c}} \int_{-a}^{a} e^{-y^2}\,dy = \sqrt{\frac{\pi}{c}}.$$

10.2 Change of Variables Formula for Triple Integrals

To state this formula, we first define the Jacobian of a transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$ — it is a simple extension of the two-variable case.

Definition 10.2.1 *Let $T : W \subset \mathbb{R}^3 \to \mathbb{R}^3$ be a C^1 function defined by the equations $x = x(u,v,w)$, $y = y(u,v,w)$, $z = z(u,v,w)$. Then the Jacobian of T, denoted $\frac{\partial(x,y,z)}{\partial(u,v,w)}$, is the determinant*

$$\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}.$$

The change of variables formula for triple integrals is given by

$$\iiint_W f(x,y,z)\,dx\,dy\,dz = \iiint_{W^*} f(T(u,v,w)) \left|\frac{\partial(x,y,z)}{\partial(u,v,w)}\right| du\,dv\,dw.$$

where $T : (u,v,w) \mapsto (x(u,v,w), y(u,v,w), z(u,v,w))$ is a one-to-one map of class C^1, except possibly on a set that is the union of graphs of functions of two variables, and where W^* is an elementary region in uvw-space such that $T(W^*) = W$.

Cylindrical Coordinates

For the cylindrical coordinate system, we compute the Jacobian for the map defining the change to cylindrical coordinates. Given

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z,$$

the Jacobian is

$$\frac{\partial(x,y,z)}{\partial(r,\theta,z)} = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r.$$

Thus, the change of variables formula for cylindrical coordinates is

$$\iiint_W f(x,y,z)\,dx\,dy\,dz = \iiint_{W^*} f(r\cos\theta, r\sin\theta, z) r\,dr\,d\theta\,dz.$$

Spherical Coordinates

For spherical coordinates,

$$x = \rho \sin\phi \cos\theta, \quad y = \rho \sin\phi \sin\theta, \quad z = \rho\cos\phi,$$

the Jacobian is

$$\frac{\partial(x,y,z)}{\partial(\rho,\theta,\phi)} = \begin{vmatrix} \sin\phi\cos\theta & -\rho\sin\phi\sin\theta & \rho\cos\phi\cos\theta \\ \sin\phi\sin\theta & \rho\sin\phi\cos\theta & \rho\cos\phi\sin\theta \\ \cos\phi & 0 & -\rho\sin\phi \end{vmatrix} = -\rho^2\sin\phi.$$

Thus, the change of variables formula for spherical coordinates is

$$\iiint_W f(x,y,z)\,dx\,dy\,dz = \iiint_{W^*} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2\sin\phi\,d\rho\,d\theta\,d\phi.$$

Example 10.2.1 Evaluate the integral

$$\iiint_W \exp\left((x^2+y^2+z^2)^{3/2}\right)\,dV,$$

where W is the unit ball in $\mathbb{R}^3$.

First, note that this function cannot be easily integrated using iterated integrals. Hence, let's try a change of variables. The transformation into spherical coordinates seems appropriate because the entire quantity $x^2+y^2+z^2$ can be replaced by one variable, ρ^2. If W^* is the region such that

$$0 \le \rho \le 1, \quad 0 \le \theta \le 2\pi, \quad 0 \le \phi \le \pi,$$

then we can apply the formula and write

$$\iiint_W \exp\left((x^2+y^2+z^2)^{3/2}\right)\,dV = \iiint_{W^*} \rho^2 e^{\rho^3}\sin\phi\,d\rho\,d\theta\,d\phi.$$

This integral equals the iterated integral

$$\begin{aligned}\int_0^1\int_0^\pi\int_0^{2\pi} \rho^3 e^{\rho^3}\sin\phi\,d\theta\,d\phi\,d\rho &= 2\pi\int_0^1\int_0^\pi e^{\rho^3}\rho^2\sin\phi\,d\phi\,d\rho \\ &= 2\pi\int_0^1 \rho^2 e^{\rho^3}\left[-\cos\phi\right]_0^\pi\,d\rho \\ &= 4\pi\int_0^1 e^{\rho^3}\rho^2\,d\rho \\ &= 4\pi\left[\frac{e^{\rho^3}}{3}\right]_0^1 = \frac{4\pi}{3}(e-1).\end{aligned}$$

Example 10.2.2 Let W be the ball of radius R and center $(0,0,0)$ in $\mathbb{R}^3$. Find the volume of W.

The volume of W is given by

$$\iiint_W dx\,dy\,dz.$$

Using spherical coordinates, we get

$$\begin{aligned}\iiint_W dx\,dy\,dz &= \int_0^{\pi}\int_0^{2\pi}\int_0^{R} \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi \\ &= 2\pi \int_0^{\pi} \left(\frac{R^3}{3}\sin\phi\right) d\phi \\ &= \frac{2\pi R^3}{3}\left[-\cos(\pi)+\cos(0)\right] = \frac{4\pi R^3}{3},\end{aligned}$$

which is the standard formula for the volume of a solid sphere.

10.3 Applications

Averages

If we want to calculate the average of a function f (for instance, the speed of a car) over an interval $[a, b]$, we can take samples of the function at n equally-spaced points $x_1, \ldots, x_n \in [a, b]$ where $\Delta x = x_i - x_{i-1} = \frac{b-a}{n}$. Then the average value of f is approximated by

$$f_{\mathrm{av}} \approx \frac{1}{n}\sum_{i=1}^{n} f(x_i).$$

Now, since $\frac{1}{n} = \frac{\Delta x}{b-a}$ we can write

$$f_{\mathrm{av}} \approx \frac{1}{b-a}\sum_{i=1}^{n} f(x_i)\Delta x.$$

As $n \longrightarrow \infty$, the sum in the right becomes the integral of f and so:

$$[f]_{\mathrm{av}} = \frac{1}{b-a}\int_a^b f(x)\,dx.$$

In the case of speed, for instance, note that $\int_a^b f(x)\,dx$ gives the total distance covered in the interval $[a, b]$, and so the average speed is just the total distance divided by the length of the time interval.

We can extend this idea for functions of two and three variables as follows:

Definition 10.3.1 (Average Values) *The average value of a function of two variables over a region D is*

$$[f]_{av} = \frac{\iint_D f(x,y)\,dx\,dy}{\iint_D dx\,dy}.$$

Similarly, the average value of a function f on a region W in 3-space is

$$[f]_{av} = \frac{\iiint_W f(x,y,z)\,dx\,dy\,dz}{\iiint_W dx\,dy\,dz}.$$

Example 10.3.1 Find the average value of $f(x, y) = x \sin^2(xy)$ on the region $D = [0, \pi] \times [0, \pi]$.

First, compute

$$\begin{aligned}\iint_D x \sin^2(xy)\, dx\, dy &= \int_0^\pi \int_0^\pi x \sin^2(xy)\, dx\, dy \\ &= \int_0^\pi \left[\int_0^\pi x \frac{1 - \cos(2xy)}{2}\, dy \right] dx \\ &= \int_0^\pi \left[\frac{\pi x}{2} - \frac{\sin(2\pi x)}{4} \right] dx \\ &= \frac{\pi^3}{4} + \frac{\cos(2\pi^2) - 1}{8\pi}.\end{aligned}$$

Thus, the average value of f is

$$[f]_{\mathrm{av}} = \frac{\pi^3/4 + [\cos(2\pi^2) - 1]/8\pi}{\pi^2} \approx 0.7839.$$

Example 10.3.2 The temperature at points in the cube $W = [-1, 1] \times [-1, 1] \times [-1, 1]$ is proportional to the square of the distance from the origin. What is the average temperature? And at which points of the cube is the temperature equal to the average temperature?

For the first question, let c be the constant of proportionality, so $T = c(x^2 + y^2 + z^2)$ and the average temperature is $[T]_{\mathrm{av}} = \frac{1}{8} \iiint_W T\, dx\, dy\, dz$. Thus,

$$[T]_{\mathrm{av}} = \frac{c}{8} \iiint_W (x^2 + y^2 + z^2)\, dx\, dy\, dz.$$

The triple integral is the sum of the integrals of x^2, y^2, and z^2. The result is

$$[T]_{\mathrm{av}} = \frac{3c}{8} \iiint_W z^2\, dx\, dy\, dz = c.$$

As for the second question, the temperature is equal to the average temperature at all points on the sphere $x^2 + y^2 + z^2 = 1$.

Centers of Mass

In one dimension, if masses $m_1, \ldots, m_n$ are placed at points $x_1, \ldots, x_n$ on the x-axis, their center of mass is defined to be

$$x = \frac{\sum m_i x_i}{\sum m_i}.$$

For a continuous mass density $\delta(x)$ along a lever, the analogue is

$$x = \frac{\int x\delta(x)\, dx}{\int \delta(x)\, dx}.$$

For a two-dimensional plate with mass density $\delta(x, y)$, the coordinates for the center of mass are given by

$$x = \frac{\iint_D x\delta(x,y)\,dx\,dy}{\iint_D \delta(x,y)\,dx\,dy},$$
$$y = \frac{\iint_D y\delta(x,y)\,dx\,dy}{\iint_D \delta(x,y)\,dx\,dy}.$$

Example 10.3.3 Find the center of mass of the rectangle $[0,1] \times [0,1]$ with mass density e^{x+y}.

First, compute the total mass:

$$\iint_D e^{x+y}\,dx\,dy = \int_0^1 \int_0^1 e^{x+y}\,dx\,dy = e^2 - 2e + 1.$$

The numerator in formula (4) for x is

$$\begin{aligned}\iint_D xe^{x+y}\,dx\,dy &= \int_0^1 \int_0^1 xe^{x+y}\,dx\,dy \\ &= \int_0^1 \left[e^{1+y} - e^y\right]\,dy = e - 1,\end{aligned}$$

so that

$$x = \frac{e-1}{e^2 - 2e + 1} \approx 0.582.$$

By symmetry, $y \approx 0.582$ as well.

For a three-dimensional region W with mass density $\delta(x,y,z)$, the coordinates for the center of mass are

$$x = \frac{\iiint_W x\delta(x,y,z)\,dx\,dy\,dz}{\iiint_W \delta(x,y,z)\,dx\,dy\,dz},$$
$$y = \frac{\iiint_W y\delta(x,y,z)\,dx\,dy\,dz}{\iiint_W \delta(x,y,z)\,dx\,dy\,dz},$$
$$z = \frac{\iiint_W z\delta(x,y,z)\,dx\,dy\,dz}{\iiint_W \delta(x,y,z)\,dx\,dy\,dz}.$$

Example 10.3.4 Find the center of mass of the hemispherical region W defined by the inequalities $x^2 + y^2 + z^2 \leq 1, z \geq 0$ (Assume that the density is unity).

By symmetry, the center of mass must lie on the z-axis, and so $x = y = 0$. To find z, we compute the numerator I in the formula for the center of mass, which is

$$I = \iiint_W z\,dx\,dy\,dz.$$

The hemisphere is an elementary region, and thus the integral becomes

$$I = \int_0^1 z \left(\iint_{\text{disk}} dx\, dy \right) dz,$$

where the inner double integral is over the disk $x^2 + y^2 \le 1 - z^2$. The area of this disk is $\pi(1 - z^2)$, hence

$$I = \int_0^1 z\pi(1 - z^2)\, dz = \pi \int_0^1 (z - z^3)\, dz = \frac{\pi}{4}.$$

The volume of the hemisphere is $\frac{2}{3}\pi$, and so the z coordinate of the center of mass is

$$z = \frac{\frac{\pi}{4}}{\frac{2}{3}\pi} = \frac{3}{8}.$$

Moments of Inertia

The concept of the moment of inertia is crucial in mechanics, particularly in studying the dynamics of rotating rigid bodies. For a solid W with uniform density δ, the moments of inertia I_x, I_y, and I_z about the x, y, and z axes, respectively, are defined as follows:

Definition 10.3.2 (Moments of Inertia About the Coordinate Axes)

$$I_x = \iiint_W (y^2 + z^2)\delta\, dx\, dy\, dz,$$
$$I_y = \iiint_W (x^2 + z^2)\delta\, dx\, dy\, dz,$$
$$I_z = \iiint_W (x^2 + y^2)\delta\, dx\, dy\, dz.$$

The moment of inertia measures a body's response to efforts to rotate it. For example, I_x measures the body's response to forces attempting to rotate it about the x-axis.

Example 10.3.5 Compute the moment of inertia I_z for the solid above the xy-plane bounded by the paraboloid $z = x^2 + y^2$ and the cylinder $x^2 + y^2 = a^2$, assuming a and the mass density to be constants.

Using cylindrical coordinates, we find

$$\begin{aligned} I_z &= \int_0^a \int_0^{2\pi} \int_0^{r^2} \delta r^2 \cdot r\, dz\, d\theta\, dr \\ &= \delta \int_0^a \left[\frac{r^3}{3} \right]_0^{r^2} 2\pi\, dr = \frac{2\pi\delta}{3} \int_0^a r^6\, dr = \frac{2\pi\delta a^6}{3}. \end{aligned}$$

Exercises

Exercise 10.1 Use a linear transformation to compute the double integral

$$\iint_S (x-y)^2 \sin^2(x+y)\, dx\, dy,$$

where S is the parallelogram with vertices $(\pi, 0)$, $(2\pi, \pi)$, $(\pi, 2\pi)$, and $(0, \pi)$.

Exercise 10.2 Compute

$$\iint_D (y-x)\, dx\, dy,$$

where D is the region in the plane defined by the lines $y = x+1$, $y = x-3$, $y = (7-x)/3$, and $y = 5 - x/3$.

Exercise 10.3 Consider the map

$$x = u + v, \quad y = v - u^2.$$

Compute:

1. The Jacobian $J(u, v)$.
2. The image S in the xy-plane of the triangle T in the uv-plane with vertices $(0, 0)$, $(2, 0)$, and $(0, 2)$.
3. The area of S.
4. The integral
$$\iint_S (x - y + 1)^{-2}\, dx\, dy.$$

Exercise 10.4 Compute the double integral

$$\iint_D \log(x^2 + y^2)\, dx\, dy,$$

where D is the region in the first quadrant defined by the curves $x^2+y^2 = 1$ and $x^2 + y^2 = 4$.

Exercise 10.5 Compute the integral of the function

$$f(x, y) = \frac{y^4}{b^4\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)\left(1 + \frac{x^2}{a^2} + \frac{y^2}{b^2}\right)} + xy^2,$$

on the region

$$D = \left\{\frac{x^2}{a^2} + \frac{y^2}{b^2} \leq 1\right\},$$

where a and b are positive constants.

Exercise 10.6 Compute the integral of the function

$$f(x, y) = \frac{x}{\sqrt{x^2 + y^2}} e^{\sqrt{x^2+y^2}},$$

on the regions:

1. $E = \{x^2 + (y-1)^2 \le 1\}$.
2. $H = \{x^2 + (y-1)^2 \le 1, x \ge 0\}$.

Exercise 10.7 Compute the integral of the function

$$h(x, y) = \frac{\sqrt{2y^2 + x^2}}{y},$$

on the region

$$R = \{(x, y) \in \mathbb{R}^2 \mid x^2 + (y-1)^2 \le 1, x \ge 0\}.$$

Exercise 10.8 Compute the integral

$$\iint_S \frac{x}{4x^2 + y^2}\, dx\, dy,$$

where S is the region in the first quadrant defined by the lines $x = 0$, $y = 0$, and the ellipses $4x^2 + y^2 = 16$, $4x^2 + y^2 = 1$.

Exercise 10.9 If R is the region defined by the plane $z = 3$ and the cone $z = \sqrt{x^2 + y^2}$, compute the integrals:

1. $\displaystyle\iiint_R \sqrt{x^2 + y^2 + z^2}\, dx\, dy\, dz.$
2. $\displaystyle\iiint_R \sqrt{9 - x^2 - y^2}\, dx\, dy\, dz.$
3. $\displaystyle\iiint_R z e^{x^2+y^2+z^2}\, dx\, dy\, dz.$

Exercise 10.10 Compute

$$\iiint_W e^{-(x^2+y^2+z^2)^{3/2}}\, dx\, dy\, dz,$$

where W is the region below the sphere $x^2 + y^2 + z^2 = 9$ and above the cone $z = \sqrt{x^2 + y^2}$.

Exercise 10.11 Compute the following areas:

1. Area limited by the curves $y = x$ and $y = 2 - x^2$.
2. Area of the region $A = \{(x, y) \in \mathbb{R}^2 : x, y > 0, a^2 y \le x^3 \le b^2 y, p^2 x \le y^3 \le q^2 x\}$, where $0 < a < b$ and $0 < p < q$.
3. Area defined by the curves $xy = 4$, $xy = 8$, $xy^3 = 5$, and $xy^3 = 15$.

Exercise 10.12 Compute the following volumes:

1. Volume defined by the intersection of the cylinder $x^2 + y^2 \le 4$ and the ball $x^2 + y^2 + z^2 \le 16$.
2. Volume of the region bounded by the cones $z = 1 - \sqrt{x^2 + y^2}$ and $z = -1 + \sqrt{x^2 + y^2}$.
3. Volume of the region bounded by the paraboloid $z = x^2 + y^2$ and the cylinder $x^2 + y^2 = 4$ for $z \ge 0$.
4. Volume of the region bounded by $x^2 + y^2 + z^2 \le 2$, $x^2 + y^2 \le z$, and $z \le \frac{6}{5}$.

Exercise 10.13 Find the volumes of the regions defined by:

1. $z = x^2 + 3y^2$, $z = 9 - x^2$.
2. $x^2 + 2y^2 = 2$, $z = 0$, $x + y + 2z = 2$.

Exercise 10.14 Compute the volume of the region limited by the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$. Consider also the particular case of the ball: $a = b = c = r$.

Exercise 10.15 Compute the volume bounded by the cylinder $\rho = 4\cos\theta$, the sphere $\rho^2 + z^2 = 16$, and the plane $z = 0$. (The equations are written in cylindrical coordinates.)

Exercise 10.16 Compute the mass of a plate corresponding to the first quadrant part of the circle $x^2 + y^2 \leq 4$, if the density at each point is proportional to the distance to the center of the circle.

Exercise 10.17 Consider the region S in the plane defined by the curves mentioned below. Compute the mass and center of mass of S assuming that the density is constant ρ:

1. $y = x^2$, $x + y = 2$.
2. $y + 3 = x^2$, $x^2 = 5 - y$.
3. $y = \sin^2 x$, $y = 0$, $x \in [0, \pi]$.
4. $y = \sin x$, $y = \cos x$, $x \in [0, \pi/4]$.

Exercise 10.18 Compute the moment of inertia about the vertical axis of the solid $V = \{x^2 + y^2 + z^2 \leq 4, z \geq \sqrt{x^2 + y^2}\}$ (assuming a constant density ρ).

Exercise 10.19 Let A be the square $[-1, 1] \times [-1, 1]$. Compute the total mass assuming that the density at a point $(x, y) \in A$ is given by the expression $|x - y|$.

Exercise 10.20 Compute the coordinates of the center of mass of the plate $B = \{(x, y) \in \mathbb{R}^2 : 1 \leq x \leq 2, 1 \leq y \leq 3\}$, where the density is given by the function $f(x, y) = xy$.

Exercise 10.21 A metal plate is given by the set of points in the plane

$$P = \{(x, y) \in \mathbb{R}^2 : |y| \leq x \leq 1\}$$

with density $f(x, y) = y^2$. Compute the center of mass and the moments of inertia about the two axes.

Exercise 10.22 Let us consider the plate:

$$Q = \{(x, y) \in \mathbb{R}^2 : (x + y)^2 \leq x - y \leq \sqrt{x + y},\ x + y \geq 0\}$$

with mass density given by the function $\rho(x, y) = x^2 - y^2$. Compute the total mass of the plate.

Exercise 10.23 1. Compute the area of the set

$$D = \{x = r\cos^3 t,\ y = r\sin^3 t,\ 0 \leq r \leq 1,\ 0 \leq t \leq \pi/2\} = \{x^{2/3} + y^{2/3} \leq 1,\ x, y \geq 0\}.$$

2. Compute the coordinates of the center of mass of D assuming constant density.

Exercise 10.24 The square Q of vertices $(0,0),(0,1),(1,0),(1,1)$ represents a plate of constant density ρ. Compute the moment of inertia with respect to the line $x = y$.

Exercise 10.25 Consider the two solids obtained by intersecting the ball $x^2 + y^2 + z^2 = c^2$ in the first octant with the plane

$$\frac{x}{a} + \frac{y}{b} = 1,$$

where $0 < a, b \leq c$. Compute the mass of the two solids using a density $\rho(x, y, z) = z$.

Exercise 10.26 The temperature at the points of the cube $[-1, 1]^3$ is proportional to the square of the distance to the origin.

1. Compute the mean temperature of the cube.
2. At which points does the temperature coincide with the mean temperature?

Exercise 10.27 Compute the center of mass of a semispherical solid of radius R where the density at a point is given by the square of the distance between this point and the origin.

Exercise 10.28 An ice cream cone is formed by a wafer cone of angle α and a semisphere of ice cream of radius R. The densities of the wafer cone and the ice cream are constant, ρ_c and ρ_h respectively. Compute the quotient ρ_c/ρ_h so that the center of mass of the whole ice cream cone is placed in the plane between the wafer and the ice cream.

Exercise 10.29 Let us consider a solid $V \subset \mathbb{R}^3$ limited by the surfaces

$$z^2 = x^2 + y^2 \quad \text{and} \quad (z-2)^2 = 9(x^2 + y^2).$$

1. Make a sketch of the solid V in cylindrical coordinates.
2. Compute the coordinates of the center of mass if the density of V is constant.

Part IV. Integrals over Curves and Surfaces

Path and Line integrals 11

11.1 Path Integrals

The concept of a path integral generalizes integrals of functions of one variable to functions of several variables. Consider a scalar function $f : \mathbb{R}^3 \to \mathbb{R}$ and a path $c : I = [a, b] \to \mathbb{R}^3$ with $c(t) = (x(t), y(t), z(t))$. The integral of f along the path c can be used to determine properties like the total mass or average temperature along a path represented by c.

Definition 11.1.1 (Path Integrals) *The path integral of $f(x, y, z)$ along the path c is defined for a C^1 class path $c : I = [a, b] \to \mathbb{R}^3$ and a continuous composite function $t \mapsto f(x(t), y(t), z(t))$ on I. The integral is given by*

$$\int_c f\, ds = \int_a^b f(x(t), y(t), z(t))\, \|c'(t)\|\, dt.$$

Sometimes $\int_c f\, ds$ is denoted as $\int_c f(x, y, z)\, ds$ or $\int_a^b f(c(t))\|c'(t)\|\, dt$.

In order to motivate the definition of a path integral, we consider "Riemann-like" sums S_N. For a C^1 class path c on an interval $I = [a, b]$, we create a partition $a = t_0 < t_1 < \cdots < t_N = b$. This leads to a decomposition of c into paths c_i defined on $[t_i, t_{i+1}]$ for $0 \le i \le N - 1$. The arc length $\|\Delta s_i\|$ of each c_i is given by $\|c(t_{i+1}) - c(t_{i-1})\|$, which, if we multiply and divide by $t_{i+1} - t_i$, is equal to

$$\|\Delta s_i\| = \left\| \frac{c(t_{i+1}) - c(t_{i-1})}{t_{i+1} - t_i} \right\| (t_{i+1} - t_i) = \|c'(h_i)\|(t_{i+1} - t_i), \qquad h_i \in (t_i, t_{i+1}).$$

As N becomes large, Δs_i becomes small, and the function $f(x, y, z)$ is approximately constant on each c_i. We consider the sums

$$S_N = \sum_{i=0}^{N-1} f(x_i, y_i, z_i)\|\Delta s_i\|,$$

where $(x_i, y_i, z_i) = c(t)$ for some $t \in [t_i, t_{i+1}]$. From Riemann sums theory, we can show that

$$\begin{aligned} \lim_{N\to\infty} S_N &= \lim_{N\to\infty} \sum_{i=0}^{N-1} f(x_i(t), y_i(t), z_i(t))\|c'(h_i)\|(t_{i+1} - t_i) \\ &= \int_a^b f(c(t))\|c'(t)\|\, dt = \int_c f(x, y, z)\, ds. \end{aligned}$$

Example 11.1.1 Let c be the helix $c : [0, 2\pi] \to \mathbb{R}^3$ defined by $t \mapsto (\cos t, \sin t, t)$, and let $f(x, y, z) = x^2 + y^2 + z^2$. Evaluate the integral

$\int_c f(x, y, z)\, ds$.

First, compute $\|c'(t)\|$:

$$\begin{aligned} \|c'(t)\| &= \sqrt{\left(\frac{d}{dt}\cos t\right)^2 + \left(\frac{d}{dt}\sin t\right)^2 + \left(\frac{d}{dt}t\right)^2} \\ &= \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}. \end{aligned}$$

Next, substitute for x, y, and z in terms of t to obtain $f(x, y, z) = \cos^2 t + \sin^2 t + t^2 = 1 + t^2$ along c. The path integral is then

$$\int_c f(x, y, z)\, ds = \int_0^{2\pi} (1 + t^2)\sqrt{2}\, dt = 2\sqrt{2}\pi \left[1 + \frac{4\pi^2}{3}\right].$$

11.2 Line Integrals

In order to motivate the definition of the line integral, consider a force field $\mathbf{F}$ in space acting upon a particle moving along a path $c : [a, b] \longrightarrow \mathbb{R}^3$. If the particle undergoes a straight-line displacement represented by the vector $\mathbf{d}$ and $\mathbf{F}$ is constant, the work done by $\mathbf{F}$ is given by the dot product $\mathbf{F} \cdot \mathbf{d}$.

For a curved path, we approximate the path by a succession of infinitesimal straight-line displacements. As t varies over a small interval $[t, t+\Delta t]$, the particle moves from $c(t)$ to $c(t + \Delta t)$, approximately represented by the displacement $\Delta\mathbf{s} \approx c'(t)\Delta t$. The work done in this small interval is approximately $\mathbf{F}(c(t)) \cdot \Delta\mathbf{s} \approx \mathbf{F}(c(t)) \cdot c'(t)\Delta t$.

If we subdivide the interval $[a, b]$ into n equal parts $a = t_0 < t_1 < \cdots < t_N = b$, with $\Delta t = t_{i+1} - t_i$, then the work done by $\mathbf{F}$ is approximately

$$\sum_{i=0}^{n-1} \mathbf{F}(c(t_i)) \cdot \Delta s \approx \sum_{i=0}^{n-1} \mathbf{F}(c(t_i)) \cdot c'(t_i)\Delta t.$$

As $n \longrightarrow \infty$ this approximation becomes better and better, and so the work done by the force field $\mathbf{F}$ on a particle moving along path c is then given by

$$\int_a^b \mathbf{F}(c(t)) \cdot c'(t)\, dt.$$

Definition 11.2.1 (Line Integral) *We define the line integral of a vector field $\mathbf{F}$ along a C^1 path $c : [a, b] \longrightarrow \mathbb{R}^3$ as follows:*

$$\int_c \mathbf{F} \cdot d\mathbf{s} = \int_a^b \mathbf{F}(c(t)) \cdot c'(t)\, dt,$$

where the integral is over the dot product of $\mathbf{F}$ with the derivative of c over the interval $[a, b]$.

Example 11.2.1 Consider a path $c(t) = (\sin t, \cos t, t)$ for $0 \le t \le 2\pi$ and a vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Compute $\int_c \mathbf{F} \cdot d\mathbf{s}$.

We have:

$$\begin{aligned} \mathbf{F}(c(t)) &= (\sin t)\mathbf{i} + (\cos t)\mathbf{j} + t\mathbf{k}, \\ c'(t) &= (\cos t)\mathbf{i} - (\sin t)\mathbf{j} + \mathbf{k}, \\ \mathbf{F}(c(t)) \cdot c'(t) &= \sin t \cos t - \cos t \sin t + t = t. \end{aligned}$$

Thus, the line integral is

$$\int_c \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} t\, dt = 2\pi^2.$$

Line integrals can also be expressed using differential forms. For a vector field $\mathbf{F}$ with components F_1, F_2, and F_3, the line integral can be written as:

$$\int_c \mathbf{F} \cdot d\mathbf{s} = \int_a^b \left(F_1 \frac{dx}{dt} + F_2 \frac{dy}{dt} + F_3 \frac{dz}{dt} \right) dt = \int_c F_1\, dx + F_2\, dy + F_3\, dz$$

This differential form corresponds to the dot product of $\mathbf{F}$ and $d\mathbf{s}$, where $d\mathbf{s}$ can be thought of as the differential form $dx\mathbf{i} + dy\mathbf{j} + dz\mathbf{k}$.

Example 11.2.2 Evaluate the line integral

$$\int_c x^2\, dx + xy\, dy + dz,$$

where $c : [0, 1] \longrightarrow \mathbb{R}^3$ is given by $c(t) = (t, t^2, 1)$.

We compute $\frac{dx}{dt} = 1$, $\frac{dy}{dt} = 2t$, and $\frac{dz}{dt} = 0$. Therefore, the line integral becomes

$$\begin{aligned} \int_c x^2\, dx + xy\, dy + dz &= \int_0^1 t^2 \cdot 1 + t \cdot t^2 \cdot 2t\, dt \\ &= \int_0^1 (t^2 + 2t^4)\, dt \\ &= \left[\frac{1}{3} t^3 + \frac{2}{5} t^5 \right]_0^1 \\ &= \frac{1}{3} + \frac{2}{5} \\ &= \frac{11}{15}. \end{aligned}$$

Example 11.2.3 Consider the force vector field $F(x, y, z) = x^3\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Parametrize the circle of radius a in the yz-plane by letting $c(\theta)$ have components:

$$x = 0, \quad y = a\cos\theta, \quad z = a\sin\theta, \quad 0 \le \theta \le 2\pi.$$

Note that $F(c(\theta)) \cdot c'(\theta) = 0$, implying that the force field F is normal to the circle at every point, so F will not do any work on a particle moving along the circle. We verify this by direct computation:

$$\begin{aligned} W &= \int_c F \cdot ds \\ &= \int_0^{2\pi} x^3 dx + y dy + z dz \\ &= \int_0^{2\pi} (0 - a^2 \cos\theta \sin\theta + a^2 \cos\theta \sin\theta) d\theta \\ &= 0. \end{aligned}$$

Reparametrizations

The line integral $\int_c \mathbf{F} \cdot d\mathbf{s}$ depends not only on the field $\mathbf{F}$ but also on the path $c : [a, b] \longrightarrow \mathbb{R}^3$. Generally, if c_1 and c_2 are different paths in $\mathbb{R}^3$, $\int_{c_1} \mathbf{F} \cdot d\mathbf{s} \neq \int_{c_2} \mathbf{F} \cdot d\mathbf{s}$. However, if c_1 is a reparametrization of c_2, then $\int_{c_1} \mathbf{F} \cdot d\mathbf{s} = \pm \int_{c_2} \mathbf{F} \cdot d\mathbf{s}$ for every vector field $\mathbf{F}$.

Definition 11.2.2 *Let $h : I \longrightarrow I_1$ be a C^1 real-valued function that maps an interval $I = [a, b]$ onto another interval $I_1 = [a_1, b_1]$. Let $c : I_1 \longrightarrow \mathbb{R}^3$ be a piecewise C^1 path. Then the composition $p = c \circ h : I \longrightarrow \mathbb{R}^3$ is called a **reparametrization** of c.*

This definition implies that h must map endpoints to endpoints, that is, either $h(a) = a_1$ and $h(b) = b_1$, or $h(a) = b_1$ and $h(b) = a_1$. Reparametrizations can be orientation-preserving or orientation-reversing depending on how they map the endpoints.

Consider a path c and its image $C = c([a_1, b_1])$. If h is orientation-preserving, then $c \circ h(t)$ goes from $c(a_1)$ to $c(b_1)$ as t goes from a to b. If h is orientation-reversing, $c \circ h(t)$ goes from $c(b_1)$ to $c(a_1)$ as t goes from a to b.

Theorem 11.2.1 (Change of Parametrization for Line Integrals) *Let $\mathbf{F}$ be a vector field continuous on the C^1 path $c : [a_1, b_1] \longrightarrow \mathbb{R}^3$, and let $p : [a, b] \longrightarrow \mathbb{R}^3$ be a reparametrization of c. If p is orientation-preserving, then*

$$\int_p \mathbf{F} \cdot d\mathbf{s} = \int_c \mathbf{F} \cdot d\mathbf{s},$$

and if p is orientation-reversing, then

$$\int_p \mathbf{F} \cdot d\mathbf{s} = -\int_c \mathbf{F} \cdot d\mathbf{s}.$$

Proof. By hypothesis, $p = c \circ h$. By the chain rule, $p'(t) = c'(h(t))h'(t)$, so

$$\int_p \mathbf{F} \cdot d\mathbf{s} = \int_a^b [\mathbf{F}(c(h(t))) \cdot c'(h(t))] h'(t) dt.$$

Changing variables with $s = h(t)$, this becomes

$$\int_{h(a)}^{h(b)} \mathbf{F}(c(s))\cdot c'(s)ds = \begin{cases} \int_{a_1}^{b_1} F(c(s)) \cdot c'(s)ds & \text{if } p \text{ is orientation-preserving,} \\ -\int_{a_1}^{b_1} F(c(s)) \cdot c'(s)ds & \text{if } p \text{ is orientation-reversing.} \end{cases}$$

□

This theorem also holds for piecewise C^1 paths, as may be seen by breaking up the intervals into segments on which the paths are of class C^1 and summing the integrals over the separate intervals.

Example 11.2.4 Let $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ and $c : [-5, 10] \to \mathbb{R}^3$ be defined by $t \mapsto (t, t^2, t^3)$. Evaluate $\int_c \mathbf{F} \cdot d\mathbf{s}$ and $\int_{\text{opposite}(c)} \mathbf{F} \cdot ds$.

For the path c, we find $dx/dt = 1$, $dy/dt = 2t$, $dz/dt = 3t^2$, and $F(c(t)) = t^5\mathbf{i} + t^4\mathbf{j} + t^3\mathbf{k}$. Therefore,

$$\int_c F \cdot ds = \int_{-5}^{10} (t^5 + 2t^5 + 3t^5)dt = [t^6]_{-5}^{10} = 984,375.$$

For the opposite path opposite(c), $t \mapsto c(5 - t)$, we find similar results but with the opposite sign, giving $-984, 375$.

Path integrals, however, are unchanged under reparametrizations:

Theorem 11.2.2 (Change of Parametrization for Path Integrals) *Let c be piecewise C^1, let f be a continuous (real-valued) function on the image of c, and let p be any reparametrization of c. Then*

$$\int_c f(x, y, z)ds = \int_p f(x, y, z)ds.$$

11.3 Line Integrals of Gradient Fields

Consider a vector field F which is a gradient vector field, meaning $F = \nabla f$ for some real-valued function f. Thus, the vector field F is given by

$$F = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}.$$

Now, consider functions g and G which are real-valued and continuous on a closed interval $[a, b]$, with G being differentiable on (a, b) and $G' = g$. By the fundamental theorem of calculus, we have

$$\int_a^b g(x)\, dx = G(b) - G(a).$$

This suggests that the integral of g depends only on the values of G at the endpoints of the interval $[a, b]$. Given ∇f represents the derivative of f, we inquire whether the line integral $\int_c \nabla f \cdot ds$ is completely determined by the values of f at the endpoints $c(a)$ and $c(b)$.

Theorem 11.3.1 (Line Integrals of Gradient Vector Fields) *Suppose $f : \mathbb{R}^3 \longrightarrow \mathbb{R}$ is of class C^1 and that $c : [a,b] \longrightarrow \mathbb{R}^3$ is a piecewise C^1 path. Then*

$$\int_c \nabla f \cdot ds = f(c(b)) - f(c(a)).$$

Proof. Apply the chain rule to the composite function $F : t \mapsto f(c(t))$ to obtain

$$F'(t) = (f \circ c)'(t) = \nabla f(c(t)) \cdot c'(t).$$

By the fundamental theorem of calculus,

$$\int_a^b F'(t)\, dt = F(b) - F(a) = f(c(b)) - f(c(a)).$$

Therefore,

$$\int_c \nabla f \cdot ds = \int_a^b \nabla f(c(t)) \cdot c'(t)\, dt = f(c(b)) - f(c(a)).$$

□

Example 11.3.1 Let c be the path $c(t) = \left(\frac{t^4}{4}, \sin^3\left(\frac{t\pi}{2}\right), 0\right)$ for $t \in [0,1]$. Evaluate $\int_c y\, dx + x\, dy$.

We recognize $y\, dx + x\, dy$, or the vector field $y\mathbf{i} + x\mathbf{j} + 0\mathbf{k}$, as the gradient of the function $f(x,y,z) = xy$. Thus,

$$\int_c y\, dx + x\, dy = f(c(1)) - f(c(0)) = \frac{1}{4} \cdot 1 - 0 = \frac{1}{4}.$$

Obviously, if we can recognize the integrand as a gradient, then evaluation of the integral becomes much easier[1] . In one-variable calculus, every integral is, in principle, obtainable by finding an antiderivative. For vector fields, however, this is not always true, because a given vector field need not be a gradient.

1: For comparison, try to work out the integral in Example 11.3.1 directly.

Exercises

Exercise 11.1 Compute the following path integrals:

1. $f(x, y) = 2xy^2$ along the first quadrant of the circle of radius R.
2. $f(x, y, z) = (x^2 + y^2 + z^2)^2$ along the helix $\mathbf{r}(t) = (\cos t, \sin t, 3t)$, from the point $(1, 0, 0)$ to the point $(1, 0, 6\pi)$.

Exercise 11.2 Compute the length and the mass of a string arranged along the parabola $y = x^2$ from the point $(0, 0)$ to the point $(2, 4)$ and whose density is $\rho(x, y) = x$.

Exercise 11.3 Compute the line integrals of the vector field $\mathbf{f}$ along the given paths:

1. $\mathbf{f}(x, y) = (x^2 - 2xy, y^2 - 2xy)$, along the parabola $y = x^2$ from $(-1, 1)$ to $(1, 1)$.
2. $\mathbf{f}(x, y) = (x^2 + y^2, x^2 - y^2)$, along the curve $y = 1 - |1 - x|$, from $(0, 0)$ to $(2, 0)$.
3. $\mathbf{f}(x, y, z) = (y^2 - z^2, 2yz, -x^2)$, along the path given by $\mathbf{r}(t) = (t, t^2, t^3)$, for $t \in [0, 1]$.
4. $\mathbf{f}(x, y, z) = (2xy, x^2 + z, y)$, along the line that connects $(1, 0, 2)$ with $(3, 4, 1)$.

Exercise 11.4 Consider the vector function $\mathbf{f}(x, y) = (x^2, y)$. Compute the line integral of $\mathbf{f}$ from $(1, 0)$ to $(-1, 0)$ along the following paths:

1. The line segment connecting both points.
2. The two possible paths of the rectangle $[-1, 1] \times [-1, 1]$.
3. The upper semi-circumference that connects both points.

Exercise 11.5 Compute:

1. $\int_g (x - y)dx + (x + y)dy$, where g is the line connecting $(1, 0)$ with $(0, 2)$.
2. $\int_C x^3 dy - y^3 dx$, where C is the unit circle.
3. $\displaystyle\int_\Gamma \frac{dx + dy}{|x| + |y|}$, where Γ is the square of vertices $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$, traversed once in a counterclockwise direction.
4. $\int_\rho (x+2y)dx+(3x-y)dy$, where ρ is the ellipse defined by $x^2+4y^2 = 4$, traversed once in a counterclockwise direction.
5. $\displaystyle\int_R \frac{y^3 dx - xy^2 dy}{x^5}$, where R is the curve $x = \sqrt{1 - t^2}$, $y = t\sqrt{1 - t^2}$, $-1 \le t \le 1$.

Exercise 11.6 Compute:

1. $\int_\gamma y\,dx - x\,dy + z\,dz$, where γ is the curve resulting from the intersection of the cylinder $x^2 + y^2 = a^2$ and the plane $z - y = a$, oriented counterclockwise.
2. $\int_\gamma \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{F}(x, y, z) = (2xy + z^2, x^2, 2xz)$ and γ is the curve resulting from the intersection of the plane $x = y$ and the sphere $x^2 + y^2 + z^2 = a^2$, oriented positively.
3. $\int_\gamma \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{F}(x, y, z) = (y, z, x)$ and γ is the intersection of $x^2 + y^2 = 2x$ with $x = z$, oriented positively.

Exercise 11.7 A particle of mass m moves along the curve

$$\mathbf{r}(t) = (t^2, \sin t, \cos t), \quad t \in [0, 1].$$

Compute the force that acts on the particle assuming $\mathbf{F}(t) = m\mathbf{r}''(t)$ (Newton's second law). Compute also the total work done by this force field.

Exercise 11.8 Find the value of $b > 0$ that minimizes the work done by the force field

$$\mathbf{F}(x, y) = (3y^2 + 2, 16x),$$

in moving a particle from $(-1, 0)$ to $(1, 0)$ along the semi-ellipse $b^2x^2+y^2 = b^2$, $y \geq 0$.

Exercise 11.9 Consider the force field $\mathbf{F}(x, y) = (cxy, x^6y^2)$, $a, b, c > 0$. Compute the value of a as a function of c so that the work done by this force field in moving a particle along the parabola $y = ax^b$ from $x = 0$ to $x = 1$ does not depend on b.

Exercise 11.10 Compute the work done by the force field

$$\mathbf{F}(r, \theta) = (-4\sin\theta, 4\sin\theta),$$

in polar coordinates, in moving a particle along the curve $r = e^{-\theta}$ from $(1, 0)$ to the origin.

Exercise 11.11 Consider the force field $\mathbf{F}(x, y, z) = (\sin y + z, x\cos y + e^z, x + ye^z)$.

1. Show that the path integral along any closed piecewise regular curve equals 0.
2. Compute a potential function for $\mathbf{F}$, i.e., find a function ϕ such that $\mathbf{F} = \nabla\phi$.

Exercise 11.12 Compute

$$\int_\gamma \mathbf{F} \cdot d\mathbf{s},$$

where $\mathbf{F}(x, y, z) = (2xze^{x^2+y^2}, 2yze^{x^2+y^2}, e^{x^2+y^2})$ and γ is the curve in $\mathbb{R}^3$ defined by $\mathbf{r}(t) = (t, t^2, t^3)$, $0 \leq t \leq 1$.

Exercise 11.13 Consider the curve in $\mathbb{R}^3$, $\gamma(t) = \left(e^{t^2} + t(1-e) - 1, \sin^5(\pi t), \cos(t^2 - t)\right)$, $t \in [0, 1]$, and the vector field

$$\mathbf{F}(x, y, z) = \left(y + z + x^4 \sin x^5, x + z + \arctan y, x + y + \sin^2 z\right).$$

1. Compute $\int_\gamma \mathbf{F} \cdot d\mathbf{s}$.
2. Is there any function f such that $\nabla f = \mathbf{F}$? Find f, if it is possible.

Exercise 11.14 Consider the curve in $\mathbb{R}^3$, $\Gamma = \{x^2 + y^2 = 1, z = y^2 - x^2\}$, and the vector field

$$\mathbf{F}(x, y, z) = (y^3, e^y, z).$$

1. Compute $\int_\Gamma \mathbf{F} \cdot d\mathbf{s}$.
2. Is there any function f such that $\nabla f = \mathbf{F}$? Find f, if it is possible.

Exercise 11.15 Find a and b such that the vector field

$$\mathbf{w}(x, y) = e^{2x+3y}\left(a \sin x + a \cos y + \cos x, b \sin x + b \cos y - \sin y\right)$$

is conservative, and compute the corresponding potential function.

Exercise 11.16 Consider the vector field

$$\mathbf{F}(x, y) = \left(\frac{\log(xy)}{x}, \frac{\log(xy)}{y}\right),$$

defined for $x > 0$, $y > 0$, and let $a > 0$, $b > 0$ be two constants.

1. Compute $\int_\gamma \mathbf{F} \cdot d\mathbf{s}$ where γ is the hyperbola $xy = a$ for $x_1 \le x \le x_2$.
2. If A is any point on the hyperbola $xy = a$, B is any point on the hyperbola $xy = b$, and γ is any curve of class C^1 contained in the first quadrant that joins A with B, show that

$$\int_\gamma \mathbf{F} \cdot d\mathbf{s} = \log(ab) \log\left(\frac{b}{a}\right).$$

Surface Integrals 12

12.1 Parametrized Surfaces

We begin by considering surfaces beyond the scope of graphs of functions. For instance, consider a surface S defined as the set of points (x, y, z) where $x - z + z^3 = 0$. This surface is not the graph of some function $z = f(x, y)$, as it doubles back on itself relative to the xy-plane. Another example is a torus, which is also not the graph of a differentiable function of two variables. These observations lead us to extend our definition of a surface.

Definition 12.1.1 (Parametrized Surfaces) *A **parametrization** of a surface is a function $\Phi : D \subset \mathbb{R}^2 \to \mathbb{R}^3$, where D is some domain in $\mathbb{R}^2$. The surface S corresponding to the function Φ is its image: $S = \Phi(D)$. We can write*

$$\Phi(u, v) = (x(u, v), y(u, v), z(u, v)).$$

If Φ is differentiable or of class C^1, meaning that $x(u, v)$, $y(u, v)$, and $z(u, v)$ are differentiable or C^1 functions of (u, v), we call S a differentiable or a C^1 surface.

Example 12.1.1 Consider the sphere of radius ρ centered at the origin, $x^2 + y^2 + z^2 = \rho^2$. A very natural parametrization would be to use spherical coordinates, where $\theta \in [0, 2\pi]$ and $\phi \in [0, \phi]$ and so:

$$\Phi(\theta, \phi) = (\rho \cos\theta \sin\phi, \rho \sin\theta \sin\phi, \rho \sin\phi).$$

Note that ρ is here a constant, as we are only considering the surface of the sphere.

For a parametrized surface Φ that is differentiable at $(u_0, v_0) \in \mathbb{R}^2$, consider the curve $t \mapsto \Phi(u_0, t)$. The tangent vector to this curve at $\Phi(u_0, v_0)$, denoted by $\mathbf{T}_v$, is given by:

$$\mathbf{T}_v = \left.\frac{\partial \Phi}{\partial v}\right|_{(u_0,v_0)} = \left.\left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right)\right|_{(u_0,v_0)}.$$

Similarly, for the curve $t \mapsto \Phi(t, v_0)$, we obtain the tangent vector $\mathbf{T}_u$ at $\Phi(u_0, v_0)$:

$$\mathbf{T}_u = \left.\frac{\partial \Phi}{\partial u}\right|_{(u_0,v_0)} = \left.\left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right)\right|_{(u_0,v_0)}.$$

A surface S is said to be **regular** or **smooth** at $\Phi(u_0, v_0)$ if the vectors $\mathbf{T}_u$ and $\mathbf{T}_v$, which are tangent to two curves on the surface at a given point, satisfy the condition $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$ at (u_0, v_0). The surface is called regular if it is regular at all points $(u_0, v_0) \in S$. The nonzero vector $\mathbf{T}_u \times \mathbf{T}_v$ is normal to S, ensuring the existence of a tangent plane. Intuitively, a smooth surface has no "corners."

Example 12.1.2 Consider the surface given by the equations $(x, y, z) = (u\cos v, u\sin v, u), \quad u \geq 0$.

This surface is a cone with a "point" at $(0, 0, 0)$. It is differentiable as each component function is differentiable in terms of u and v. However, it is not regular at $(0, 0, 0)$. To verify this, we compute $\mathbf{T}_u$ and $\mathbf{T}_v$ at $(0, 0) \in \mathbb{R}^2$:

$$\mathbf{T}_u = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right)\bigg|_{(0,0)} = (\cos 0, \sin 0, 1) = (1, 0, 1)\,,$$

$$\mathbf{T}_v = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right)\bigg|_{(0,0)} = (0, 0, 0).$$

Since $\mathbf{T}_u \times \mathbf{T}_v = \mathbf{0}$, the surface is not regular at $(0, 0, 0)$.

Example 12.1.3 For the sphere we have

$$\mathbf{T}_\theta = (-\rho\sin\theta\sin\phi, \rho\cos\theta\sin\phi, 0),$$

$$\mathbf{T}_\phi = (\rho\cos\theta\cos\phi, \rho\cos\theta\cos\phi, -\rho\sin\phi),$$

and therefore

$$\mathbf{T}_\theta \times \mathbf{T}_\phi = -\rho^2\sin\phi(\rho\cos\theta\sin\phi, \rho\sin\theta\sin\phi, \rho\sin\phi)$$

or

$$\mathbf{T}_\theta \times \mathbf{T}_\phi = -\rho^2\sin\phi(x, y, z).$$

That is, the normal vector to the sphere is proportional to the vector that goes from the origin to the surfce of the sphere. In this case, the normal vector points inwards of the sphere, and we will discuss later how this depends on the choice of parametrization.

Note also that, if $\phi = 0$ or $\phi = \pi$ the normal vector is $(0, 0, 0)$. Does that mean that the sphere is not regular at these points? No: in this case, this depends entirely on the parametrization, as we can very easily picture a tangent plane to the sphere at these points. In the case of the cone, however, there is no way to choose a parametrization such that the point $(0, 0, 0)$ has a meaningful tangent plane.

12.2 Integrals of Scalar Functions Over Surfaces

A regular surface (loosely speaking) is one that has no corners or breaks. For the remainder of this course, we shall consider only piecewise regular surfaces that are unions of images of parametrized surfaces $\Phi_i : D_i \to \mathbb{R}^3$ for which:

1. D_i is an elementary region in the plane;
2. Φ_i is of class C^1 and one-to-one, except possibly on the boundary of D_i; and
3. S_i, the image of Φ_i, is regular, except possibly at a finite number of points.

Now we are ready to define the integral of a scalar function f over a surface S:

Definition 12.2.1 (Integral of a Scalar Function Over a Surface) *If $f(x, y, z)$ is a real-valued continuous function defined on a parametrized surface S, we define the integral of f over S to be*

$$\iint_S f(x, y, z)\, dS = \iint_D f(\Phi(u, v))\|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv.$$

If f is identically 1, we obtain the area of the surface S. This is quite analogous to considering the path integral as a generalization of arc length. The surface integral is independent of the particular parametrization used.

We can gain some intuitive knowledge about this integral by considering it as a limit of sums. Let D be a rectangle partitioned into n^2 rectangles. Let R_{ij} be the ijth rectangle in the partition with area $\Delta u \Delta v$, with vertices (u_i, v_j), (u_{i+1}, v_j), (u_i, v_{j+1}), and (u_{i+1}, v_{j+1}), $0 \le i \le n-1$, $0 \le j \le n-1$. Denote the values of $\mathbf{T}_u$ and $\mathbf{T}_v$ at (u_i, v_j) by $\mathbf{T}_{u_i}$ and $\mathbf{T}_{v_j}$. We can think of the vectors $\Delta u \mathbf{T}_{u_i}$ and $\Delta v \mathbf{T}_{v_j}$ as tangent to the surface at $\Phi(u_i, v_j) = (x_{ij}, y_{ij}, z_{ij})$. Then these vectors form a parallelogram P_{ij} that lies in the plane tangent to the surface at (x_{ij}, y_{ij}, z_{ij}). We thus have a "patchwork cover" of the surface by the P_{ij}. For n large, the area of P_{ij} is a good approximation to the area of $\Phi(R_{ij})$.

Let $S_{ij} = \Phi(R_{ij})$ be the portion of the surface $\Phi(D)$ corresponding to R_{ij}, and let $A(S_{ij})$ be the area of this portion of the surface. For large n, f will be approximately constant on S_{ij}, and we form the sum

$$S_n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} f(\Phi(u_i, v_j))A(S_{ij}).$$

where $(u_i, v_j) \in R_{ij}$. Because the area of the parallelogram spanned by two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is $\|\mathbf{v}_1 \times \mathbf{v}_2\|$, we can approximate $A(S_{ij})$ by

$$A(S_{ij}) \approx \|\mathbf{T}_{u_i} \times \mathbf{T}_{v_j}\|\Delta u \Delta v$$

and therefore

$$S_n \approx \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} f(\Phi(u_i, v_j))\|\mathbf{T}_{u_i} \times \mathbf{T}_{v_j}\|\Delta u \Delta v,$$

As $n \to \infty$,

$$\lim_{n\to\infty} S_n = \iint_S f\, dS.$$

If S is a union of parametrized surfaces S_i, $i = 1, \ldots, N$, that do not intersect except possibly along curves defining their boundaries, then the integral of f over S is defined by

$$\iint_S f\, dS = \sum_{i=1}^{N} \iint_{S_i} f\, dS,$$

as we should expect. For example, the integral over the surface of a cube may be expressed as the sum of the integrals over the six sides.

Example 12.2.1 A helicoid is defined by $\Phi : D \longrightarrow \mathbb{R}^3$, where $\Phi(r,\theta) = (r\cos\theta, r\sin\theta, \theta)$ and D is the region where $0 \le \theta \le 2\pi$ and $0 \le r \le 1$. Let

$$f(x,y,z) = \sqrt{x^2+y^2+1}.$$

Find $\iint_S f\,dS$.

We have

$$\|\mathbf{T}_r \times \mathbf{T}_\theta\| = \sqrt{r^2+1}$$

and $f(r\cos\theta, r\sin\theta, \theta) = \sqrt{r^2+1}$. Therefore,

$$\iint_S f(x,y,z)\,dS = \int_0^{2\pi}\int_0^1 (r^2+1)\,dr\,d\theta = \frac{8\pi}{3}.$$

Example 12.2.2 Let D be the region determined by $0 \le \theta \le 2\pi$, $0 \le r \le 1$ and let the function $\Phi : D \longrightarrow \mathbb{R}^3$, given by $\Phi(r,\theta) = (r\cos\theta, r\sin\theta, r)$ be a parametrization of a cone S. Find its surface area.

We have

$$\mathbf{T}_r \times \mathbf{T}_\theta = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \cos\theta & \sin\theta & 1 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (-r\cos\theta, -r\sin\theta, r)$$

so the area integrand is

$$\|\mathbf{T}_r \times \mathbf{T}_\theta\| = \sqrt{r^2 + r^2\cos^2\theta + r^2\sin^2\theta} = r\sqrt{2}.$$

Clearly, $\|\mathbf{T}_r \times \mathbf{T}_\theta\|$ vanishes for $r = 0$, but $\Phi(0,\theta) = (0,0,0)$ for any θ. Thus, $(0,0,0)$ is the only point where the surface is not regular. We have

$$\iint_D \|\mathbf{T}_r \times \mathbf{T}_\theta\|\,dr\,d\theta = \int_0^{2\pi}\int_0^1 \sqrt{2}r\,dr\,d\theta = \sqrt{2}\pi.$$

To confirm that this is the area of $\Phi(D)$, we must verify that Φ is one-to-one (for points not on the boundary of D). Let D' be the set of (r,θ) with $0 < r < 1$ and $0 < \theta < 2\pi$. Hence, D' is D without its boundary. To see that $\Phi : D' \longrightarrow \mathbb{R}^3$ is one-to-one, assume that $\Phi(r,\theta) = \Phi(r',\theta')$ for (r,θ) and (r',θ') in D'. Then

$$r\cos\theta = r'\cos\theta',$$

$$r\sin\theta = r'\sin\theta',$$

$$r = r'.$$

From these equations it follows that $\cos\theta = \cos\theta'$ and $\sin\theta = \sin\theta'$. Thus, either $\theta = \theta'$ or $\theta = \theta' + 2\pi n$. But the second case is impossible for n a nonzero integer, because both θ and θ' belong to the open interval $(0, 2\pi)$ and thus cannot be more than 2π radians apart. This proves that off the boundary, Φ is one-to-one.

Example 12.2.3 Evaluate

$$\iint_S z^2\, dS,$$

where S is the unit sphere $x^2 + y^2 + z^2 = 1$.

For this problem, it is convenient to use spherical coordinates and to represent the sphere parametrically by the equations $x = \cos\theta \sin\phi$, $y = \sin\theta \sin\phi$, $z = \cos\phi$, over the region D in the $\theta\phi$ plane given by the inequalities $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$. We get

$$\iint_S z^2\, dS = \iint_D (\cos\phi)^2 \|\mathbf{T}_\theta \times \mathbf{T}_\phi\|\, d\theta\, d\phi.$$

A little computation shows that

$$\|\mathbf{T}_\theta \times \mathbf{T}_\phi\| = \sin\phi.$$

(Note that for $0 \le \phi \le \pi$, we have $\sin\phi \ge 0$.) Thus,

$$\iint_S z^2\, dS = \int_0^{2\pi} \int_0^{\pi} \cos^2\phi \sin\phi\, d\phi\, d\theta = \frac{4\pi}{3}.$$

12.3 Surface Integrals of Vector Fields

The goal of this section is to develop the notion of the integral of a vector field over a surface. Recall that the definition of the line integral of a vector field was motivated by the fundamental physical notion of work. Similarly, there is a basic physical notion of flux that motivates the definition of the surface integral of a vector field.

For example, if the vector field is the velocity field of a fluid (perhaps the velocity field of a flowing river), and we put an imagined mathematical surface into the fluid, we can ask: "What is the rate at which fluid is crossing the given surface (measured in, say, cubic meters per second)?" The answer is given by the surface integral of the fluid velocity vector field over the surface.

We shall come back to the physical interpretation shortly and reconcile it with the formal definition that we give first.

Definition 12.3.1 (The Surface Integral of Vector Fields) *Let* $\mathbf{F}$ *be a vector field defined on* S*, the image of a parametrized surface* Φ*. The surface integral of* $\mathbf{F}$ *over* Φ*, denoted by*

$$\iint_S \mathbf{F} \cdot d\mathbf{S},$$

is defined by

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)\, du\, dv.$$

The geometric and physical significance of the surface integral can be understood by expressing it as a limit of Riemann sums. For simplicity, we assume D is a rectangle. Fix a parametrization ϕ of S that preserves orientation and partition the region D into n^2 pieces D_{ij}, where $0 \leq i \leq n-1$ and $0 \leq j \leq n-1$. We let Δu denote the length of the horizontal side of D_{ij} and Δv denote the length of the vertical side of D_{ij}. Let (u, v) be a point in D_{ij}, and $(x, y, z) = \phi(u, v)$ the corresponding point on the surface. We consider the parallelogram with sides $\Delta u \mathbf{T}_u$ and $\Delta v \mathbf{T}_v$ lying in the plane tangent to S at (x, y, z) and the parallelepiped formed by $\mathbf{F}$, $\Delta u \mathbf{T}_u$, and $\Delta v \mathbf{T}_v$. The volume of the parallelepiped is the absolute value of the triple product

$$\mathbf{F} \cdot (\Delta u \mathbf{T}_u \times \Delta v \mathbf{T}_v) = \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) \Delta u \Delta v.$$

The vector $\mathbf{T}_u \times \mathbf{T}_v$ is normal to the surface at (x, y, z) and points away from the outside of the surface. Thus, the number $\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)$ is positive when the parallelepiped lies on the outside of the surface.

In general, the parallelepiped lies on that side of the surface away from which $\mathbf{F}$ is pointing. If we think of $\mathbf{F}$ as the velocity field of a fluid, $\mathbf{F}(x, y, z)$ is pointing in the direction in which fluid is moving across the surface near (x, y, z). Moreover, the number

$$|\mathbf{F} \cdot (\Delta u \mathbf{T}_u \times \Delta v \mathbf{T}_v)|$$

measures the amount of fluid that passes through the tangent parallelogram per unit time. Because the sign of $\mathbf{F} \cdot (\Delta u \mathbf{T}_u \times \Delta v \mathbf{T}_v)$ is positive if the vector $\mathbf{F}$ is pointing outward at (x, y, z) and negative if $\mathbf{F}$ is pointing inward, $i, j\mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v) \Delta u \Delta v$ is an approximate measure of the net quantity of fluid to flow outward across the surface per unit time. (Remember that "outward" or "inward" depends on our choice of parametrization. Hence, the integral

$$\int_S \mathbf{F} \cdot d\mathbf{S}$$

is the net quantity of fluid to flow across the surface per unit time, that is, the rate of fluid flow. This integral is also called the flux of $\mathbf{F}$ across the surface.

In the case where $\mathbf{F}$ represents an electric or a magnetic field, $\int_S \mathbf{F} \cdot d\mathbf{S}$ is also commonly known as the flux. The reader may be familiar with physical laws (such as Faraday's law) that relate flux of a vector field to a circulation (or current) in a bounding loop. This is the historical and physical basis of Stokes' theorem, which we will discuss in Chapter 13. The corresponding principle in fluid mechanics is called Kelvin's circulation theorem.

Example 12.3.1 Let D be the rectangle in the $\theta\phi$ plane defined by

$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi,$$

and let the surface S be defined by the parametrization $\Phi : D \to \mathbb{R}^3$ given by

$$x = \cos\theta \sin\phi, \quad y = \sin\theta \sin\phi, \quad z = \cos\phi.$$

Thus, θ and ϕ are the angles of spherical coordinates, and S is the unit sphere parametrized by Φ). Let $\mathbf{r}$ be the position vector $\mathbf{r}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Compute $\iint_S \mathbf{r} \cdot d\mathbf{S}$.

First, we find

$$\mathbf{T}_\theta = (-\sin\phi \sin\theta)\mathbf{i} + (\sin\phi\cos\theta)\mathbf{j},$$

$$\mathbf{T}_\phi = (\cos\theta\cos\phi)\mathbf{i} + (\sin\theta\cos\phi)\mathbf{j} - (\sin\phi)\mathbf{k},$$

and hence

$$\mathbf{T}_\theta \times \mathbf{T}_\phi = (-\sin^2\phi\cos\theta)\mathbf{i} - (\sin^2\phi\sin\theta)\mathbf{j} - (\sin\phi\cos\phi)\mathbf{k}.$$

Then we evaluate

$$\mathbf{r} \cdot (\mathbf{T}_\theta \times \mathbf{T}_\phi) = -\sin\phi.$$

Thus,

$$\iint_D \mathbf{r} \cdot d\mathbf{S} = \int_0^{2\pi} \int_0^{\pi} (-\sin\phi)\, d\phi\, d\theta = (-2)\int_0^{2\pi} d\theta = -4\pi.$$

12.4 Oriented Surfaces

It is not evident that both surface integrals given above are independent of the parametrization. In order to study this, we need the concept of oriented surfaces.

Definition 12.4.1 (Oriented Surfaces) *An oriented surface is a two-sided surface with one side specified as the outside or positive side; we call the other side the inside or negative side. At each point $(x, y, z) \in S$ there are two unit normal vectors $\mathbf{n}_1$ and $\mathbf{n}_2$, where $\mathbf{n}_1 = -\mathbf{n}_2$. Each of these two normals can be associated with one side of the surface. Thus, to specify a side of a surface S, at each point we choose a unit normal vector $\mathbf{n}$ that points away from the positive side of S at that point.*

This definition assumes that our surface does have two sides. In fact, this is necessary, because there are examples of surfaces with only one side![1]

Let $\Phi : D \to \mathbb{R}^3$ be a parametrization of an oriented surface S and suppose S is regular at $\Phi(u_0, v_0)$, $(u_0, v_0) \in D$; thus, the vector $(\mathbf{T}_{u_0} \times \mathbf{T}_{v_0})/\|\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}\|$ is defined. If $\mathbf{n}(\Phi(u_0, v_0))$ denotes the unit normal to S at $\Phi(u_0, v_0)$, it follows that

$$\frac{\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}}{\|\mathbf{T}_{u_0} \times \mathbf{T}_{v_0}\|} = \pm\mathbf{n}(\Phi(u_0, v_0)).$$

The parametrization Φ is said to be orientation-preserving if $\frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} = \mathbf{n}(\Phi(u, v))$ at all $(u, v) \in D$ for which S is smooth at $\Phi(u, v)$. In other words, Φ is orientation-preserving if the vector $\mathbf{T}_u \times \mathbf{T}_v$ points to the outside of the surface. If $\mathbf{T}_u \times \mathbf{T}_v$ points to the inside of the surface at all points $(u, v) \in D$ for which S is regular at $\Phi(u, v)$, then Φ is said to be orientation-reversing. Using the preceding notation, this condition corresponds to the choice $\frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} = -\mathbf{n}(\Phi(u, v))$.

1: The first known example of such a surface was the Möbius strip, named after the German mathematician and astronomer A. F. Möbius discovered it in 1858 along with the mathematician J. B. Listing.

Thus, any one-to-one parametrized surface for which $\mathbf{T}_u \times \mathbf{T}_v$ never vanishes can be considered as an oriented surface with a positive side determined by the direction of $\mathbf{T}_u \times \mathbf{T}_v$.

Example 12.4.1 We can give the unit sphere $x^2 + y^2 + z^2 = 1$ in $\mathbb{R}^3$ an orientation by selecting the unit vector $\mathbf{n}(x, y, z) = \mathbf{r}$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, which points to the outside of the surface. This choice corresponds to our intuitive notion of outside for the sphere (You can obtain this by understanding the sphere as the level surface of a function $f(x, y, z) = x^2 + y^2 + z^2$ and finding ∇f, which we know will be perpendicular to $f = 1$).

Now that the sphere S is an oriented surface, consider the parametrization Φ of S given in Example 1. The cross product of the tangent vectors $\mathbf{T}_\theta$ and $\mathbf{T}_\phi$—that is, a normal to S is given by

$$(-\sin\phi)[(\cos\theta\sin\phi)\mathbf{i} + (\sin\theta\sin\phi)\mathbf{j} + (\cos\phi)\mathbf{k}] = -\mathbf{r}\sin\phi.$$

Because $-\sin\phi \leq 0$ for $0 \leq \phi \leq \pi$, this normal vector points inward from the sphere. Thus, the given parametrization Φ is orientation-reversing. By swapping the order of θ and ϕ, we would get an orientation-preserving parametrization.

We can show that the absolute value of the surface integral does not change depending on parametrizations. Just as we did for Line and Path integrals, the proof depends on the change of variable formula:

Theorem 12.4.1 (Independence of Surface Integrals on Parametrizations) *Let S be an oriented surface and let Φ_1 and Φ_2 be two regular orientation-preserving parametrizations, with* $\mathbf{F}$ *a continuous vector field defined on S. Then*

$$\iint_{\Phi_1} \mathbf{F}\cdot d\mathbf{S} = \iint_{\Phi_2} \mathbf{F}\cdot d\mathbf{S}.$$

If Φ_1 is orientation-preserving and Φ_2 orientation-reversing, then

$$\iint_{\Phi_1} \mathbf{F}\cdot d\mathbf{S} = -\iint_{\Phi_2} \mathbf{F}\cdot d\mathbf{S}.$$

If f is a real-valued continuous function defined on S, and if Φ_1 and Φ_2 are parametrizations of S, then

$$\iint_{\Phi_1} f\, d\mathbf{S} = \iint_{\Phi_2} f\, d\mathbf{S}.$$

Note that if $f = 1$, we obtain

$$A(S) = \iint_{\Phi_1} d\mathbf{S} = \iint_{\Phi_2} d\mathbf{S},$$

thus showing that area is independent of parametrization.

12.5 Relation with Scalar Integrals

For an oriented smooth surface S and an orientation-preserving parametrization ϕ of S, we can express the surface integral $\int_S \mathbf{F} \cdot d\mathbf{S}$ as an integral of a real-valued function f over the surface. Let $\mathbf{n} = \frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|}$ be the unit normal pointing to the outside of S. Then,

$$
\begin{aligned}
\int_S \mathbf{F} \cdot d\mathbf{S} &= \int_D \mathbf{F} \cdot (\mathbf{T}_u \times \mathbf{T}_v)\, du\, dv \\
&= \int_D \mathbf{F} \cdot \left(\frac{\mathbf{T}_u \times \mathbf{T}_v}{\|\mathbf{T}_u \times \mathbf{T}_v\|} \right) \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv \\
&= \int_D (\mathbf{F} \cdot \mathbf{n}) \|\mathbf{T}_u \times \mathbf{T}_v\|\, du\, dv = \int_S (\mathbf{F} \cdot \mathbf{n})\, dS = \int_S f\, dS,
\end{aligned}
$$

where $f = \mathbf{F} \cdot \mathbf{n}$. We have thus proved the following theorem:

Theorem 12.5.1 (Surface Integral of Vector Fields) *Let S be an oriented surface and $\mathbf{F}$ be a vector field defined over S. The surface integral of $\mathbf{F}$ over S, denoted as $\iint_S \mathbf{F} \cdot d\mathbf{S}$, is equal to the integral of the normal component of $\mathbf{F}$ over the surface. In short,*

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S \mathbf{F} \cdot \mathbf{n}\, dS.$$

Example 12.5.1 We can redo Example 12.3.1 by taking into account the relationship between scalar and vectorial surface integrals: remember that $\mathbf{r}(x, y, z) = (x, y, z)$ and note that the unit "inward" normal vector for the sphere is $\mathbf{n}(x, y, z) = -x\mathbf{i} - y\mathbf{j} - z\mathbf{k}$ (see Example 12.4.1). Therefore:

$$\mathbf{r} \cdot \mathbf{n} = -x^2 - y^2 - z^2 = -1,$$

and so

$$\iint_S \mathbf{r} \cdot d\mathbf{S} = \iint_S (\mathbf{r} \cdot n)\, dS = -\iint_S dS = -4\pi,$$

since $\iint_S dS$ is the area of the unit sphere, equal to 4π.

Exercises

Exercise 12.1 Compute the area of the following surfaces:

1. Sphere of radius R.
2. Circular cone parametrized by $\mathbf{r}(u, v) = (u \cos v, u \sin v, u)$, where $0 \leq u \leq a, 0 \leq v \leq 2\pi$.
3. Piece of the paraboloid $z = x^2 + y^2$ which lies within the cylinder $x^2 + y^2 = a^2$.
4. Piece of the cylinder $x^2 + z^2 = 16$ bounded by the cylinder $x^2 + y^2 = 16$.

Exercise 12.2 Find the area of the surface of the sphere $x^2 + y^2 + z^2 = a^2$ lying outside the cylinders $x^2 + y^2 = \pm ax$.

Exercise 12.3
1. Deduce the formula for the area of a surface of revolution obtained by rotating the graph of the function $y = f(x)$, $0 < a \leq x \leq b$, around the vertical axis:
$$A = 2\pi \int_a^b x\sqrt{1 + \left(f'(x)\right)^2}\, dx,$$
for the parametrization $\mathbf{s}(r, \theta) = (r \cos\theta, r \sin\theta, f(r))$, where $a \leq r \leq b$ and $0 \leq \theta \leq 2\pi$.
2. Obtain the area of the surface of the torus obtained by rotating the graph $(x - R)^2 + y^2 = c^2$, $0 < c < R$.
3. Deduce the corresponding parametrization for an analogous formula in the case where the graph $y = f(x)$, $a \leq x \leq b$, is rotated along the horizontal axis.

Exercise 12.4 Consider a subset of $\mathbb{R}^3$ given by $W = \{1 \leq z \leq (x^2 + y^2)^{-1/2}\}$. Show that the volume of W is finite and that its boundary has infinite area.

Exercise 12.5 Compute the moment of inertia with respect to a diameter of a homogeneous spherical surface of mass m and radius a.

Integral theorems of Vector Calculus (I): Green and Stokes Theorem

13

13.1 Green's Theorem

Green's theorem relates a line integral along a closed curve C in the plane $\mathbb{R}^2$ to a double integral over the region enclosed by C. This important result will be generalized in the following sections to curves and surfaces in $\mathbb{R}^3$. We shall be referring to line integrals around curves that are the boundaries of elementary regions.

Theorem 13.1.1 (Green's Theorem) *Let $D \subset \mathbb{R}^2$ be a simple region and let C be its boundary. Suppose $P : D \to \mathbb{R}$ and $Q : D \to \mathbb{R}$ are of class C^1. Then*

$$\int_{C^+} P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy.$$

The correct (positive) orientation for the boundary curves of region D can be remembered by the following device: If you walk along the curve C with the correct orientation, the region D will be on your left.

Green's theorem actually applies to any "decent" region in $\mathbb{R}^2$. For instance, Green's theorem applies to regions that are not simple, but that can be broken up into pieces, each of which is simple.

Example 13.1.1 Verify Green's theorem for $P(x, y) = x$ and $Q(x, y) = xy$, where D is the unit disc $x^2 + y^2 \leq 1$.

We do this by evaluating both sides in Green's theorem directly. The boundary of D is the unit circle parametrized by $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$, and so

$$\begin{aligned}\int_{\partial D} P\,dx + Q\,dy &= \int_0^{2\pi} [(\cos t)(-\sin t) + \cos t \sin t \cos t]\ dt \\ &= \int_0^{2\pi} \frac{\cos^2 t}{2} + \frac{\cos^3 t}{3}\ dt = 0.\end{aligned}$$

On the other hand,

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy = \iint_D y\,dx\,dy,$$

which is also zero by symmetry. Thus, Green's theorem is verified in this case.

We can use Green's theorem to obtain a formula for the area of a region bounded by a simple closed curve.

Theorem 13.1.2 (Area of a Region) *If C is a simple closed curve that* *bounds a region to which Green's theorem applies, then the area of the region D bounded by $C = \partial D$ is*

$$A = \frac{1}{2}\int_{\partial D} x\,dy - y\,dx.$$

Proof. Let $P(x, y) = -y$, $Q(x, y) = x$; then by Green's theorem we have

$$\iint_D \left(\frac{\partial(x)}{\partial x} - \frac{\partial(-y)}{\partial y}\right) dx\,dy = \iint_D 2\,dx\,dy = 2A.$$

□

Example 13.1.2 Compute the area of the region enclosed by the hypocycloid defined by $x^{2/3} + y^{2/3} = a^{2/3}$ using the parametrization $x = a\cos^3\theta$, $y = a\sin^3\theta$, $0 \le \theta \le 2\pi$.

From the preceding theorem, and using the trigonometric identities $\cos^2\theta + \sin^2\theta = 1$, $\sin 2\theta = 2\sin\theta\cos\theta$, and $\sin^2\theta = (1 - \cos 2\theta)/2$, we get

$$\begin{aligned}
A &= \frac{1}{2}\int_{\partial D} x\,dy - y\,dx \\
&= \frac{1}{2}\int_0^{2\pi} (a\cos^3\theta)(3a\sin^2\theta\cos\theta) - (a\sin^3\theta)(-3a\cos^2\theta\sin\theta)\,d\theta \\
&= \frac{3a^2}{2}\int_0^{2\pi} \sin^2\theta\cos^2\theta\,d\theta = \frac{3a^2}{8}\int_0^{2\pi}\sin^2 2\theta\,d\theta \\
&= \frac{3a^2}{16}\int_0^{2\pi}(1 - \cos 4\theta)\,d\theta = \frac{3\pi a^2}{8}.
\end{aligned}$$

13.2 Stokes' Theorem

Stokes' theorem relates the line integral of a vector field around a simple closed curve C in $\mathbb{R}^3$ to an integral over a surface S for which C is the boundary. In this regard, it is very much like Green's theorem.

Suppose $\sigma : D \longrightarrow \mathbb{R}^3$ is a parametrization of a surface S and $c(t) = (u(t), v(t))$ is a parametrization of ∂D. We might be tempted to define ∂S as the curve parametrized by $t \mapsto \sigma(u(t), v(t))$. However, with this definition, ∂S might not be the boundary of S in any reasonable geometric sense.

For example, consider the unit sphere S parametrized by spherical coordinates in $\mathbb{R}^3$. In this case, the boundary of S would be half of the great circle on S lying in the xz-plane, which clearly does not match the geometric intuition of a boundary for a smooth surface like a sphere.

We can get around this difficulty by assuming that σ is one-to-one on all of D. Then the image of ∂D under σ, namely, $\sigma(\partial D)$, will be the geometric boundary of $S = \sigma(D)$. If $c(t) = (u(t), v(t))$ is a parametrization of ∂D in the positive direction, we define ∂S to be the oriented simple closed curve that is the image of the mapping $p : t \mapsto \sigma(u(t), v(t))$, with the orientation of ∂S induced by p.

Theorem 13.2.1 (Stokes' Theorem for Parametrized Surfaces) *Let S be an oriented surface defined by a one-to-one parametrization $\sigma : D \subset \mathbb{R}^2 \to S$, where D is a region to which Green's theorem applies. Let ∂S denote the oriented boundary of S and let $\mathbf{F}$ be a C^1 vector field on S. Then*

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

If S has no boundary, and this includes surfaces such as the sphere, then the integral on the left is zero.

Example 13.2.1 Let S be a portion of a sphere sitting on top of the circle $x^2 + y^2 = 1$, oriented outwards. It does not include the disc $x^2 + y^2 < 1$ in the xy plane. Let $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + e^{xz}\mathbf{k}$. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$.

This surface could be parametrized using spherical coordinates based at the center of the sphere. However, we need not explicitly find σ in order to solve this problem.

By Theorem 13.2.1, $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}$, and so if we parametrize ∂S by $x(t) = \cos t$, $y(t) = \sin t$, $0 \le t \le 2\pi$, we determine

$$\int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_0^{2\pi} \left[-\sin^2 t - \cos^2 t\right] dt = -2\pi.$$

Therefore, $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = -2\pi$.

If S is the graph of a function, i.e. we can express it as $z = f(x, y)$, then the trivial parametrization is $x = u, y = v, z = f(u, v)$ and so

$$T_u = (1, 0, z_u), \quad T_v = (0, 1, z_v) \implies T_u \times T_v = (-z_v, -z_u, 1).$$

(Note that $z_u = z_x, z_v = z_y$).

Stokes theorem relates the line integral $\int_{\partial S} \mathbf{F} \cdot d\mathbf{S}$ with

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_D (\nabla \times \mathbf{F}) \cdot (-z_x, -z_y, 1)\, dx\, dy.$$

In particular, if $\mathbf{F}$ has zero $\mathbf{k}$ component, then the last expression becomes $\iint_D (\text{curl } F \cdot \mathbf{k})\, dx\, dy$ and we recover Green's theorem.

Example 13.2.2 Use Stokes' theorem to evaluate the line integral

$$\int_C (-y^3\, dx + x^3\, dy - z^3\, dz),$$

where C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $x + y + z = 1$, and the orientation on C corresponds to counterclockwise motion in the xy plane.

The curve C bounds the surface S parametrized by $\Phi(x, y) = (x, y, 1 - x - y)$ for (x, y) in the set $D = \{x^2 + y^2 \le 1\}$. We set $\mathbf{F} = -y^3\mathbf{i} + x^3\mathbf{j} - z^3\mathbf{k}$, which has curl $\nabla \times \mathbf{F} = (3x^2 + 3y^2)\mathbf{k}$. Then, by Stokes' theorem, the line integral is equal to the surface integral

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_D (3x^2 + 3y^2)\, dx\, dy.$$

This integral can be evaluated by changing to polar coordinates. Doing this, we get

$$3\iint_D (x^2+y^2)\, dx\, dy = \int_0^1 \int_0^{2\pi} r^2 \cdot r\, d\theta\, dr = \int_0^1 6\pi r^3\, dr = \frac{6\pi}{4} = \frac{3\pi}{2}.$$

Let us verify this result by directly evaluating the line integral

$$\int_C (-y^3\, dx + x^3\, dy - z^3\, dz).$$

We can parametrize the curve ∂D by the equations

$$x = \cos t, \quad y = \sin t, \quad z = 0, \quad 0 \le t \le 2\pi.$$

The curve C is therefore parametrized by the equations

$$x = \cos t, \quad y = \sin t, \quad z = 1 - \sin t - \cos t, \quad 0 \le t \le 2\pi.$$

Thus,

$$\int_C -y^3\, dx + x^3\, dy - z^3\, dz =$$

$$= \int_0^{2\pi} (\cos^4 t + \sin^4 t)\, dt - \int_0^{2\pi} (1 - \sin t - \cos t)^3 (-\cos t + \sin t)\, dt.$$

The second integrand is zero. Using $\cos^2 2t = (1 + \cos 2t)/2$ and $\sin^2 2t = (1 - \cos 2t)/2$, we are left with

$$\int_0^{2\pi} (\cos^4 t + \sin^4 t)\, dt = \int_0^{2\pi} \frac{1 + \cos^2 2t}{2}\, dt = \pi + \int_0^{2\pi} \frac{1}{2}\cos^2 2t\, dt$$

$$\begin{aligned} &= \pi + \int_0^{2\pi} \frac{1 + \cos 4t}{4}\, dt \\ &= \pi + \frac{\pi}{2} + \frac{1}{4}\sin 4t\Big|_0^{2\pi} = \frac{3\pi}{2}. \end{aligned}$$

13.3 The Curl as Circulation per Unit Area

Let us now use Stokes' theorem to justify the physical interpretation of $\nabla \times \mathbf{F}$ as the measure of rotation in a vector field.

$$\iint_S (\operatorname{curl} \mathbf{F}) \cdot \mathbf{n}\, dS = \iint_S (\operatorname{curl} \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s} = \int_{\partial S} \mathbf{F}_T\, ds,$$

where $\mathbf{F}_T$ is the tangential component of $\mathbf{F}$. This says that the integral of the normal component of the curl of a vector field over an oriented surface S is equal to the line integral of $\mathbf{F}$ along ∂S, which in turn is equal to the path integral of the tangential component of $\mathbf{F}$ over ∂S.

Suppose $\mathbf{V}$ represents the velocity vector field of a fluid. Consider a point P and a unit vector $\mathbf{n}$. Let S_ρ denote the disc of radius ρ and center P, which is perpendicular to $\mathbf{n}$. By Stokes' theorem,

$$\iint_{S_\rho} (\text{curl}\,\mathbf{V}) \cdot d\mathbf{S} = \iint_{S_\rho} (\text{curl}\,\mathbf{V}) \cdot \mathbf{n}\, dS = \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s},$$

where ∂S_ρ has the orientation induced by $\mathbf{n}$.

By the mean-value theorem for integrals there is a point Q in S_ρ such that

$$\iint_{S_\rho} (\text{curl}\,\mathbf{V}) \cdot \mathbf{n}\, dS = [\text{curl}\,\mathbf{V}(Q) \cdot \mathbf{n}] A(S_\rho),$$

where $A(S_\rho) = \pi\rho^2$ is the area of S_ρ and curl $\mathbf{V}(Q)$ is the value of curl $\mathbf{V}$ at Q. Thus,

$$\begin{aligned} \lim_{\rho \to 0} \frac{1}{A(S_\rho)} \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s} &= \lim_{\rho \to 0} \frac{1}{A(S_\rho)} \iint_{S_\rho} (\text{curl}\,\mathbf{V}) \cdot d\mathbf{S} \\ &= \lim_{\rho \to 0} \text{curl}\,\mathbf{V}(Q) \cdot \mathbf{n} = (\text{curl}\,\mathbf{V}(P)) \cdot \mathbf{n}. \end{aligned}$$

Thus,

$$(\text{curl}\,\mathbf{V}(P)) \cdot \mathbf{n} = \lim_{\rho \to 0} \frac{1}{A(S_\rho)} \int_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s}.$$

Let us pause to consider the physical meaning of $\oint_C \mathbf{V} \cdot d\mathbf{s}$ when $\mathbf{V}$ is the velocity field of a fluid. Suppose, for example, that $\mathbf{V}$ points in the direction tangent to the oriented curve C. Then clearly $\oint_C \mathbf{V} \cdot d\mathbf{s} > 0$, and particles on C tend to rotate counterclockwise. If $\mathbf{V}$ is pointing in the opposite direction, then $\oint_C \mathbf{V} \cdot d\mathbf{s} < 0$ and particles tend to rotate clockwise. If $\mathbf{V}$ is perpendicular to C, then particles do not rotate on C at all and $\oint_C \mathbf{V} \cdot d\mathbf{s} = 0$.

In general, $\oint_C \mathbf{V} \cdot d\mathbf{s}$, being the integral of the tangential component of $\mathbf{V}$, represents the net amount of turning of the fluid in a counterclockwise direction around C. We therefore refer to $\oint_C \mathbf{V} \cdot d\mathbf{s}$ as the circulation of $\mathbf{V}$ around C.

These results allow us to see just what $\nabla \times \mathbf{V}$ means for the motion of a fluid. The circulation $\oint_{\partial S_\rho} \mathbf{V} \cdot d\mathbf{s}$ is the net velocity of the fluid around ∂S_ρ, so that $(\nabla \times \mathbf{V}) \cdot \mathbf{n}$ represents the turning or rotating effect of the fluid around the axis $\mathbf{n}$.

Vector forms of Green's Theorem

We can write Green's theorem using the curl:

Theorem 13.3.1 (Vector Form of Green's Theorem) *Let $D \subset \mathbb{R}^2$ be a region to which Green's theorem applies, let ∂D be its (positively oriented) boundary, and let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ be a C^1 vector field on D. Then*

$$\int_{\partial D} \mathbf{F} \cdot d\mathbf{s} = \iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dA = \iint_D (\mathit{curl}\,\mathbf{F}) \cdot \mathbf{k}\, dA$$

The result comes from the fact that curl $\mathbf{F} \cdot \mathbf{k} = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

Example 13.3.1 Let $\mathbf{F} = (xy^2, y + x)$. Integrate $(\nabla \times \mathbf{F}) \cdot \mathbf{k}$ over the region in the first quadrant bounded by the curves $y = x^2$ and $y = x$.

Method 1. We first compute the curl

$$\nabla \times \mathbf{F} = \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\mathbf{k} = (1 - 2xy)\mathbf{k}.$$

Thus, $(\nabla \times \mathbf{F}) \cdot \mathbf{k} = 1 - 2xy$. This can be integrated over the given region D using an iterated integral as follows:

$$\begin{aligned}\iint_D (\nabla \times \mathbf{F}) \cdot \mathbf{k}\, dx\, dy &= \int_0^1 \int_{x^2}^{x} (1 - 2xy)\, dy\, dx \\ &= \int_0^1 [y - xy^2]_{y=x^2}^{y=x}\, dx \\ &= \int_0^1 \left(x - x^3 - x^2 + x^5\right)\, dx = \frac{1}{12}.\end{aligned}$$

Method 2. The line integral of $\mathbf{F}$ along the curve $y = x$ from left to right is

$$\int_0^1 F_1\, dx + F_2\, dy = \int_0^1 (x^3 + 2x)\, dx = \frac{5}{4}.$$

Along the curve $y = x^2$ we get

$$\int_0^1 F_1\, dx + F_2\, dy = \int_0^1 x^5\, dx + \int_0^1 (x + x^2)(2x\, dx) = \frac{1}{6} + \frac{2}{3} + \frac{1}{2} = \frac{4}{3}.$$

As the integral along $y = x$ is to be taken from right to left,

$$\int_{\partial D} \mathbf{F} \cdot d\mathbf{s} = \frac{4}{3} - \frac{5}{4} = \frac{1}{12}.$$

There is another form of Green's theorem that can be generalized to $\mathbb{R}^3$.

Theorem 13.3.2 (Divergence Theorem in the Plane) *Let $D \subset \mathbb{R}^2$ be a region to which Green's theorem applies and let ∂D be its boundary. Let $\mathbf{n}$ denote the outward unit normal to ∂D. If $c : [a, b] \to \mathbb{R}^2$, $t \mapsto c(t) = (x(t), y(t))$ is a positively oriented parametrization of ∂D, $\mathbf{n}$ is given by*

$$\mathbf{n} = \frac{(y'(t), -x'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

Let $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ be a C^1 vector field on D. Then

$$\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D \mathit{div}\, \mathbf{F}\, dA.$$

Proof. Recall that $\mathbf{c}'(t) = (x'(t), y'(t))$ is tangent to ∂D, and note that $\mathbf{n} \cdot \mathbf{c}' = 0$. Thus, $\mathbf{n}$ is normal to the boundary. The sign of $\mathbf{n}$ is chosen to make it correspond to the outward (rather than the inward) direction. By the definition of the line integral

$$\begin{aligned}\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds &= \int_a^b \left[\frac{P(x(t), y(t))y'(t) - Q(x(t), y(t))x'(t)}{\sqrt{x'(t)^2 + y'(t)^2}} \right] \sqrt{x'(t)^2 + y'(t)^2}\, dt \\ &= \int_{\partial D} P\, dy - Q\, dx.\end{aligned}$$

By Green's theorem, this equals

$$\iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) dx\, dy = \iint_D \operatorname{div} \mathbf{F}\, dA.$$

□

Example 13.3.2 Let $\mathbf{F} = y^3\mathbf{i} + x^5\mathbf{j}$. Compute the integral of the normal component of $\mathbf{F}$ around the unit square.

This can be done using the divergence theorem. Indeed,

$$\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D \operatorname{div} \mathbf{F}\, dA.$$

But div $\mathbf{F} = 0$, and so the integral is zero.

Exercises

Exercise 13.1 Compute

$$\int_\gamma (5 - xy - y^2)dx - (2xy - x^2)dy,$$

where γ is the square with vertices $(0,0)$, $(1,0)$, $(1,1)$, and $(0,1)$ in two ways: directly and by applying Green's Theorem.

Exercise 13.2 Let f be a differentiable function in $\mathbb{R}$ and consider

$$P(x,y) = e^{x^2} - \frac{y}{3 + e^{xy}}, \quad Q(x,y) = f(y).$$

If γ is the boundary of the square $[0,1] \times [0,1]$ traversed positively, compute

$$\int_\gamma P\,dx + Q\,dy.$$

Exercise 13.3 Consider the functions $P(x,y) = \frac{y}{x^2+y^2}$ and $Q(x,y) = \frac{-x}{x^2+y^2}$. Let $C = \partial D$ be a closed, piecewise regular curve that does not pass through the origin.

1. Show that $\frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y}$.
2. If $(0,0) \in D$, prove that

$$\int_C P\,dx + Q\,dy = \pm 2\pi.$$

3. If $(0,0) \notin D$, compute

$$\int_C P\,dx + Q\,dy.$$

Exercise 13.4 Compute

$$\int_\gamma \frac{-y\,dx + (x-1)\,dy}{(x-1)^2 + y^2},$$

where γ is a closed, simple, piecewise regular curve containing the point $(1,0)$ in its interior.

Exercise 13.5
1. Let A be the area of the region D, bounded by a closed, simple, piecewise regular curve C which is positively oriented. Show that in Cartesian coordinates

$$A = \frac{1}{2}\int_C -y\,dx + x\,dy,$$

and that in polar coordinates

$$A = \frac{1}{2}\int_C r^2(\theta)\,d\theta.$$

2. Compute the area contained in the loop parametrized by $\mathbf{s}(t) = (t^2 - 1, t^3 - t)$, $t \in [0,1]$.

3. Compute the area of the cardioid given in polar coordinates by $r(\theta) = a(1 - \cos\theta)$, $0 \le \theta \le 2\pi$.

Exercise 13.6 1. Compute

$$\int_D (x + 2y)\, dx\, dy,$$

where D is the region bounded by the interval $[0, 2\pi]$ and the arc of the cycloid $x = t - \sin t$, $y = 1 - \cos t$, for $0 \le t \le 2\pi$.

2. Compute

$$\int_D xy^2\, dx\, dy,$$

where D is the region bounded by the astroid $x = \cos^3 t$, $y = \sin^3 t$, $0 \le t \le \pi/2$ and the two axes.

3. Compute $\int_D y^2 dxdy$, where D is region bounded by the curve $x = a(t - \sin^2 t)$, $y = a\sin^2 t$, $0 \le t \le \pi$, and the line connecting the end points.

Exercise 13.7 Use Stokes' Theorem to compute $\int_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n}\, dS$ in the following cases, where the orientation of S is given by the unit outer normal to S:

1. $\mathbf{F}(x, y, z) = (x^2y^2, yz, xy)$ and S is the paraboloid $z = a^2 - x^2 - y^2$, $z \ge 0$.
2. $\mathbf{F}(x, y, z) = ((1 - z)y, ze^x, x\sin z)$ and S is the upper semi-sphere of radius a.
3. $\mathbf{F}(x, y, z) = (x^3 + z^3, e^{x+y+z}, x^3 + y^3)$ and $S = \{x^2 + y^2 + z^2 = 1, y \ge 0\}$.

Exercise 13.8 Consider the vector field $\mathbf{F}(x, y, z) = \left(y, x^2, \left(x^2 + y^4\right)^{3/2} \sin(e^{\sqrt{xyz}})\right)$. Compute

$$\int_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n}\, dS,$$

where $\mathbf{n}$ is the unit inner normal of the semi-ellipsoid

$$S = \{(x, y, z) : 4x^2 + 9y^2 + 36z^2 = 36, z \ge 0\}.$$

Exercise 13.9 Consider the vector field $\mathbf{F}(x, y, z) = (2y, 3z, x)$ and the triangle of vertices $A(0, 0, 0)$, $B(0, 2, 0)$, and $C(1, 1, 1)$, denoted by T.

1. Choose an orientation for the surface of the triangle T and the corresponding induced orientation for its boundary.
2. Compute the path integral of the field $\mathbf{F}$ along the boundary of T.

Exercise 13.10 Verify Stokes' Theorem in the following cases:

1. $\mathbf{F}(x, y, z) = (y^2, xy, xz)$, over the paraboloid $z = a^2 - x^2 - y^2$, $z \ge 0$.
2. $\mathbf{F}(x, y, z) = (-y^3, x^3, z^3)$, over $S = \{z = y, y \ge 0, x^2 + y^2 \le 1\}$.

Exercise 13.11 Let S be the square with vertices $(0, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$, and $(0, 1, 1)$, oriented with the upper unit normal (i.e., positive first coordinate). Consider the vector field

$$\mathbf{F}(x, y, z) = (xy^2, 2y^2z, 3z^2x).$$

Compute $\int_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n} \, dS$ in two different ways using Stokes' Theorem.

Integral theorems of Vector Calculus (II): Gauss' Theorem

14

The boundary of an elementary region in $\mathbb{R}^3$ is a surface made up of a finite number (at most six, at least two) of surfaces that can be described as graphs of functions from $\mathbb{R}^2$ to $\mathbb{R}$. This kind of surface is called a closed surface. The surfaces $S_1, S_2, \dots, S_N$ composing such a closed surface are called its faces.

Example 14.0.1 The cube is an elementary region, and in fact a symmetric elementary region, with six rectangles composing its boundary. The sphere is the boundary of a solid ball, which is also a symmetric elementary region.

Closed surfaces can be oriented in two ways. The outward orientation makes the normal point outward into space, and the inward orientation makes the normal point into the bounded region.

Suppose $S = \sum_i S_i$ is a closed surface oriented in one of these two ways and $\mathbf{F}$ is a vector field on S. Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \sum_i \iint_{S_i} \mathbf{F} \cdot d\mathbf{S}.$$

If S is given the outward orientation, the integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ measures the total flux of $\mathbf{F}$ outward across S. That is, if we think of $\mathbf{F}$ as the velocity field of a fluid, $\iint_S \mathbf{F} \cdot d\mathbf{S}$ indicates the amount of fluid leaving the region bounded by S per unit time. If S is given the inward orientation, the integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ measures the total flux of $\mathbf{F}$ inward across S.

Remember that these integrals can also be written as

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n})\, dS,$$

that is, the integral of the normal component of $\mathbf{F}$ over S. In the remainder of this section, if S is a closed surface enclosing a region W, we adopt the default convention that $S = \partial W$ is given the outward orientation, with outward unit normal $\mathbf{n}(x, y, z)$ at each point $(x, y, z) \in S$. Furthermore, we denote the surface with the opposite (inward) orientation by ∂W_{op}. Then the associated unit normal direction for this orientation is $-\mathbf{n}$. Thus,

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_S (\mathbf{F} \cdot \mathbf{n})\, dS = -\iint_{\partial W_{\text{op}}} [\mathbf{F} \cdot (-\mathbf{n})]\, dS = -\iint_{\partial W_{\text{op}}} \mathbf{F} \cdot d\mathbf{S}.$$

14.1 Gauss' Theorem

Gauss' theorem states that the flux of a vector field out of a closed surface equals the integral of the divergence of that vector field over the volume enclosed by the surface. The result parallels Stokes' theorem and Green's theorem in that it relates an integral over a closed geometric object (curve or surface) to an integral over a contained region (surface or volume).

Theorem 14.1.1 (Gauss' Divergence Theorem) *Let W be an elementary region in space. Denote by ∂W the oriented closed surface that bounds W. Let $\mathbf{F}$ be a smooth vector field defined on W. Then*

$$\iiint_W (\nabla \cdot \mathbf{F})\, dV = \iint_{\partial W} \mathbf{F} \cdot d\mathbf{S},$$

or, alternatively,

$$\iiint_W (div\, \mathbf{F})\, dV = \iint_{\partial W} (\mathbf{F} \cdot \mathbf{n})\, dS.$$

Example 14.1.1 Consider $\mathbf{F} = 2x\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$. Let S be the unit sphere defined by $x^2 + y^2 + z^2 = 1$. Evaluate $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$.

By Gauss' theorem,

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_W (\mathrm{div}\, \mathbf{F})\, dV,$$

where W is the ball bounded by the sphere. The integral on the right is

$$\iiint_W (2 + 2y + 2z)\, dV = 2 \iiint_W dV + 2 \iiint_W y\, dV + 2 \iiint_W z\, dV.$$

By symmetry, we can argue that

$$\iiint_W y\, dV = \iiint_W z\, dV = 0$$

Thus, because a sphere of radius R has volume $\frac{4\pi R^3}{3}$, it follows that

$$2 \iiint_W (1 + y + z)\, dV = 2 \iiint_W 1\, dV = \frac{8\pi}{3}.$$

You can convince yourself that direct computation of $\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$ proves unwieldy.

Example 14.1.2 Use the divergence theorem to evaluate

$$\iint_{\partial W} (x^2 + y + x)\, dS,$$

where W is the solid ball $x^2 + y^2 + z^2 \le 1$.

To apply Gauss' divergence theorem, we find a vector field $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ on W with $\mathbf{F} \cdot \mathbf{n} = x^2 + y + z$. At any point $(x, y, z) \in \partial W$, the outward unit normal $\mathbf{n}$ to ∂W is

$$\mathbf{n} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k},$$

because on ∂W, $x^2 + y^2 + z^2 = 1$ and the radius vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

is normal to the sphere ∂W.

Therefore, if $\mathbf{F}$ is the desired vector field, then

$$\mathbf{F} \cdot \mathbf{n} = F_1 x + F_2 y + F_3 z.$$

We set

$$F_1 x = x^2, \quad F_2 y = y, \quad F_3 z = z,$$

and solve for F_1, F_2, and F_3 to find that $\mathbf{F} = x\mathbf{i} + \mathbf{j} + \mathbf{k}$. Computing $\operatorname{div} \mathbf{F}$, we get

$$\operatorname{div} \mathbf{F} = 1 + 0 + 0 = 1.$$

Thus, by Gauss' divergence theorem,

$$\int_{\partial W} (x^2 + y + z)\, dS = \int_W dV = \text{volume}\,(W) = \frac{4}{3}\pi.$$

Divergence as the flux per unit Volume

The physical meaning of divergence is that at a point P, $\operatorname{div} \mathbf{F}(P)$ is the rate of net outward flux at P per unit volume. This follows from Gauss' theorem and the mean-value theorem for integrals: If W_ρ is a ball in $\mathbb{R}^3$ of radius ρ centered at P, then there is a point $Q \in W_\rho$ such that

$$\iint_{\partial W_\rho} \mathbf{F} \cdot \mathbf{n}\, dS = \iiint_{W_\rho} \operatorname{div} \mathbf{F}\, dV = \operatorname{div} \mathbf{F}(Q) \cdot \text{volume}(W_\rho),$$

and so

$$\operatorname{div} \mathbf{F}(P) = \lim_{\rho \to 0} \operatorname{div} \mathbf{F}(Q) = \lim_{\rho \to 0} \frac{1}{V(W_\rho)} \iint_{\partial W_\rho} \mathbf{F} \cdot \mathbf{n}\, dS.$$

This is analogous to the limit formulation of the curl given at the end of Section 13.3. Thus, if $\operatorname{div} \mathbf{F}(P) > 0$, we consider P to be a source, for there is a net outward flow near P. If $\operatorname{div} \mathbf{F}(P) < 0$, P is called a sink for $\mathbf{F}$.

A C^1 vector field $\mathbf{F}$ defined on $\mathbb{R}^3$ is said to be divergence-free if $\operatorname{div} \mathbf{F} = 0$. If $\mathbf{F}$ is divergence-free, then $\iint_S \mathbf{F} \cdot d\mathbf{S} = 0$ for all closed surfaces S. The converse can also be demonstrated readily using Gauss' theorem: If $\iint_S \mathbf{F} \cdot d\mathbf{S} = 0$ for all closed surfaces S, then $\mathbf{F}$ is divergence-free. If $\mathbf{F}$ is divergence-free, we thus see that the flux of $\mathbf{F}$ across any closed surface S is 0, so that if $\mathbf{F}$ is the velocity field of a fluid, the net amount of fluid that flows out of any region will be 0. Thus, exactly as much fluid must flow into the region as flows out (in unit time). A fluid with this property is therefore described as incompressible.

Example 14.1.3 Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = xy^2\mathbf{i} + x^2y\mathbf{j} + y\mathbf{k}$ and S is the surface of the cylinder $x^2 + y^2 = 1$, bounded by the planes $z = 1$ and $z = -1$, and including the portions $x^2 + y^2 \leq 1$ when $z = \pm 1$.

We can compute this integral directly, but it is easier to use the divergence theorem. Now S is the boundary of the region W given by

$x^2+y^2 \leq 1, -1 \leq z \leq 1$. Thus, $\iint_S \mathbf{F}\cdot d\mathbf{S} = \iiint_W (\operatorname{div}\mathbf{F})\, dV$. Moreover,

$$\iiint_W (\operatorname{div}\mathbf{F})\, dV = \iiint_W (x^2+y^2)\, dx\, dy\, dz = 2\iint_{x^2+y^2\leq 1} (x^2+y^2)\, dx\, dy.$$

Before evaluating the double integral, we note that the surface integral satisfies

$$\iint_{\partial W} \mathbf{F}\cdot\mathbf{n}\, dS = \iint_{x^2+y^2\leq 1} (x^2+y^2)\, dx\, dy > 0.$$

This means that $\iint_{\partial W} \mathbf{F}\cdot dS$, the net flux of $\mathbf{F}$ out of the cylinder, is positive.

We change variables to polar coordinates to evaluate the double integral:

$$x = r\cos\theta, \quad y = r\sin\theta, \quad 0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi.$$

Hence, the Jacobian is r and $x^2+y^2 = r^2$. Thus,

$$\iint_{x^2+y^2\leq 1} (x^2+y^2)\, dx\, dy = \int_0^{2\pi}\int_0^1 r^2 r\, dr\, d\theta = \frac{1}{2}\pi.$$

Therefore,

$$\iiint_W \operatorname{div}\mathbf{F}\, dV = \frac{\pi}{2}.$$

Let's see a final example, where we connect Stokes' and Gauss' theorems:

Example 14.1.4 Consider the surface $S = \{(x,y,z) \in \mathbb{R}^3 : x^2+y^2+z^2 = 1, y \geq 0\}$ with the normal vector field $\mathbf{n}$ outer to the unit sphere and the function $\mathbf{F}(x,y,z) = (x+z, y+z, 2z)$. Calculate (i) $\iint_S \mathbf{F}\cdot\mathbf{n}\, dS$ and (ii) $\iint_S \nabla\times\mathbf{F}\cdot\mathbf{n}\, dS$.

For (i), S is not closed, so how can we apply the divergence theorem? Consider the surface $S_2 = \{x^2+z^2 \leq 0, y = 0\}$, then

$$\partial W = S \cup S_2,$$

where $W = \{x^2+y^2+z^2 \leq 1, y \geq 0\}$. Then $\iint_{\partial W} \mathbf{F}\cdot\mathbf{n}\, dS = \iint_S \mathbf{F}\cdot\mathbf{n}\, dS + \iint_{S_2} \mathbf{F}\cdot\mathbf{n}\, dS$ or

$$\iint_S \mathbf{F}\cdot\mathbf{n}\, dS = \iint_{\partial W} \mathbf{F}\cdot\mathbf{n}\, dS - \iint_{S_2} \mathbf{F}\cdot\mathbf{n}\, dS.$$

Now, because ∂W is a closed surface, we can apply Gauss' theorem and

$$\iint_{\partial W} \mathbf{F}\cdot\mathbf{n}\, dS = \iiint_W \operatorname{div}\mathbf{F}\, dV = 4\iiint_W dV = \frac{8\pi}{3}.$$

As for S_2, the unit outer normal vector is $(0,-1,0)$ and

$$\iint_{S_2} (x+z, y+z, 2z)\cdot(0,-1,0)\, dS = \iint_{S_2} (y+z)\, dS = \iint_{S_2} z\, dS,$$

since $y = 0$ on S_2. We can argue that this integral is zero due to symmetry in S_2, or we can solve it using the change of variable $x = r\cos\theta, y = 0, z = r\sin\theta$ wih $r \in [0,1], \theta \in [0, 2\pi]$. Finally,

$$\iint_S \mathbf{F}\cdot\mathbf{n}\, dS = \frac{8\pi}{3}.$$

Note that we can solve this integral directly using spherical coordinates $x = \sin\phi\cos\theta, y = \sin\phi\sin\theta, z = \cos\phi$, with both ϕ and θ ranging from 0 to π. The outer normal vector is $T_\phi \times T_\theta = (\sin^2\phi\cos\theta, \sin^2\phi\sin\theta, \sin\phi\cos\theta)$. The integral becomes

$$\int_0^\pi\int_0^\pi [\sin\phi + \sin^2\phi\cos\phi(\sin\theta + \cos\theta) + \sin\phi\cos^2\phi]\, d\phi\, d\theta = \frac{8\pi}{3}.$$

As for (ii), we can solve it using either Stokes' or Gauss' theorems. In order to use Stokes', note that $\partial S = \{x^2 + z^2 = 1, y = 0\}$ and so

$$\iint_S \nabla\times\mathbf{F}\cdot\mathbf{n}\, dS = \int_{\partial S}\mathbf{F}\cdot d\mathbf{s}.$$

Now, we can use the parametrization of ∂S given by $x = \cos\theta, y = 0, z = \sin\theta$, with θ going from 0 to 2π. Note that this means moving in the negative orientation with respect to S, and so we need to consider the negative integral:

$$\int_{\partial S}\mathbf{F}\cdot d\mathbf{s} = -\int_0^{2\pi}(\sin\theta\cos\theta - \sin^2\theta)\, d\theta = \int_0^{2\pi}\frac{1-\cos 2\theta}{2} = \pi$$

In order to use Gauss' theorem, we need to use the same trick of closing the boundary as before, and so

$$\iint_S \nabla\times\mathbf{F}\cdot\mathbf{n}\, dS = \iint_{\partial W}\nabla\times\mathbf{F}\cdot\mathbf{n}\, dS - \iint_{S_2}\nabla\times\mathbf{F}\cdot\mathbf{n}\, dS,$$

with ∂W and S_2 as in the previous part. Then

$$\iint_{\partial W}\nabla\times\mathbf{F}\cdot\mathbf{n}\, dS = \iiint_W \nabla\cdot(\nabla\times\mathbf{F})\, dV = 0,$$

as the divergence of a curl is zero (Theorem 5.4.2).

For S_2, curl $\mathbf{F} = (-1, 1, 0)$ and $\mathbf{n} = (0, -1, 0)$, so

$$\iint_{S_2}\nabla\times\mathbf{F}\cdot\mathbf{n}\, dS = -\iint_{S_2} dS = -\pi,$$

which is the area of the unit circle. Finally,

$$\iint_S \nabla\times\mathbf{F}\cdot\mathbf{n}\, dS = \iint_{\partial W}\nabla\times\mathbf{F}\cdot\mathbf{n}\, dS - \iint_{S_2}\nabla\times\mathbf{F}\cdot\mathbf{n}\, dS = \pi,$$

as before.

Exercises

Exercise 14.1 Consider the vector field $\mathbf{F}(x, y, z) = \big(y \sin(x^2+y^2), -x \sin(x^2+y^2), z(3-2y)\big)$ and the region $W = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 \leq 1, z \geq 0\}$. Compute

$$\int_{\partial W} \mathbf{F} \cdot \mathbf{n}\, dS.$$

Exercise 14.2 Compute the integral $\int_S \mathbf{F} \cdot \mathbf{n}\, dS$ for the following cases:

1. $\mathbf{F}(x, y, z) = (18z, -12, 3y)$, where S is the region of the plane $2x + 3y + 6z = 12$ in the first octant.
2. $\mathbf{F}(x, y, z) = (x^3, x^2y, x^2z)$, where S is the closed surface bounded by the cylinder $x^2 + y^2 = a^2$, $0 \leq z \leq b$, and the upper and lower covers.
3. $\mathbf{F}(x, y, z) = (4xz, -y^2, yz)$, where S is the surface bounded by the cube $0 \leq x, y, z \leq 1$.
4. $\mathbf{F}(x, y, z) = (x, y, z)$, where S is a bounded simple surface.

Exercise 14.3 Consider the surface $S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1, y \geq 0\}$ with normal vector field $\mathbf{n}$ outer to the unit sphere and the function $\mathbf{F}(x, y, z) = (x + z, y + z, 2z)$.

1. Compute $\int_S \mathbf{F} \cdot \mathbf{n}\, dS$.
2. Compute $\int_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n}\, dS$.

Exercise 14.4 Compute the flux of the vector field $\mathbf{F}(x, y, z) = (y^2, yz, xz)$ across the surface of the tetrahedron bounded by $x = 0$, $y = 0$, $z = 0$, $x + y + z = 1$, with orientation given by the unit outer normal to the surface.

Exercise 14.5 Assume that the temperature in $\mathbb{R}^3$ is proportional to the square of the distance to the vertical axis and consider the region $V = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \leq 2z, z \leq 2\}$.

1. Compute the volume of V.
2. Compute the mean temperature on V.
3. Compute the (outward) flux of the gradient of the temperature across ∂V.

Exercise 14.6 Compute $\int_S \mathbf{F} \cdot \mathbf{n}\, dS$ in the following cases, where $\mathbf{n}$ is the unit outer normal in items (1), (3), (4), and the unit upper normal (i.e., the third component is positive) in item (ii):

1. $\mathbf{F}(x, y, z) = (x^2, y^2, z^2)$ and S is the boundary of the cube $0 \leq x, y, z \leq 1$.
2. $\mathbf{F}(x, y, z) = (xy, -x^2, x + z)$ and S is the piece of the plane $2x + 2y + z = 6$ in the first octant.
3. $\mathbf{F}(x, y, z) = (xz^2, x^2y - z^2, 2xy + y^2z)$ and S is the upper semi-sphere $z = \sqrt{a^2 - x^2 - y^2}$.
4. $\mathbf{F}(x, y, z) = (2x^2 + \cos yz, 3y^2z^2 + \cos(x^2 + z^2), e^{y^2} - 2yz^3)$ and S is the surface of the solid defined by the intersection of the cone $z \geq \sqrt{x^2 + y^2}$ with the sphere $x^2 + y^2 + z^2 \leq 1$.

Exercise 14.7 Consider S, the sphere of radius a oriented with respect to its outer normal, and let $\mathbf{F}(x, y, z) = (\sin yz + e^z, x\cos z + \log(1 + x^2 + z^2), e^{x^2+y^2+z^2})$ be a vector field. Compute $\int_S \mathbf{F} \cdot \mathbf{n}\, dS$.

Exercise 14.8 Consider the union $S = S_1 \cup S_2$, where S_1 and S_2 are the surfaces

$$S_1 = \{x^2 + y^2 = 1, 0 \le z \le 1\}, \quad S_2 = \{x^2 + y^2 + (z-1)^2 = 1, z \ge 1\},$$

and let $\mathbf{F}(x, y, z) = (zx + z^2y + x, z^3yx + y, z^4x^2)$ be a vector field.

1. Compute $\int_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n}\, dS$ using Stokes' Theorem.
2. Compute the same integral using Gauss' Theorem.

Exercise 14.9 Consider the differentiable function $h : \mathbb{R}^2 \to \mathbb{R}$ and the vector field

$$\mathbf{F}(x, y, z) = \left(e^{y^2+z^2} + \int_0^x \frac{e^{t^2+y^2}}{\sqrt{t^2 + y^2}}\, dt, \sin(x^2 + e^z), h(x, y) \right),$$

and the region

$$\Omega = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \le 1, 0 \le z \le \sqrt{x^2 + y^2}, x \ge 0, y \ge 0\}.$$

Compute $\int_{\partial\Omega} \mathbf{F} \cdot \mathbf{n}\, dS$, where $\mathbf{n}$ is the unit inner normal with respect to $\partial\Omega$.

Exercise 14.10 Consider the vector field

$$\mathbf{F}(x, y, z) = \left(ye^z, \int_0^x e^{-t^2+\cos z}\, dt, z(x^2 + y^2) \right),$$

and the region

$$\Omega = \{(x, y, z) : x^2 + y^2 + z^2 < a^2, x^2 + y^2 < z^2\}.$$

Compute $\int_{\partial\Omega} \mathbf{F} \cdot \mathbf{n}\, dS$, where $\mathbf{n}$ is the normal outer to the boundary of Ω.

Appendix

A Solutions to Exercises

A.1 Topology of $\mathbb{R}^n$. Functions of several variables

Exercise 1.1 We will use A° to refer to the *interior* of A and ∂A to refer to its *boundary*. The interior of A is the largest open set that is contained in A, or alternatively, it is equal to $A - \partial A$.

1. $A = \{(x, y) \mid -1 \leq x \leq 1,\ y < 0\}$
   Neither open nor closed.

$$A^\circ = \{(x, y) \mid -1 < x \leq 1,\ y < 0\}$$
$$\partial A = \{(x, y) \mid |x| = 1,\ y \leq 0\} \cup \{(x, y) \mid -1 \leq x \leq 1,\ y = 0\}$$

2. $B = \{(x, y) \mid x = 1,\ 1 < y < 2\}$
   Neither open nor closed.

$$B^\circ = \emptyset$$
$$\partial B = B \cup \{(1, 1), (1, 2)\}$$

3. $C = \{(x, y) \mid x \geq 1,\ 0 \leq y \leq 2x\}$
   Closed (not bound).

$$C^\circ = \{(x, y) \mid x > 1,\ 0 < y < 2x\}$$
$$\partial C = \{(x, y) \mid x \geq 1, y = 0\} \cup \{(x, y) \mid x \geq 1, y = 2x\} \cup \{(x, y) \mid x = 1, 0 \leq y \leq 2\}$$

4. $D = \{(x, y) \mid 2x + 3y \leq 5,\ -5 \leq 0\}$
   Closed.

$$D^\circ = \emptyset$$
$$\partial D = D$$

5. $E = \{(x, y) \mid x^2 + 3y^2 \leq 2\}$
   Closed (compact).

$$E^\circ = F$$
$$\partial E = G$$

6. F is the interior of E. *Open.*

$$F^\circ = F$$
$$\partial F = G$$

7. $G = \{(x, y) \mid x^2 - 3y = 2\}$
   Closed.

$$G^\circ = \emptyset$$

$$\partial G = G$$

8. $H = \{(x, y) \mid y > x^2\}$
   Open.

$$H^\circ = H$$

$$\partial H = \{(x, y) \mid y = x^2\}$$

Exercise 1.2 A set is **compact** if it is open and bounded at the same time.

1. Open,
2. Open,
3. None of the three possibilities,
4. Closed and compact,
5. Open,
6. Closed,
7. Closed and compact,
8. Open.

Exercise 1.3 1.

$$A^\circ = A$$

$$\partial A = \{(5, y) : 0 \le y \le 1\} \cup \{(7, y) : 0 \le y \le 1\} \cup \{(x, 0) : 5 \le x \le 7\} \cup \{(x, 1) : 5 \le x \le 7\}$$

$$\bar{A} = D$$

2.

$$B^\circ = B$$

$$\partial B = \{x = 0,\ y \in \mathbb{R}\}$$

$$\bar{B} = B \cup \partial B$$

3.

$$C^\circ = A$$

$$\partial C = \partial A$$

$$\bar{C} = A \cup \partial A = D$$

4.

$$D^\circ = A$$

$$\partial D = \partial A$$

$$\bar{D} = D$$

5.

$$E^\circ = E$$

$$\partial E = \{x \mid \|x\| = 3\}$$

$$\bar{E} = E \cup \partial E = F$$

6.

$$F^\circ = E$$

$$\partial F = \partial E$$

$$\bar{F} = F$$

7.

$$G^\circ = \emptyset$$

$$\partial G = G$$
$$\bar{G} = G$$

8.
$$H^\circ = H$$
$$\partial H = \partial E$$
$$\bar{H} = H \cup \partial H = \{x \mid \|x\| \leq 3\}$$

Exercise 1.4 1.

$$f(1, y/x) = \frac{2 \cdot 1 \cdot (y/x)}{1^2 + (y/x)^2} = \frac{2y}{x^2 + y^2} = f(x, y).$$

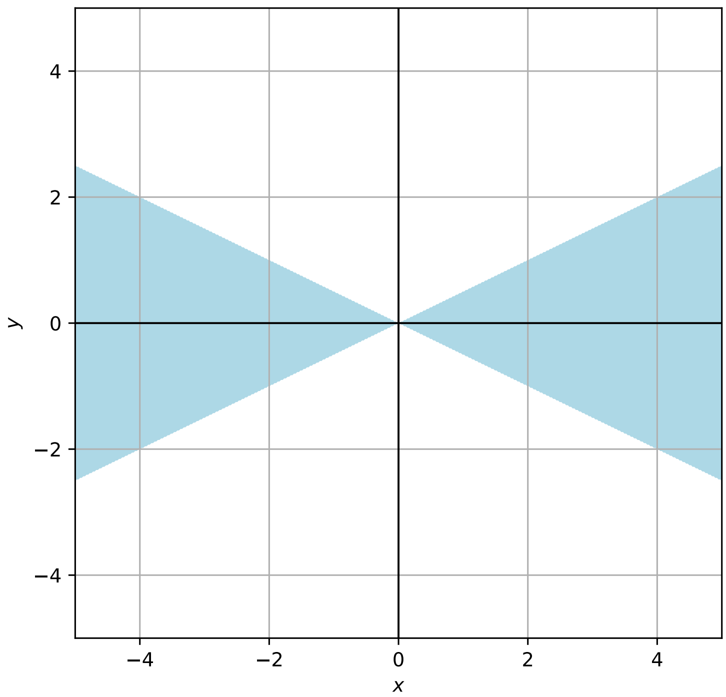
2. Calling
$$u = x + y, \quad v = y/x,$$
we need to find $x(u, v)$ and $y(u, v)$ so that we can substitute into the expression for $f(u, v)$. Solving, we find
$$y = xv \implies u = x+xv = x(1+v) \implies x = \frac{u}{1+v} \implies y = \frac{uv}{1+v}.$$
Substituting back:
$$f(u, v) = \frac{u^2(1 - v^2)}{(1 + v)^2} = u^2\frac{1 - v}{1 + v}$$
and, therefore,
$$f(x, y) = x^2\frac{1 - y}{1 + y}$$

3. Given
$$\begin{cases} f(x - y) + g(x, y) = x + y \\ g(x, 0) = x^2 \end{cases}$$
we can make $y = 0$ in the first equation to obtain $f(x) = x - x^2$. Plugging $f(x - y) = (x - y) - (x - y)^2$ into the first equation, we obtain
$$g(x, y) = 2y + (x - y)^2.$$

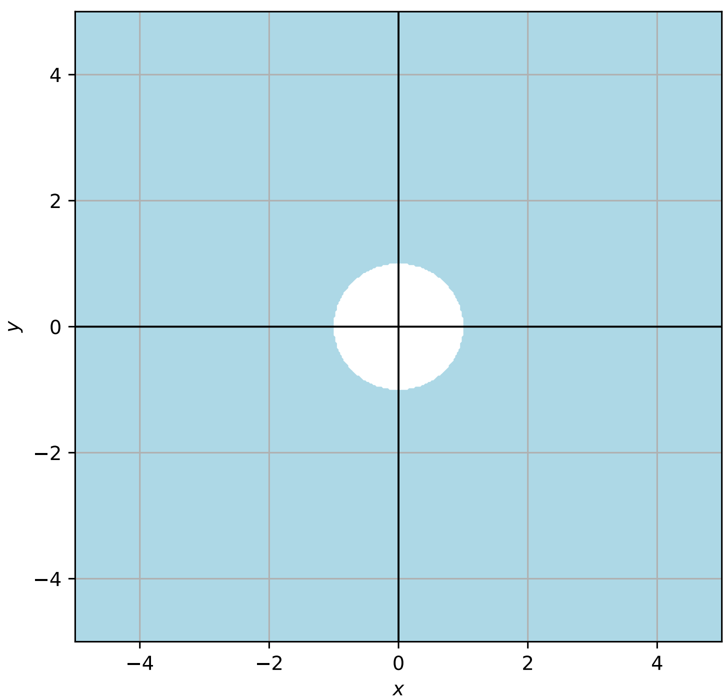
4. Given
$$\begin{cases} f(\sqrt{x} - 1) + g(x, y) = \sqrt{y} \\ g(x, 1) = x \end{cases}$$
we make $y = 1$ in the first equation to obtain
$$f(\sqrt{x} - 1) = 1 - x,$$
so calling $u = \sqrt{x} - 1$ we obtain $x = (u + 1)^2$ and so
$$f(x) = 1 - (x + 1)^2 = -x^2 - 2x.$$
Finally:
$$g(x, y) = \sqrt{y} + (\sqrt{x} - 1)^2 + 2(\sqrt{x} - 1) = x + \sqrt{y} - 1.$$

Exercise 1.5 1. $D = \mathbb{R}^2$

2. $D = \{(x, y) \in \mathbb{R}^2 : x^2 - 4y^2 \geq 0\}$
   This is equivalent to the condition $|x| \pm |2y|$ (remember, $\sqrt{x^2} = |x|$) which is pictured below:

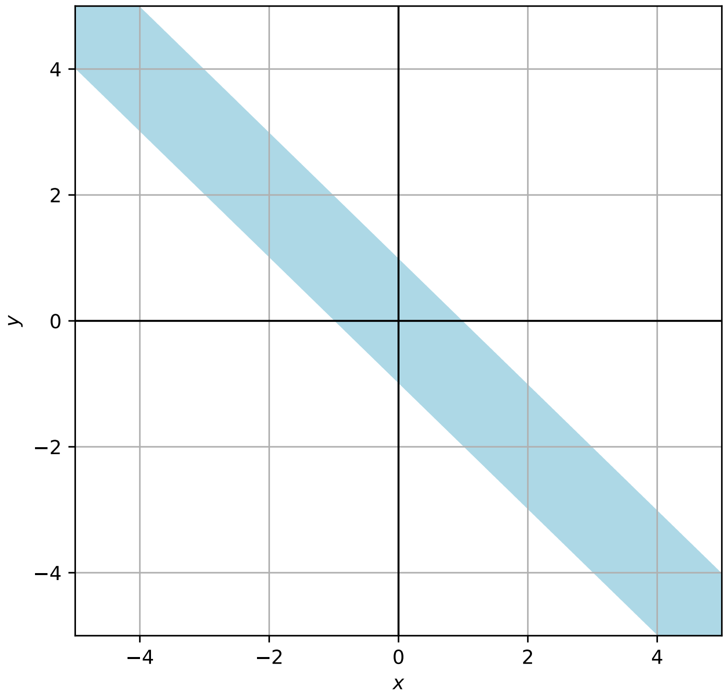


3. $D = \mathbb{R}^2 \setminus \{x = y\}$
4. $D = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \geq 1\}$



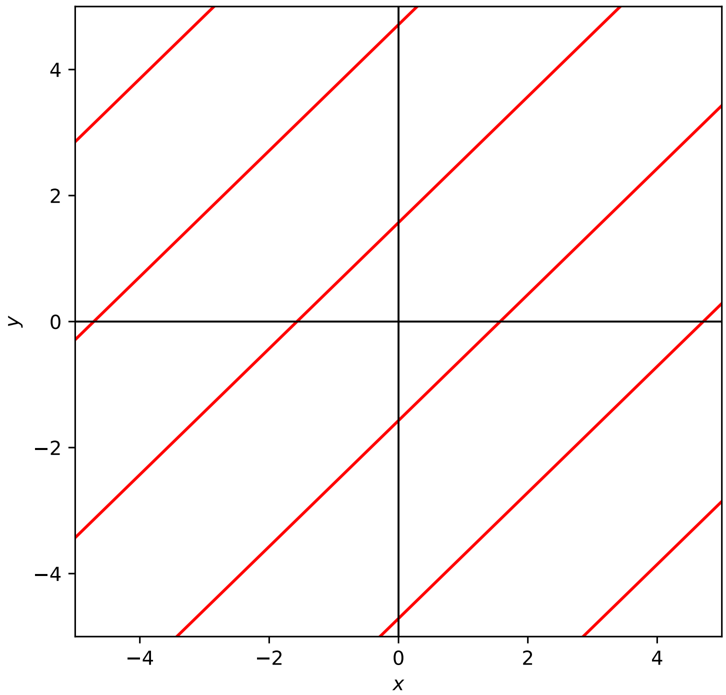
5. $D = \mathbb{R}^2 \setminus \{x = 0 \text{ or } y = 0\}$

6. $D = \{(x, y) \in \mathbb{R}^2 : -1 \leq x + y \leq 1\}$

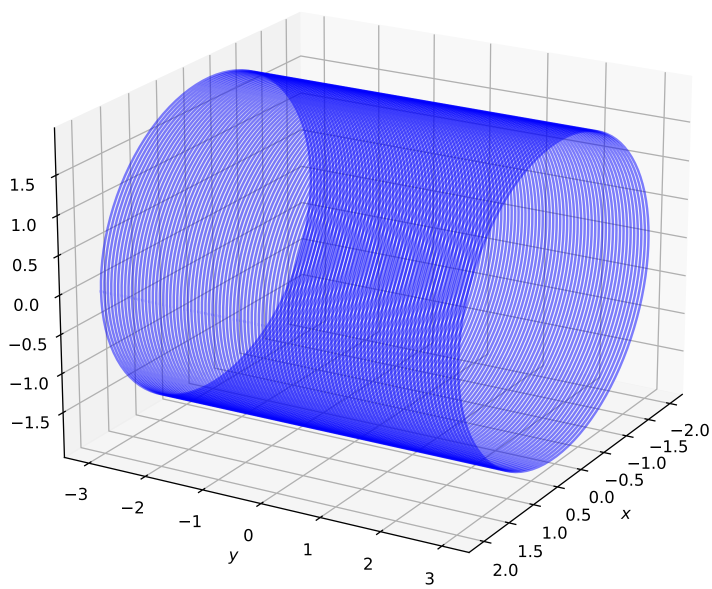


7. $D = \mathbb{R}^2 \setminus \{y = 0\}$
8. $D = \{xy > 0\}$
9. $D = \{\cos(x - y) \neq 0\}$
   This condition implies that

$$x - y \neq \frac{(2k+1)\pi}{2}, \quad k \in \mathbb{Z}$$



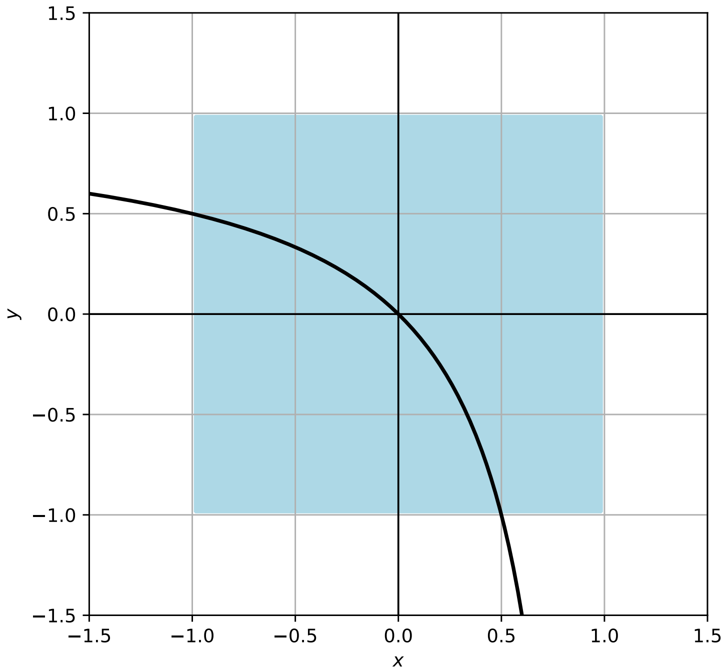
10. $D = \mathbb{R}^2 \setminus \{x = y\}$
11. $D = \{9 - y^2 \geq 0, 4 - (x^2 + z^2) \geq 0\}$
Now, $9 \geq y^2$ is equivalent to $|y| \leq 3$ and $x^2 + z^2 \leq 4$ is the circle of radius 2 in the xz plane. Combining the two conditions, we obtain a cylinder.



12. The conditions that have to be fulfilled are:

$$\begin{aligned} x &> -1, \\ y &> -1, \\ x &< 1, \\ y &< 1, \\ (1-x)(1-y) \neq 1 \implies x + y &\neq xy. \end{aligned}$$

The last condition is fulfilled except in the curve $y = x/(x-1)$. Therefore, $D = \{|x| \leq 1, |y| \leq 1, y \neq x/(x-1)\}$



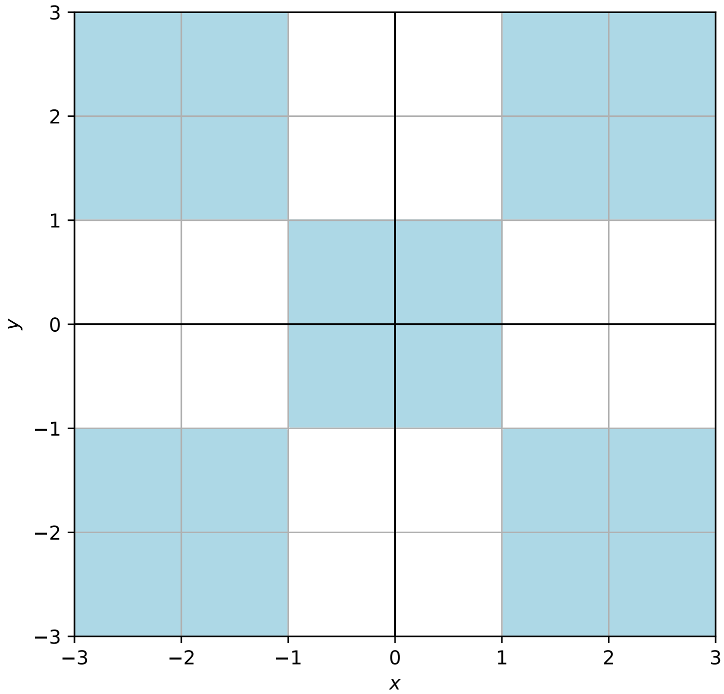
13. The argument of the logarithm has to be positive,

$$\frac{(1+x)(1+y)}{(1-x)(1-y)} > 0$$

This is fulfilled if:

$$\begin{aligned} |x| &< 1, |y| < 1 \\ x &< -1, y < -1 \\ x &< -1, y > 1 \\ x &> 1, y < -1 \\ x &> 1, y > 1 \end{aligned}$$

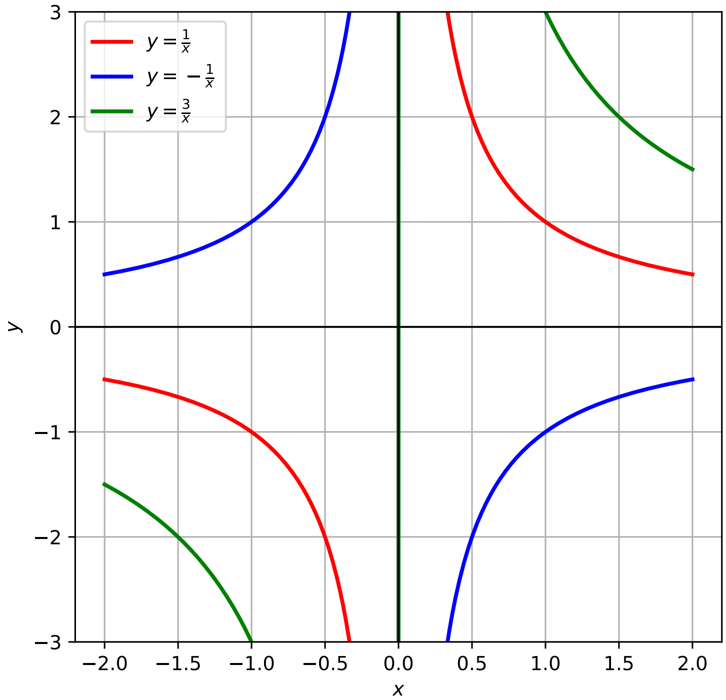
In summary, $D = \{(x, y) \in \mathbb{R}^2 : |x|, |y| < 1\} \cup \{(x, y) \in \mathbb{R}^2 : |x|, |y| > 1\}$



Exercise 1.6 1. $\mathbb{R}$
2. $\mathbb{R}^+ \cup \{0\}$
3. $\mathbb{R}$
4. $\mathbb{R}^+ \cup \{0\}$
5. $\mathbb{R} \setminus \{0\}$

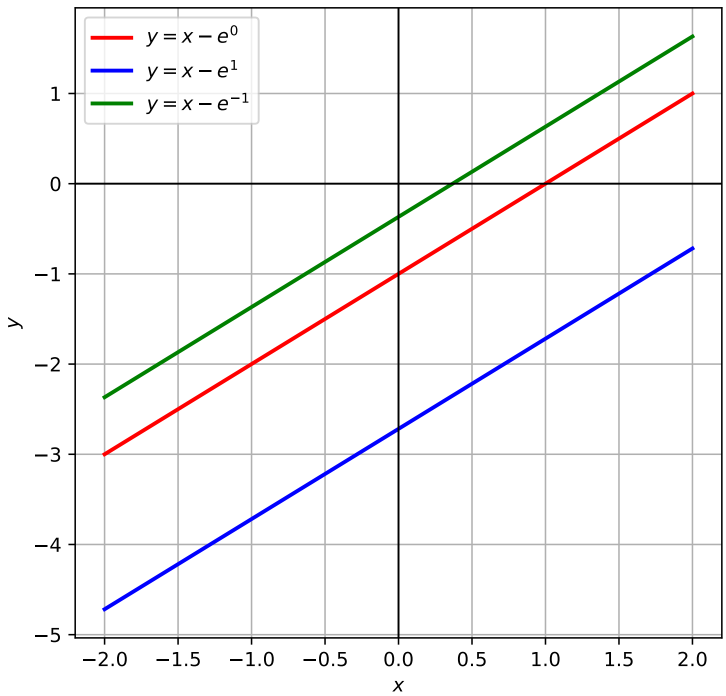
Exercise 1.7 1. $f(x, y) = xy$, with $c = 1, -1, 3$.

$$xy = 1 \quad \Rightarrow \quad y = \frac{1}{x},$$
$$xy = -1 \quad \Rightarrow \quad y = -\frac{1}{x},$$
$$xy = 3 \quad \Rightarrow \quad y = \frac{3}{x}.$$



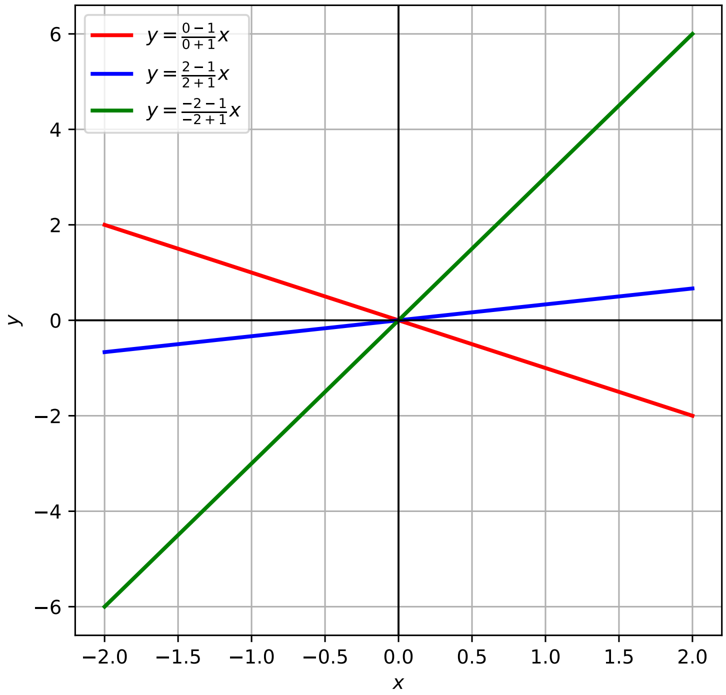
2. $f(x, y) = \log(x - y)$, with $c = 0, 1, -1$.

$$\begin{aligned}
\log(x-y) = 0 \quad &\Rightarrow \quad x - y = 1 \quad \Rightarrow \quad y = x - 1, \\
\log(x-y) = 1 \quad &\Rightarrow \quad x - y = e \quad \Rightarrow \quad y = x - e, \\
\log(x-y) = -1 \quad &\Rightarrow \quad x - y = \frac{1}{e} \quad \Rightarrow \quad y = x - \frac{1}{e}.
\end{aligned}$$

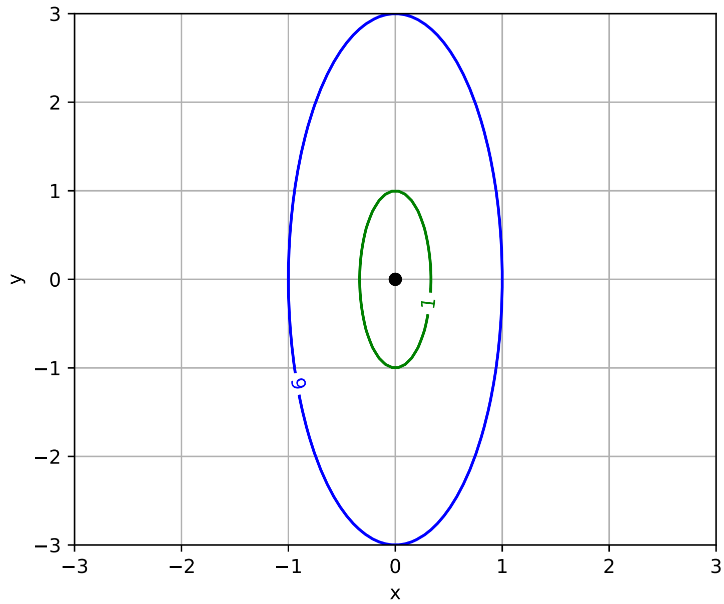


3. $f(x, y) = \frac{x+y}{x-y}$, with $c = 0, 2, -2$.

$$\frac{x+y}{x-y} = 0 \quad\Longrightarrow\quad x+y=0 \quad\Longrightarrow\quad y=-x,$$
$$\frac{x+y}{x-y} = 2 \quad\Longrightarrow\quad x+y=2(x-y) \quad\Longrightarrow\quad x+y=2x-2y \quad\Longrightarrow\quad y=3x,$$
$$\frac{x+y}{x-y} = -2 \quad\Longrightarrow\quad x+y=-2(x-y) \quad\Longrightarrow\quad x+y=-2x+2y \quad\Longrightarrow\quad y=-3x.$$

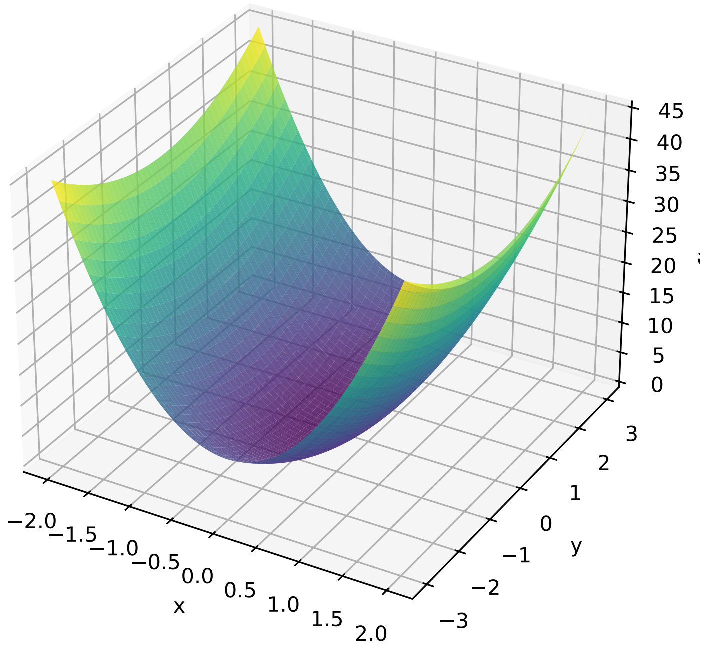


Exercise 1.8 1. The level curves are $(0,0)(c=0)$ (a point), $9x^2+y^2=1(c=1)$ and $9x^2+y^2=9(c=9)$ (two ellipses).



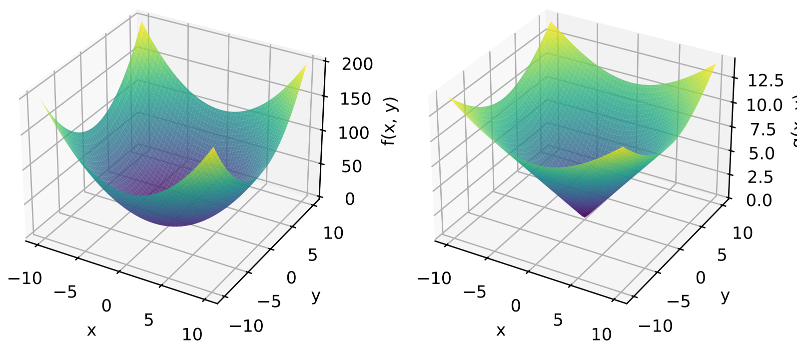
2. The sections are $f(-1,y) = f(0,y) = 9+y^2$ and $f(0,y) = y^2$, parabolas in the y axis.

3. The sections are $f(0, -1) = f(0, 1) = 9x^2 + 1$ and $f(x, 0) = x^2$, parabolas in the x axis.
4. Putting everything together, we obtain:



Exercise 1.9 Both families of level curves are circumferences centered at the origin, but they differ in the radius. So, for $c = 4$, the level curve for f is $x^2 + y^2 = 4$, a circumference with radius 2. However, for g, the level curve is $x^2 + y^2 = 2$, a circumference of radius $\sqrt{2}$.

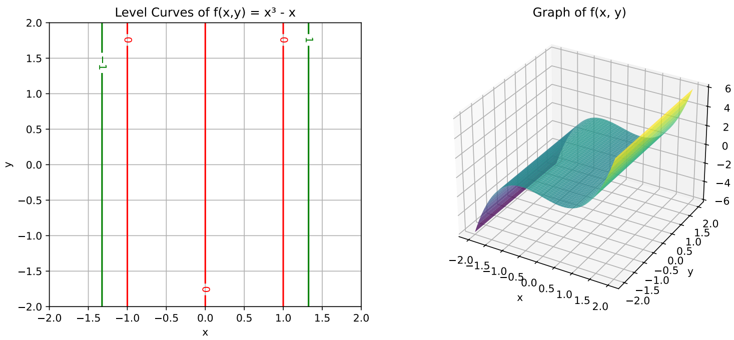
These two graphs differ in their sections: if we look at these functions in polar coordinates, $x = r\cos\theta, y = r\sin\theta$, then $f(r, \theta) = r^2$, while $g(r, \theta) = r$: the former is a parabola in radial directions (a paraboloid), while the latter is linear in r for all θ, i.e., it is a cone.



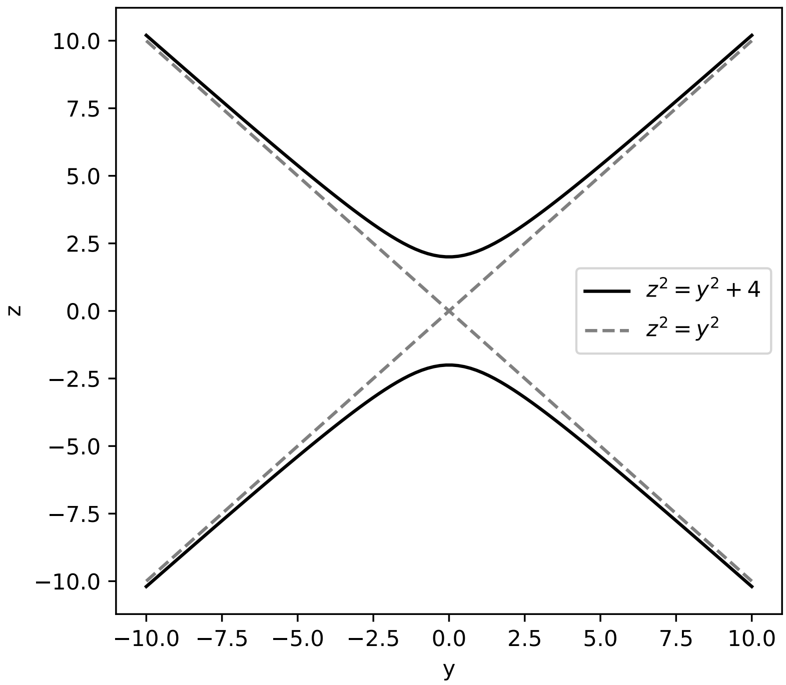
Exercise 1.10 1. There are infinite solutions to this question but, for instance, the function $f(x, y, z) = x^2y^6 - 2z$ has S as a level surface for the value $c = 3$.

2. Solving for z, we obtain $z = (x^2y^6 - 3)/2$ and so the function $g(x, y) = (x^2y^6 - 3)/2$ has S as its graph.

Exercise 1.11 1. The level curves are empty if $c < 1$, are equal to a single point (the origin) if $c = 1$ and are concentric cirumferences of radius $\sqrt{c-1}$ if $c > 1$.
2. The level curves are empty if $c > 1$, are equal to a single point (the origin) if $c = 1$ and are concentric cirumferences of radius $\sqrt{1-c}$ if $c < 1$.
3. The equation $x^3 - x = c$ has solutions for all c, and so $\{(x, y) \in \mathbb{R}^2 : x^3 - x = c\}$ are straight lines with a fixed x and y free.



Exercise 1.12 The equation $z^2 - y^2 = 4$ is the graph of two functions, $f_\pm = \pm\sqrt{y^2+4}$ which, for large values of y, are very similar to the straight lines $z = \pm y$, but which has a slight curvature at $y \approx 0$. The graph is flat in the direction of the x axis.



The graph of $z = x^2$ is a parabola in the xz plane, but flat in the y axis.

Exercise 1.13 1. Since the image of $\arctan x$ is $(-\pi/2, \pi/2)$, the image of f is $[0, \pi/2)$ as $x^2 + y^2 \geq 0$. Yes, it is bounded.

2. For $c = 0$, the level curve is a single point, the origin. For $c = \pi/4$, we have

$$\arctan(x^2 + y^2) = \pi/4 \implies x^2 + y^2 = 1,$$

and so the level curve is a circumference. It is easy to see that this function is equivalent to $\arctan|x|$ in any radial direction.

Exercise 1.14 For $c = 1$ we have

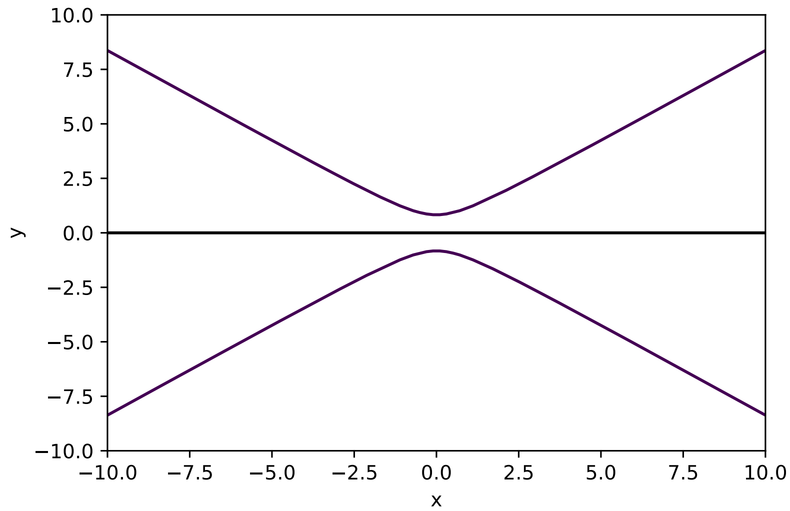
$$1 = e^{y^2/(1+x^2)} \implies \frac{y^2}{1+x^2} = \log 1 = 0 \implies y = 0,$$

and so the level curve is the line $y = 0$ (in black).

For $c = 2$, we have

$$2 = e^{y^2/(1+x^2)} \implies \frac{y^2}{1+x^2} = \log 2 \implies y^2 - x^2 \log 2 = \log 2,$$

which is a hyperbola, that crosses the y axis at $\pm\sqrt{\log 2}$, and doesn't cross the x axis (in purple).



A.2 Limits and Continuity

Exercise 2.1 i) The limit doesn't exist, as

$$\lim_{x\to 0} f(x,x) = \lim_{x\to 0} \frac{1}{2x^2}$$

doesn't exist.

ii)

$$\lim_{(x,y)\to(2,4)} f(x,y) = -3,$$

as f is continuous in $(2,4)$.

iii) Trying

$$\lim_{x\to 0} f(x,\lambda x) = \lim_{x\to 0} \frac{\lambda^2 x^3}{(1+\lambda^2)x^2} = \frac{\lambda^2}{1+\lambda^2} \lim_{x\to 0} x = 0,$$

which does not depend on λ. So we try to show that $|f(x,y)-0| \to 0$ as $(x,y) \to (0,0)$ using the sandwich theorem. Since $y^2 \leq x^2+y^2$, we have

$$0 \leq \left|\frac{xy^2}{x^2+y^2}\right| \leq \left|\frac{xy^2}{y^2}\right| \leq |x|,$$

which goes to 0 as $x \to 0$. This proves the result. Alternatively, we could have reasoned that $y^2/(x^2+y^2) \leq 1$ for all (x,y) to obtain the same result.

Using polar coordinates, the limit becomes

$$\lim_{r\to 0} \frac{r^3 \cos\theta \sin^2\theta}{r^2} = \lim_{r\to 0} r\cos\theta\sin^2\theta = 0,$$

since $|\cos\theta\sin^2\theta| \leq 1$. Importatntly, this includes all possible directions where θ is a function of r, since no matter what we do the trigonometric product is bounded.

iv) Trying

$$\lim_{x\to 0} f(x,\lambda x) = \lim_{x\to 0} \frac{\tan x}{\lambda x} = \frac{1}{\lambda},$$

since $\tan x \sim x$ as $x \to 0$. Since the limit depends on λ, it doesn't exist.

v) Trying

$$\lim_{x\to 0} f(x,\lambda x) = \lim_{x\to 0} \frac{\lambda x^2}{(1+\lambda^2)x^2} = \frac{\lambda}{1+\lambda^2}.$$

Since the limit depends on λ, it doesn't exist.

vi) Trying

$$\lim_{x\to 0} f(x,\lambda x) = \lim_{x\to 0} \frac{\lambda^3 x^3}{(1+\lambda^2)^{3/2} x^2} = \frac{\lambda^3}{(1+\lambda^2)^{3/2}}.$$

Since the limit depends on λ, it doesn't exist.

vii) Trying

$$\lim_{x\to 0} f(x,\lambda x) = \lim_{x\to 0} \frac{\lambda^2 x^2}{(1-\lambda^2)x^2} = \frac{\lambda^2}{1-\lambda^2}.$$

Since the limit depends on λ, it doesn't exist.

viii) Making the change of variable $t = xy, u = y$, the limit becomes

$$\lim_{t\to 0} \frac{\sin t}{t} = 1.$$

ix) We have

$$\lim_{y\to 0} f(0, y) = -1, \quad \lim_{x\to 0} f(x, 0) = 0,$$

so the limit does not exist.

x) We have

$$\lim_{y\to 0} f(0, y) = 0, \quad \lim_{x\to x} f(x, x) = 2,$$

so the limit does not exist.

Exercise 2.2 It is impossible to extend the function by continuity at any of the two points.

As for $(0, 0)$, we have

$$\lim_{y\to 0} f(0, y) = \infty$$

and so the limit does not exist.

For $(1, 1)$, we have

$$\lim_{y\to 1} f(1, y) = \lim_{y\to 1} \frac{1-y}{1-y^3} = \frac{1}{3},$$

using L'Hôpital's rule, but

$$\lim_{x\to 0} f(x, 1) = 1.$$

Hence neither limit exists.

Exercise 2.3 i) Continuous on $\mathbb{R}^2 \setminus \{(0, 0)\}$, as

$$\lim_{x\to 0} f(x, \lambda x) = \lim_{x\to 0} \frac{\lambda x^2}{(1+\lambda^2)x^2} = \frac{\lambda}{1+\lambda^2}$$

depends on λ.

ii) Continuous on $\mathbb{R}^2$, as

$$0 \le \left|\frac{xy}{\sqrt{x^2+y^2}}\right| \le \left|\frac{xy^2}{y}\right| \le |x|,$$

which goes to 0 as $(x, y \longrightarrow (0, 0)$.
Alternatively, using polar coordinates,

$$\lim_{r\to 0} \frac{r^2 \cos\theta \sin\theta}{r} = \lim_{r\to 0} r\cos\theta\sin\theta = 0,$$

and this is valid for any value of θ.

iii) Continuous on $\mathbb{R}^2$, since

$$0 \le \left|\frac{x^4+x^2y-y^3}{\sqrt{x^2+y^2}}\right| \le \left|\frac{x^4}{\sqrt{x^2+y^2}}\right| + \left|\frac{x^2y}{\sqrt{x^2+y^2}}\right| + \left|\frac{y^3}{\sqrt{x^2+y^2}}\right| \le \left|\frac{x^4}{x}\right| + \left|\frac{x^2y}{x}\right| + \left|\frac{y^3}{y}\right|,$$

which goes to 0 as $(x, y) \longrightarrow (0, 0)$. Here we have used that $x^2 \le x^2 + y^2$ or $y^2 \le x^2 + y^2$, depending on which one was more useful.

Alternatively, using polar coordinates,

$$\lim_{r\to 0} \frac{r^4\cos^4\theta + r^3\cos^2\theta\sin\theta - r^3\sin^3\theta}{r} = \lim_{r\to 0} r^2(r\cos^4\theta{+}\cos^2\theta\sin\theta{-}\sin^3\theta) = 0,$$

since the term in parenthesis is bounded.

iv) Continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$, since

$$\lim_{x\to 0} f(x,0) = 1, \quad \lim_{y\to 0} f(0,y) = 0,$$

so the limit does not exist at the origin.

v) Continuous on $\mathbb{R}^2$, since

$$0 \le \left|\frac{x^4 - y^4}{x^2+y^2}\right| \le \left|\frac{x^4}{x^2}\right| + \left|\frac{y^4}{y^2}\right| \le |x|^2 + |y|^2,$$

which goes to 0 as $(x, y \to (0,0)$. Here we have used the inequality $|x-y| \le |x| + |y|$ and the fact that $x^2+y^2 \ge x^2$, $x^2+y^2 \ge y^2$. Alternatively, using polar coordinates,

$$\lim_{r\to 0} \frac{r^4(\cos^4\theta - \sin^4\theta)}{r^2} = \lim_{r\to 0} r^2(\cos^4\theta - \sin^4\theta) = 0,$$

since the term in parenthesis is bounded.

vi) Continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$, since

$$\lim_{x\to 0} f(x,0)\text{''} \lim_{x\to 0} \frac{1}{x}$$

does not exist.

vii) Continuous on $\mathbb{R}^2$. Here we have a fraction of two continuous functions, and so the only points where it could be discontinuous are those where the denominator is zero. In this case, this implies $e^x + e^y = 0$, but this is impossible since $e^x \ge 0$ for all $x \in \mathbb{R}$.

viii) Continuous on $\mathbb{R}^2 \setminus \{(x_0, y_0) : x_0 = y_0 \neq 0\}$. The function $\sin(x - y)/(e^x - e^y)$ is the fraction of two continuous functions and so it is continuous everywhere except on $x = y$, where the denominator is zero. Now, as we approach some point (y_0, y_0) on the line $x = y$, we can make the change of variable $t = x - y$ and consider the limit

$$\lim_{(t,y)\to(0,y_0)} \frac{\sin t}{e^{t+y} - e^y} = \lim_{(t,y)\to(0,y_0)} \frac{1}{e^y}\frac{\sin t}{e^t - 1} = \frac{1}{e^{y_0}},$$

since $\lim_{t\to 0} \sin t/(e^t - 1) = 1$. Now, this limit depends on y_0 and is only equal to 1 when $y_0 = 0$.

ix) Continuous on $\mathbb{R}^2 \setminus \{(x_0, y_0) : y_0 = \pm x_0^2\}$. If $x_0 \neq 0$, we have

$$\lim_{(x_0,y)\to(x_0,\pm x_0^2)} \frac{x_0^2 y}{x_0^4 - y^2}$$

does not exist since the numerator is different than zero and the denominator goes to zero. For $x_0 = 0$,

$$\lim_{x\to 0} f(x, \lambda x^2) = \lim_{x\to 0} \frac{\lambda x^4}{x^4 - \lambda^2 x^4} = \frac{\lambda}{1-\lambda^2}$$

depends on λ.

x) Continuous on $\mathbb{R}^2 \setminus \{(x_0, y_0) : x_0 y_0 = 0\} \cup \{(0,0)\} \cup \left\{\left(\frac{1}{k\pi}, 0\right), \left(0, \frac{1}{k\pi}\right) : k \in \mathbb{Z} \setminus \{0\}\right\}$.
The function $(x+y)\sin(1/x)\sin(1/y)$ is defined and continuous everywhere except on the axes.
The limit at the origin is zero because $\sin(1/x)\sin(1/y)$ is bounded, and $x + y \longrightarrow 0$ as $(x, y) \longrightarrow (0, 0)$.
For $x = x_0 \neq 0$, taking the limit

$$\lim_{(x,y)\to(x_0,0)} f(x) = x_0 \sin(1/x_0) \lim_{y\to 0} \sin(1/y),$$

does not exist unless $\sin(1/x_0)$ is zero, and that happens whenever $x_0 = (k\pi)^{-1}, k \in \mathbb{Z} \setminus \{0\}$. The analysis for $y = y_0 \neq 0$ is identical since $f8x)$ is symmetrical with respect to x and y.

Exercise 2.4 i)

$$\lim_{x\to 0} f(x, \lambda x) = \lim_{x\to 0} \frac{\lambda^3 x^4}{x^6 + \lambda^2 x^2} = \lim_{x\to 0} \frac{\lambda^3 x^4}{\lambda^2 x^2} = 0,$$

since $x^6 \ll x^2$ when $x \longrightarrow 0$ and we can neglect it when calculating the limit.

ii)

$$\lim_{x\to 0} f(x, x^3) = \lim_{x\to 0} \frac{x^6}{x^6 + x^6} = \frac{1}{2}.$$

iii) It is not continuous because $\lim_{x\to 0} f(x, \lambda x^3)$ depends on λ.

Exercise 2.5

i) We can check that $\lim_{x\to\infty} f(x, \lambda x) = 0$ for all λ, and so this suggest the limit, if it exists, is zero. Since $x^2 - xy + y^2 = (x-y)^2 + xy \geq xy$,

$$0 \leq \left|\frac{x+y}{x^2 - xy + y^2}\right| \leq \left|\frac{x+y}{xy}\right| \leq \frac{1}{|y|} + \frac{1}{|x|} \longrightarrow 0 \text{ as } (x, y) \longrightarrow (\infty, \infty).$$

Alternatively, we can take $x = r\cos\theta, y = r\sin\theta$ and study the limit

$$\lim_{r\to\infty} \frac{r(\cos\theta + \sin\theta)}{r^2(1 - \sin\theta\cos\theta)} = \lim_{r\to\infty} \frac{1}{r} \frac{\cos\theta + \sin\theta}{1 - \sin\theta\cos\theta} = 0,$$

where the last limit exists because the denominator is never zero: $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta < 1$ always, and therefore the whole fraction is bounded.

ii) We have

$$0 \leq \left|\frac{x^2 + y^2}{x^4 + y^4}\right| \leq \left|\frac{x^2}{x^4 + y^4}\right| + \left|\frac{y^2}{x^4 + y^4}\right| \leq \left|\frac{x^2}{x^4}\right| + \left|\frac{y^2}{y^4}\right| \longrightarrow 0 \text{ as } (x, y) \longrightarrow (\infty, \infty).$$

iii) We can rewrite

$$\frac{\sin xy}{x} = y \frac{\sin xy}{xy} \text{ if } y \neq 0.$$

Since

$$\lim_{x\to 0, y\to a} \frac{\sin xy}{xy} = \lim_{t\to 0} \frac{\sin t}{t} = 1,$$

we have

$$\lim_{x\to 0, y\to a} y\frac{\sin xy}{xy} = a.$$

(you can check that when $a = 0$ the limit is satisfied trivially).

iv) Since $e^{x+y} \geq e^x$ for $y \geq 1$, we have

$$0 \leq \left|\frac{x^2+y^2}{e^{x+y}}\right| \leq \left|\frac{x^2}{e^{x+y}}\right| + \left|\frac{y^2}{e^{x+y}}\right| \leq \left|\frac{x^2}{e^x}\right| + \left|\frac{y^2}{e^y}\right|,$$

which goes to 0 as $(x, y) \to (\infty, \infty)$ (If you have trouble understanding this, remember that $e^x = \sum_{k=0}^{\infty} x^k/k!$ and so when $x \to \infty$ it will grow faster than any finite polynomial.

Alternatively, we can use polar coordinates, noting that because both $x \to infty$ and $y \to \infty$ we can consider $0 < \theta < \pi/2$:

$$\lim_{r\to\infty} r^2 e^{-r(\cos\theta + \sin\theta)} = \lim_{r\to\infty} r^2 e^{-\alpha r} = 0,$$

since $\alpha = \cos\theta + \sin\theta > 0$ in $(0, \pi/2)$.

v) Considering all radial directions $\lambda > 0$ (since both $x \to \infty$ and $y \to \infty$), we have

$$\lim_{x\to\infty} f(x, \lambda x) = \lim_{x\to\infty} \left(\frac{\lambda x^2}{(1+\lambda^2)x^2}\right)^{x^2} = \lim_{x\to\infty} \left(\frac{\lambda}{(1+\lambda^2)}\right)^{x^2} = 0,$$

as $\lambda < 1 + \lambda^2$ for all $\lambda > 0$. This suggests that the limit, if it exists, is zero.

In order to prove this, we will use the sandwich theorem. As $(x-y)^2 \geq 0$ we can conclude $x^2 + y^2 \geq 2xy$ and so

$$0 \leq \left|\left(\frac{xy}{x^2+y^2}\right)^{x^2}\right| \leq \left|\left(\frac{xy}{2xy}\right)^{x^2}\right| \left|\left(\frac{1}{2}\right)^{x^2}\right|,$$

which goes to zero as $x \to \infty$.

Alternatively, using polar coordinates,

$$\lim_{r\to\infty} \left(\frac{r^2 \sin\theta\cos\theta}{r^2}\right)^{r^2\cos^2\theta} = \lim_{r\to\infty} (\sin\theta\cos\theta)^{r^2\cos^2\theta} = 0,$$

since $\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta < 1$ for all θ and $r^2\cos^2\theta \to \infty$ as $r \to \infty$.

Exercise 2.6 i) Using polar coordinates,

$$\lim_{r\to 0} \exp\left(\frac{r\cos\theta}{r^2}\right) < \infty$$

only if $\cos\theta \leq 0$. Otherwise the argument of the exponential is positive and the limit is divergent. That is, the limit is finite for $\theta \in [\pi/2, 3\pi/2]$: it is 1 if $\theta = \pi/2$ or $3\pi/2$ and 0 if $\theta \in (\pi/2, 3\pi/2)$.

ii) In this case, $\lim_{r\to\infty} \sin 2r^2 \cos\theta \sin\theta$ does not exist unless $\theta = 0, \pi/2, \pi$ or $3\pi/2$, where it is zero. In the remaining directions, we need $x^2 - y^2 < 0$ (in this case $e^{x^2-y^2} \to 0$). Using polar coordinates, this is equal to

$$\cos^2\theta - \sin^2\theta = \cos 2\theta < 0,$$

which happens if $\theta \in (\pi/4, 3\pi/4)$.

iii) Taking polar coordinates,

$$\lim_{r\to 0}(r^2)^{r^4\cos^2\theta\sin^2\theta} = \lim_{r\to 0}\exp(r^4 2\log r\cos^2\theta\sin^2\theta) = 1,$$

since $\lim_{r\to 0} r\log r = 0$ (you can check this using L'Hôpital's rule, for instance).
So the limit is finite in every direction.

A.3 Derivatives in Higher Dimensions

Exercise 3.1 1. $\nabla f = \big(6x - y, 1 - x\big)$
2. $\nabla f = \big(3x^2 e^{-y}, -x^3 e^{-y}\big)$
3. $\nabla f = \left(\dfrac{2x}{x^2 + y^4}, \dfrac{4y^3}{x^2 + y^4}\right)$
4. $\nabla f = \big(2g(x)g'(x)h(y), g(x)^2 h'(y)\big)$
5. $\nabla f = \left(\dfrac{-x}{\sqrt{1 - (x^2 + y^2)}}, \dfrac{-y}{\sqrt{1 - (x^2 + y^2)}}\right)$
6. $\nabla f = \left(e^{y^2}, 2xye^{y^2} + e^z, ye^z\right)$
7. $\nabla f = \big(z\cos x + \sin y, x\cos y + \sin z, y\cos z + \sin x\big)$
8. $\nabla f = \big((y+1)\cos(xy + x + z^2), x\cos(xy + x + z^2), 2z\cos(xy + x + z^2)\big)$
9. $\nabla f = \left(y^2 z^{xy^2}\log z, 2xyz^{xy^2}\log z, xy^2 z^{xy^2 - 1}\right)$
10. $\nabla f = \left(2g(x,y)\dfrac{\partial g}{\partial x}(x,y)h(x,z)^3 + 3g(x,y)^2\dfrac{\partial h}{\partial x}(x,z)h(x,z)^2, 2g(x,y)\dfrac{\partial g}{\partial y}(x,y)h(x,z)^3, 3g(x,y)^2\dfrac{\partial h}{\partial z}(x,y)h(x,z)^2\right)$
11. $\nabla f = \big(2r\sin\theta, r^2\cos\theta - 2\sin\theta\cos\theta\big)$
12. $\nabla f = \big(3\rho^2\sin\phi\cos\theta, \rho^3\cos\phi\cos\theta, -\rho^3\sin\phi\sin\theta\big)$

Exercise 3.2 1. We compute the partial derivatives using limits:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0}\frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0}\frac{h^2 \cdot 0}{h^4 + 0^4} = 0.$$

It is the same for y since f is symmetrical with respect to x and y.
2. To check continuity, we compute the limit:

$$\lim_{(x,y)\to(0,0)} f(x,y).$$

Evaluating along the path $y = x$, we obtain:

$$f(x,x) = \frac{x^4}{x^4 + x^4} = \frac{x^4}{2x^4} = \frac{1}{2}.$$

Since this limit is not equal to $f(0,0) = 0$, we conclude that f is not continuous at $(0,0)$.

Exercise 3.3 All functions have continuous partial derivatives in the indicated sets:

1. We compute the gradient of $f(x,y)$:

$$\nabla f = \big(2x, 2y\big)$$

Since f has continuous partial derivatives, it is differentiable.
2. We compute the gradient of $f(x,y)$:

$$\nabla f = \big(y, x\big)$$

Since f has continuous partial derivatives, it is differentiable.
3. We compute the gradient of $f(x,y,z)$:

$$\nabla f = \big(2x, 2y, 2z\big)$$

Since f has continuous partial derivatives, it is differentiable.

4. We compute the gradient of $f(x, y, z)$:

$$\nabla f = \big(\cos(x + y + z), \cos(x + y + z), \cos(x + y + z)\big)$$

Since f has continuous partial derivatives, it is differentiable.

5. We compute the gradient of $f(x, y)$:

$$\nabla f = \big(e^x \sin y, e^x \cos y\big)$$

Since f has continuous partial derivatives, it is differentiable.

6. We compute the gradient of $f(x, y)$:

$$\nabla f = \big(2xe^{-xy} - y(x^2 + y^2)e^{-xy}, -x(x^2 + y^2)e^{-xy} + 2ye^{-xy}\big)$$

Since f has continuous partial derivatives, it is differentiable.

7. We compute the gradient of $f(x, y)$:

$$\nabla f = \left(\frac{y^2 - x^2}{(x^2 + y^2)^2}, -\frac{2xy}{(x^2 + y^2)^2}\right)$$

Since f has continuous partial derivatives for $(x, y) \neq (0, 0)$, it is differentiable in its domain.

Exercise 3.4 1. We compute the partial derivatives using limits:

$$\frac{\partial f}{\partial x}(0, 0) = \lim_{h \to 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \to 0} \frac{2h \cdot 0}{h^2 + 0^2} = 0.$$

It is the same for y since f is symmetrical with respect to x and y.

2. To check continuity, we compute the limit of $\frac{\partial f}{\partial x}$ as $(x, y) \to (0, 0)$. For $(x, y) \neq 0$,

$$\frac{\partial f}{\partial x} = \frac{2y^3 - 2x^2 y}{(x^2 + y^2)^2}$$

and if we approach the origin from the $(0, y)$ axis,

$$\lim_{y \to 0} \frac{\partial f}{\partial x}(0, y) = \lim_{0, y} \frac{2y^3}{y^4},$$

which does not exist. So f does not have continuous partial derivatives at the origin.

3. To check differentiability, we analyze the limit:

$$\lim_{(x,y) \to (0,0)} \frac{f(x, y) - (f(0, 0) + \frac{\partial f}{\partial x}(0, 0)x + \frac{\partial f}{\partial y}(0, 0)y)}{\sqrt{x^2 + y^2}} = \lim_{(x,y) \to (0,0)} \frac{2xy}{(x^2 + y^2)^{3/2}}.$$

Choosing the path $y = x$, we get:

$$\lim_{x \to 0} \frac{2x^2}{\sqrt{8x^3}} = \frac{1}{\sqrt{2x}}.$$

Since this limit is not zero, f is not differentiable at $(0, 0)$. Alternatively, we could have shown that f is not continuous at the origin, which implies it can't be differentiable, and also that it cannot have continuous partial derivatives.

Exercise 3.5 Using the definition,

$$\frac{\partial f}{\partial x}(0,0) = \lim_{x\to 0} \frac{f(x,0)-f(0,0)}{x} = \lim_{x\to 0} \frac{\sqrt{x^2}}{x} = \lim_{x\to 0} \frac{|x|}{x},$$

which does not exist.

Exercise 3.6

1. Since

   $$\lim_{x\to 0} f(x,\lambda x) = \lim_{x\to 0} \frac{\lambda^4 x^6}{(1+\lambda^2)x^2} = 0$$

   independently of λ, we seek to prove that $|f(x,y)| \to 0$ as $(x,y) \to (0,0)$ using the sandwich theorem. Using the inequality $x^2+y^2 \geq x^2$ we have

   $$0 \leq \left| \frac{x^2y^4}{x^2+y^2} \right| \leq \left| \frac{x^2y^4}{x^2} \right| \leq |y^4|,$$

   which goes to 0 as $(x,y) \to (0,0)$. Hence, f is continuous at the origin.
   Alternatively, using polar coordinates, we have

   $$\lim_{r\to 0} \frac{r^6 \cos^2\theta \sin^4\theta}{r^2} = 0,$$

   independently of θ.
2. We compute the partial derivatives using limits:

   $$\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{0}{h} = 0.$$

   Similarly, for y we obtain

   $$\frac{\partial f}{\partial y}(0,0) = 0.$$

   Thus, both partial derivatives exist and are equal to 0.
   To check differentiability, we analyze the limit:

   $$\lim_{(x,y)\to(0,0)} \frac{f(x,y) - (\frac{\partial f}{\partial x}(0,0)x + \frac{\partial f}{\partial y}(0,0)y)}{\sqrt{x^2+y^2}} = \lim_{(x,y)\to(0,0)} \frac{x^2y^4}{(x^2+y^2)^{3/2}}.$$

   Using the inequality $x^2+y^2 \geq y^2$ we have

   $$0 \leq \left| \frac{x^2y^4}{(x^2+y^2)^{3/2}} \right| \leq \left| \frac{x^2y^4}{y^3} \right| \leq |x^2y|,$$

   which goes to 0 as $(x,y) \to (0,0)$. Hence, f is differentiable at the origin.
   Alternatively, using polar coordinates, we have

   $$\lim_{r\to 0} \frac{r^6 \cos^2\theta \sin^4\theta}{r^3} = 0,$$

   independently of θ.

Exercise 3.7 1. g is continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$, because

$$\lim_{y\to 0} g(0,y) = \lim_{y\to 0} \frac{y}{|y|}$$

which does not exist.

2. The partial derivatives do not exist at $(0,0)$, because

$$\frac{\partial g}{\partial x}(0,0) = \lim_{h\to 0}\frac{h}{|h|}, \quad \frac{\partial g}{\partial y}(0,0) = \lim_{h\to 0}\frac{1}{|h|}$$

3. g cannot be differentiable at $(0,0)$ because the partial derivatives do not exist.

Exercise 3.8 1. The partial derivatives at the origin, assuming $\alpha > 0$, are zero. Therefore, we need to study whether

$$\lim_{(x,y)\to(0,0)} \frac{|xy|^\alpha - 0 - 0 - 0}{(x^2+y^2)^{1/2}}$$

is zero or not. Using polar coordinates, we get

$$\lim_{r\to 0}\frac{r^{2\alpha}|\sin\theta\cos\theta|^\alpha}{r} = \lim_{r\to 0} r^{2\alpha-1}\left|\frac{1}{2}\sin 2\theta\right|,$$

which is zero if and only if $\alpha > 1/2$. This also solves part 3. Alternatively, we could use the sandwich theorem with the bound $x^2+y^2 \geq 2xy$ (which comes from the fact that $(x-y)^2 \geq 0$):

$$0 \leq \left|\frac{|xy|^\alpha}{(x^2+y^2)^{1/2}}\right| \leq \left|\frac{|xy|^\alpha}{\sqrt{2}|xy|^{1/2}}\right| \leq |xy|^{\alpha-1/2},$$

which goes to zero if $\alpha > 1/2$. Note that this way of solving the problem does not prove that f is not differentiable if $a \leq 1/2$.

2. For any value of $\alpha > 0$,

$$\frac{\partial f}{\partial x}(0,0) = 0, \quad \frac{\partial f}{\partial y}(0,0) = 0.$$

3. The limit

$$\lim_{(x,y)\to(0,0)} \frac{|xy|^\alpha - 0 - 0 - 0}{(x^2+y^2)^{1/2}}$$

does not exist for $\alpha = 1/2$. In particular, taking the direction $y = x$,

$$\lim_{x\to 0}\frac{|x|}{\sqrt{2}|x|} = \frac{1}{\sqrt{2}} \neq 0,$$

so f is not differentiable if $\alpha = \frac{1}{2}$.

Exercise 3.9 1. We have $f(1,3) = 0, \nabla f(x,y) = (1,-1), \nabla f(1,3) = (1,-1)$, and so

$$z = 0 + (x-1) - (y-3) \implies z = x - y + 2.$$

2. We have $f(2,-1) = 8, \nabla f(x,y) = (2x, 8y), \nabla f(2,-1) = (4,-8)$,

and so

$$z = 8 + 4(x-2) - 8(y+1) \implies z = 4x - 8y - 8.$$

3. We have $f(1,0) = 1, \nabla f(x,y) = ((x+1)^{-1} + \cos y, (x+y)^{-1} - x\sin y), \nabla f(1,0) = (2,1)$, and so

$$z = 1 + 2(x-1) + (y-0) \implies z = 2x + y - 1.$$

4. We have $f(1,1/2) = \sqrt{5}/2, \nabla f(x,y) = (x,y)/\sqrt{x^2+y^2}, \nabla f(1,1/2) = (2,1)/\sqrt{5}$, and so

$$z = \frac{\sqrt{5}}{2} + \frac{2}{\sqrt{5}}(x-1) + \frac{1}{\sqrt{5}}(y-1/2) \implies \sqrt{5}z = 2x + y.$$

Exercise 3.10 The gradient of f is

$$\nabla f(x,y) = \left(\frac{2x}{1+(x^2+y^2)^2}, \frac{2y}{1+(x^2+y^2)^2}\right)$$

which, evaluated at $(1,0)$, yields $\nabla f(1,0) = (1,0)$. Since $f(1,0) = \pi/4$ we have

$$z = \frac{\pi}{4} + x - 1.$$

Exercise 3.11 1.

$$DA = \begin{pmatrix} yx^{y-1} & x^y \log x & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2.

$$DB = \begin{pmatrix} \sin y \cos(x \sin y) & x \cos y \cos(x \sin y) \end{pmatrix}$$

3.

$$DC = \begin{pmatrix} 1 + e^x \\ 2x \\ -\sin x \end{pmatrix}$$

4.

$$DD = \begin{pmatrix} e^y & xe^y - \sin y \\ 1 & 0 \\ 1 & e^y \end{pmatrix}$$

5.

$$DE = \begin{pmatrix} 1 & 1 & e^z \\ 2xy & x^2 & 0 \end{pmatrix}$$

6.

$$DF = \begin{pmatrix} ye^{xy} + xy^2e^{xy} & xe^{xy} + x^2ye^{xy} & 0 \\ \sin y & x \cos y & 0 \\ 5y^2 & 10xy & 0 \end{pmatrix}$$

7.

$$DG = \begin{pmatrix} 2x \log t + \sqrt{z} & -t & \frac{x^2}{2\sqrt{z}} & \frac{x^2}{t} - y \end{pmatrix}$$

Exercise 3.12 1.

$$DT = \begin{pmatrix} 1 & -3 & 2 \\ 2 & 1 & -2 \end{pmatrix}$$

2. Recalling that $T_A(\mathbf{x}) = A\mathbf{x}$ we get

$$\lim_{\mathbf{x}\to\mathbf{a}} \frac{\|T_A(\mathbf{x}) - T_A(\mathbf{a}) - DT_A(\mathbf{a})(\mathbf{x}-\mathbf{a})\|}{\|\mathbf{x}-\mathbf{a}\|} = \lim_{\mathbf{x}\to\mathbf{a}} \frac{\|A\mathbf{x} - A\mathbf{a} - A(\mathbf{x}-\mathbf{a})\|}{\|\mathbf{x}-\mathbf{a}\|} = 0.$$

Exercise 3.13 1. Since the transformation between polar and Cartesian coordinates is $T_1 : \mathbb{R}^2 \to \mathbb{R}^2, T_1(r,\theta) = (x(r,\theta), y(r,\theta)) = (r\cos\theta, r\sin\theta)$, we have:

$$J_{T_1}(r,\theta) = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix} \implies \det J_{T_1} = r$$

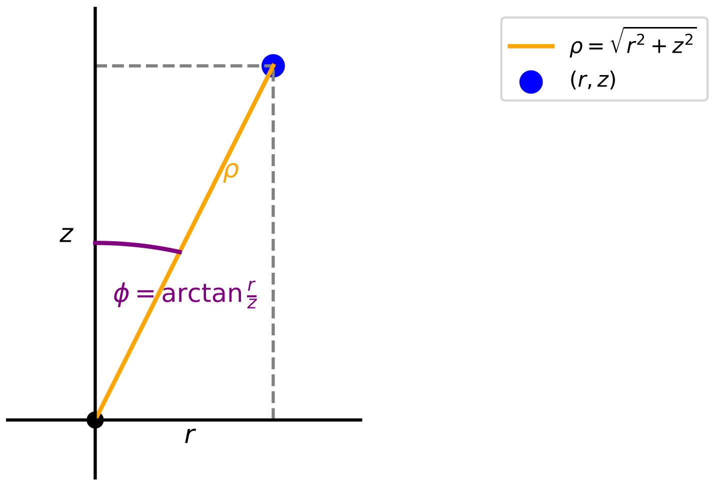
2. Since the transformation between cylindrical and Cartesian coordinates is $T_2 : \mathbb{R}^3 \to \mathbb{R}^3, T_2(r,\theta,z) = (x(r,\theta,z), y(r,\theta,z), z(r,\theta,z)) = (r\cos\theta, r\sin\theta, z)$, we have:

$$J_{T_2}(r,\theta,z) = \begin{pmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \implies \det J_{T_2} = r$$

3. Since the transformation between spherical and Cartesian coordinates is $T_3 : \mathbb{R}^3 \to \mathbb{R}^3, T_2(\rho,\phi,\theta) = (x(\rho,\phi,\theta), y(\rho,\phi,\theta), z(\rho,\phi,\theta)) = (\rho\sin\phi, \cos\theta, \rho\sin\phi\sin\theta, \rho\cos\theta)$, where ϕ is the angle that a vector (x,y,z) makes with the vertical z axis, we have:

$$J_{T_3}(\rho,\phi,\theta) = \begin{pmatrix} \sin\phi\cos\theta & \rho\cos\phi\cos\theta & -\rho\sin\phi\sin\theta \\ \sin\phi\sin\theta & \rho\cos\phi\sin\theta & \rho\sin\phi\cos\theta \\ \cos\phi & -\rho\sin\phi & 0 \end{pmatrix} \implies \det J_{T_3} = \rho^2\sin\phi$$

4. To go from cylindrical coordinates (r,θ,z) to spherical coordinates (ρ,ϕ,θ), we need to use some geometric arguments. First, we need to express ρ in terms of r,θ,z. Using Pythagoras' theorem, $\rho^2 = r^2 + z^2$. As for ϕ, we have that it is the angle of a right triangle with sides r and z (see figure). Therefore, $\phi = \arctan r/z$. Finally, $\theta = \theta$.



The transformation is $T_4(r,\theta,z) = (\sqrt{r^2+z^2}, \arctan r/z, \theta)$, and so

$$J_{T_4}(r,\theta,z) = \begin{pmatrix} \frac{r}{\sqrt{r^2+z^2}} & \frac{z}{\sqrt{r^2+z^2}} & 0 \\ \frac{z}{r^2+z^2} & \frac{-r}{r^2+z^2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \implies \det J_{T_4} = \frac{-1}{\sqrt{r^2+z^2}}.$$

A.4 Properties of the Derivative. The Chain Rule. Directional Derivatives

Exercise 4.1 Since all functions are differentiable at the desired points (they are compositions of differentiable functions), we can find the directional derivative as $f'_{\mathbf{v}}(\mathbf{x}) = \nabla f(\mathbf{x}) \cdot \mathbf{v}$, where $\mathbf{v}$ is a unti vector. With this, we have:

1. The gradient is $\nabla f(x, y) = (2x, 3y^2)$. Therefore, since $\mathbf{x} = (1, 1)$, $\mathbf{v} = (1/\sqrt{2}, -1/\sqrt{2})$ and $\nabla f(1, 1) = (2, 3)$, the directional derivative is
$$f'_{\mathbf{v}}(\mathbf{x}) = -\frac{1}{\sqrt{2}}.$$
2. The gradient is $\nabla f(x, y) = (1 + \cos(x + y), \cos(x + y))$. Therefore, since $\mathbf{x} = (0, 0)$, $\mathbf{v} = (2/\sqrt{5}, 1/\sqrt{5})$ and $\nabla f(0, 0) = (2, 1)$, the directional derivative is
$$f'_{\mathbf{v}}(\mathbf{x}) = \sqrt{5}.$$
3. The gradient is $\nabla f(x, y) = (e^y - ye^x, xe^y - e^x)$. Therefore, since $\mathbf{x} = (1, 0)$, $\mathbf{v} = (3/5, 4/5)$ and $\nabla f(1, 0) = (1, 1 - e)$, the directional derivative is
$$f'_{\mathbf{v}}(\mathbf{x}) = \frac{7 - 4e}{5}.$$
4. The gradient is $\nabla f(x, y) = (-3y, 3x)/(x - y)^2$. Therefore, since $\mathbf{x} = (1, 0)$, $\mathbf{v} = (1/2, -\sqrt{3}/2)$ and $\nabla f(1, 0) = (0, 3)$, the directional derivative is
$$f'_{\mathbf{v}}(\mathbf{x}) = -\frac{3\sqrt{3}}{2}.$$
5. The gradient is $\nabla f(x, y, z) = (2xy + z^2, x^2 + 2yz, y^2 + 2xz)$. Therefore, since $\mathbf{x} = (1, -1, 1)$, $\mathbf{v} = (1/\sqrt{6}, -1/\sqrt{6}, 2/\sqrt{6})$ and $\nabla f(1, -1, 1) = (-1, -1, 3)$, the directional derivative is
$$f'_{\mathbf{v}}(\mathbf{x}) = \sqrt{6}.$$
6. The gradient is $\nabla f(x, y) = (2xy + y^2, x^2 + 2xy)$. Therefore, since $\mathbf{x} = (1, 1)$, $\mathbf{v} = (-1/\sqrt{10}, 3/\sqrt{10})$ and $\nabla f(1, 1) = (3, 3)$, the directional derivative is
$$f'_{\mathbf{v}}(\mathbf{x}) = \frac{6}{\sqrt{10}}.$$
7. The gradient is $\nabla f(x, y, z) = (x, y, z)/(x^2+y^2+z^2)$. Therefore, since $\mathbf{x} = (2, 0, 1)$, $\mathbf{v} = (1/\sqrt{5}, 2/\sqrt{5}, 0)$ and $\nabla f(2, 0, 1) = (2/5, 0, 1/5)$, the directional derivative is
$$f'_{\mathbf{v}}(\mathbf{x}) = \frac{2\sqrt{5}}{25}.$$
8. The gradient is $\nabla f(x, y, z) = (e^x \cos yz, -ze^x \sin yz, -ye^x \sin yz)$. Therefore, since $\mathbf{x} = (0, 0, 0)$, $\mathbf{v} = (2/3, 1/3, -2/3)$ and $\nabla f(0, 0, 0) = (1, 0, 0)$, the directional derivative is
$$f'_{\mathbf{v}}(\mathbf{x}) = \frac{2}{3}.$$
9. The gradient is $\nabla f(x, y) = (6x^2y, 2x^3 - 6y)$. Therefore, since $\mathbf{x} =$

$(2,1)$, $\mathbf{v} = (a, \sqrt{1-a^2})$ and $\nabla f(2,1) = (24,10)$, the directional derivative is

$$f'_{\mathbf{v}}(\mathbf{x}) = 24a + 10\sqrt{1-a^2}.$$

Now, this is a differentiable function of a and so in order to find its maximum we need to calculate its derivative and equate it to zero:

$$24 - \frac{10a}{\sqrt{1-a^2}} = 0 \implies 10a = 24\sqrt{1-a^2} \implies a^2 = \frac{576}{676} \implies a = \pm\frac{12}{13}.$$

Now, the value $a = -12/13$ does not satisfy the original equation (check it), so the only possible value is $a = 12/13$.
Alternatively, we can use the fact that the direction of maximal growth is parallel to the gradient of f. Since $(a, \sqrt{1-a^2}$ is a unit vector, it has to be equal to

$$\frac{\nabla f}{\|\nabla f\|} = \left(\frac{24}{26}, \frac{10}{26}\right) = \left(\frac{12}{13}, \frac{5}{13}\right),$$

which yields the result $a = 12/13$ as before.

10. The gradient is $\nabla f(x,y,z) = (y+z, x+z, x+y)$. Therefore, since $\mathbf{x} = (1,1,2)$, $\mathbf{v} = (10/\sqrt{105}, -1/\sqrt{105}, 2/\sqrt{105})$ and $\nabla f(1,1,2) = (3,3,2)$, the directional derivative is

$$f'_{\mathbf{v}}(\mathbf{x}) = \frac{31}{\sqrt{105}}.$$

Exercise 4.2 The gradient of f is $\nabla f(x,y) = (e^{x+2y}, 2e^{x+2y})$, and $\mathbf{v} = (4/5, 3/5)$, so the directional derivative is

$$f'_{\mathbf{v}}(x,y) = 2e^{x+2y}.$$

The values that make this directional derivative equal to $2e$ are

$$2e^{x+2y} = 2e \implies x + 2y = 1,$$

which is a straight line in $\mathbb{R}^2$.

Exercise 4.3 Here the gradient is $\nabla f(x,y) = (3\cos(3x+y), \cos(3x+y))$, which at $(0,0)$ is $\nabla f(0,0) = (3,1)$. Considering a unit vector $\mathbf{v} = (u_1, u_2)$ such that $u_1^2 + u_2^2 = 1$, we have

$$\begin{cases} f'_{\mathbf{v}}(0,0) = 3u_1 + u_2 = 1 \\ u_1^2 + u_2^2 = 1 \end{cases} \implies u_1^2 + (1-3u_1)^2 = 1 \implies 10u_1^2 - 6u_1 = 0,$$

wo $u_1 = 0$ or $u_1 = 3/5$. So the two directions are $(0,1)$ and $(3/5, -4/5)$.

Exercise 4.4 The gradient is

$$\nabla f(x,y) = \left(\frac{4xy^2}{(x^2+y^2)^2}, \frac{-4x^2y}{(x^2+y^2)^2}\right)$$

1. For a given direction $\mathbf{v} = (u_1, u_2)$, considering that $\nabla f(1,1) = (1,-1)$, the directional derivative at $(1,1)$ is

$$f'_{\mathbf{v}}(1,1) = u_1 - u_2,$$

and so the direction that makes this zero is $u_1 = u_2$, or $(1/\sqrt{2}, 1/\sqrt{2})$ (since $u_1^2 + u_2^2 = 1$). Note that this direction is perpendicular to the gradient at $(1, 1)$.

2. For a general point (x_0, y_0) in the first quadrant, we can repeat the same analysis with a general vector $\mathbf{v} = (v_1, v_2)$ such that $v_1^2 + v_2^2 = 1$. In this case, the directional derivative is

$$f'_{\mathbf{v}} = \frac{4x_0 y_0^2 v_1 - 4x_0^2 y_0 v_2}{(x_0^2 + y_0^2)^2} = 0 \implies y_0 v_1 = x_0 v_2,$$

since $x_0, y_0 > 0$. That is, $\mathbf{v} = \alpha(x_0, y_0)$, where $\alpha = (x_0^2 + y_0^2)^{-1/2}$. In other words, the vector points in the same line as the one that joins the point (x_0, y_0) to the origin.

3. As a result of part 2, we can conclude that the level curves of f are straight lines throuhg the origin. For instance, if, starting in the point $(1, 1)$, we move in the direction $(1, 1)$ towards $(1 + \epsilon, 1 + \epsilon)$ (that is, we stay on the line $y = x$), the value of f will remain constant, as the directional derivative in that direction is zero. Now, at $(1 + \epsilon, 1 + \epsilon)$, the directional derivative is still zero in the same direction. And this is true for all points in the line. So connecting the dots, we conclude that there is no change in f along the line $y = x$ (you can check that $f = 0$ in this line).

Exercise 4.5 1. We can consider the surface as the level surface of the function $f(x, y, z) = x^2 + y^2 + z^2$ with level 3. In this case, $\nabla f(x, y, z) = (2x, 2y, 2z)$ will be perpendicular to the surface at the point (x, y, z) and so the tangent plane to the surface at $(1, 1, 1)$ is

$$\nabla f(1, 1, 1)\cdot(x-1, y-1, z-1) = 0 \implies (2, 2, 2)\cdot(x-1, y-1, z-1) = 0,$$

which yields the plane $x + y + z = 3$.
Alternatively, we can consider the surface as the graph of the functions $g_1(x, y) = \sqrt{3 - x^2 - y^2}$ and $g_2(x, y) = -\sqrt{3 - x^2 - y^2}$. In this case, the point $(1, 1, 1)$ belongs to the graph of g_1. Knowing that $\nabla g_1(x, y) = (-x, -y)/\sqrt{3 - x^2 - y^2}$, the tangent plane to the graph of $g_1(x, y)$ at the point $(1, 1, 1)$ is

$$z = g_1(1, 1) + \nabla g_1(1, 1)\cdot(x-1, y-1) = 1 + (-1, -1)\cdot(x-1, y-1) = 3 - x - y,$$

which yields $x + y + z = 3$, as expected.

2. Here the surface can be understood as the level surface of $f(x, y, z) = x^3 - 2y^3 + z^3$ with level 0, and since $\nabla f(x, y, z) = (3x^2, -6y^2, 3z^2)$ the tangent plane at $(1, 1, 1)$ is

$$\nabla f(1, 1, 1)\cdot(x-1, y-1, z-1) = 0 \implies (3, -6, 3)\cdot(x-1, y-1, z-1) = 0,$$

yielding $x - 2y + z = 0$.
Alternatively, we can consider the surface as the graph of the function $g(x, y) = (2y^3 - x^3)^{1/3}$. Knowing that $\nabla g(x, y) = \frac{1}{3}(2y^3 - x^3)^{-2/3}(-3x^2, 6y^2)$, the tangent plane to the graph of $g(x, y)$ at the point $(1, 1, 1)$ is

$$z = g(1, 1) + \nabla g(1, 1)\cdot(x-1, y-1) = 1 + (-1, 2)\cdot(x-1, y-1) = x + 2y,$$

which yields the plane $x - 2y + z = 0$ as expected.

3. Here the surface can be understood as the level surface of $f(x, y, z) = e^z \cos x \cos y$ with level 0, and since

$$\nabla f(x, y, z) = (-e^z \sin x \cos y, -e^z \cos x \sin y, e^z \cos x \cos y)$$

the tangent plane at $(\pi/2, 1, 0)$ is

$$\nabla f(\pi/2, 1, 0)\cdot(x - \pi/2, y - 1, z) = 0 \implies (-1, 0, 0)\cdot(x - \pi/2, y - 1, z) = 0,$$

so the tangent plane is $x = \frac{\pi}{2}$.
In this case we cannot consider the surface as the graph of any function, since we cannot solve for any of the variables in the equation.

4. Here the surface can be understood as the level surface of $f(x, y, z) = e^{xyz}$ with level 1, and since $\nabla f(x, y, z) = (yze^{xyz}, xze^{eyz}, xye^{xyz})$ the tangent plane at $(1, 2, 0)$ is

$$\nabla f(1, 2, 0) \cdot (x - 1, y - 2, z) = 0 \implies (0, 0, 2) \cdot (x - 1, y - 2, z) = 0,$$

hence the tangent plane is $z = 0$.
Again, we cannot understand this surface as the graph of any function, since we cannot solve for any of the three variables in the equation.

Exercise 4.6 1. We have

$$Df(x, y) = \begin{pmatrix} -(1 + \tan^2(y/x))\frac{y}{x^2} + 1 & (1 + \tan^2(y/x))\frac{y}{x} + 1 \\ -\frac{1}{x} & \frac{1}{y+1} \end{pmatrix},$$

$$Dg(t, s) = \begin{pmatrix} \cos s & -t \sin s \\ e^t & 0 \\ -2 & 1 \end{pmatrix},$$

$$Dh(u, v, w) = \begin{pmatrix} (v^2 e^w & 2uve^w & uv^2 e^w \end{pmatrix}.$$

As for F, the chain rule says that $DF(x, y) = Dh(g(f(x, y)))Dg(f(x, y))Df(x, y)$. For $(x, y) = (1, 0)$ we have $f(x, y) = (-1, 0)$ and $g(f(x, y)) = (-1, e^{-1}, 2)$. Particularizing the Jacobian matrices we have

$$Df(1, 0) = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix},$$

$$Dg(-1, 0) = \begin{pmatrix} 1 & 0 \\ e^{-1} & 0 \\ -2 & 1 \end{pmatrix},$$

$$Dh(-1, e^{-1}, 2) = \begin{pmatrix} 1 & -2e & -1 \end{pmatrix},$$

and therefore

$$DF(1, 0) = \begin{pmatrix} 1 & -2e & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ e^{-1} & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \end{pmatrix}.$$

2. Since $F(1, 0) = h(-1, e^{-1}, 2) = -1$, the tangent plane is

$$z = -1 + \nabla F(1, 0) \cdot (x - 1, y) = -1 + y \implies z = y - 1.$$

3. Using $\mathbf{v} = (3/5, -4/5)$, we have

$$f'_{\mathbf{v}} = (0,1) \cdot (3/5, .4/5) = -\frac{4}{5}.$$

Exercise 4.7 The gradient is

$$\nabla f(x,y) = (e^x \cos y - e^y \sin x, -e^x \sin y + e^y \cos x) \implies \nabla f(0,0) = (1,1).$$

1. The direction of fastest increase is parallel to the gradient, so any vector of the form (λ, λ) with $\lambda > 0$ signals the direction of fastest increase.
2. The direction of fastest decrease is parallel to minus the gradient, so any vector of the form $(-\lambda, -\lambda)$ with $\lambda > 0$ signals the direction of fastest decrease.

Exercise 4.8 The gradient is

$$\nabla \rho(x,y,z) = (-2x, -2y, -2z)ke^{-(x^2+y^2+z^2)}.$$

1. Since the direction of fastest increase is proportional to the gradient, we can say that any vector proportional to $(-x, -y, -z)$ will signal the direction of fastest increase.
2. Doing $\rho(x,y,z) = c$ with $c > 0$ we obtain

$$ke^{-x^2-y^2-z^2} = c \implies x^2 + y^2 + z^2 = \log \frac{k}{c},$$

so the level surfaces are spheres centered at the origin.

Exercise 4.9 1. The gradient is $\nabla h(x,y) = (-4xe^{-x^2}, -6ye^{-3y^2})$ and so the direction of fastest increase of the height is $\nabla h(1,0) = (-4e, 0)$, which points to the West if we consider the positive y axis to mark the North.
2. The gradient is $\nabla T(x,y,z) = (-e^{-x}, -2e^{-2y}, -3e^{-3z})$ and so the direction of fastest decrease at the point $(1,1,1)$ is $-\nabla T(1,1,1) = (e^{-1}, 2e^{-2}, 3e^{-3})$.

Exercise 4.10 1. Since

$$DF(x,y) = \begin{pmatrix} 2x & 0 \\ 0 & 2y \end{pmatrix}, \quad DG(u,v) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 2v \end{pmatrix},$$

the derivative of $G \circ F$ at $(1,1)$ will be given, by the chain rule, as

$$D(G \circ F)(1,1) = DG(F(1,1))DF(1,1) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 0 \\ 0 & 4 \end{pmatrix},$$

since $F(1,1) = (2,1)$.
We could have obtained the same result directly, since $G(F(x,y)) = (x^2 + y^2 + 1, x^2 + 1, y^4)$ and so

$$D(G \circ F)(x,y) = \begin{pmatrix} 2x & 2y \\ 2x & 0 \\ 0 & 4y^3 \end{pmatrix} \implies D(G \circ F)(1,1) = \begin{pmatrix} 2 & 2 \\ 2 & 0 \\ 0 & 4 \end{pmatrix}.$$

2. **a)** Since this may be confusing, it may be easier if we write $h(x) = f(g(x))$, where $g(x) = (x, u(x), v(x))$. Therefore, the chain rule says

$$h'(x) = Df(g(x))Dg(x) = \nabla f(g(x)) \cdot g'(x),$$

and since $\nabla f(g(x)) = (\partial f/\partial x, \partial f/\partial u, \partial f/\partial v)$ and $g'(x) = (1, u'(x), v'(x))$, we have

$$h'(x) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u}u'(x) + \frac{\partial f}{\partial v}v'(x).$$

b) As in a),

$$\frac{\partial h}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} + \frac{\partial f}{\partial w}w'(x).$$

c) Note that, here,

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial w}\frac{\partial w}{\partial z} \quad \text{and} \quad \frac{\partial v}{\partial z} = \frac{\partial v}{\partial w}\frac{\partial w}{\partial z} + \frac{\partial v}{\partial z},$$

so

$$\begin{aligned}\frac{\partial h}{\partial z} &= \frac{\partial f}{\partial u}\frac{\partial u}{\partial z} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial z} \\ &= \frac{\partial f}{\partial u}\frac{\partial u}{\partial w}\frac{\partial w}{\partial z} + \frac{\partial f}{\partial v}\left(\frac{\partial v}{\partial w}\frac{\partial w}{\partial z} + \frac{\partial v}{\partial z}\right).\end{aligned}$$

3. Writing $g(x, y) = x + y$ and $h(x, y) = x - y$ we have $z(x, y) = f(g(x, y)) + f(h(x, y))$ and therefore

$$\begin{aligned}\frac{\partial z}{\partial x} &= f'(g(x, y))\frac{\partial g}{\partial x} + f'(h(x, y))\frac{\partial h}{\partial x} \\ &= f'(x + y) + f'(x - y).\end{aligned}$$

Similarly,

$$\frac{\partial z}{\partial y} = f'(x + y) - f'(x - y).$$

4.

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u}(u(x, y), v(x, y))\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}(u(x, y), v(x, y))\frac{\partial v}{\partial x} \\ &= [v(x, y)e^{u(x,y)v(x,y)}]\frac{1}{x + y} + [u(x, y)e^{u(x,y)v(x,y)}]\frac{y}{x^2 + y^2} \\ &= \arctan(x/y)(x + y)^{\arctan(x/y)-1} + \log(x + y)(x + y)^{\arctan(x/y)}\frac{y}{x^2 + y^2}.\end{aligned}$$

Similarly,

$$\begin{aligned}\frac{\partial z}{\partial y} &= \frac{\partial z}{\partial u}(u(x, y), v(x, y))\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}(u(x, y), v(x, y))\frac{\partial v}{\partial y} \\ &= [v(x, y)e^{u(x,y)v(x,y)}]\frac{1}{x + y} + [u(x, y)e^{u(x,y)v(x,y)}]\left(-\frac{x}{x^2 + y^2}\right) \\ &= \arctan(x/y)(x + y)^{\arctan(x/y)-1} - \log(x + y)(x + y)^{\arctan(x/y)}\frac{x}{x^2 + y^2}.\end{aligned}$$

5. We have

$$Df(x, y, z) = \begin{pmatrix} \frac{2x}{(1+y^2)(1+x^2+2z^2)} & -\frac{2y\log(1+x^2+2x^2)}{1+y^2} & \frac{4z}{(1+y^2)(1+x^2+2z^2)} \end{pmatrix}$$

and

$$s'(t) = (1, -2t, \cos t) \implies Ds(t) = \begin{pmatrix} 1 \\ -2t \\ \cos t \end{pmatrix}.$$

Since $s(0) = (1, 1, 0)$ the chain rules gives

$$h'(0) = Df(1,1,0)Ds(0) = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2}\log 2 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{2}.$$

Exercise 4.11 We need to write $\partial u/\partial x$ and $\partial u/\partial y$ in terms of r, θ, since $x = r\sin\theta$ and $y = r\cos\theta$. That is,

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial x},$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial u}{\partial \theta}\frac{\partial \theta}{\partial y}.$$

In order to do this, we need to find the expressions for r and θ in terms of x and y. The expression for r is simple, as $r = \sqrt{x^2 + y^2}$ and therefore

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} = \cos\theta, \quad \frac{\partial r}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} = \sin\theta.$$

Since $\tan\theta = y/x$ we have $\theta = \arctan y/x$ and so

$$\frac{\partial \theta}{\partial x} = -\frac{y}{x^2+y^2} = -\frac{\sin\theta}{r}, \quad \frac{\partial \theta}{\partial y} = \frac{x}{x^2+y^2} = \frac{\cos\theta}{r}.$$

Finally,

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial r}\cos\theta - \frac{\partial u}{\partial \theta}\frac{\sin\theta}{r},$$

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial r}\sin\theta + \frac{\partial u}{\partial \theta}\frac{\cos\theta}{r},$$

and therefore

$$x\frac{\partial u}{\partial y} - y\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \theta}.$$

Exercise 4.12 Since $h = (V \circ c)(t) = \left(e^{t^2}e^{-t^2}\right))^2 = 1$, we have $h'(t) = 0$.

Now, using the chain rule,

$$h'(t) = DV(c(t))Dc(t),$$

where

$$DV(x,y,z) = \begin{pmatrix} 2xy^2 & 2x^2y & 0 \end{pmatrix} \implies DV(c(t)) = \begin{pmatrix} 2e^{-t^2} & 2e^{r^2} & 0 \end{pmatrix},$$

and

$$Dc(t) = \begin{pmatrix} 2te^{t^2} \\ -2te^{-t^2} \\ \frac{2t}{1+t^2} \end{pmatrix}.$$

Multiplying, we get

$$h'(t) = DV(c(t))Dc(t) = 4t - 4t = 0.$$

Exercise 4.13 1. Showing that $E(t)$ is constant is equivalent to showing that $E'(t) = 0$, that is

$$E'(t) = \frac{m}{2}[\|\mathbf{r}'(t)\|^2]' + [V(\mathbf{r}(t))]'.$$

Now, $\|\mathbf{r}'(t)\|^2 = (x'(t))^2 + (y'(t))^2 + (z'(t))^2$ and so its derivative is

$$[\|\mathbf{r}'(t)\|^2]' = 2x'(t)x''(t) + 2y'(t)y''(t) + 2z'(t)z''(t) = 2\mathbf{r}'(t) \cdot \mathbf{r}''(t).$$

As for $V(\mathbf{r}(t))$, the chain rule gives

$$[V(\mathbf{r}(t))]' = \nabla V(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = -F \cdot \mathbf{r}'(t).$$

Putting everything together, and remembering that $F = m\mathbf{r}''(t)$, we have

$$E'(t) = \frac{m}{2}2\mathbf{r}'(t) \cdot \mathbf{r}''(t) - F \cdot \mathbf{r}'(t) = 0.$$

2. If $\mathbf{r}(t) \subset S$, where S is a level surface for V, then $V(\mathbf{r}(t)) = k$ with $k \in \mathbb{R}$ a constant. Since $E(t)$ is also a constant, it must be the case that $\|\mathbf{r}'(t)^2\|$ is constant as well.

Exercise 4.14 Here we have to show that $s(t)$ are solutions to the differential equations $\mathbf{x}' = F(\mathbf{x})$. Finding solutions to a differential equation is hard, but checking that a given function is a solution to a given differential equation is easy:

1. The differential equation here is

$$\begin{cases} x'(t) = y + 1 \\ y'(t) = 2 \\ z'(t) = \frac{1}{2z} \end{cases}$$

The proposed solution $s(t)$ must satisfy that $s'(t) = F(s(t))$. Here,

$$s'(t) = \left(2t, 2, \frac{1}{2\sqrt{t}}\right),$$

$$F(s(t)) = \left(2t - 1 + 1, 2, \frac{1}{2\sqrt{t}}\right),$$

which are equal.

2. The differential equation here is

$$\begin{cases} x'(t) = 2x \\ y'(t) = -y^2 \\ z'(t) = y \end{cases}$$

Here,

$$s'(t) = \left(2e^{2t}, -\frac{1}{t^2}, \frac{1}{t}\right),$$

$$F(s(t)) = \left(2e^{2t}, -\frac{1}{t^2}, \frac{1}{t}\right).$$

3. The differential equation here is

$$\begin{cases} x'(t) = x^2 \\ y'(t) = \frac{x}{x-1} \\ z'(t) = z(1+x) \end{cases}$$

Here,

$$s'(t) = \left(\frac{1}{(1-t)^2}, \frac{1}{t}, \frac{e^t(1-t)+e^t}{(1-t)^2}\right),$$

$$F(s(t)) = \left(\frac{1}{(1-t)^2}, \frac{1}{1-(1-t)}, \frac{e^t(2-t)}{(1-t)^2}\right).$$

A.5 Higher-order derivatives. Differential Operators

Exercise 5.1 1.

$$\frac{\partial f}{\partial x} = \frac{y(x^4+4x^2y^2-y^4)}{(x^2+y^2)^2}, \quad \frac{\partial f}{\partial y} = \frac{-x(y^4+4x^2y^2-x^4)}{(x^2+y^2)^2},$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{(x^2-y^2)(x^4+10x^2y^2+y^4)}{(x^2+y^2)^3}, \quad \text{for } (x,y) \neq (0,0);$$

2. The limit does not exist because

$$\lim_{x\to 0} \frac{\partial^2 f}{\partial x \partial y}(x, \lambda x) = \lim_{x\to 0}(1-\lambda^2)(1+10\lambda^2+\lambda^4)$$

depends on λ.

3. Since

$$\frac{\partial f}{\partial x}(0,0) = \frac{\partial f}{\partial y}(0,0) = 0,$$

we have

$$\frac{\partial^2 f}{\partial x \partial y}(0,0) = \lim_{x\to 0} \frac{\frac{\partial f}{\partial y}(x,0) - \frac{\partial f}{\partial y}(0,0)}{x} = \lim_{x\to 0} \frac{x}{x} = 1,$$

and

$$\frac{\partial^2 f}{\partial y \partial x}(0,0) = \lim_{y\to 0} \frac{\frac{\partial f}{\partial x}(0,y) - \frac{\partial f}{\partial x}(0,0)}{x} = \lim_{x\to 0} \frac{-x}{x} = -1.$$

This is explained because, as we have proven, f is not C^2 on any neighborhood of the origin.

Exercise 5.2 1. Due to the equality of the mixed partial derivatives,

$$\operatorname{curl}(\nabla f) = (\partial_y \partial_z f - \partial_z \partial_y f, -(\partial_x \partial_z f - \partial_z \partial_x f), \partial_x \partial_y f - \partial_y \partial_x f) = (0,0,0);$$

2. Again due to the equality of the mixed partial derivatives,

$$\operatorname{div}(\operatorname{curl} \mathbf{F}) = \partial_x(\partial_y F_3 - \partial_z F_2) + \partial_y(\partial_z F_1 - \partial_x F_3) + \partial_z(\partial_x F_2 - \partial_y F_1) = 0;$$

3. Almost immediate, since

$$\frac{\partial (fg)}{\partial x_i} = f\frac{\partial g}{\partial x_i} + g\frac{\partial f}{\partial x_i};$$

4. Since $\mathbf{F} = (F_1, \dots, F_n)$

$$\operatorname{div}(f\mathbf{F}) = \sum_{i=1}^n \frac{\partial (fF_i)}{\partial x_i} = \sum_{i=1}^n \left(f\frac{\partial F_i}{\partial x_i} + F_i \frac{\partial f}{\partial x_i}\right),$$

which yields the result after realizing that

$$\sum_{i=1}^n F_i \frac{\partial f}{\partial x_i} = \mathbf{F} \cdot \nabla f.$$

5. Using the result in the previous part,

$$\operatorname{div}(f\nabla g - g\nabla f) = f\operatorname{div}(\nabla g) + \nabla f\cdot\nabla g - g\operatorname{div}(\nabla f) - \nabla g\cdot\nabla f,$$

which yields the result after noting that

$$\operatorname{div}(\nabla g) = \sum_{i=1}^{n}\frac{\partial}{\partial x_i}\left(\frac{\partial g}{\partial x_i}\right) = \Delta g.$$

Exercise 5.3 1. Using the chain rule, since $f(r(\mathbf{x}))$, we have

$$D(f\circ r)(\mathbf{x}) = Df(r)Dr(\mathbf{x}),$$

or

$$\nabla f(\mathbf{x}) = (r^k)'\nabla r = kr^{k-1}\frac{\mathbf{x}}{r} = kr^{k-2}\mathbf{x},$$

since

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{\sqrt{x_1^2+\cdots+x_n^2}} = \frac{x_i}{r}.$$

2. Using part 4 from the previous problem:

$$\operatorname{div}\mathbf{F} = r^k\operatorname{div}\mathbf{x} + \nabla(r^k)\cdot(\mathbf{x}) = r^k n + kr^{k-2}\mathbf{x}\cdot\mathbf{x} = (k+n)r^k,$$

since $\operatorname{div}\mathbf{x} = n$ and $\mathbf{x}\cdot\mathbf{x} = r^2$.

3. From part 1, we can write

$$r^k\mathbf{x} = \nabla f$$

with

$$f = \frac{r^{k+2}}{k+2},\quad \text{if } k\neq -2,\quad \text{and } f = \log r\quad \text{if } k=-2.$$

Using now part 1 from the previous problem,

$$\operatorname{curl}\mathbf{F} = \mathbf{0}.$$

Alternatively, we can calculate it directly, remembering that $\partial_{x_i} r^k = kr^{k-2}x_i$:

$$\operatorname{curl}\mathbf{F} = \begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ \partial_x & \partial_y & \partial_z\\ r^k x & r^k y & r^k z\end{vmatrix} = kr^{k-2}\left[(yz-zy)\mathbf{i} + (zx-xz)\mathbf{j} + (xy-yx)\mathbf{k}\right] = \mathbf{0}.$$

4. Using parts 1 and 2:

$$\Delta f = \operatorname{div}(\nabla f) = \operatorname{div}(kr^{k-2}\mathbf{x}) = k(k+n-2)r^{k-2}.$$

Exercise 5.4 1. Since $\mathbf{F} = ms''(t)$ we have

$$\mathbf{F}(t) = m(e^t, e^{-t}, -\cos t) \implies F(0) = (3,3,-3).$$

2. The tangent line is $r(t) = s(1) + s'(1)(t-1)$ and so

$$r(t) = (e, e^{-1}, \cos 1) + (e, -e^{-1}, -\sin 1)(t-1) \implies r(2) = (2e, 0, \cos 1 - \sin 1).$$

Exercise 5.5 1.

$$\operatorname{div}\mathbf{F}(x,y) = 6xy + 3y^2; \quad \operatorname{curl}\mathbf{F}(x,y,z) = (0,0,0);$$

Since $\operatorname{curl}\mathbf{F} = \mathbf{0}$ this means that $\mathbf{F} = \nabla f$ and so

$$\partial_x f = 3x^2 y, \quad \partial_y f = x^3 + y^3.$$

From inspection we see that $f = x^3 y + y^4/4 + c$.

2.

$$\operatorname{div}\mathbf{F}(x,y,z) = 0; \quad \operatorname{curl}\mathbf{F}(x,y,z) = (0,0,0);$$

Again, $\mathbf{F} = \nabla f$, where $f = xyz + c$. Since $\operatorname{div}\mathbf{F} = 0$ this means that $\mathbf{F} = \operatorname{curl}\mathbf{G}$ for some unknown vector field $\mathbf{G}$. However, finding the curl from the system of differential equations that results is not an easy task in general. In this case, one possible solution is

$$\mathbf{G} = \left(xz^2 - \frac{xy^2}{2}, yx^2 - \frac{yz^2}{2}, zy^2 - \frac{zx^2}{2}\right).$$

3.

$$\operatorname{div}\mathbf{F}(x,y,z) = \frac{-2xyz}{(x^2+y^2+z^2)^2};$$

$$\operatorname{curl}\mathbf{F}(x,y,z) = \left(\frac{2x^3}{(x^2+y^2+z^2)^2}, \frac{2y(x^2-z^2)}{(x^2+y^2+z^2)^2}, \frac{-2z^3}{(x^2+y^2+z^2)^2}\right).$$

Exercise 5.6 1.

$$\operatorname{div}\mathbf{v}(x,y,z) = 3; \quad \operatorname{curl}\mathbf{v}(x,y,z) = (0,0,0);$$

and you can check that $\mathbf{v} = \nabla V$, where $V = \frac{x^2+y^2+z^2}{2} + c$.

2.

$$\operatorname{div}\mathbf{v}(x,y,z) = 6; \quad \operatorname{curl}\mathbf{v}(x,y,z) = (0,0,0);$$

and you can check that $\mathbf{v} = \nabla V$, where $V = \frac{x^2+2y^2+3z^2}{2} + c$.

3. Writing $r^2 = x^2 + y^2 + z^2$, and since

$$\partial_x \frac{x}{r^2} = \frac{-x^2+y^2+z^2}{r^4}$$

we have

$$\operatorname{div}\mathbf{v}(x,y,z) = \frac{-x^2+y^2+z^2+x^2-y^2+z^2+x^2+y^2-z^2}{r^4} = \frac{1}{r^2}.$$

Now, for the curl,

$$\operatorname{curl}\mathbf{v}(x,y,z) = (0,0,0).$$

You can check that $\mathbf{v} = \nabla V$ where $V = \log r^2$.

Exercise 5.7 1.

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\frac{2xy}{(x^2+y^2)^2} + \frac{2xy}{(x^2+y^2)^2} = 0.$$

2.
$$\Delta u = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2} + \frac{2(x^2 - y^2)}{(x^2 + y^2)^2} = 0.$$

Exercise 5.8 1. Since $\partial_x r = x/r$, using the chain rule on the function $(f \circ r)(\mathbf{x})$ we have

$$\frac{\partial u}{\partial x} = f'(r)\frac{x}{r}, \quad \frac{\partial u}{\partial y} = f'(r)\frac{y}{r},$$

Now, for the second-order partial derivatives, we apply the chain rule again to obtain

$$\frac{\partial^2 u}{\partial x^2} = f''(r)\frac{x^2}{r^2} + f'(r)\frac{r^2 - x^2}{r^3},$$

$$\frac{\partial^2 u}{\partial y^2} = f''(r)\frac{y^2}{r^2} + f'(r)\frac{r^2 - y^2}{r^3}.$$

Summing everything:

$$\Delta u = f''(r)\left(\frac{x^2 + y^2}{r^2}\right) + f'(r)\left(\frac{2r^2 - x^2 - y^2}{r^3}\right) = f''(r) + \frac{1}{r}f'(r).$$

2. Note that

$$\partial_x \theta = \partial_x \arctan(y/x) = -\frac{y}{x^2 + y^2}, \quad \partial_y \theta = \partial_y \arctan(y/x) = \frac{x}{x^2 + y^2},$$

and so

$$\frac{\partial u}{\partial x} = \frac{\partial f}{\partial r}\frac{x}{r} - \frac{\partial f}{\partial \theta}\frac{y}{r^2}, \quad \frac{\partial u}{\partial y} = \frac{\partial f}{\partial r}\frac{y}{r} + \frac{\partial f}{\partial \theta}\frac{x}{r^2}.$$

Remember that

$$\partial_{xx}\theta = \partial_{xx} \arctan(y/x) = -\frac{2xy}{(x^2 + y^2)^2}, \quad \partial_{yy}\theta = \partial_{yy} \arctan(y/x) = \frac{2xy}{(x^2 + y^2)^2},$$

and so the second partial derivatives are

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 f}{\partial r^2}\left(\frac{x}{r}\right)^2 + \frac{\partial f}{\partial r}\frac{r^2 - x^2}{r^3} + \frac{\partial^2 f}{\partial \theta^2}\left(-\frac{y}{r^2}\right)^2 + \frac{\partial f}{\partial \theta}\left(\frac{-2xy}{r^4}\right),$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 f}{\partial r^2}\left(\frac{y}{r}\right)^2 + \frac{\partial f}{\partial r}\frac{r^2 - y^2}{r^3} + \frac{\partial^2 f}{\partial \theta^2}\left(\frac{x}{r^2}\right)^2 + \frac{\partial f}{\partial \theta}\frac{2xy}{r^4}.$$

Summing:

$$\Delta u = \frac{\partial^2 f}{\partial r^2}\frac{x^2}{r^2} + \frac{1}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2}.$$

3. Since $u(x, y) = \log(x^2 + y^2 = 2\log r$ we can apply the first formula to obtain

$$\Delta u = -\frac{2}{r^2} + \frac{1}{r}\frac{2}{r} = 0.$$

As for $u(x, y) = \arctan(y/x) = \theta$, we can apply the second formula:

$$\Delta u = 0.$$

Exercise 5.9 1. From part 4 in Exercise 5.3, we have that

$$\Delta(r^k) = k(k+n-2)r^{k-2}.$$

In this case, $u_1(x, y, z) = r^{-1}$ where $n = 3$ and so

$$\Delta u_1 = -(-1+3-2)r^{-3} = 0.$$

As for $u_2(x, y, z, w) = r^{-2}$ with $n = 4$ we have

$$\Delta u_2 = -2(-2+4-2)r^{-4} = 0.$$

Alternatively, we can differentiate to obtain

$$\frac{\partial^2 u_1}{\partial x^2} = \frac{2x^2 - y^2 - z^2}{(x^2+y^2+z^2)^{5/2}}, \quad \frac{\partial^2 u_2}{\partial x^2} = \frac{2(x^2 - y^2 - z^2 - w^2)}{(x^2+y^2+z^2+w^2)^3}.$$

which leads to the result.

2. On $\mathbb{R}^n$, $r = \sqrt{x_1^2 + \cdots + x_n^2}$ and so

$$\frac{\partial u}{\partial x_j} = f'(r)\frac{x_j}{r},$$

and

$$\frac{\partial^2 u}{\partial x_j^2} = f''(r)\frac{x_j^2}{r^2} + f'(r)\frac{r^2 - x_j^2}{r^3}.$$

Summing:

$$\Delta u = f''(r)\sum_{j=1}^{n}\frac{x_j^2}{r^2} + f'(r)\sum_{j=1}^{n}\frac{r^2 - x_j^2}{r^3} = f''(r) + \frac{n-1}{r}f'(r).$$

3. Again using $\Delta(r^k) = k(k+n-2)r^{k-2}$ we have

$$\Delta u = k(k+n-2)r^{k-2} = 0 \quad \text{for } k = 2-n, \text{ provided } n \neq 2.$$

Exercise 5.10 Since

$$\frac{\partial^2}{\partial t^2}f(x,t) = -\omega^2\cos(kx - \omega t), \quad \frac{\partial^2}{\partial x^2}f(x,t) = -k^2\cos(kx - \omega t),$$

we have that, in order to fulfill the equations, the following conditions must be met:

1.
$$\omega^2 = c^2k^2;$$

2.
$$\omega^2 = c^2k^2 + h^2.$$

Exercise 5.11 Since

$$\frac{\partial u}{\partial t} = -\lambda u, \quad \frac{\partial^2 u}{\partial x^2} = -k^2 u,$$

we must have:

$$\lambda = \mu k^2.$$

A.6 Taylor expansions. Maxima and minima

Exercise 6.1 1.

$$\begin{pmatrix} \frac{2y^3-2x^2}{(x^2+y^3)^2} & -\frac{6xy^2}{(x^2+y^3)^2} \\ -\frac{6xy^2}{(x^2+y^3)^2} & \frac{6x^2y-3y^4}{(x^2+y^3)^2} \end{pmatrix};$$

2.

$$\begin{pmatrix} 2\cos y - y^2 \sin x & -2x\sin y + 2y\cos x \\ -2x\sin y + 2y\cos x & -x^2\cos y + 2\sin x \end{pmatrix};$$

3.

$$\begin{pmatrix} y(y-1)x^{y-2} & x^{y-1}(1+y\log x) \\ x^{y-1}(1+y\log x) & x^y \log^2 x \end{pmatrix};$$

4.

$$\begin{pmatrix} \frac{-2xy^3}{(1+x^2y^2)^2} & \frac{1-x^2y^2}{(1+x^2y^2)^2} \\ \frac{1-x^2y^2}{(1+x^2y^2)^2} & \frac{-2xy}{(1+x^2y^2)^2} \end{pmatrix};$$

5.

$$\begin{pmatrix} \frac{-2xy}{(x^2+y^2)^2} & \frac{x^2-y^2}{(x^2+y^2)^2} \\ \frac{x^2-y^2}{(x^2+y^2)^2} & \frac{2xy}{(x^2+y^2)^2} \end{pmatrix};$$

6.

$$\begin{pmatrix} 4 & -6 & 1 \\ -6 & -2 & 1 \\ 1 & 1 & 6 \end{pmatrix};$$

7.

$$\begin{pmatrix} 2y+2z & 2x+2y+2z & 2x+2y+2z \\ 2x+2y+2z & 2x+2y+2z & 2x+2y+2z \\ 2x+2y+2z & 2x+2y+2z & 2x+2y \end{pmatrix}.$$

Exercise 6.2 1.

$$\nabla f = (2x, 4y-4) = \mathbf{0} \quad \text{at } (0,1);$$

$$Hf(0,1) = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix},$$

so f has a local minimum at $(0,1)$ with value $f(0,1) = -2$;

2.

$$\nabla g = (2x - y + 2, -x + 2y + 2) = \mathbf{0} \quad \text{at } (-2,-2);$$

$$Hg(-2,-2) = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

so g has a local minimum at $(-2,-2)$ with value $g(-2,-2) = -10$;

3.

$$\nabla h = (6x + 2y + 2, 2x + 2y - 1) = \mathbf{0} \quad \text{at } (-3/4, 5/4);$$

$$Hh(-3/4, 5/4) = \begin{pmatrix} 6 & 2 \\ 2 & 2 \end{pmatrix},$$

so h has a local minimum at $(-3/4, 5/4)$ with value $h(-3/4, 5/4) = \frac{21}{8}$.

4.

$$\nabla k = (24x^2 - 24y, -24x + 3y^2) = \mathbf{0} \quad \text{at } (0,0), (2,4);$$

$$Hk(x,y) = \begin{pmatrix} 48x & -24 \\ -24 & 6y \end{pmatrix},$$

and therefore

$$Hk(0,0) = \begin{pmatrix} 0 & -24 \\ -24 & 0 \end{pmatrix} \implies k \text{ has a saddle point at } (0,0).$$

and

$$Hk(2,4) = \begin{pmatrix} 96 & -24 \\ -24 & 24 \end{pmatrix} \implies k \text{ has a local minimum at } (2,4) \text{ with value } k(2,4) = -64.$$

5.

$\nabla m = (3x^2+6x, 3y^2-36y+81) = \mathbf{0}$ at $(0,3), (0,9), (-2,3), (-2,9)$;

$$Hm(x,y) = \begin{pmatrix} 6x+6 & 0 \\ 0 & 6y-36 \end{pmatrix},$$

and therefore, since the matrix is diagonal, we can read the eigenvalues off it:

$$Hm(0,3) = \begin{pmatrix} 6 & 0 \\ 0 & -18 \end{pmatrix} \implies m \text{ has a saddle point at } (0,3).$$

$$Hm(0,9) = \begin{pmatrix} 6 & 0 \\ 0 & 18 \end{pmatrix} \implies m \text{ has a local minimum at } (0,9) \text{ with value } m(0,9) = 5.$$

$$Hm(-2,3) = \begin{pmatrix} -6 & 0 \\ 0 & -18 \end{pmatrix} \implies m \text{ has a local maximum at } (-2,3) \text{ with value } m(-2,3) = 107.$$

$$Hm(-2,9) = \begin{pmatrix} -6 & 0 \\ 0 & 18 \end{pmatrix} \implies m \text{ has a saddle point at } (-2,9).$$

Exercise 6.3

$$\nabla f = e^{-x^2+\varepsilon y^2}(-2x, 2\varepsilon y);$$

If $\varepsilon = 0$, the gradient is null at the points $(0, y_0)$ for all $y_0 \in \mathbb{R}$, and if $\varepsilon \neq 0$, it is null only at the origin. The Hessian matrix is

$$Hf(x,y) = e^{-x^2+\varepsilon y^2} \begin{pmatrix} -2+4x^2 & -4\varepsilon xy \\ -4\varepsilon xy & 2\varepsilon + 4\varepsilon^2 y^2 \end{pmatrix},$$

For $\varepsilon = 0$, we have:

$$Hf(0,y_0) = \begin{pmatrix} -2 & 0 \\ 0 & 0 \end{pmatrix},$$

f reaches its global maximum at those points, with value $f(0, y_0) = 1$.

Notice that one eigenvalue is -2 and the other is 0, thus making the determinant null and preventing us from using the usual method. However, when $\varepsilon = 0$ the function becomes $f(x,y) = e^{-x^2}$, which is constant for all values of y. That is, the value at $(0, y_0)$ does not change with y, and therefore we have a degenerate maximum: in the x direction there is a maximum, and in the y direction the function is constant.

If $\varepsilon = -1$, then:

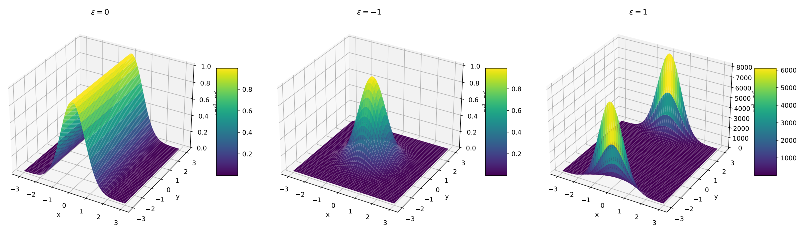
$$Hf(0,0) = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix},$$

and $(0,0)$ is a global maximum with value 1. Note that in this case we are multiplying two Gaussian functions together, one is a function of x and the other a function of y.

For $\varepsilon = 1$, we compute:

$$Hf(0,0) = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix},$$

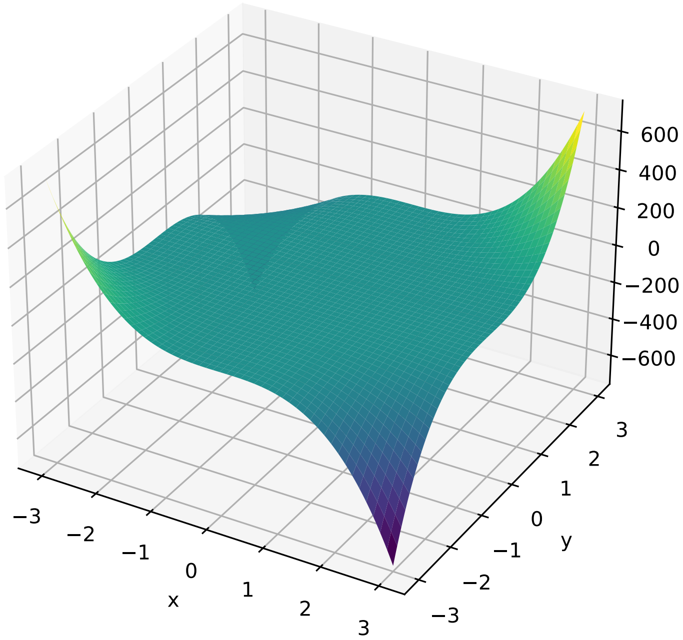
and $(0,0)$ is a saddle point.



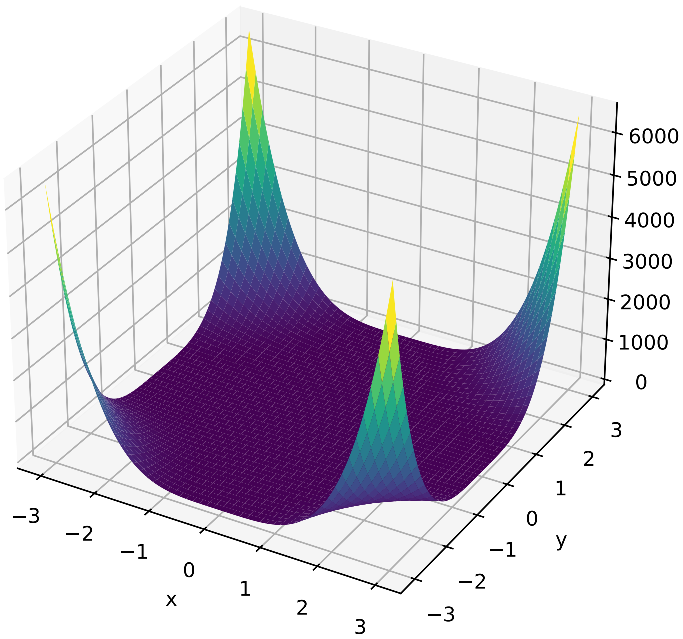
Exercise 6.4 1.

$$\nabla f = (3x^2y^3, 3x^3y^2) = \mathbf{0} \quad \text{when } xy = 0 \implies Hf(x,0) = Hf(0,y) = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

The points of the form $(x,0)$ and $(0,y)$ are saddle points because f is null there and it is positive in the first and third quadrants while it is negative in the second and fourth quadrants: f has neither maxima nor minima.
Alternatively, we can fix a value of $y = y_0$ and notice that in that case $f(x, y_0) = cx^3$ ($c \neq 0$ a constant) and so, as we move through the y axis, the function is a cubic curve that has an inflection point at zero. And identically with fixing a value of x and moving in the y direction.



2. As before we find that the points $(x, 0)$ and $(0, y)$ are critical points with the Hessian matrix identical to the zero matrix in all of them. In this case, f reaches the global minimum at the points of the form $(x, 0)$ and $(0, y)$ with value 0. Just like before, we can see that $f > 0$ anywhere again from the axes, and so they are all minima. Alternatively, fixing a value of $y = y_0$ we can see that $f(x, y_0) = cx^4$ ($c \neq 0$ a constant), which has a minimum at $x = 0$.



3. $\nabla f(x, y) = (2x + y - 3, x + 2y - 6) = (0, 0)$ when $x = 0, y = 3$. The

Hessian matrix is

$$Hf(0,3) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

and so $(0,3)$ is a minimum.

4.

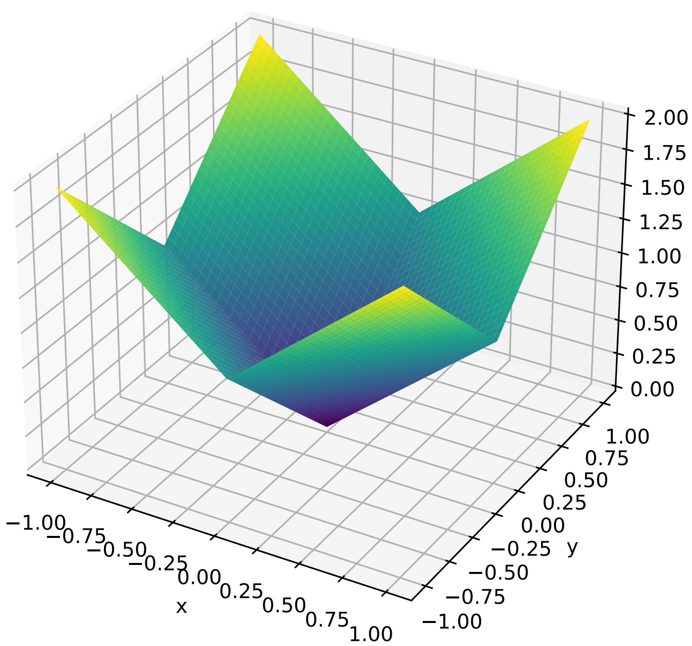
$$\nabla f = \frac{1}{(1+x^2+y^2)^2}(y^2 - x^2 + 2xy + 1, y^2 - x^2 - 2xy - 1) = \mathbf{0}$$

if $x = \frac{1}{\sqrt{2}} = -y$, or $x = -\frac{1}{\sqrt{2}} = -y$.

$$Hf(1/\sqrt{2}, -1/\sqrt{2}) = \begin{pmatrix} -1/\sqrt{2} & 0 \\ 0 & -1/\sqrt{2} \end{pmatrix} \implies \text{this is a local maximum with value } 1/\sqrt{2}.$$

$$Hf(-1/\sqrt{2}, 1/\sqrt{2}) = \begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \end{pmatrix} \implies \text{this is a local minimum with value } -1/\sqrt{2}.$$

Exercise 6.5 1. f has the global minimum at $(0,0)$ with value 0 and global maxima at $(\pm 1, \pm 1)$ with value 2.
We can see that f is symmetric with respect to the origin and so we can focus only on the first quadrant. Here $f(x,y) = x + y$ which has a minimum at $(0,0)$ and a maximum at $(1,1)$. Due to symmetry, we obtain all other maxima.



2. $f(x,y) = (x+y)^2$,

$$\nabla f = (2(x+y), 2(x+y)) = \mathbf{0} \quad \text{if } x + y = 0;$$

since $f(x_0, -x_0) = 0$ and $f(x,y) \geq 0$, the points of the form $(x_0, -x_0)$ are global minima on B with value 0.

We have to study the boundary: ∂B consists of four segments:

$$\partial B = \{x = 2, -1 \leq y \leq 1\} \cup \{-2 \leq x \leq 2, y = 1\} \cup \{x = -2, -1 \leq y \leq 1\} \cup \{-2 \leq x \leq 2, y = -1\}$$

$$= \gamma_1 \cup \gamma_2 \cup \gamma_3 \cup \gamma_4.$$

For example, on γ_1, we have the function:

$$g_1(y) = f(2, y) = (2 + y)^2,$$

with maximum value of 9 at $y = 1$, and minimum value of 1 at $y = -1$. Similarly, on γ_2 we have

$$g_2(x) = f(x, 1) = (x + 1)^2,$$

with maximum value of 3 at $x = 2$ and minimum value of 1 at $x = -2$. The analysis for the other two segments is identical due to symmetry.
Therefore, $(2, 1)$ and $(-2, -1)$ are the global maxima on B, and there f has the value 9.

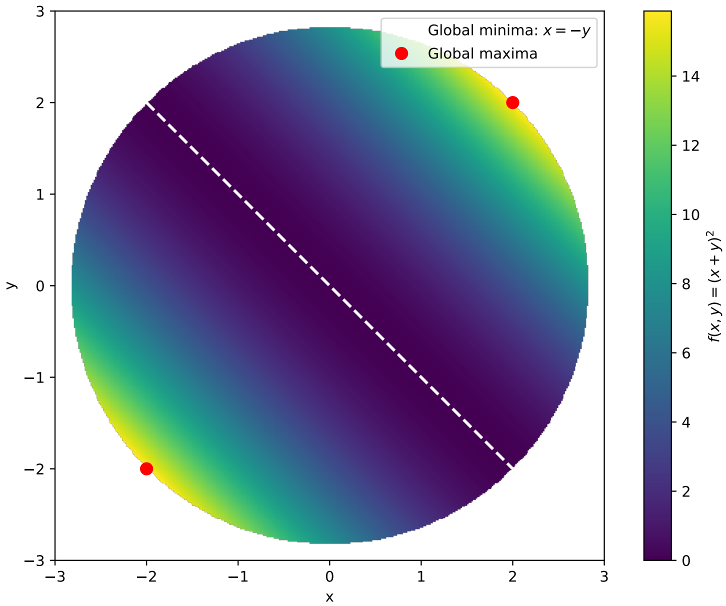
3. Here the function is the same as in the previous part, and its global minimum is still 0 at $(x_0, -x_0)$. Now, for the boundary, we can parametrize it with the function $c(\theta) = (r\cos\theta, r\sin\theta)$ where $r = 2\sqrt{2}$, and $\theta \in [0, 2\pi]$. Now,

$$f(c(\theta)) = (1 + 2\sin\theta\cos\theta)r^2 = (1 + \sin 2\theta)r^2,$$

and its derivative is

$$f'(c(\theta)) = 2\cos 2\theta = 0 \implies \cos 2\theta = 0 \implies \theta = \pi/4 \text{ or } \theta = 5\pi/4.$$

Therefore, the maxima are at the points $(r\cos pi/4, r\sin\pi/4) = (2, 2)$ and $(r\cos 5pi/4, r\sin 5\pi/4) = (-2, -2)$ with value 16.



Exercise 6.6 1.

$$\nabla f(x,y) = (4x^3 + 2y^3 + 2y, 6xy^2 + 4y^3 + 2x) = \mathbf{0} \text{ if } (x,y) = (0,0),$$

at that point the Hessian matrix is

$$Hf(0,0) = \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix},$$

so $(0,0)$ is a saddle point.

2.

$$\nabla f(x,y) = (y + y^3 \sin(x/y) + xy^2 \cos(x/y), x + 3xy^2 \sin(x/y) - x^2 y \cos(x/y)) = \mathbf{0} \text{ if } (x,y) = (0,0),$$

The Hessian matrix, away from $y = 0$ is

$$Hf(x,y) = \begin{pmatrix} 2y^2 \cos\left(\frac{x}{y}\right) - xy \sin\left(\frac{x}{y}\right) & x^2 \sin\left(\frac{x}{y}\right) + 3y^2 \sin\left(\frac{x}{y}\right) + xy \cos\left(\frac{x}{y}\right) + 1 \\ x^2 \sin\left(\frac{x}{y}\right) + 3y^2 \sin\left(\frac{x}{y}\right) + xy \cos\left(\frac{x}{y}\right) + 1 & -4x^2 \cos\left(\frac{x}{y}\right) + 6xy \sin\left(\frac{x}{y}\right) - \frac{x^3}{y} \sin\left(\frac{x}{y}\right) \end{pmatrix}.$$

Now, in order to calculate the Hessian matrix at $(0,0)$ we need to use the definition of partial derivatives (check that f is of class C^1, that is, its partial derivatives are continuous):

$$\frac{\partial^2 f}{\partial x^2}(0,0) = \lim_{x\to 0} \frac{\partial f(x,0)/\partial x - \partial f(0,0)/\partial x}{x} = \lim_{x\to 0} \frac{0-0}{x} = 0,$$

$$\frac{\partial^2 f}{\partial x \partial y}(0,0) = \lim_{x\to 0} \frac{\partial f(x,0)/\partial y - \partial f(0,0)/\partial y}{x} = \lim_{x\to 0} \frac{x-0}{x} = 1,$$

$$\frac{\partial^2 f}{\partial y^2}(0,0) = \lim_{y\to 0} \frac{\partial f(0,y)/\partial y - \partial f(0,0)/\partial y}{y} = \lim_{x\to 0} \frac{0-0}{y} = 0,$$

and so

$$Hf(0,0) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

so $(0,0)$ is a saddle point. Notice also that $f(x,x) = x^2 + x^4 \sin 1$ has a minimum at 0 but that $f(x,-x) = -x^2 - x^4 \sin(-1)$ has a maximum.

Exercise 6.7

$$\nabla f(m,b) = \left(-2 \sum_{i=1}^{N} (y_i - mx_i - b)x_i, -2 \sum_{i=1}^{N} (y_i - mx_i - b) \right) = \mathbf{0}$$

when

$$\begin{aligned} \sum x_i y_i - m \sum x_i^2 - b \sum x_i &= 0 \\ \sum y_i - m \sum x_i - Nb &= 0. \end{aligned}$$

Calling

$$A = \sum x_i, \quad B = \sum y_i, \quad C = \sum x_i y_i, \quad D = \sum x_i^2,$$

the system becomes

$$\begin{cases} Dm + Ab = C \\ Am + Nb = B \end{cases}$$

in the variables m and b. We have:

$$m = \frac{AB - NC}{A^2 - ND} = \frac{\sum_{i=1}^{N} x_i \sum_{i=1}^{N} y_i - N \sum_{i=1}^{N} x_i y_i}{\left(\sum_{i=1}^{N} x_i\right)^2 - N \sum_{i=1}^{N} x_i^2},$$

$$b = \frac{B - Am}{N} = \frac{1}{N}\sum_{i=1}^{N} y_i - \frac{1}{N}\sum_{i=1}^{N} x_i m.$$

Now,

$$\sum_{i=1}^{N} x_i \sum_{i=1}^{N} y_i - N \sum_{i=1}^{N} x_i y_i = -N^2 \operatorname{Cov}(x, y),$$

the covariance of x and y, and

$$\left(\sum_{i=1}^{N} x_i\right)^2 - N \sum_{i=1}^{N} x_i^2 = -N^2 \operatorname{Var}(x),$$

the variance of x. Finally,

$$\frac{1}{N}\sum_{i=1}^{N} y_i = \mathbf{E}(y),$$

the average value of y, so we can rewrite the solution as

$$m = \frac{\operatorname{Cov}(x, y)}{\operatorname{Var}(x)}$$

$$b = \mathbf{E}(y) - m\mathbf{E}(x).$$

Exercise 6.8 1. $\nabla f(x, y) = (\cos x \cos y, -\sin x \sin y) = \mathbf{0}$ when $x = (2k_1 + 1)\frac{\pi}{2}, k_1 \in \mathbb{Z}$ and $y = k_2\pi, \in k_2 \in \mathbb{Z}$ or vice versa.

Now, the Hessian matrix is

$$Hf(x, y) = \begin{pmatrix} -\sin x \cos y & -\cos x \sin y \\ -\cos x \sin y & -\sin x \cos y \end{pmatrix}.$$

At the points $\left((2k_1 + 1)\frac{\pi}{2}, k_2\pi\right)$ the Hessian is

$$Hf(k_1, k_2) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

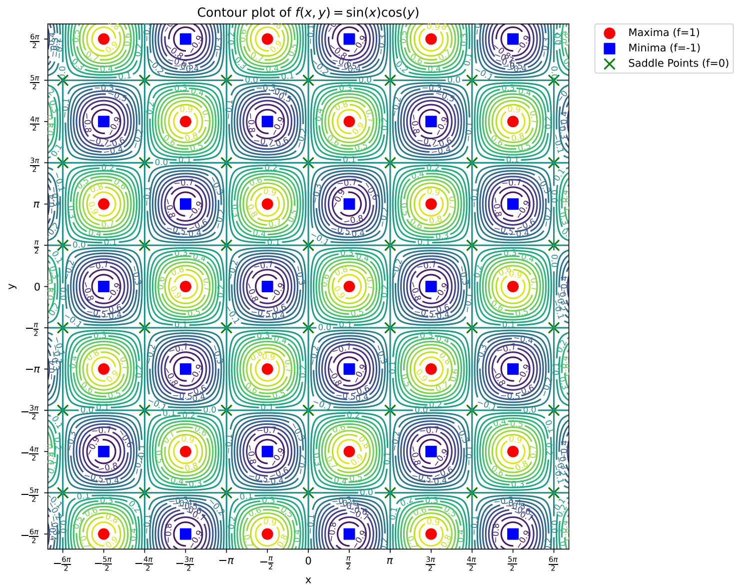
if k_1 is even (and therefore f has a minimum there), and

$$Hf(k_1, k_2) = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

if k_1 is odd (and therefore f has a maximum there).
Moreover, at the points $\left(k_1\pi, (2k_2 + 1)\frac{\pi}{2}\right)$ the Hessian is

$$Hf(k_1, k_2) = \begin{pmatrix} 0 & \pm 1 \\ \pm 1 & 0 \end{pmatrix}$$

and so these points are saddle points.



2. $\nabla f(x, y) = (2x\cos x, -2y\cos y^2) = \mathbf{0}$ when $x = 0$ or $x = \pm\sqrt{(2k+1)\frac{\pi}{2}}$, and the same with y. Therefore, we have four types of critical points. Now, the Hessian matrix is

$$Hf(x, y) = \begin{pmatrix} 2\cos x^2 - 4x^2\sin x^2 & 0 \\ 0 & -2\cos y^2 + 4y^2\sin y^2 \end{pmatrix}.$$

At $(0, 0)$ it is

$$Hf(0, 0) = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} \implies \text{saddle point.}$$

At the points $\left(0, \pm\sqrt{(2k+1)\frac{\pi}{2}}\right)$ the Hessian is

$$Hf\left(0, \pm\sqrt{(2k+1)\frac{\pi}{2}}\right) = \begin{pmatrix} 2 & 0 \\ 0 & \pm\alpha(k) \end{pmatrix},$$

where $\alpha(k) = 2(2k+1)\pi$ and so we have minima or saddle points depending on the value k. In particular, at $(0, \pm\sqrt{\pi/2})$ we have a minimum, at $(0, \pm\sqrt{3\pi/2})$ a saddle point, and so on alternating between the two.
At the points $\left(0, \pm\sqrt{(2k+1)\frac{\pi}{2}}\right)$ the Hessian is

$$Hf\left(0, \pm\sqrt{(2k+1)\frac{\pi}{2}}\right) = \begin{pmatrix} \mp\alpha(k) & 0 \\ 0 & -2 \end{pmatrix},$$

so we have maxima or saddle points depending on the value of k. In particular, at $(\pm\sqrt{\pi/2}, 0)$ we have a maximum, at $(\pm\sqrt{3\pi/2}, 0)$ a saddle point, and so on alternating between the two.
At the points $\left(\pm\sqrt{(2k+1)\frac{\pi}{2}}, \pm\sqrt{(2k+1)\frac{\pi}{2}}\right)$ the Hessian is

$$Hf\left(\pm\sqrt{(2k_1+1)\frac{\pi}{2}}, \pm\sqrt{(2k_2+1)\frac{\pi}{2}}\right) = \begin{pmatrix} \mp\alpha(k_1) & 0 \\ 0 & \pm\alpha(k_2) \end{pmatrix},$$

and so it all depends on the values of k_1, k_2. Therefore, we'll have saddle points at:

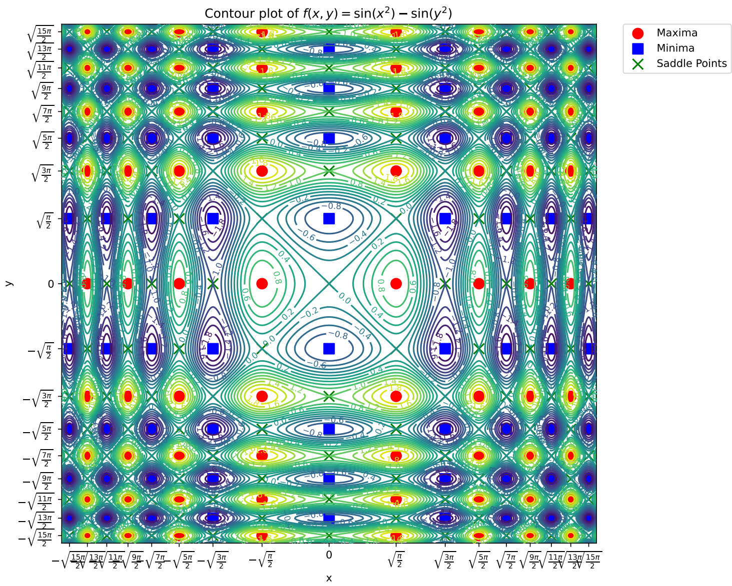
$$x(k_1, k_2) = \left(\pm\sqrt{(2k_1+1)\frac{\pi}{2}}, \pm\sqrt{(2k_2+1)\frac{\pi}{2}}\right), \quad k_1, k_2 \in \mathbb{Z},\ \text{if } k_1{+}k_2 \text{ is even},$$

maxima at:

$$x(k_1, k_2) = \left(\pm\sqrt{(2k_1+1)\frac{\pi}{2}}, \pm\sqrt{(2k_2+1)\frac{\pi}{2}}\right), \quad k_1, k_2 \in \mathbb{Z} \text{ if } (k_1 \text{ is even}, k_2 \text{ is odd}).$$

and minima at:

$$x(k_1, k_2) = \left(\pm\sqrt{(2k_1+1)\frac{\pi}{2}}, \pm\sqrt{(2k_2+1)\frac{\pi}{2}}\right), \quad k_1, k_2 \in \mathbb{Z} \text{ if } (k_1 \text{ is odd}, k_2 \text{ is even}).$$

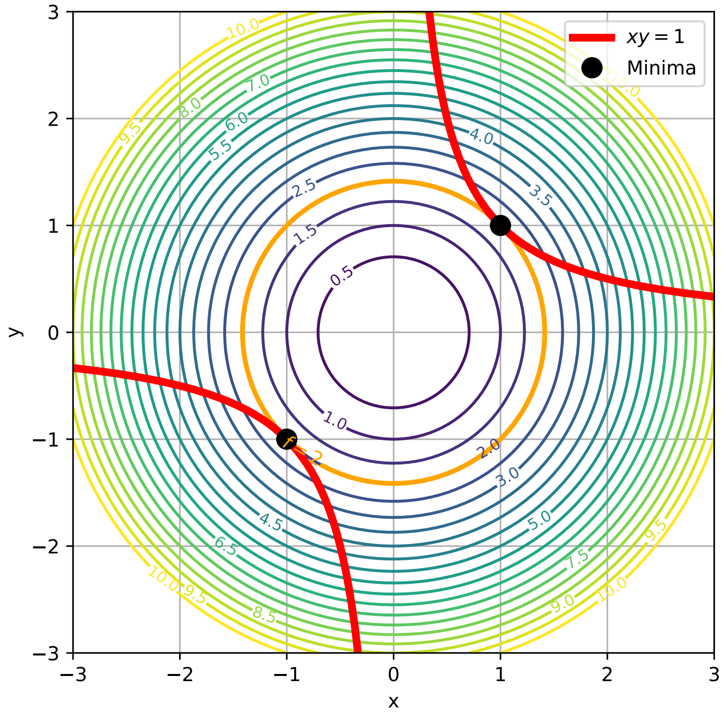


A.7 Lagrange multipliers

Exercise 7.1 1. We solve the system:

$$\begin{cases} 2x = \lambda y \\ 2y = \lambda x \\ xy = 1 \end{cases}$$

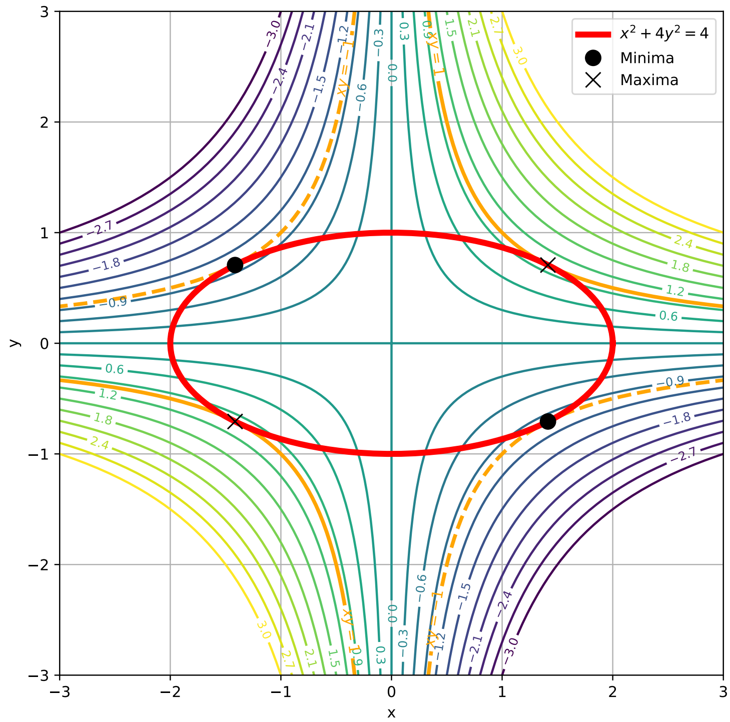
and obtain $x = y = \pm 1$, which give us the critical points $(1, 1)$ and $(-1, -1)$. In order to know if they are maxima or minima we could calculate the values of f at points to their left and right: we'd see that they are indeed, global minima. Alternatively, we can reason that, since f has a global minimum at $(0, 0)$ and its level curves are circumferences around the origin, the points that are closest to the origin in $xy = 1$ will be the global minima. These are $(\pm 1, \pm 1)$. This function does not have global maxima, since the restriction is not bounded and the function is increasing towards infinity in it.



2. We solve:

$$\begin{cases} y = 2\lambda x \\ x = 8\lambda y \\ x^2 + 4y^2 = 1 \end{cases}$$

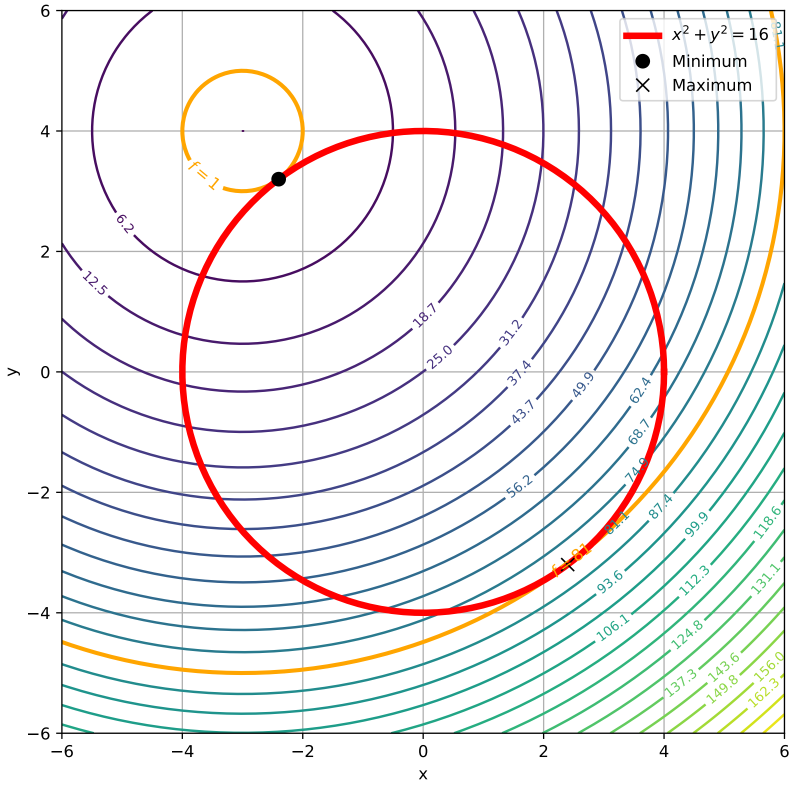
and obtain $x = \pm 2y = \pm\sqrt{2}$, which give us four critical points $(\sqrt{2}, 1/\sqrt{2})$, $(\sqrt{2}, -1/\sqrt{2})$, $(-\sqrt{2}, 1/\sqrt{2})$ and $(-\sqrt{2}, -1/\sqrt{2})$. In order to know whether they are maxima or minima, we evaluate their values: since the restriction is closed and bounded, f must obtain a global maximum and a global minimum in it. And, since f is differentiable at every point in the restriction, these maxima and minima must be among the critical points found using Lagrange multipliers. Evaluating the values of f at the critical points, we find that $(\sqrt{2}, 1/\sqrt{2})$ and $(-\sqrt{2}, -1/\sqrt{2})$ are global maxima, while $(-\sqrt{2}, 1/\sqrt{2})$, $(\sqrt{2}, -1/\sqrt{2})$ are global minima.



Exercise 7.2 First, we find the critical points of f, namely those that satisfy $\nabla f = \mathbf{0}$. This occurs when $x = -3$, $y = 4$, but this point is outside the region of interest, as $3^2 + 4^2 > 16$. We solve the system:

$$\begin{cases} 2x + 6 = 2\lambda x \\ 2y - 8 = 2\lambda y \\ x^2 + y^2 = 16 \end{cases}$$

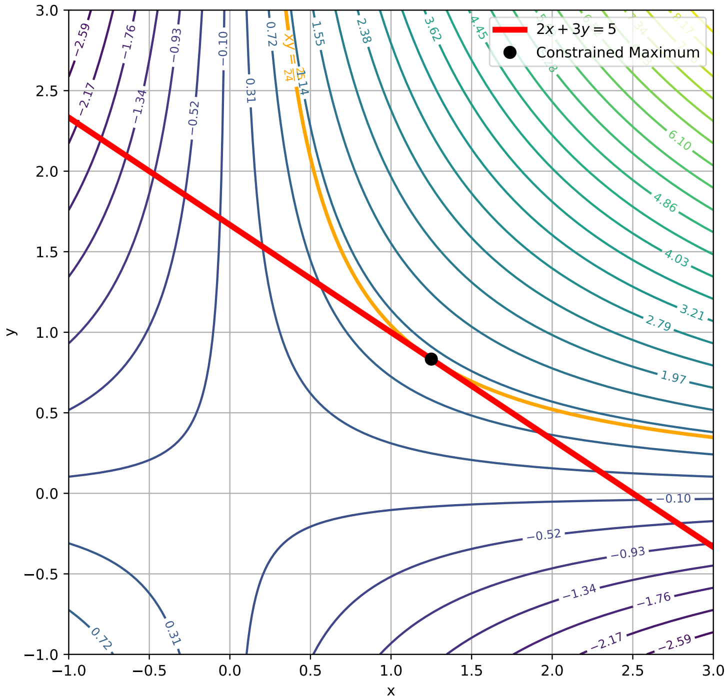
and obtain $x = -3y/4 = \pm 12/5$. Since the restriction is closed and bounded, f must reach a global maximum and a minimum, and since f is differentiable everywhere, they must be reached at the critical points found using Lagrange multipliers. Evaluating f at these points, we find the global minimum $(-12/5, 16/5)$ and the global maximum $(12/5, -16/5)$.



Exercise 7.3 1. We solve the system:

$$\begin{cases} y = 2\lambda \\ x = 3\lambda \\ 2x + 3y - 5 \end{cases}$$

and obtain $x = 5/4$, $y = 5/6$, which gives us the global maximum $(5/4, 5/6)$. There is no minimum: the function decreases towrads $-\infty$ as we move on the restriction away from the critical point.



2. We solve the system:

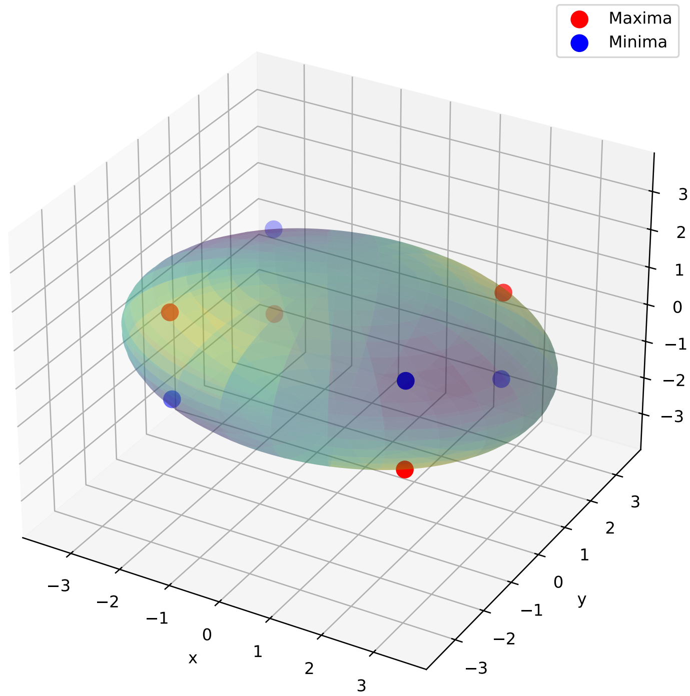
$$\begin{cases} yz = 2\lambda x/a^2 \\ xz = 2\lambda y/b^2 \\ xy = 2\lambda z/c^2 \\ x^2/a^2 + y^2/b^2 + z^2/c^2 = 1 \end{cases}$$

and obtain $x^2 = a^2/3$, $y^2 = b^2/3$, $z^2 = c^2/3$, which make eight critical points. As in previous exercises, the restriction is closed and bounded, and so f, being differentiables, reaches its global maxima and minima among the points found using lagrange multipliers. From this, the global maxima are:

$$\frac{1}{\sqrt{3}}(|a|,|b|,|c|), \quad \frac{1}{\sqrt{3}}(-|a|,-|b|,|c|), \quad \frac{1}{\sqrt{3}}(-|a|,|b|,-|c|), \quad \frac{1}{\sqrt{3}}(|a|,-|b|,-|c|);$$

and the global minima:

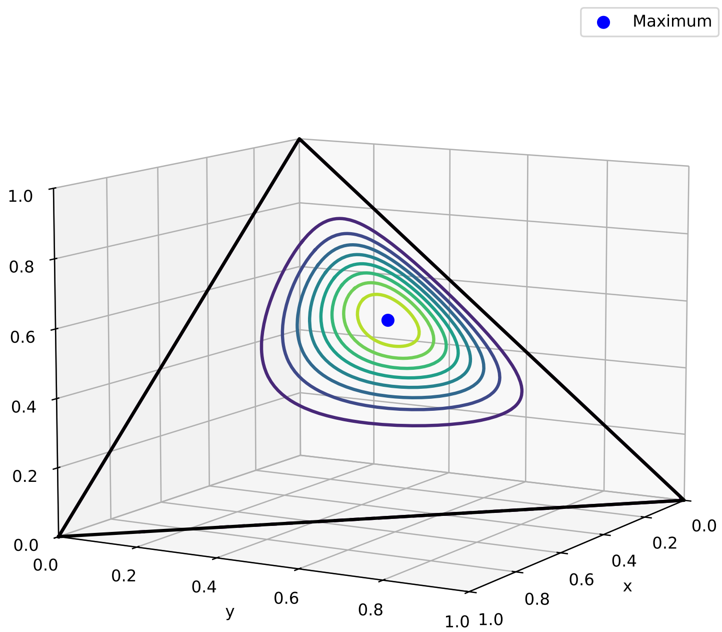
$$\frac{1}{\sqrt{3}}(-|a|,|b|,|c|), \quad \frac{1}{\sqrt{3}}(-|a|,-|b|,|c|), \quad \frac{1}{\sqrt{3}}(|a|,|b|,-|c|), \quad \frac{1}{\sqrt{3}}(-|a|,-|b|,-|c|).$$



3. We solve:

$$\begin{cases} 2xy^4z^6 = \lambda \\ 4x^2y^3z^6 = \lambda \\ 6x^2y^4z^5 = \lambda \\ x + y + z = 1 \end{cases}$$

and, since $x, y, z > 0$ we obtain $x = 1/6$, $y = 1/3$, $z = 1/2$. Now, f reaches a minimum value of 0 at the edges of the triangle formed by the planes $x + y = 1$, $x + z = 1$, $y + z = 1$ and $x + y + z = 1$. f is continuous in this triangle, so it reaches a maximum and a minimum value. Since inside the triangle f is differentiable, the maximum must be the critical point found by Lagrange multipliers, at $(1/6, 1/3, 1/2)$.



Exercise 7.4 1. To find the critical points in the interior, we solve:

$$\nabla f = (2x - 2, 2y - 2) = 0$$

and obtain $x = y = 1$. Notice that $f(x, y) = (x-1)^2 + (y-1)^2$, so $(1, 1)$ is a global minimum. At the boundary, we use the method of Lagrange multipliers for each piece:

i) For the left boundary $y = 2x$, so the restriction is $g_1(x, y) = 0$ with $g_1 = 2x - y$. We solve

$$\begin{cases} 2x - 2 = 2\lambda \\ 2y - 2 = -\lambda \\ 2x = y \end{cases}$$

to obtain a minimum at $(3/5, 6/5)$. We also need to check the extremes of this segment, as f restricted to the line $y = 2x$ does not have critical points there. At $(0, 0)$ the function equals 2, while at $(1, 2)$ it equals 1.

ii) For the lower boundary $y = 0$, so the restriction is $g_1(x, y) = 0$ with $g_1 = y$. We solve

$$\begin{cases} 2x - 2 = 0 \\ 2y - 2 = \lambda \\ y = 0 \end{cases}$$

to obtain a minimum at $(1, 0)$. We also need to check the extremes of this segment, as f restricted to the line $y = 0$ does not have critical points there. At $(0, 0)$ the function equals 2, while at $(3, 0)$ it equals 5.

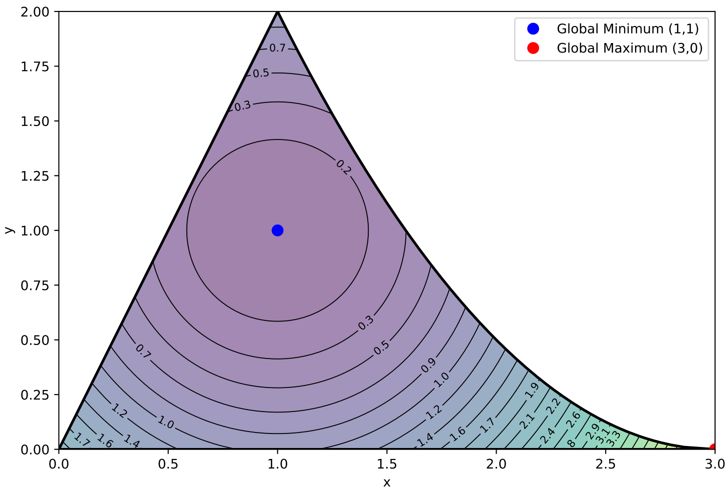
iii) For the right boundary $y = (3 - x)^2/2$, so the restriction is

$g_2(x, y) = 0$ with $g_1 = y - (3-x)^2/2$. We solve

$$\begin{cases} 2x - 2 = (3-x)\lambda \\ 2y - 2 = \lambda \\ y = \frac{(3-x)^2}{2} \end{cases}$$

to obtain a minimum at $(3 - \sqrt[3]{4}, \sqrt[3]{16}/2)$. We have already checked the extremes of this segment.

After considering all points, the global minimum is at $(1, 1)$ and the global maximum at $(3, 0)$.



2. We solve the system:

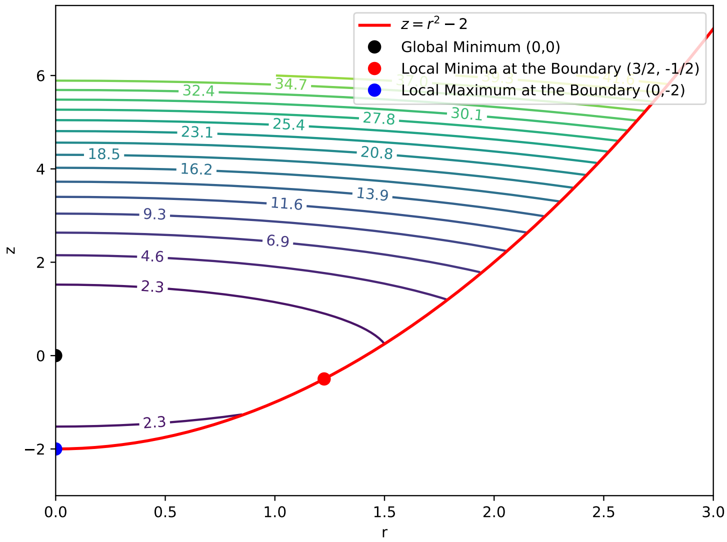
$$\begin{cases} 1 = 4\lambda x \\ 1 = 6\lambda y \\ 1 = 12\lambda z \\ 2x^2 + 3y^2 + 6z^2 = 1 \end{cases}$$

and obtain $x = \pm 1/2$, $y = \pm 1/3$, $z = \pm 1/6$, so eight critical points. We evaluate the function at these points and obtain the global maximum at $(1/2, 1/3, 1/6)$ and the global minimum at $(-1/2, -1/3, -1/6)$.
3. This function has a global minimum at $(0, 0, 0)$, which belongs to U. At the boundary, the restriction is $g(x, y, z) = 2$, where $g(x, y, z) = x^2 + y^2 - z$. We solve

$$\begin{cases} 2x = 2\lambda x \\ 2y = 2\lambda y \\ 2z = -\lambda \\ x^2 + y^2 - z = 2 \end{cases}$$

since $\lambda \neq 0$, then unless $x = 0$ and $y = 0$ simultaneously the system yields $\lambda = 1$ and therefore $z = -1/2$. Substituting into the restriction we obtain $x^2 + y^2 = 3/2$, which satisfies the equations. At this circumference of critical points, the function has the (minimal) value $7/4$.

If $\lambda \neq 0$ but $x = y = 0$, we have $z = -2$, which yields $f(0,0,-2) = 4$. However, this is not a maximum, as U is not bounded and f grows with z.
An easier way to visualize this is with cylindrical coordinates: $r^2 = x^2 + y^2$ and so $f(r,z) = r^2 + z^2$, with $U = \{z \geq r^2 - 2\}$. The set U is now the interior of a parabola where f grows unbounded. The points found with Lagrange multipliers become the point $(3/2, -1/2)$ (local minimum) and the point $(0,-2)$ (local maximum).



Exercise 7.5 We solve the system:

$$\begin{cases} yz = 2\lambda x \\ xz = 2\lambda y \\ xy = 2\lambda z \\ x^2 + y^2 + z^2 = 1 \end{cases}$$

to obtain the solution $x^2 = y^2 = z^2 = 1/3$. The points that maximize the value of f are

$$\frac{1}{\sqrt{3}}(1,1,1), \quad \frac{1}{\sqrt{3}}(-1,-1,1), \quad \frac{1}{\sqrt{3}}(1,-1,-1), \quad \frac{1}{\sqrt{3}}(-1,1,-1),$$

with a maximum value of $M = \sqrt{3}/9$.

Exercise 7.6 We solve the system:

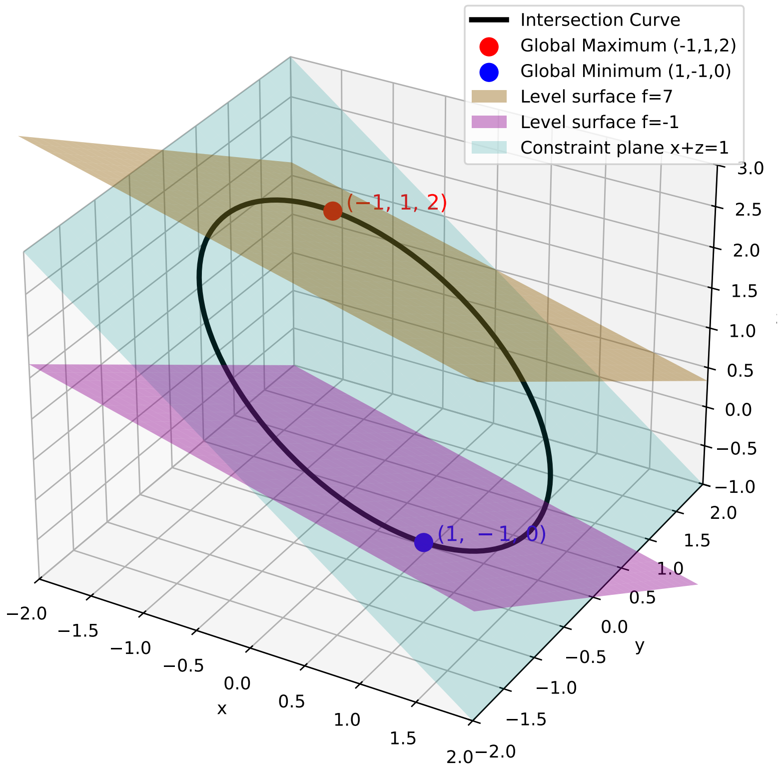
$$\begin{cases} 1 = 2\lambda x \\ 2 = 2\lambda y \\ 4 = 2\lambda z \\ x^2 + y^2 + z^2 = 1 \end{cases}$$

to obtain the solution $x^2 = 1/21, y^2 = 4/21, z^2 = 16/21$. The global minimum of $-\sqrt{21}$ is obtained at $\frac{-1}{\sqrt{21}}(1,2,4)$.

Exercise 7.7 The two restrictions intersect to form an ellipsoidal curve in $\mathbb{R}^3$. We solve the system:

$$\begin{cases} 1 = 2\lambda_1 x + \lambda_2 \\ 2 = 2\lambda_1 y \\ 3 = \lambda_2 \\ x^2 + y^2 = 2 \\ x + z = 1 \end{cases}$$

and obtain $x = \pm 1 = -y = 1 - z$, so we have two critical points, $(-1, 1, 2)$ and $(1, -1, 0)$. This yields the maximum $M = f(-1, 1, 2) = 7$ and the minimum $m = f(1, -1, 0) = -1$. Note that the level surfaces of f are tangent to the curve at these points:



Exercise 7.8

1. The required distance d is the distance from the point to the center of the sphere $(1, 0, -2)$ minus its radius. The distance to the center is

$$\|P - C\| = \sqrt{(4-1)^2 + (4-0)^2 + (10+2)^2} = 13,$$

so the distance to the surface is $d = 8$.

2. We have to minimize the function (squared distance):

$$f(x, y, z) = (x-4)^2 + (y-4)^2 + (z-10)^2$$

on the restriction $h(x, y, z) = (x-1)^2 + y^2 + (z+2)^2 - 25 = 0$ with Lagrange multipliers.

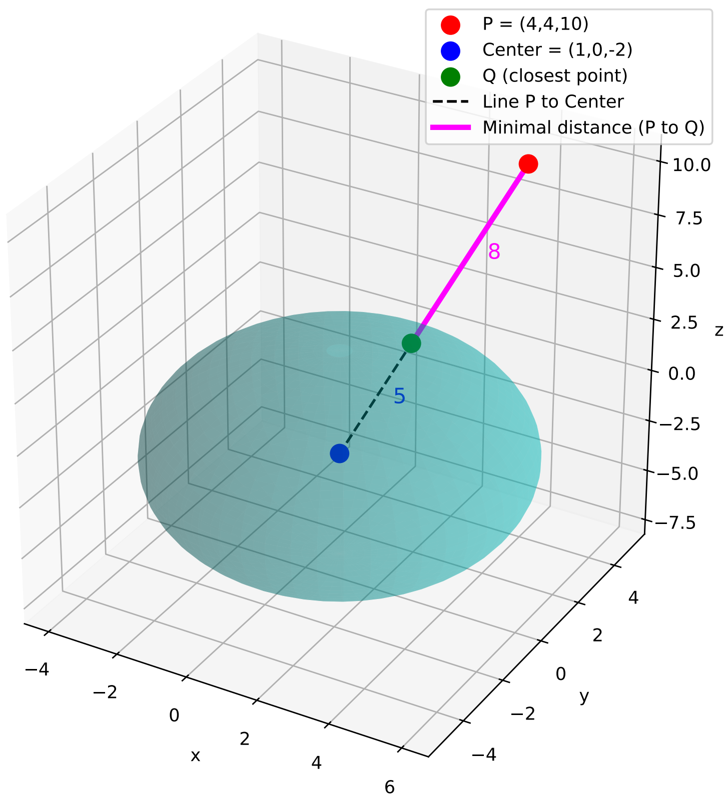
Note that the distance function to the point $(4, 4, 10)$ is $g(x, y, z) = \sqrt{(x-4)^2 + (y-4)^2 + (z-10)^2}$, but the minimal values of f and g are the same (the square root is an injective function and it does not introduce new extrema), and it is easier to operate with f. The system is

$$\begin{cases} 2(x-4) = 2\lambda(x-1) \\ 2(y-4) = 2\lambda y \\ 2(z-10) = 2\lambda(z+2) \\ (x-1)^2 + y^2 + (z+2)^2 = 25 \end{cases}$$

and we obtain two solutions:

$$P_1 = \left(\frac{28}{13}, \frac{20}{13}, \frac{34}{13}\right), \quad P_2 = \left(\frac{-2}{13}, \frac{-20}{13}, \frac{-86}{13}\right),$$

with $f(P_1) = 64$ and $f(P_2) = 324$, which correspond to the closest and the farthest points of the sphere to the given point, with distances 8 and 18, respectively.



Exercise 7.9 Minimize $S(a, b, c, d) = a + b + c + d$ with the restriction $g(a, b, c, d) = abcd - A = 0$. We solve

$$\begin{cases} 1 = \lambda bcd \\ 1 = \lambda acd \\ 1 = \lambda abd \\ 1 = \lambda abc \\ abcd = A \end{cases}$$

and obtain

$$a = b = c = d = A^{1/4}.$$

By giving values to a, b, c, d subject to $abcd = A$ (which is an unbounded set) we can see that any other point yields a larger value of S. For instance, by fixing $c = d = 1$ and choosing $b = A/a$, we can make a (and therefore S) as large as we want while still satisfying the restriction. Therefore, the point $(A^{1/4}, A^{1/4}, A^{1/4}, A^{1/4})$ is a global minimum.

Exercise 7.10 The restriction is $Kp + Lq = B$, so we have to solve

$$\begin{cases} \alpha K^{\alpha-1}L^{1-\alpha} = \lambda p \\ (1-\alpha)K^{\alpha}L^{-\alpha} = \lambda q \\ Kp + Lq = B \end{cases}$$

which yields

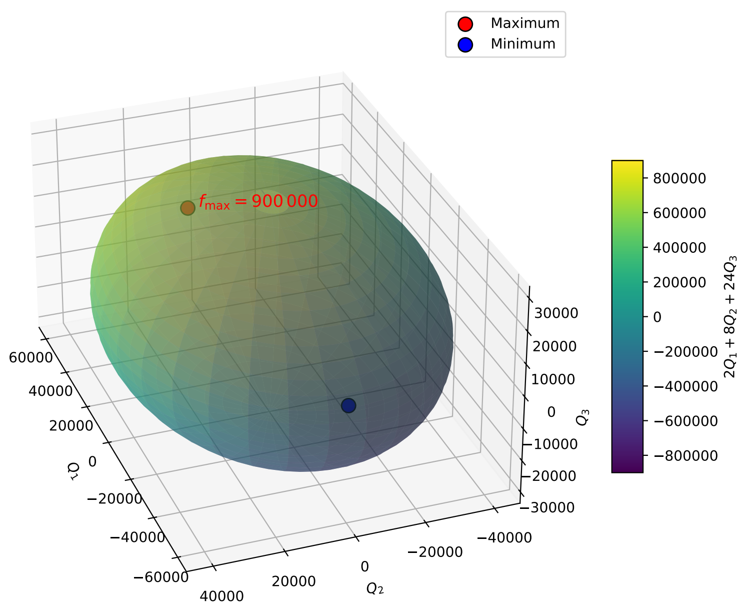
$$\frac{K}{L} = \frac{\alpha q}{(1-\alpha)p}.$$

Exercise 7.11 We solve:

$$\begin{cases} 2 = 2\lambda Q_1 \\ 8 = 4\lambda Q_2 \\ 24 = 8\lambda Q_3 \\ Q_1^2 + 2Q_2^2 + 4Q_3^2 = C \end{cases}$$

where $C = 4.5 \times 10^9$, and obtain two points: $(10^4, 2 \cdot 10^4, 3 \cdot 10^4)$ and $(-10^4, -2 \cdot 10^4, -3 \cdot 10^4)$. Knowing that P is differentiable and that the set $Q_1^2 + 2Q_2^2 + 4Q_3^2 = C$ is closed and bounded, we deducae that one point must be the maximum, and the other the minimum. Evaluating the function, we obtain that the values Q_1, Q_2, Q_3 that maximize production are

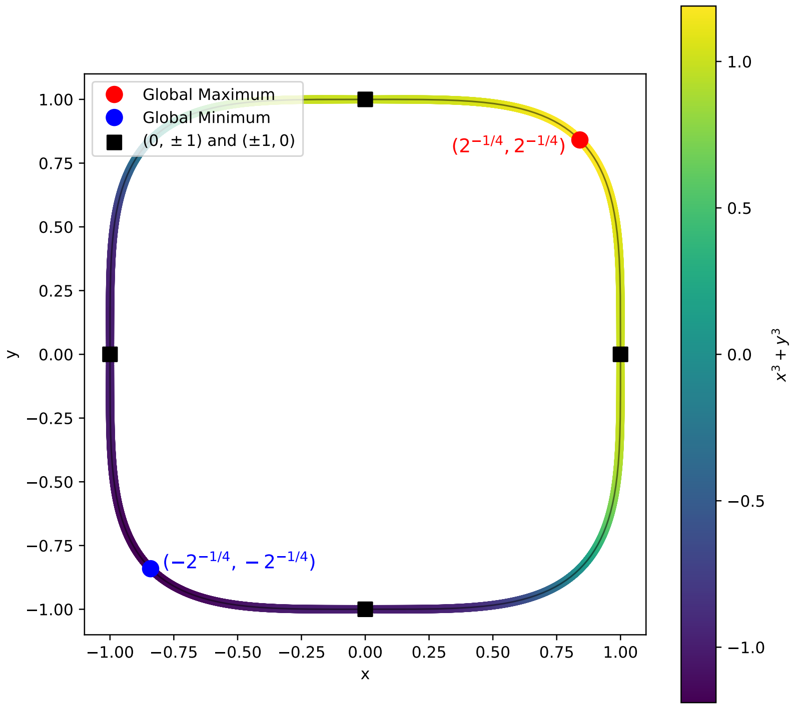
$$Q_1 = \pm 10^4, \quad Q_2 = 2 \pm Q_1, \quad Q_3 = 3Q_1.$$



Exercise 7.12 We solve the system

$$\begin{cases} 3x^2 = 4\lambda x^3 \\ 3y^3 = 4\lambda y^3 \\ x^4 + y^4 - 1 = 0 \end{cases}$$

and obtain $x = 0, y = \pm 1$, or $x = \pm 1, y = 0$ or $x = y = \pm 2^{-1/4}$. Evaluating the function at these six points, we find the global maximum $M = 2^{1/4}$ at $(2^{-1/4}, 2^{-1/4})$ and the global minimum $m = -2^{1/4}$ at $(-2^{-1/4}, -2^{-1/4})$. The points $(0, \pm 1)$ and $(\pm 1, 0)$ are inflection points of f restricted to $x^4 + y^4 = 1$.

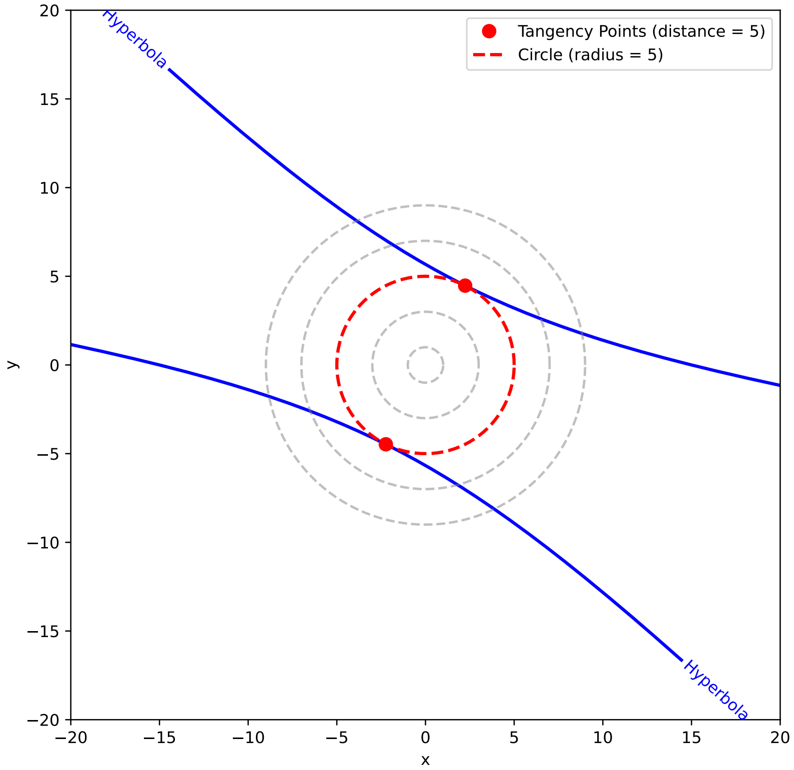


Exercise 7.13 We have to minimize the function $f(x, y) = x^2 + y^2$ (see Exercise 7.8 above) restricted to the hyperbola, so we solve the system

$$\begin{cases} 2x = \lambda(2x + 8y) \\ 2y = \lambda(8x + 14y) \\ x^2 + 8xy + 7y^2 = 225 \end{cases}$$

and obtain $x = -2y$ which does not satisfy the restriction, or $x = y/2$, which gives the minimal distance $d = 5$ at

$$(\sqrt{5}, 2\sqrt{5}) \quad \text{and} \quad (-\sqrt{5}, -2\sqrt{5}).$$


Tangency Points (distance = 5)
Circle (radius = 5)
Hyperbola
Hyperbola
x
y

A.8 Calculating volumes: Double Integrals

Exercise 8.1 1.

$$\int_0^1 \int_0^1 xy(x+y)\,dy\,dx = \int_0^1 \left(x\frac{y^2}{2} + \frac{y^3}{3}\right)\Bigg|_0^1 dx = \int_0^1 \left(\frac{x^2}{2} + \frac{x}{3}\right) dx = \frac{x^3}{6} + \frac{x^2}{6}\Bigg|_0^1 = \frac{1}{3}.$$

2.

$$\int_0^1 \int_0^1 (x^3+3x^2y+y^3)\,dy\,dx = \int_0^1 \left(x^3y + \frac{3x^2y^2}{2} + \frac{y^4}{4}\right)\Bigg|_0^1 dx = \int_0^1 x^3+\frac{3}{2}x^2+\frac{1}{4}\,dx = \frac{x^4}{4}+\frac{x^3}{2}+\frac{x}{4}\Bigg|_0^1 = 1.$$

3. Since $f(x,y) = \sin^2 x \sin^2 y = g(x)g(y)$ where $g(x) = \sin^2 x$ we can write

$$\int_0^\pi \int_0^\pi (\sin^2 x \sin^2 y)\,dy\,dx = \left(\int_0^\pi \sin^2 x dx\right)^2 = \left(\frac{1}{2}\int_0^\pi (1-\cos 2x)\,dx\right)^2 = \frac{\pi^2}{4}.$$

4.

$$\int_0^{\pi/2} \int_0^{\pi/2} \sin(x+y)\,dy\,dx = \int_0^{\pi/2} (\cos x - \cos(x+\pi/2))\,dx = \sin x - \sin(y+\pi/2)\big|_0^{\pi/2} = 2.$$

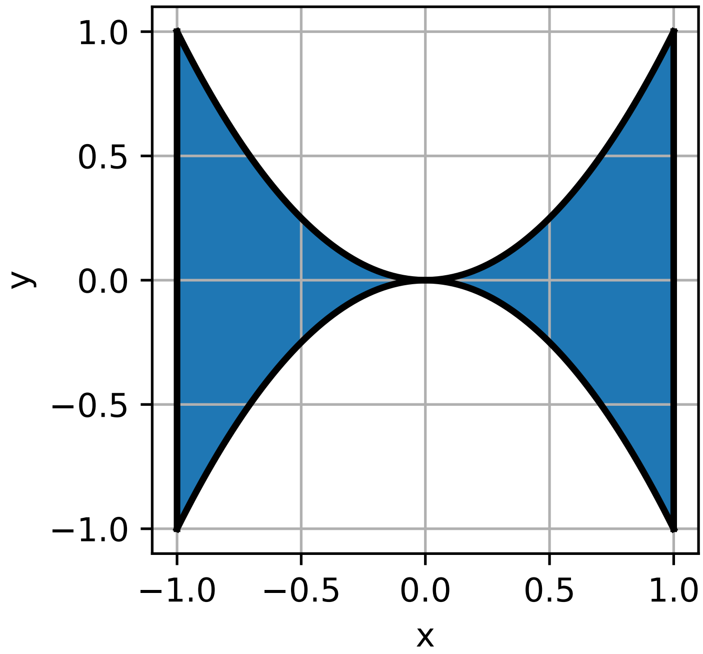
5.

$$\int_{-1}^1 \int_0^{\pi/2} x\sin y - ye^x\,dy\,dx = \int_{-1}^1 \left(-\frac{\pi^2}{8}e^x + x\right) dx = -\frac{\pi^2(e-e^{-1})}{8} = -\frac{\pi^2}{4}\sinh 1.$$

Exercise 8.2 1.

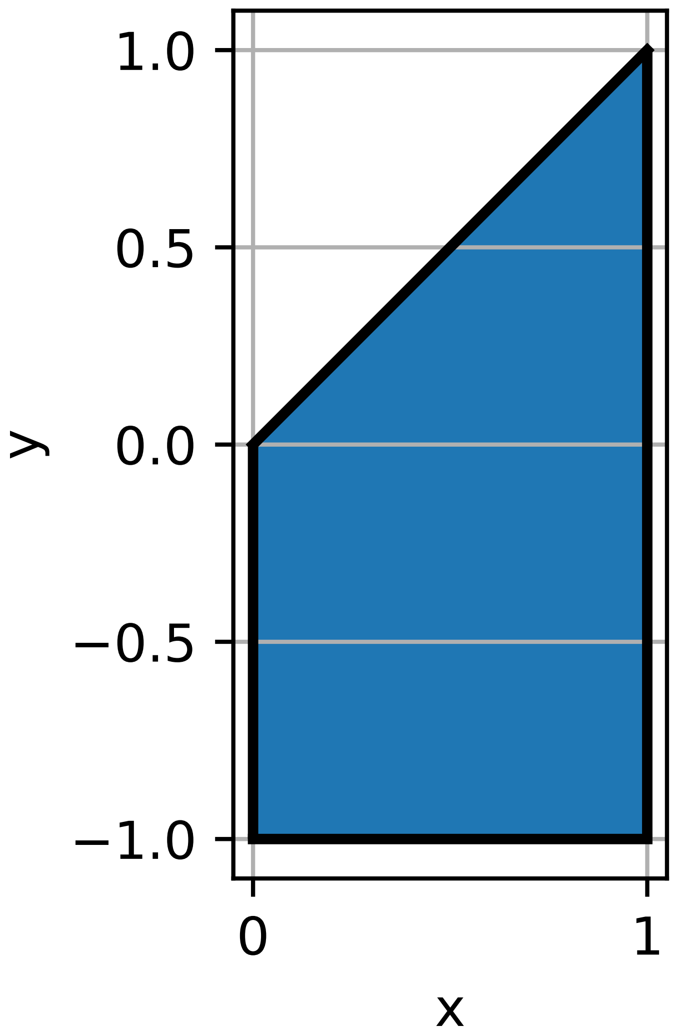
$$\int_{-1}^1 \int_{-x^2}^{x^2} (x^2-y)\,dy\,dx = \int_{-1}^1 x^2y - \frac{y^2}{2}\Big|_{-x^2}^{x^2} dx = \int_{-1}^1 2x^4\,dx = \frac{4}{5}.$$

The region is



2.
$$\int_0^1 \int_0^x (xy - x^3)\, dy\, dx = \int_0^1 \left(x^4 + \frac{x^3}{2} + \frac{x}{2} \right) dx = \frac{-23}{40};$$

The region is



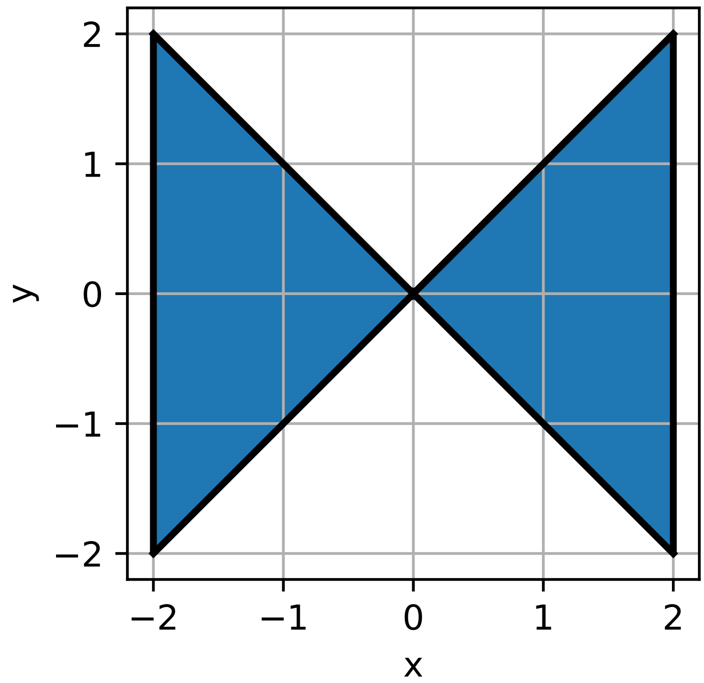
3. The integral is

$$I = \int_{-2}^{0}\int_{x}^{-x}(2x - \sin(x^2 y))\,dy\,dx + \int_{0}^{2}\int_{-x}^{x}(2x - \sin(x^2 y))\,dy\,dx.$$

Therefore,

$$\begin{aligned} I &= \int_{-2}^{0}\left(2xy + \frac{\cos x^2 y}{x^2}\right)\Big|_{x}^{-x} dx + \int_{0}^{2}\left(2xy + \frac{\cos x^2 y}{x^2}\right)\Big|_{-x}^{x} dx \\ &= -\int_{-2}^{0} 4x^2\,dx + \int_{0}^{2} 4x^2\,dx = 0. \end{aligned}$$

The region of integration is

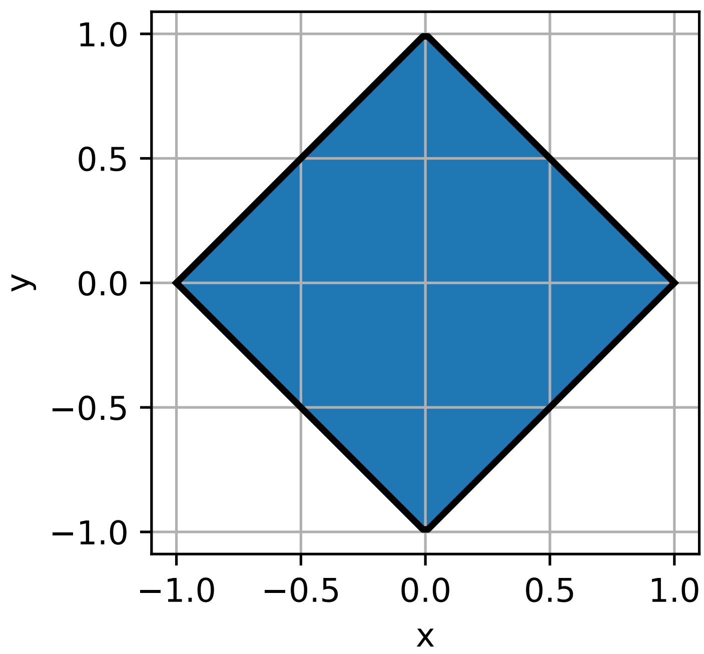


We could also argue that $2x$ is an odd function of x and the region of integration is symmetric with respect to the y axis. Similarly, $\sin(x^2 y)$ is and odd function of y and the region of integration is symmetric with respect to y. So the integral must be zero.

4.

$$\int_{-1}^{1}\int_{-1+|x|}^{1-|x|} y \sin x \, dy \, dx = \frac{1}{2}\int_{-1}^{1} \sin x[(1-|x|)^2-(-1+|x|)^2] \, dx = 0.$$

The region of integration is



We could also argue that $\sin x$ is an odd function of x and the region of integration is symmetric with respect to the y axis. Similarly, y is and odd function of y and the region of integration is symmetric with respect to y. So the integral must be zero.

Exercise 8.3 1. Since f is bounded by its maximum and minimum values in D:

$$m \leq f \leq M,$$

where

$$M = \max_D f = 5, \quad m = \min_D f = 1,$$

the integral of f in D will also be bounded by the integrals of the constant functions m and M:

$$\iint_D m\, dA \leq \iint_D f(x)\, dA \leq \iint_D M\, dA$$

Now,

$$\iint_D m\, dA = m \iint_D dA = mA(D) = 4\pi,$$

since $\iint_D dA$ is equal to the area of D and $A(D) = 4\pi$. Similarly,

$$\iint_D M\, dA = MA(D) = 20\pi,$$

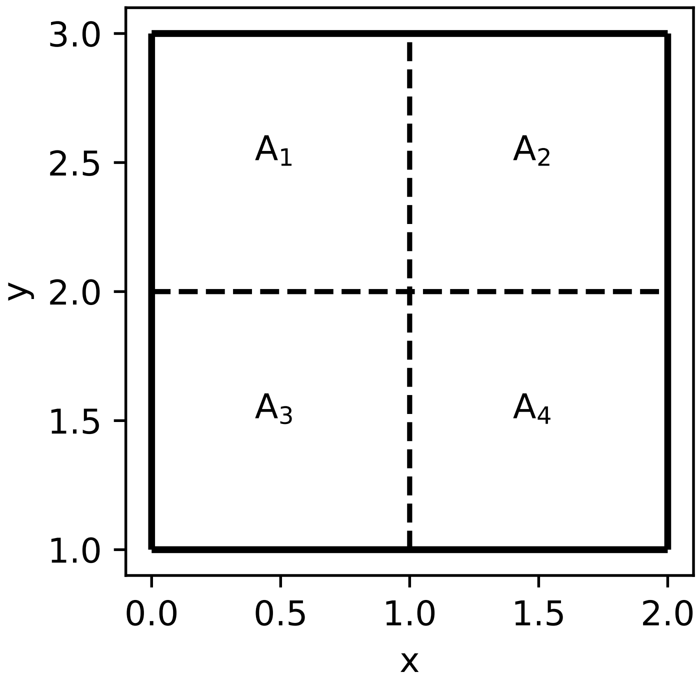
which yields the result.
We can use polar coordinates to calculate the true value of the integral, obtaining 12π.

2. In this case, $M = 12$, $m = 0$, $A(A) = 4$, and applying the previous result we obtain the result. For comparison, the true value of the integral is $32/3$.
3. We split A as:

$$A = A_1 \cup A_2 \cup A_3 \cup A_4$$

where:

$$A_1 = [0,1]\times[1,2], \quad A_2 = [1,2]\times[1,2], \quad A_3 = [0,1]\times[2,3], \quad A_4 = [1,2]\times[2,3].$$



The integral becomes

$$\iint_A x^2y\,dxdy = \iint_{A_1} x^2y\,dxdy + \iint_{A_2} x^2y\,dxdy + \iint_{A_3} x^2y\,dxdy + \iint_{A_4} x^2y\,dxdy,$$

and so we can bound each separately.
Evaluating f in each square, we have:

$$m_1 = 0, \quad M_1 = 2, \quad m_2 = 1, \quad M_2 = 8, \quad m_3 = 0, \quad M_3 = 3, \quad m_4 = 2, \quad M_4 = 12,$$

with $A(A_k) = 1$ for all $k = 1, 2, 3, 4$. So:

$$0 \le \iint_{A_1} x^2y\,dxdy \le 2,$$

$$1 \le \iint_{A_2} x^2y\,dxdy \le 8,$$

$$0 \le \iint_{A_3} x^2y\,dxdy \le 3,$$

$$2 \le \iint_{A_4} x^2y\,dxdy \le 12,$$

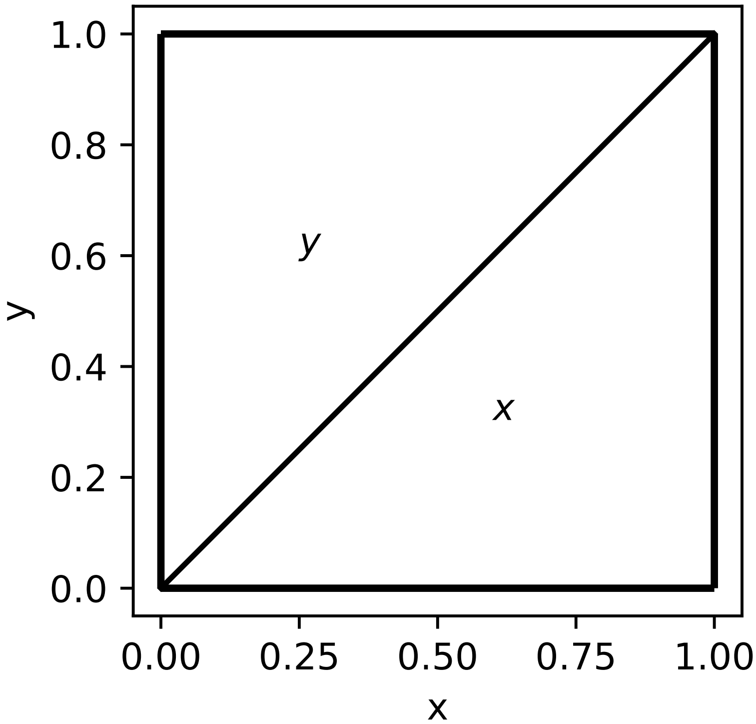
which leads to

$$0 + 1 + 0 + 2 \le \iint_A f\,dA \le 2 + 8 + 3 + 12,$$

or

$$3 \le \iint_A f\,dA \le 25.$$

Exercise 8.4 The function takes different values below and above the $y = x$ line:



Now, the integrals $\int_0^1 \int_0^x x \, dy dx$ and $\int_0^1 \int_x^1 y \, dy dx$ are equal, as we can see if we change the order of integration in the second integral:

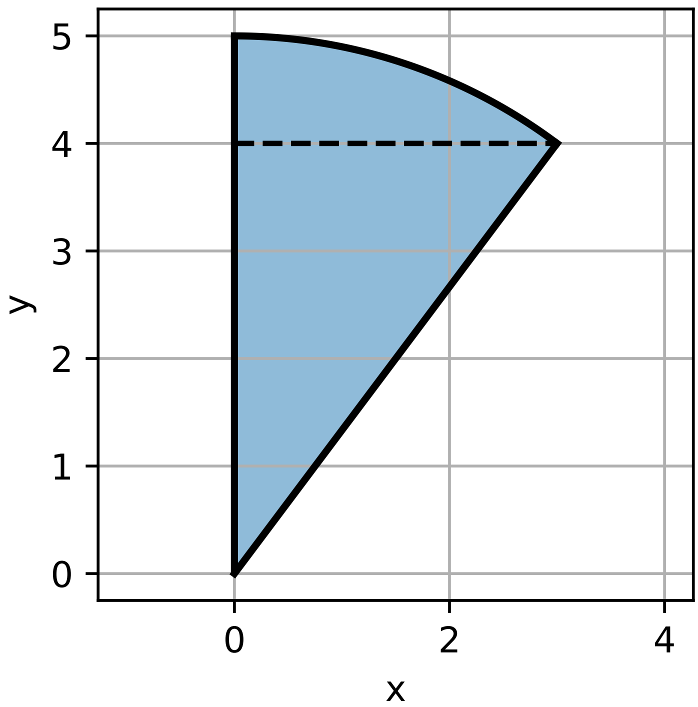
$$\int_0^1 \int_x^1 y \, dy dx = \int_0^1 \int_0^y y \, dx dy = \int_0^1 \int_0^x x \, dy dx.$$

Finally,

$$I = 2 \int_0^1 \int_0^x x \, dy \, dx = 2 \int_0^1 x^2 \, dx = \frac{2}{3}.$$

A.9 Changing the order of integration and Triple Integrals

Exercise 9.1 1. The region of integration is



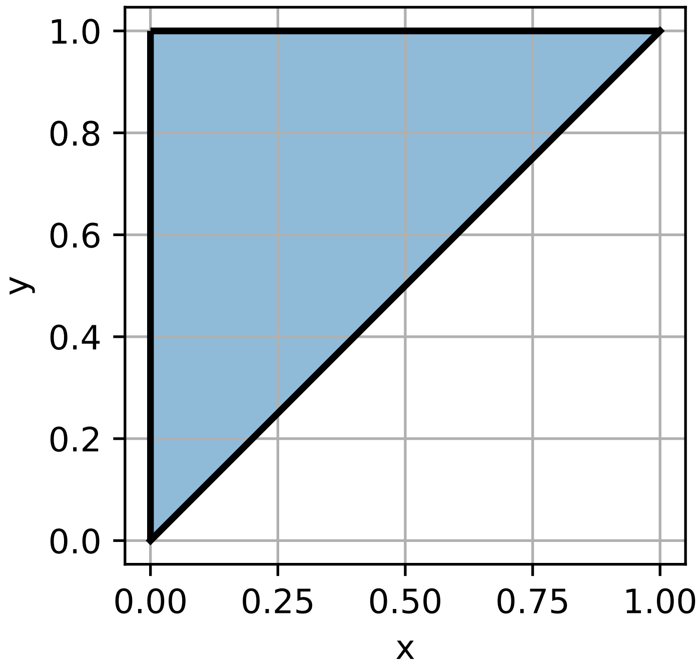
Now,

$$0 \leq x \leq 3, \qquad \frac{4x}{3} \leq y \leq \sqrt{25 - x^2},$$

which, if we change the order of integration, while the left limit is always $x = 0$, the right limit of integration is formed by two functions, $x = \sqrt{25 - y^2}$ and $x = 3y/4$. So we need to separate the integral in two:

$$\int_0^3 \int_{4x/3}^{\sqrt{25-x^2}} f(x,y)\,dy\,dx = \int_0^4 \int_0^{3y/4} f(x,y)\,dx\,dy + \int_4^5 \int_0^{\sqrt{25-y^2}} f(x,y)\,dx\,dy.$$

2. The region of integration is



The integration limits are

$$0 \le y \le 1, \qquad 0 \le x \le y,$$

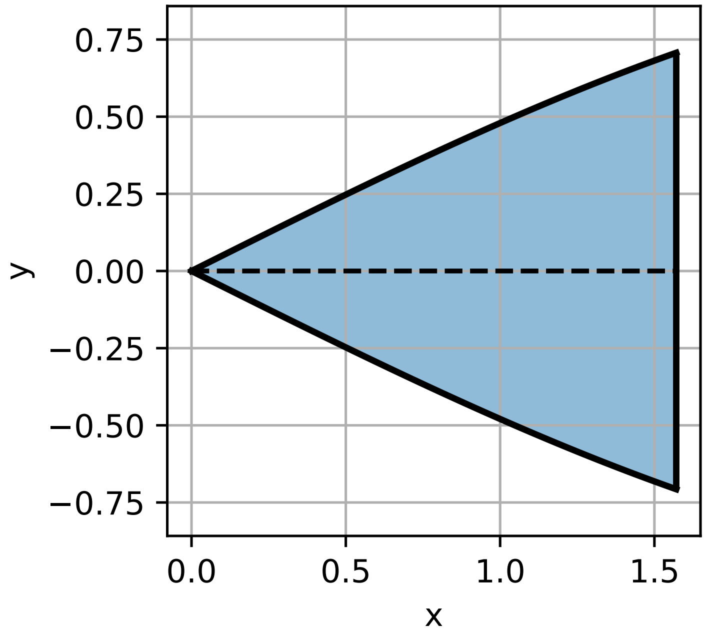
which, changing the order of integration, become

$$0 \le x \le 1, \qquad x \le y \le 1.$$

Therefore,

$$\int_0^1 \int_0^y f(x,y)\,dx\,dy = \int_0^1 \int_x^1 f(x,y)\,dy\,dx.$$

3. The region of integration is



Now,

$$0 \leq x \leq \frac{\pi}{2}, \qquad -\sin(x/2) \leq y \leq\in (x/2),$$

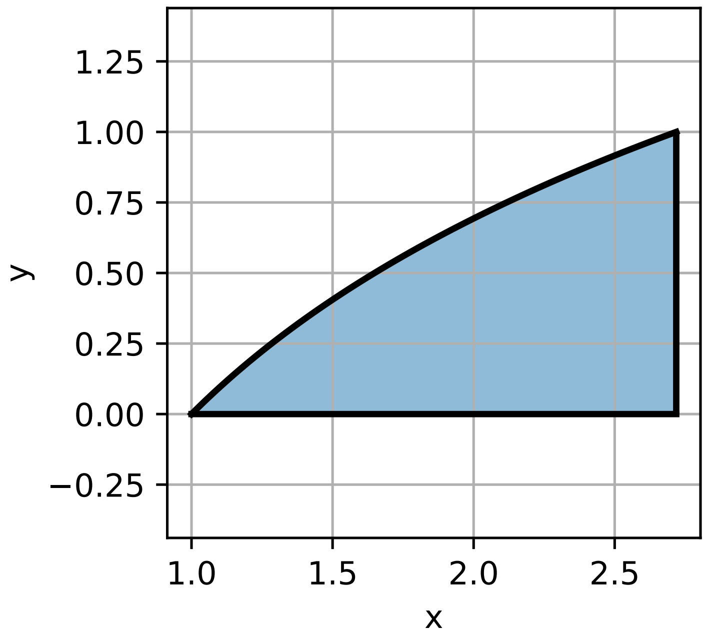
which, if we change the order of integration, while the right limit is always $x = \pi/2$, the left limit of integration is formed by two functions, $x = 2\arcsin y$ (in the positive part of the axis) and $x = -2\arcsin y$ (in the negative part of the axis). Note that

$$y = \pm\sin(x/2) \implies \arcsin(\pm y) = \frac{x}{2} \implies \pm 2\arcsin y = x,$$

since $\arcsin y$ is an odd function of y. So we need to separate the integral in two:

$$\int_0^{\pi/2}\int_{-\sin(x/2)}^{\sin(x/2)} f(x,y)\,dy\,dx = \int_{-\sqrt{2}/2}^{0}\int_{-2\arcsin y}^{\pi/2} f(x,y)\,dx\,dy + \int_0^{\sqrt{2}/2}\int_{2\arcsin y}^{\pi/2} f(x,y)\,dx\,dy.$$

4. The region of integration is



Now, since

$$1 \le x \le e, \qquad 0 \le y \le \log x,$$

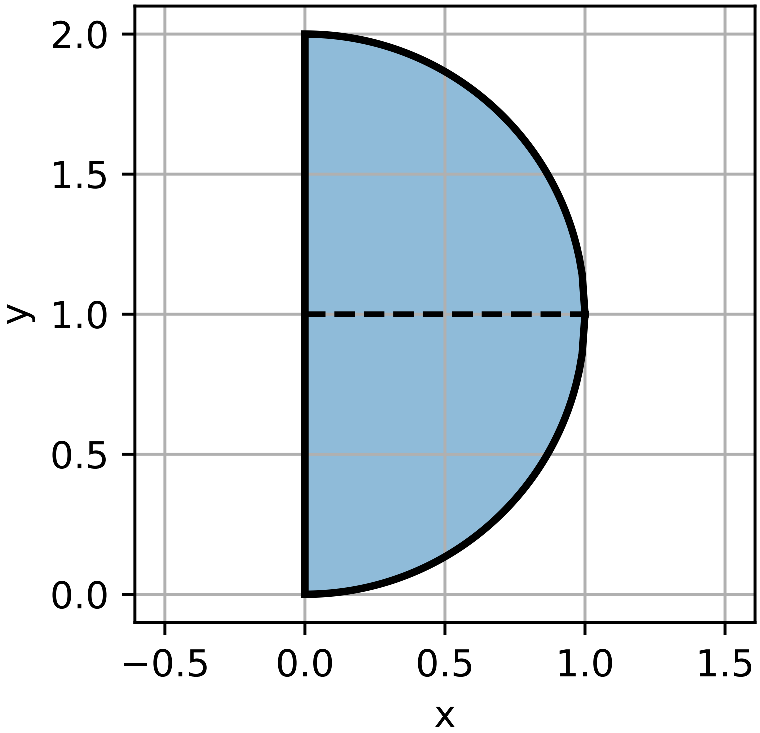
we can invert the relationship to write

$$0 \le y \le 1, \qquad e^y \le x \le e.$$

Therefore,

$$\int_1^e \int_0^{\log x} f(x,y)\, dy\, dx = \int_0^1 \int_{e^y}^e f(x,y)\, dx\, dy.$$

Exercise 9.2 The region of integration is half a circle of radius 1 centered at $(0, 1)$:



We can see that $0 \le x \le 1$ and that $1 - \sqrt{1-x^2} \le y \le 1 + \sqrt{1-x^2}$, so

$$\iint_R f(x,y)\,dA = \int_0^1 \int_{1-\sqrt{1-x^2}}^{1+\sqrt{1-x^2}} f(x,y)\,dy\,dx.$$

Alternatively, we can say that $0 \le y \le 2$ and $0 \le x \le \sqrt{1-(y-1)^2}$ and so

$$\iint_R f(x,y)\,dA = \int_0^2 \int_0^{\sqrt{1-(y-1)^2}} f(x,y)\,dx\,dy.$$

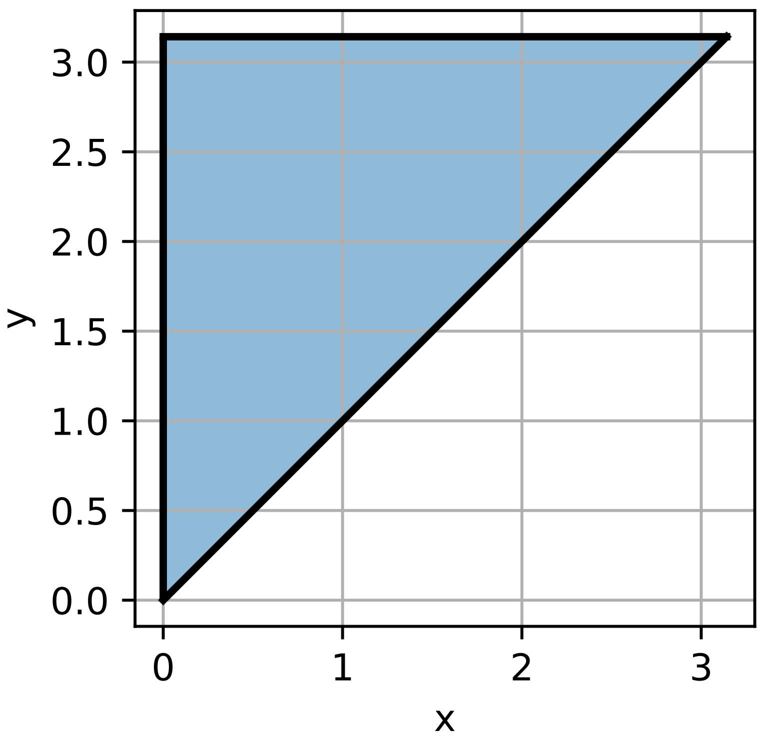
In both cases, for $f(x,y) = 1/\sqrt{1-x^2}$ and $g(x,y) = \sin(y-1)$ it is easier to integrate first with respect to y and then with respect to x:

$$\begin{aligned}\iint_R \frac{1}{\sqrt{1-x^2}}\,dA &= \int_0^1 \int_{1-\sqrt{1-x^2}}^{1+\sqrt{1-x^2}} \frac{1}{\sqrt{1-x^2}}\,dy\,dx \\ &= \int_0^1 \frac{1+\sqrt{1-x^2}-1+\sqrt{1-x^2}}{\sqrt{1-x^2}}\,dx = 2.\end{aligned}$$

$$\begin{aligned}\iint_R \sin(y-1)\,dA &= \int_0^1 \int_{1-\sqrt{1-x^2}}^{1+\sqrt{1-x^2}} \sin(y-1)\,dy\,dx \\ &= \int_0^1 [-\cos(1+\sqrt{1-x^2}-1) + \cos(1-\sqrt{1-x^2}-1)]\,dx = \\ &= \int_0^1 [-\cos(\sqrt{1-x^2}) + \cos(-\sqrt{1-x^2})]\,dx = 0\end{aligned}$$

since $\cos x$ is an even function of x. Alternatively, we could have reasoned that the integral would be zero since $\sin(y-1)$ is an odd function with respect to $y = 1$ and the region is symmetric with respect to $y = 1$.

Exercise 9.3 The function $f(y) = \frac{\sin y}{y}$ is called the cardinal sine or sinc y, which does not have a primitive. Therefore, we need to change the order of integration in order to calculate the integral. The region is



$$\int_0^\pi \int_x^\pi \frac{\sin y}{y}\, dy\, dx = \int_0^\pi \int_0^y \frac{\sin y}{y}\, dx\, dy = \int_0^\pi \sin y\, dy = 2.$$

Exercise 9.4 1. We can write the triple integral

$$\iiint_{[0,1]\times[0,1]\times[0,1]} (x^2 + y^2 + z^2)\, dx\, dy\, dz$$

as

$$\int_0^1 \int_0^1 \int_0^1 x^2\, dx\, dy\, dz + \int_0^1 \int_0^1 \int_0^1 y^2\, dx\, dy\, dz + \int_0^1 \int_0^1 \int_0^1 z^2\, dx\, dy\, dz.$$

Now these three integrals are identical, and they are equal to

$$\int_0^1 \int_0^1 \int_0^1 x^2\, dx\, dy\, dz = \int_0^1 x^2\, dx = \frac{1}{3}.$$

Finally,

$$\iiint_{[0,1]\times[0,1]\times[0,1]} (x^2+y^2+z^2)\, dx\, dy\, dz = 3\int_0^1 \int_0^1 \int_0^1 x^2\, dx\, dy\, dz = 1.$$

2.

$$\begin{aligned}
\iiint_{[0,1]\times[0,1]\times[0,1]} (x+y+z)^2\,dx\,dy\,dz &= \frac{1}{3}\int_0^1\int_0^1 (x+y+z)^3\Big|_0^1\,dx\,dy\\
&= \frac{1}{3}\int_0^1\int_0^1 (x+y+1)^3-(x+y)^3\,dx\,dy\\
&= \frac{1}{12}\int_0^1 (x+y+1)^4-(x+y)^4\Big|_0^1\,dx\\
&= \frac{1}{12}\int_0^1 (x+2)^4-2(x+1)^4+x^4\,dx\\
&= \frac{1}{60}[(x+2)^5-2(x+1)^5-x^5]_0^1\\
&= \frac{1}{60}[3^5-2\cdot 2^5-1] = \frac{5}{2}.
\end{aligned}$$

Exercise 9.5 1.

$$\iiint_W x^3\,dV = \int_0^1 x^3\,dx = \frac{1}{4}.$$

2.

$$\begin{aligned}
\iiint_W e^{-xy}y\,dV &= \int_0^1\int_0^1 ye^{-yx}\,dx\,dy\\
&= \int_0^1 [-e^{-yx}]_0^1\,dy\\
&= \int_0^1 [1-e^{-y}]\,dy\\
&= 1+e^{-y}|_0^1 = 1+e^{-1}-1 = e^{-1}.
\end{aligned}$$

3.

$$\begin{aligned}
\iiint_W (2x+3y+z)\,dV &= \int_1^2\int_{-1}^1 \left(2x+3y+\frac{1}{2}\right)dy\,dx\\
&= \int_1^2 (4x+1)\,dx = 2x^2\Big|_1^2+1 = 8-2+1 = 7.
\end{aligned}$$

4.

$$\begin{aligned}
\iiint_W ze^{x+y}\,dV &= \int_0^1 z\,dz\int_0^1 e^x\,dx\int_0^1 e^y\,dy\\
&= \frac{1}{2}(e-1)^2
\end{aligned}$$

Exercise 9.6

$$\begin{aligned}
\iiint_W x^2\cos x\,dV &= \int_0^1\int_0^{1-x}\int_0^{\pi} x^2\cos x\,dz\,dy\,dx\\
&= \pi\int_0^1\int_0^{1-x} x^2\cos x\,dy\,dx\\
&= \pi\int_0^1 (1-x)x^2\cos x\,dx = \pi\int_0^1 x^2\cos x\,dx - \int_0^1 x^3\cos x\,dx.
\end{aligned}$$

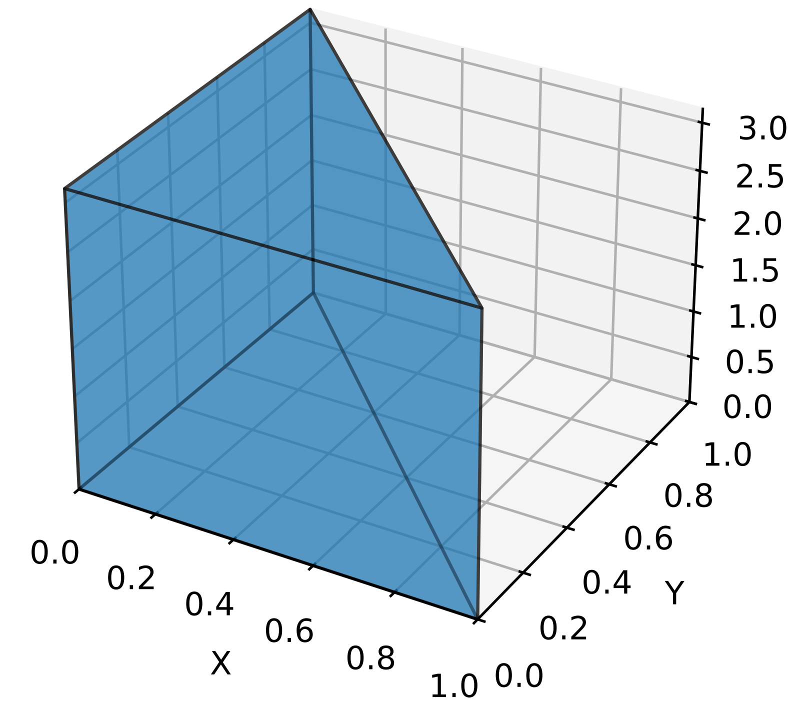
Now, using integration by parts we can prove that

$$\int_0^1 x^n\cos x\,dx = \sin 1 - n\int_0^1 x^{n-1}\sin x\,dx,$$

$$\int_0^1 x^n \sin x \, dx = -\cos 1 + n \int_0^1 x^{n-1} \cos x \, dx.$$

Substituting in the integral,

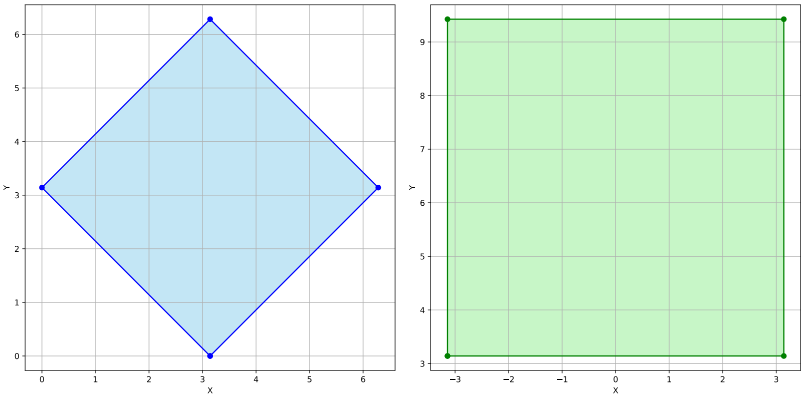
$$\begin{aligned}
\iiint_W x^2 \cos x \, dV &= \pi \int_0^1 x^2 \cos x \, dx - \int_0^1 x^3 \cos x \, dx \\
&= \pi \left[\cancel{\sin 1} - 2 \int_0^1 x \sin x \, dx \cancel{- \sin 1} + 3 \int_0^1 x^2 \sin x \, dx \right] \\
&= \pi \left[2 \cos 1 - 2 \int_0^1 \cos x \, dx - 3 \cos 1 + 6 \int_0^1 x \cos x \, dx \right] \\
&= \pi \left[-\cos 1 - 2 \sin 1 + 6 \sin 1 - 6 \int_0^1 \sin x \, dx \right] \\
&= \pi \left[5 \cos 1 + 4 \sin 1 - 6 \right].
\end{aligned}$$



A.10 Change of Variables

Exercise 10.1 The transformation $T(x, y) = (u(x, y), v(x, y)) = (x - y, x + y)$ transforms S into the rectangle $R = [-\pi, \pi] \times [-\pi, 3\pi]$ in the uv-plane. That is, $R = T(S)$, or $S = T^{-1}(R)$ and therefore

$$\iint_S (x - y)^2 \sin^2(x + y)\, dx\, dy = \iint_R u^2 \sin^2 v\, |J_{T^{-1}}|\; du\, dv.$$



Now,

$$|J_{T^{-1}}| = \frac{1}{|J_T|} = \frac{1}{2},$$

since

$$|J_T| = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & -1 \\ 1 & 1 \end{vmatrix} = 2.$$

Finally,

$$\begin{aligned} \iint_S (x - y)^2 \sin^2(x + y)\, dx\, dy &= \frac{1}{2} \int_{-\pi}^{\pi} u^2\, du \int_{-\pi}^{3\pi} \sin^2 v\, dv \\ &= \frac{1}{4} \left[\frac{u^3}{3} \right]_{-\pi}^{\pi} \int_{-\pi}^{3\pi} (1 - \cos 2v)\, dv \\ &= \frac{1}{4} \frac{2\pi^3}{3} [1 - \cos 2v]_{-\pi}^{3\pi} \\ &= \frac{\pi^4}{3}. \end{aligned}$$

Exercise 10.2 The transformation $T(x, y) = (u(x, y), v(x, y))$ given by

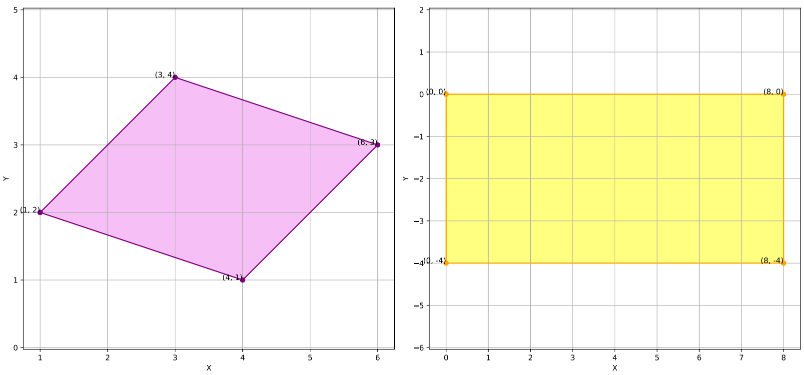
$$\begin{cases} u = x + 3y - 7 \\ v = -x + y - 1 \end{cases}$$

transforms the region D into the rectangle $D^* = [0, 8] \times [-4, 0]$ in the uv-plane. That is, $D^* = T(D)$, or $D = T^{-1}(D^*)$ and therefore

$$\iint_D (y - x)\, dx\, dy = \iint_{D^*} (v + 1)\, |J_{T^{-1}}|\; du\, dv.$$

Since the Jacobian of T is equal to 4, we have $|J_{T^{-1}}| = 1/4$ and so

$$I = \frac{1}{4}\int_{-4}^{0}\int_{0}^{8}(v+1)\,du\,dv = 2\frac{(v+1)^2}{2}\Big|_{-4}^{4} = -8.$$



Exercise 10.3 1. $J(u,v) = \begin{vmatrix} 1 & 1 \\ -2u & 1 \end{vmatrix} = 1 + 2u$. Note that for the integrals we would need to use $|1+2u|$ but, since in this example $u > 0$ there is no need to consider the negative case.

2. We have to see how each side in T is transformed by the transformation $(x(,uv), y(u,v))$. The vertical side $\{u = 0, 0 \le v \le 2\}$ gets transformed into the straight line $\{y = x, 0 \le x \le 2\}$. The horizontal line $\{v = 0, 0 \le u \le 2\}$ is transformed into a curved parabola $\{y = -x^2, 0 \le x \le 2\}$. Finally, the diagonal line $\{u + v = 2, 0 \le u \le 2\}$ is transformed into the vertical line $\{x = 2, -4 \le y \le 2\}$.
3. We can do this eaither directly in the xy-plane or using the change of variable formula. Directly, we have

$$A = \int_0^2\int_{-x^2}^{x} dy\,dx = \int_0^2 x + x^2\,dx = \frac{x^2}{2} + \frac{x^3}{3}\Big|_0^2 = \frac{14}{3}.$$

Alternatively, using the change of variable formula, we have

$$A = \int_0^2\int_0^{2-v}(1+2u)\,du\,dv = \frac{1}{4}\int_0^2[(5-2v)^2 - 1]\,dv = \frac{14}{3}.$$

4.

$$\iint_S \frac{1}{(x-y+1)^2}\,dx\,dy = \int_0^{2-v}\frac{1+2u}{(1+u+u^2)^2}\,du\,dv = \int_0^2 -\frac{1}{1+u+u^2}\Big|_0^{2-v} = 2+\int_0^2\frac{1}{v^2-5v+7}\,dv.$$

Now,

$$\int_0^2\frac{1}{v^2-5v+7}\,dv = -\frac{4}{3}\int_0^2\frac{dv}{1+\left(\frac{2v-5}{\sqrt{3}}\right)^2} = -\frac{2}{\sqrt{3}}\int_{-5/\sqrt{3}}^{-1/\sqrt{3}}\frac{dt}{1+t^2},$$

using the change of variable $t = (2v-5)/\sqrt{3}$. Finally,

$$-\frac{2}{\sqrt{3}}\arctan t\Big|_{-5/\sqrt{3}}^{-1/\sqrt{3}} = \frac{\sqrt{3}}{9}\left(\pi - 6\arctan\left(\frac{5}{\sqrt{3}}\right)\right),$$

where we have used that $\arctan(-1/\sqrt{3}) = \pi/6$ and that $\arctan(-x) = -\arctan x$ (i.e. it is an odd function). The whole integral is then

$$\iint_S \frac{1}{(x-y+1)^2}\,dx\,dy = 2 + \frac{\sqrt{3}}{9}\left(\pi - 6\arctan\left(\frac{5}{\sqrt{3}}\right)\right).$$

Exercise 10.4

$$\int_0^{\pi/2}\int_1^2 r\log(r^2)\,dr\,d\theta = 2\pi(\log 2 - 3/8).$$

Exercise 10.5 We define:

$$\begin{cases} x = ar\cos t \\ y = br\sin t \end{cases}$$

The Jacobian is $J = abr$, and the integral is:

$$\int_0^{2\pi}\int_0^1 \left(\frac{\sin^4 t\, r^2}{1+r^2} + ab^2r^3\cos t\sin^2 t\right) abr\,dr\,dt = \frac{3}{8}\pi ab(1-\log 2).$$

Exercise 10.6 Since f is odd in x and E is symmetric in that variable, we have:

$$\int_E f = 0; \quad \int_H f = \int_0^{\pi/2}\int_0^{2\sin\theta} \cos\theta\, r\,dr\,d\theta = 2.$$

Exercise 10.7

$$\int_0^{\pi/2}\int_0^{2\sin\theta} \frac{r\sqrt{1+\sin^2\theta}}{\sin\theta}\,dr\,d\theta = 1 + \frac{\pi}{2}.$$

Exercise 10.8 Putting:

$$\begin{cases} x = \frac{r}{2}\cos t \\ y = \frac{r}{2}\sin t \end{cases}$$

the Jacobian is $J = \frac{r}{2}$, and the integral is:

$$\int_0^{\pi/2}\int_1^4 \frac{1}{4}\cos t\, r\,dr\,dt = \frac{3}{4}.$$

Exercise 10.9 1.

$$\int_0^{2\pi}\int_0^3\int_r^3 r\sqrt{r^2+z^2}\,dz\,dr\,d\theta = \frac{27}{2}\pi(2\sqrt{2}-1);$$

2.

$$\int_0^{2\pi}\int_0^3\int_0^z r\sqrt{9-r^2}\,dr\,dz\,d\theta = 54\pi - \frac{81}{8}\pi^2;$$

3.

$$\int_0^{2\pi}\int_0^3\int_0^z rze^{r^2+z^2}\,dr\,dz\,d\theta = \frac{\pi}{4}(e^9-1)^2.$$

Exercise 10.10

$$\int_0^{2\pi}\int_0^{\pi/4}\int_0^3 e^{-\rho^2}\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = \frac{\pi}{3}(2-\sqrt{2})(1-e^{-27}).$$

Exercise 10.11 1.

$$\int_{-2}^{1}\int_{x}^{2-x^2} dy\,dx = \frac{9}{2};$$

2. We define

$$\begin{cases} u = \frac{x^3}{y}, \\ v = \frac{y^3}{x}, \end{cases}$$

the Jacobian is $J = \frac{1}{8\sqrt{uv}}$, and the integral is:

$$\int_{a^2}^{b^2}\int_{p^2}^{q^2}\frac{1}{8\sqrt{uv}}\,dv\,du = \frac{1}{2}(b-a)(q-p);$$

3. We define

$$\begin{cases} u = xy \\ v = xy^3 \end{cases}$$

the Jacobian is $J = \frac{1}{2v}$, and the integral is:

$$\int_4^8\int_5^{15}\frac{1}{2v}\,dv\,du = 2\log 3.$$

Exercise 10.12 1.

$$2\int_0^{2\pi}\int_0^2\int_0^{\sqrt{16-r^2}} r\,dz\,dr\,d\theta = \frac{32}{3}\pi(8-3\sqrt{3});$$

2.

$$2\int_0^{2\pi}\int_0^1\int_0^{1-r} r\,dz\,dr\,d\theta = \frac{2\pi}{3};$$

3.

$$2\int_0^{2\pi}\int_0^2\int_0^{r^2} r\,dz\,dr\,d\theta = 8\pi;$$

4.

$$\int_0^{2\pi}\int_0^1\int_0^{\sqrt{z}} r\,dr\,dz\,d\theta + \int_0^{2\pi}\int_1^{6/5}\int_0^{\sqrt{2-z^2}} r\,dr\,dz\,d\theta = \frac{493\pi}{750}.$$

Exercise 10.13 1. Observe first that the intersection of the surfaces $9-x^2 = x^2+3y^2$ is the ellipse $2x^2+3y^2 = 9$. The volume is then, by symmetry,

$$4\int_0^{3/\sqrt{2}}\int_0^{\sqrt{(9-2x^2)/3}}\int_0^{9-x^2} dz\,dy\,dx = \frac{27}{4}\pi\sqrt{6};$$

2.

$$\int_{-\sqrt{2}}^{\sqrt{2}}\int_{-\sqrt{1-x^2/2}}^{\sqrt{1-x^2/2}}\int_0^{(2-x-y)/2} dz\,dy\,dx = \pi\sqrt{2}.$$

Exercise 10.14 By symmetry, the volume is:

$$8\int_0^a \int_0^{b\sqrt{1-x^2/a^2}} \int_0^{c\sqrt{1-x^2/a^2-y^2/b^2}} dz\, dy\, dx = \frac{4}{3}\pi abc;$$

We can also use adapted spherical coordinates:

$$x = a\rho\cos\theta\sin\varphi, \quad y = b\rho\sin\theta\sin\varphi, \quad z = c\rho\cos\varphi,$$

with Jacobian:

$$J = abc\rho^2\sin\varphi,$$

and volume:

$$\int_0^{2\pi}\int_0^{\pi}\int_0^1 abc\rho^2\sin\varphi\, d\rho\, d\varphi\, d\theta = \frac{4}{3}\pi abc.$$

In the case of the ball of radius R, we obtain:

$$\frac{4}{3}\pi R^3.$$

Exercise 10.15

$$\int_{-\pi/2}^{\pi/2}\int_0^{4\cos\theta}\int_0^{\sqrt{16-r^2}} r\, dz\, dr\, d\theta = \frac{64}{9}(3\pi - 4).$$

Exercise 10.16

$$\int_0^{\pi/2}\int_0^2 kr^2\, dr\, d\theta = \frac{4k\pi}{3}.$$

Exercise 10.17 1.

$$M = \int_{-2}^{2}\int_x^{2-x^2} \rho\, dy\, dx = \frac{9\rho}{2}, \quad x_{CM} = \frac{2}{9\rho}\int_{-2}^{2}\int_x^{2-x^2} x\, dy\, dx = \frac{1}{2},$$

$$y_{CM} = \frac{2}{9}\int_{-2}^{2}\int_x^{2-x^2} y\, dy\, dx = \frac{8}{5};$$

2.

$$M = 2\int_0^2\int_x^{\sqrt{5-x^2}} \rho\, dy\, dx = \frac{64\rho}{3}, \quad x_{CM} = 0 \text{ (by symmetry)},$$

$$y_{CM} = \frac{3}{32}\int_0^2\int_x^{\sqrt{5-x^2}} y\, dy\, dx = \frac{8}{5};$$

3.

$$M = \int_0^{\pi}\int_0^{\sin^2 x} \rho\, dy\, dx = \frac{\pi\rho}{2}, \quad x_{CM} = \frac{2}{\pi}\int_0^{\pi} x\, dx = \frac{\pi}{2} \text{ (by symmetry)},$$

$$y_{CM} = \frac{2}{\pi}\int_0^{\pi}\int_0^{\sin^2 x} y\,dy\,dx = \frac{3}{8};$$

4.

$$M = \int_0^{\pi/4}\int_{\sin x}^{\cos x} \rho\,dy\,dx = (\sqrt{2}-1)\rho;$$

$$x_{CM} = \frac{1}{\sqrt{2}-1}\int_0^{\pi/4}\int_{\sin x}^{\cos x} x\,dy\,dx = \frac{\pi}{4}(2+\sqrt{2}) - \sqrt{2} - 1;$$

$$y_{CM} = \frac{1}{\sqrt{2}-1}\int_0^{\pi/4}\int_{\sin x}^{\cos x} y\,dy\,dx = \frac{\sqrt{2}+1}{4}.$$

Exercise 10.18 In spherical coordinates, since $x^2 + y^2 = \rho^2\sin^2\varphi$, the moment of inertia is:

$$\int_0^{2\pi}\int_0^{\pi/4}\int_0^2 \alpha\rho^2\sin^2\varphi\sin\varphi\,d\rho\,d\varphi\,d\theta = \frac{16}{15}\pi\alpha(8-5\sqrt{2}).$$

Exercise 10.19 By symmetry, the mass is:

$$2\int_{-1}^1\int_x^1 (y-x)\,dy\,dx = \frac{8}{3}.$$

Exercise 10.20

$$M = \int_1^2\int_1^3 xy\,dy\,dx = 6; \quad x_{CM} = \frac{1}{6}\int_1^2\int_1^3 x^2y\,dy\,dx = \frac{14}{9}, \quad y_{CM} = \frac{1}{6}\int_1^2\int_1^3 xy^2\,dy\,dx = \frac{13}{6}.$$

Exercise 10.21

$$M = \int_0^1\int_{-x}^x y^2\,dy\,dx = \frac{1}{6}; \quad x_{CM} = \frac{6}{1}\int_0^1\int_{-x}^x xy^2\,dy\,dx = \frac{4}{5}, \quad y_{CM} = 6\int_0^1\int_{-x}^x y^3\,dy\,dx = 0 \quad \text{(by symmetry)};$$

$$I_x = \int_0^1\int_{-x}^x y^4\,dy\,dx = \frac{1}{15}, \quad I_y = \int_0^1\int_{-x}^x x^2y^2\,dy\,dx = \frac{1}{9}.$$

Exercise 10.22 Clearing the variable y, the set can be described as

$$Q = \left\{0 \le x \le 1, \quad \frac{1}{2}(2x+1-\sqrt{8x+1}) \le y \le \frac{1}{2}(\sqrt{8x+1}-(2x+1))\right\};$$

so, the mass is:

$$M = \int_0^1\int_{\frac{1}{2}(2x+1-\sqrt{8x+1})}^{\frac{1}{2}(\sqrt{8x+1}-(2x+1))} (x^2-y^2)\,dy\,dx.$$

However, this integral is not easy. On the other hand, the definition suggests the change of variables:

$$\begin{cases} u = x+y \\ v = x-y \end{cases},$$

by which the set is described as:

$$Q^\star = \left\{0 \le u \le 1, \quad u^2 \le v \le \sqrt{u}\right\}$$

and the density is $\rho = uv$. The Jacobian is $J = \frac{1}{2}$, and the mass is:

$$M = \frac{1}{2}\int_0^1 \int_{u^2}^{\sqrt{u}} uv\, dv\, du = \frac{1}{24}.$$

Exercise 10.23 1.

$$\int_0^1 \int_0^{(1-x^{2/3})^{3/2}} dy\, dx = \int_0^1 (1-x^{2/3})^{3/2}\, dx = \int_0^{\pi/2} 3\sin^2 u \cos^4 u\, du = \frac{3\pi}{32},$$

using the change $x^{1/3} = \sin u$.
We can also use the two-variable transformation of the original integral

$$\begin{cases} x = r\cos^3 t \\ y = r\sin^3 t \end{cases}$$

suggested by the description of the set. The Jacobian is

$$J = 3r\sin^2 t\cos^2 t,$$

and the area is

$$\int_0^1 \int_0^{\pi/2} 3r\sin^2 t\cos^2 t\, dt\, dr = \frac{3\pi}{32}.$$

2. With the previous transformation and the symmetry:

$$x_{CM} = y_{CM} = \frac{32}{3\pi}\int_0^1 \int_0^{\pi/2} 3r^2\sin^2 t\cos^5 t\, dt\, dr = \frac{256}{315\pi}.$$

Exercise 10.24 The distance from a point $P = (x, y)$ to the straight line $r = y = x$ is

$$d(P, r) = \frac{|x - y|}{\sqrt{2}},$$

so the moment of inertia is

$$I_r = \int_0^1 \int_0^1 \frac{(x-y)^2}{2}\rho\, dx\, dy = \frac{\rho}{12}.$$

Exercise 10.25

$$M_1 = \int_0^a \int_0^{b(1-x/a)} \int_0^{\sqrt{c^2-x^2-y^2}} z\, dz\, dy\, dx = \frac{1}{24}ab(6c^2 - a^2 - b^2);$$

The mass of the complete first octant is (in spherical coordinates):

$$M = \int_0^{\pi/2} \int_0^{\pi/2} \int_0^c \rho\sin\varphi\, \rho^2\cos\varphi\, d\rho\, d\varphi\, d\theta = \frac{\pi c^4}{16};$$

Finally:

$$M_2 = M - M_1.$$

Exercise 10.26 1. $T(x, y, z) = \alpha(x^2 + y^2 + z^2)$, then:

$$T_m = \frac{1}{8}\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} \alpha(x^2 + y^2 + z^2)\, dx\, dy\, dz = \alpha;$$

2. $T(x, y, z) = \alpha$ at the unit sphere $x^2 + y^2 + z^2 = 1$.

Exercise 10.27

$$M = \int_0^{2\pi}\int_0^{\pi/2}\int_0^{R} \rho^4 \sin\varphi\, d\rho\, d\varphi\, d\theta = \frac{2\pi R^5}{5}; \quad \text{by symmetry} \quad x_{CM} = y_{CM} = 0;$$

$$z_{CM} = \frac{5}{2\pi R^5}\int_0^{2\pi}\int_0^{\pi/2}\int_0^{R} \rho^5 \sin\varphi \cos\varphi\, d\rho\, d\varphi\, d\theta = \frac{5R}{12}.$$

Exercise 10.28 Let us consider the plane $z = 0$ as the separating plane between the ice cream and the wafer cone. In cylindrical coordinates, the ice cream is described as the set

$$H = \left\{0 \le r \le R,\ 0 \le z \le \sqrt{R^2 - r^2}\right\},$$

while the wafer cone is

$$C = \{0 \le r \le R,\ \lambda(r - R) \le z \le 0\},$$

where $\lambda = \frac{1}{\tan\alpha}$ indicates the cone opening. Then the z-coordinate of the mass center is

$$z_{CM} = 0 = \frac{1}{M}\left(\int_0^{2\pi}\int_0^{R}\int_0^{\sqrt{R^2-r^2}} \rho_h r z\, dz\, dr\, d\theta + \int_0^{2\pi}\int_0^{R}\int_{\lambda(r-R)}^{0} \rho_c r z\, dz\, dr\, d\theta\right).$$

This implies:

$$\frac{\rho_c}{\rho_h} = \frac{\int_0^R \int_0^{\sqrt{R^2-r^2}} rz\, dz\, dr}{\int_0^R \int_{\lambda(r-R)}^{0} rz\, dz\, dr} = \frac{\pi R^4/4}{\pi R^4 \lambda^2/12} = 3\tan^2\alpha.$$

Exercise 10.29 1. The region is the set between $z = \pm r$ and $z = 2 \pm 3r$, that is:

$$\left\{0 \le r \le \frac{1}{2},\ r \le z \le 2 - 3r\right\};$$

2. The volume is:

$$\text{Volume} = \int_0^{2\pi}\int_0^{1/2}\int_r^{2-3r} r\, dz\, dr\, d\theta = \frac{\pi}{6};$$

(alternatively, as volume of two cones with radii $\frac{1}{2}$ and heights $\frac{1}{2}$ and $\frac{3}{2}$, respectively):

$$V = \frac{1}{3}\pi \left(\frac{1}{2}\right)^2 \left(\frac{1}{2} + \frac{3}{2}\right) = \frac{\pi}{6};$$

By symmetry:

$$x_{CM} = y_{CM} = 0, \quad z_{CM} = \frac{6}{\pi} \int_0^{2\pi} \int_0^{1/2} \int_r^{2-3r} rz\, dz\, dr\, d\theta = \frac{3}{4}.$$

A.11 Path and Line integrals

Exercise 11.1 1.

$$\int_0^{\pi/2} 2R^4 \cos t \sin^2 t\, dt = \frac{2}{3}R^4;$$

2.

$$\sqrt{10}\int_0^{2\pi} (1+9t^2)^2\, dt = \frac{2\sqrt{10}\pi}{5}(5+120\pi^2+1296\pi^4).$$

Exercise 11.2

$$L = \int_0^2 \sqrt{1+4x^2}\, dx = \sqrt{17}+\frac{1}{4}\log(4+\sqrt{17}); \quad M = \int_0^2 x\sqrt{1+4x^2}\, dx = \frac{17^{3/2}-1}{12}.$$

Exercise 11.3 1. $\gamma(x) = (x, x^2),\ -1 \le x \le 1,$

$$I = \int_{-1}^1 (x^2-2x^3, x^4-2x^3)\cdot(1,2x)\, dx = -\frac{14}{15};$$

2.

$$\gamma(x) = \begin{cases} (x, & 0 \le x \le 1) \\ 2-x, & 1 \le x \le 2 \end{cases}, \quad I = \int_0^1 (2x^2,0)\cdot(1,1)\, dx + \int_1^2 (x^2+(2-x)^2, x^2-(2-x)^2)\cdot(1,-1)\, dx = \frac{4}{3};$$

3.

$$I = \int_0^1 (t^4-t^6, 2t^5, -t^2)\cdot(1,2t,3t^2)\, dt = \frac{1}{35};$$

4. $\gamma(t) = (1+2t, 4t, 2-t),\ 0 \le t \le 1,$

$$I = \int_0^1 (8t(1+2t), (1+2t)^2+2-t, 4t)\cdot(2,4,-1)\, dt = 40.$$

Exercise 11.4 1. $\gamma_1(t) = (-t, 0),\ -1 \le t \le 1,$

$$I = \int_{-1}^1 (t^2,0)\cdot(-1,0)\, dt = -\frac{2}{3};$$

2.

$$\gamma_2(t) = \begin{cases} (1,t), & 0 \le t \le 1 \\ (2-t,1), & 1 \le t \le 3 \\ (1,4-t), & 3 \le t \le 4 \end{cases}, \quad I = \int_0^1 (1,t)\cdot(0,1)\, dt + \int_1^3 ((2-t)^2,1)\cdot(-1,0)\, dt + \int_3^4 (1,4-t)\cdot(0,1)\, dt = -\frac{2}{3};$$

3. $\gamma_4(t) = (\cos t, \sin t),\ 0 \le t \le \pi,$

$$I = \int_0^\pi (\cos^2 t, \sin t)\cdot(-\sin t, \cos t)\, dt = -\frac{2}{3};$$

Exercise 11.5 1.

$$\int_0^1 (1-3t, 1+t)\cdot(-1,2)\, dt = \frac{7}{2};$$

2.

$$\int_0^{2\pi} (-\sin^3 t, \cos^3 t) \cdot (-\sin t, \cos t)\, dt = \frac{3\pi}{2};$$

3.

$$\gamma(t) = \begin{cases} (1-t, t), & 0 \le t \le 1 \\ (1-t, 2-t), & 1 \le t \le 2 \\ (t-3, 2-t), & 2 \le t \le 3 \\ (t-3, t-4), & 3 \le t \le 4 \end{cases}, \quad I = \int_0^1 (1,1)\cdot(-1,-1)\, dt + \int_1^2 (1,1)\cdot(1,-1)\, dt + \int_2^3 (1,1)\cdot(-1,-1)\, dt + \int_3^4 (1,1)\cdot(1,1$$

4. $\gamma(t) = (2\cos t, \sin t),\ 0 \le t \le 2\pi,$

$$I = \int_0^{2\pi} (6\cos^2 t - 4\sin^2 t - 5\sin t \cos t)\, dt = 2\pi;$$

5.

$$\int_0^1 \frac{-2t^2}{\sqrt{1-t^2}}\, dt = -\frac{\pi}{2}.$$

Exercise 11.6 1.

$$\gamma(t) = (a\cos t, a\sin t, a{+}a\sin t), \quad 0 \le t \le 2\pi, \quad I = a^2 \int_0^{2\pi} (\cos t{+}\sin t)\cos t{-}1\, dt = -2\pi a^2;$$

2.

$$\gamma(t) = \left(\frac{a}{\sqrt{2}}\cos t, \frac{a}{\sqrt{2}}\cos t, a\sin t\right), \quad I = \frac{a}{2\sqrt{2}} \int_0^{2\pi} (-2\sin^2 t\cos t{-}2\sin^3 t{+}(2\sqrt{2}{-}1)\sin t\cos^2 t)\, dt = 0;$$

3.

$$\gamma(t) = (1{+}\cos t, \sin t, 1{+}\cos t), \quad I = \int_0^{2\pi} (-\sin^2 t{+}\cos^2 t{-}2\sin t{+}2\cos t)\, dt = 0.$$

Exercise 11.7

$$\mathbf{F}(t) = m(2, -\sin t, -\cos t); \quad T = m\int_0^1 4t\, dt = 2m.$$

Exercise 11.8

$$T(b) = -\int_0^{\pi} (3b^2\sin^2 t{+}2, 16\cos t)\cdot(-\sin t, b\cos t)\, dt = 4b^2{-}8\pi b{+}4; \quad \text{Minimum work is } 4(1{-}\pi^2), \text{ at } b = \pi.$$

Exercise 11.9

$$T(a,b,c) = \int_0^1 (act^{b+1}, a^2t^{2b+6})\cdot(1, abt^{b-1})\, dt = \frac{a^3b + 3ac}{3b+6} = \frac{a^3}{3}\cdot\frac{b + 3c/a^2}{b+2};$$

It is independent of b if $a = \sqrt{3c/2}$.

Exercise 11.10

$$\gamma(\theta) = (e^{-\theta}\cos\theta, e^{-\theta}\sin\theta), \quad 0 \le \theta < \infty; \quad T = 8\int_0^{\infty} e^{-\theta}\sin\theta\cos\theta\, d\theta = \frac{8}{5}.$$

Exercise 11.11 1. curl $\mathbf{F} = 0$ and $\mathbf{F} \in C^1$ over $\mathbb{R}^3$, a simply connected set;

2.

$$\frac{\partial\phi}{\partial x} = \sin y + z \Rightarrow \phi = x\sin y + xz + g(y,z), \quad \frac{\partial\phi}{\partial y} = x\cos y + e^z \Rightarrow g = ye^z + h(z),$$

$$\frac{\partial\phi}{\partial z} = x + ye^z \Rightarrow h = \text{const}; \quad \text{so } \phi(x,y,z) = x(\sin y + z) + ye^z + \text{cte}.$$

Exercise 11.12 The potential of $\mathbf{F}$ is $\phi = xe^{x^2+y^2}$, and

$$\int_\gamma \mathbf{F} = \phi(\mathbf{r}(1)) - \phi(\mathbf{r}(0)) = e^2.$$

Exercise 11.13 1. The vector field is conservative and the curve is closed, so the integral is 0;

2.

$$f(x,y,z) = xy + xz + yz - \frac{1}{5}\cos x^5 + y\arctan y - \frac{1}{2}\log(1+y^2) + \frac{z}{2} - \frac{1}{4}\sin 2z.$$

Exercise 11.14 1.

$$\gamma(t) = (\cos t, \sin t, \sin^2 t - \cos^2 t), \quad 0 \le t \le 2\pi; \quad \int_\Gamma \mathbf{F} = \int_0^{2\pi} (-\sin^4 t + \cos t\, e^{\sin t} + 4\sin^3 t\cos t - 4\sin t\cos^3 t)\, dt = -\frac{3\pi}{4}$$

2. There is no potential since the vector field is not conservative; curve is closed and previous integral is not zero (alternatively you can compute the curl).

A.12 Surface Integrals

Exercise 12.1 1. $T(\theta, \varphi) = (R\cos\theta\sin\varphi, R\sin\theta\sin\varphi, R\cos\varphi)$, $0 \leq \theta \leq 2\pi$, $0 \leq \varphi \leq \pi$,

$$\mathbf{n} = (R^2\cos\theta\sin^2\varphi, R^2\sin\theta\sin^2\varphi, R^2\sin\varphi\cos\varphi), \quad \|\mathbf{n}\| = R^2\sin\varphi,$$

$$A = \int_0^{2\pi}\int_0^{\pi} R^2\sin\varphi\, d\varphi\, d\theta = 4\pi R^2;$$

2. $T(u, v) = (u\cos v, u\sin v, u)$, $0 \leq u \leq a$, $0 \leq v \leq 2\pi$,

$$\mathbf{n} = (-u\cos v, -u\sin v, u), \quad \|\mathbf{n}\| = u\sqrt{2},$$

$$A = \sqrt{2}\int_0^{2\pi}\int_0^{a} u\, du\, dv = \pi a^2\sqrt{2};$$

3. $T(r, \theta) = (r\cos\theta, r\sin\theta, r^2)$, $0 \leq r \leq a$, $0 \leq \theta \leq 2\pi$,

$$\mathbf{n} = (-2r^2\cos\theta, -2r^2\sin\theta, r), \quad \|\mathbf{n}\| = r\sqrt{1+4r^2},$$

$$A = \int_0^{2\pi}\int_0^{a} r\sqrt{1+4r^2}\, dr\, d\theta = \frac{\pi}{6}\left((1+4a^2)^{3/2} - 1\right);$$

4. $T(r, \theta) = (r\cos\theta, r\sin\theta, \sqrt{16 - r^2\cos^2\theta})$, $0 \leq r \leq 4$, $0 \leq \theta \leq \pi/2$ (by symmetry we only consider one octant; in the figure at the right we consider the upper part, not the side part, multiplied by 8),

$$\mathbf{n} = \left(\frac{r^2\cos\theta}{\sqrt{16 - r^2\cos^2\theta}}, 0, r\right), \quad \|\mathbf{n}\| = \frac{4r}{\sqrt{16 - r^2\cos^2\theta}},$$

$$A = 32\int_0^{\pi/2}\int_0^{4} \frac{r}{\sqrt{16 - r^2\cos^2\theta}}\, dr\, d\theta = 128.$$

Exercise 12.2 We compute the area of the part of the sphere inside one of the cylinders (the upper part), then we multiply by 4 and subtract from the total area of the sphere:

$$T(r, \theta) = (r\cos\theta, r\sin\theta, \sqrt{a^2 - r^2}), \quad 0 \leq r \leq a\cos\theta, \quad -\pi/2 \leq \theta \leq \pi/2,$$

$$\mathbf{n} = \left(\frac{r^2\cos\theta}{\sqrt{a^2 - r^2}}, \frac{r^2\sin\theta}{\sqrt{a^2 - r^2}}, r\right), \quad \|\mathbf{n}\| = \frac{ar}{\sqrt{a^2 - r^2}},$$

$$A_1 = 4\int_{-\pi/2}^{\pi/2}\int_0^{a\cos\theta} \frac{ar}{\sqrt{a^2 - r^2}}\, dr\, d\theta = 4a^2(\pi - 2),$$

finally

$$A = 4\pi a^2 - A_1 = 8a^2.$$

Exercise 12.3 1. We just calculate $\|\mathbf{n}\| = r\sqrt{1 + (f'(r))^2}$ and integrate on θ;

2. $f(x) = \sqrt{c^2 - (x - R)^2}$ (the upper half), $A = 4\pi\int_{R-c}^{R+c} x\sqrt{1 + \frac{(x-R)^2}{c^2 - (x-R)^2}}\, dx = 4\pi c\int_{R-c}^{R+c} \frac{x}{\sqrt{c^2 - (x-R)^2}}\, dx = 4\pi c\int_{-\pi/2}^{\pi/2}(R + c\sin w)\, dw = 4\pi^2 cR$;

3. $s(x, \theta) = (x, f(x)\cos\theta, f(x)\sin\theta)$, $\|\mathbf{n}\| = |f(x)|\sqrt{1 + (f'(x))^2}$, $A = 2\pi\int_a^b |f(x)|\sqrt{1 + (f'(x))^2}\, dx$.

Exercise 12.4 $V = 2\pi \int_0^1 \left(\frac{1}{r} - 1\right) r\, dr = \pi;\quad A = 2\pi \int_0^1 r\sqrt{1 + \frac{1}{r^4}}\, dr \geq 2\pi \int_0^1 \frac{1}{r}\, dr = \infty.$

Exercise 12.5 With the standard parametrization of the sphere (problem 4.2.1.i), we have $\|\mathbf{n}\| = a^2 \sin\varphi$, $d^2 = a^2 \sin^2\varphi$, so that

$$I_z = \int_0^{2\pi} \int_0^{\pi} \sigma a^4 \sin^3\varphi\, d\varphi\, d\theta = \frac{8}{3}\pi\sigma a^4;$$

since $\sigma = \frac{m}{4\pi a^2}$, we finally get

$$I_z = \frac{2}{3} m a^2.$$

A.13 Integral Theorems of Vector calculus

Exercise 13.1 $\gamma(t) = \begin{cases} (t,0), & 0 \le t \le 1 \\ (1,t-1), & 1 \le t \le 2 \\ (3-t,1), & 2 \le t \le 3 \\ (0,4-t), & 3 \le t \le 4 \end{cases}$, $I = \int_0^1 (5,t^2)\cdot(1,0)\,dt +$ $\int_1^2 (5-(t-1)-(t-1)^2, 1-2(t-1))\cdot(0,1)\,dt + \int_2^3 (5-(3-t)-1,(3-t)^2-2(3-t))\cdot(-1,1)\,dt + \int_3^4 (4,0)\cdot(0,-1)\,dt = \frac{3}{2}$;

by Green's Theorem, $I = \int_0^1\int_0^1 3x\,dxdy = \frac{3}{2}$.

Exercise 13.2 $\int_\gamma P dx + Q dy = \int_0^1\int_0^1 \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,dxdy = \int_0^1\int_0^1 \frac{1}{3+e^{xy}}\,dxdy =$ $\frac{1}{3}\left(1-\log(e+3)+\log 4\right)$.

Directly, without Green's Theorem, the integral would be:

$$\int_0^1 (e - \frac{t}{3+e^t}\cdot f(t))\cdot(0,1)\,dt = \int_0^1 (e^t - \frac{1}{3+e^t}\cdot f(1))\cdot(1,0)\,dt - \int_0^1 (1-\frac{t}{4}\cdot f(t))\cdot(0,1)\,dt = \int_0^1 e^t\,dt + \int_0^1 f(t)\,dt - \int_0^1 (e^{t^2} - \frac{1}{3+e^t}$$

Exercise 13.3 1. $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} = \frac{x^2-y^2}{(x^2+y^2)^2}$;

2. We cannot apply Green's Theorem directly and must use the previous item to deduce the integral is 0, since P and Q are not C^1 on D. Let $D_\varepsilon = D - B_\varepsilon$, where $B_\varepsilon = \{x^2+y^2 \le \varepsilon^2\}$, then P and Q are C^1 on D_ε and:

$$\int_{D_\varepsilon}\left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)dxdy = 0 \Rightarrow \int_C P dx + Q dy = \int_{\partial B_\varepsilon} P dx + Q dy,$$

if both boundaries are positively oriented. But the last integral is:

$$\int_0^{2\pi}\left(\frac{\sin\theta}{\varepsilon}, -\frac{\cos\theta}{\varepsilon}\right)\cdot(-\varepsilon\sin\theta, \varepsilon\cos\theta)\,d\theta = -2\pi;$$

3. Now P and Q are C^1 on D, so by Green's Theorem the integral is 0.

Exercise 13.4 With $x-1=z$ the region is symmetric with respect to the line $x=1$, and the integral is $\pm 2\pi$, depending on the orientation of γ.

Exercise 13.5 1. By Green's Theorem, $\int_C -y\,dx + x\,dy = 2\int_D dxdy = 2A$;
in polar coordinates $x = r(\theta)\cos\theta$, $y = r(\theta)\sin\theta$, the area is:

$$A = \frac{1}{2}\int_C [-r(\theta)\sin\theta\cdot r'(\theta)\cos\theta - r(\theta)\sin\theta\cdot r(\theta)\cos\theta]\,d\theta = \frac{1}{2}\int_0^2 r^2(\theta)\,d\theta;$$

2.

$$A = \frac{1}{2}\int_0^1 [(t-t^3)2t + (t^2-1)(3t^2-1)]\,dt = \frac{15}{8}; \quad \frac{1}{2}\int_0^{2\pi} a^2(1-\cos^2\theta)^2\,d\theta = \frac{3\pi a^2}{4}.$$

Exercise 13.6 1. We define $\mathbf{F} = (P,Q)$, with $P = 0$, $Q = \frac{1}{2}x^2 + 2xy$, and obtain

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = x + 2y;$$

then, since the boundary consists of two curves (be careful with the orientation), by Green's Theorem

$$\int_D (x+2y)dxdy = \int_{\partial D} Q\,dy = -\int_0^{2\pi}\left[\frac{1}{2}(t-\sin t)^2 + 2(t-\sin t)(1-\cos t)\right]\sin t\,dt + \int_0^{2\pi} 0\,dt = \pi(3\pi+5);$$

(other possibility, $P = -y^2$, $Q = \frac{1}{2}x^2$, etc.);

2. we take now $Q = \frac{1}{2}x^2y^2$, and the integral becomes

$$\int_0^{\pi/2} \frac{1}{2}\cos^6 t\sin^6 t 3\sin^2 t\cos t\,dt - \int_0^1 0\,dt + \int_0^1 0\,dt = \frac{8}{2145};$$

3. with $Q = xy^2$ the integral is

$$-\int_0^{\pi} 2a^4(t-\sin^2 t)\sin^5 t\cos t\,dt + \int_0^{\pi} 0\,dt = 0.$$

Exercise 13.7 1. The boundary of S can be described as $\gamma(t) = (a\cos t, a\sin t, 0)$, $0 \le t \le 2\pi$,

$$I = -\int_0^{2\pi} a^5\sin^3 t\cos^2 t\,dt = 0;$$

2. the boundary is the same as in the previous item,

$$I = -\int_0^{2\pi} a^2\sin^2 t\,dt = -\pi a^2;$$

3. here the boundary is $\gamma(t) = (\cos t, 0, \sin t)$, $0 \le t \le 2\pi$, but traveled in the opposite sense,

$$I = \int_0^{2\pi}(\cos^3 t\sin t + \sin^4 t - \cos^4 t)\,dt = 0.$$

Exercise 13.8 The boundary of S is parametrized as $\gamma(t) = (3\cos t, 2\sin t, 0)$, $0 \le t \le 2\pi$,

$$I = -\int_0^{2\pi}(-6\sin^2 t + 18\cos^3 t)\,dt = 6\pi.$$

Exercise 13.9 1. It is contained in the plane $x = z$, $\mathbf{n} = (1, 0, -1)$; we move along the boundary from A to B, from B to C and from C to A;

2. $T = \{(x, y, x),\ 0 \le x \le 1,\ x \le y \le 2-x\}$, curl $\mathbf{F} = (-3, -1, -2)$,

$$I = -\int_0^1\int_x^{2-x} dydx = -1.$$

Exercise 13.10 1. $\int_S \operatorname{curl}\mathbf{F} = \int_0^{2\pi}\int_0^a (0, r^2-a^2, -r\sin\theta)\cdot(2r^2\cos\theta, 2r^2\sin\theta, r)drd\theta =$ 0; the boundary is a circumference of radius a,

$$\int_{\partial S}\mathbf{F} = \int_0^{2\pi}(a^2\sin^2\theta, a^2\sin\theta\cos\theta, 0)\cdot(-a\sin\theta, a\cos\theta, 0)\,d\theta = 0;$$

2. $\displaystyle\int_S \operatorname{curl} \mathbf{F} = \int_0^{\pi} \int_0^1 (0,0,3r^2) \cdot (0,r,r)\, dr d\theta = \frac{3\pi}{4}$; the boundary is a half circumference plus a line segment,

$$\int_{\partial S} \mathbf{F} = \int_0^{\pi} (-\sin^3\theta, \cos^3\theta, \sin^3\theta)\cdot(-\sin\theta, \cos\theta, 0)\, d\theta + \int_{-1}^{1} (0, t^3, 0)\cdot(1,0,0)\, dt = \frac{3\pi}{4} + 0.$$

Exercise 13.11 The square is $S = \{(0,y,z),\ 0 \le y,\ z \le 1\}$,

$$\int_S \operatorname{curl} \mathbf{F} \cdot \mathbf{n} = \int_0^1 \int_0^1 (-2y^2, -3z^2, 0) \cdot (1,0,0)\, dy\, dz = -\frac{2}{3};$$

the boundary of the square consists of 4 segments,

$$\int_{\partial S} \mathbf{F} = \int_0^1 (0,0,0)\cdot(0,1,0)\, dt + \int_0^1 (0,2t,0)\cdot(0,0,1)\, dt + \int_0^1 (0,2(1-t)^2,0)\cdot(0,-1,0)\, dt + \int_0^1 (0,0,0)\cdot(0,0,-1)\, dt = 0 + 0 - \frac{2}{3} + 0.$$

A.14 Integral Theorems of Vector calculus

Exercise 14.1 $\operatorname{div}\mathbf{F} = 3 - 2y$, using Gauss' Theorem and spherical coordinates, the integral is

$$\int_0^{2\pi}\int_0^{\pi/2}\int_0^1 (3 - 2\sin\theta\sin\varphi)\rho^2\sin\varphi d\rho d\varphi d\theta = 2\pi.$$

Exercise 14.2 1.

$$\int_0^6\int_0^{\frac{12-2x}{3}} (36 - 6x - 9y,\ -12,\ 3y)\cdot\left(\frac{1}{3},\frac{1}{2},1\right) dy\,dx = 24;$$

2. $\operatorname{div}\mathbf{F} = 5x^2$, using Gauss' Theorem and cylindrical coordinates, the integral is

$$\int_0^{2\pi}\int_0^a\int_0^b 5r^3\cos^2\theta\,dz\,dr\,d\theta = \frac{5}{4}\pi a^4 b;$$

3. $\operatorname{div}\mathbf{F} = 4z - y$, using Gauss' Theorem the integral is

$$\int_0^1\int_0^1\int_0^1 (4z - y)\,dx\,dy\,dz = \frac{3}{2};$$

4. $\operatorname{div}\mathbf{F} = 3$, using Gauss' Theorem the integral is

$$\int_\Omega 3 = 3|\Omega|, \quad \text{where } S = \partial\Omega.$$

Exercise 14.3 1. If W is the half unitary hemisphere with positive y coordinate, we have $\partial W = S \cup S_2$, with $S_2 = \{x^2 + z^2 \le 1,\ y = 0\}$; since $\div\mathbf{F} = 4$, by Gauss' Theorem the integral is

$$\iint_S \mathbf{F}\cdot\mathbf{n} = \iiint_W 4 = \iint_{S_2}\mathbf{F}\cdot\mathbf{n} = 4|W| - 0 = \frac{8\pi}{3},$$

because $\iint_{S_2}\mathbf{F}\cdot\mathbf{n} = \iint_0^{2\pi}\int_0^1 (r\cos\theta + r\sin\theta,\ r\sin\theta,\ 2r\sin\theta)\cdot(0, r, 0)\,dr d\theta = 0$;

2. Using Stokes' Theorem, since the boundary of S is the unitary circumference on the plane $y = 0$, the integral is

$$-\int_0^{2\pi}(\cos\theta + \sin\theta,\ \sin\theta,\ 2\sin\theta)\cdot(-\sin\theta, 0, \cos\theta)\,d\theta = \pi.$$

Exercise 14.4 Since $\operatorname{div}\mathbf{F} = x + z$, by Gauss' Theorem the flux is

$$\iiint_0^1\int_0^{1-x}\int_0^{1-x-y}(x + z)\,dz\,dy\,dx = \frac{1}{12}.$$

Exercise 14.5 1. $|V| = \int_0^{2\pi}\int_{r^2/2}^2 r\,dz\,dr\,d\theta = 4\pi$;

2. $T_m = \frac{1}{4\pi}\int_0^{2\pi}\int_{r^2/2}^2 \alpha r^3\,dz\,dr\,d\theta = \frac{4\alpha}{3}$;

3. By Gauss' Theorem the integral of the flux of the gradient is

$$\iint_{\partial V} \nabla T \cdot \mathbf{n} = \iiint_V \Delta T = \iiint_V 4\alpha = 4\alpha|V| = 16\pi\alpha.$$

Exercise 14.6 1. By Gauss' Theorem,

$$\iiint_0^1 \int_0^1 \int_0^{1-x} (2x + 2y + 2z)\, dz\, dy\, dx = 3;$$

2. $\int_0^3 \int_0^{3-x} x(xy - x^2, x + 6 - 2x - 2y) \cdot (2, 2, 1)\, dy\, dx = \frac{27}{4}$;
3. S is not closed, in order to apply Gauss' Theorem we consider the lid $\{x^2 + y^2 \le a^2,\ z = 0\}$, since $\operatorname{div} \mathbf{F} = x^2 + y^2 + z^2$, the integral is

$$\int_0^{2\pi} \int_0^{\pi/2} \int_0^a \rho^4 \sin\varphi\, d\rho\, d\varphi\, d\theta - \int_0^{2\pi} \int_0^a (0, r^3 \cos^2\theta \sin\theta,\ 2r^2 \cos\theta \sin\theta) \cdot (0, 0, -r)\, dr\, d\theta = \frac{2}{5}\pi a^5 - 0;$$

4. $\operatorname{div} \mathbf{F} = 4x$, the integral is zero by symmetry (also, in spherical coordinates, the integral in θ vanishes),

$$\int_0^{2\pi} \int_0^{\pi/4} \int_0^1 4\rho^3 \cos\theta \sin^2\varphi\, d\rho\, d\varphi\, d\theta = 0.$$

Exercise 14.7 $\operatorname{div} \mathbf{F} = 2xe^{x^2+y^2+z^2}$, the integral is zero by symmetry (alternatively, we may see in spherical coordinates that the integral in φ vanishes),

$$\int_0^{2\pi} \int_0^{\pi} \int_0^a 2\rho^3 \cos\varphi \sin\varphi e^{\rho^2}\, d\rho\, d\varphi\, d\theta = 0.$$

Exercise 14.8 1. The boundary is the unitary circumference on the plane $z = 0$, the integral is then

$$\int_0^{2\pi} (\cos\theta, \sin\theta, 0) \cdot (-\sin\theta, \cos\theta, 0)\, d\theta = 0;$$

2. In order to apply Gauss' Theorem we add up the lid, the unitary circle on the plane $z = 0$; then the integral is

$$\int_0^{2\pi} \int_0^1 (0, 0, -r)\, dr\, d\theta = 0.$$

Exercise 14.9 Since $\operatorname{div} \mathbf{F} = \frac{e^{x^2+y^2}}{\sqrt{x^2+y^2}}$, by Gauss' Theorem and using cylindrical coordinates, the integral is

$$\int_0^{\pi/2} \int_0^1 \int_0^r \frac{e^{r^2}}{r} r\, dz\, dr\, d\theta = \frac{1}{4}(1 - e).$$

Exercise 14.10 The set Ω is formed by two conic parts of the ball of radius a (see figure in Problem 3.3.8, that represents a half of the set); since $\operatorname{div} \mathbf{F} = x^2 + y^2$, by Gauss' Theorem and spherical coordinates, using symmetry, the integral is

$$2\int_0^{2\pi}\int_0^{\pi/4}\int_0^{a}\rho^4\sin^3\varphi\cdot\rho\,d\rho\,d\varphi\,d\theta=\frac{1}{15}(8-5\sqrt{2})\pi a^5.$$